BUILD GUIDE

Detailed assembly - revision 0.8

Based on the Hybrid model opened in Fusion.
Landscape steps in the Hypernova guide format.

MECHANICAL REVIEW EDITION
Assembly sequence and CAD illustrations for review. Hardware specifications, belt termination, toolhead fitment, wiring and firmware require the confirmations listed at the end.

Source model: HyperDrive Galaxy One hybrid
Updated 11 November 2026

Use the Hybrid parts

This guide follows the Hybrid assembly selected in Fusion. Do not substitute the earlier Galaxy One current v97 assembly, CNC parts, or similarly named plates without checking the revision and hole pattern.

Read each stage before fastening

Sort mirrored parts and identify the front, rear and electronics bay. Dry-fit assemblies before final tightening. Use the specified fastener head style, thread and length; component names alone do not establish every hardware specification.

What this edition establishes

Frame sequence is adapted from the existing Hybrid frame documentation. The remaining images are new exports from the selected Fusion assembly. Instructions after the frame section are a proposed mechanical sequence and need a physical build review.

Release limits

This is not a complete electrical or firmware manual. No wiring diagram, pin assignments, heater settings, motor current, torque values or complete purchase BOM were verified. Resolve the open items on the final pages before releasing this as a finished customer build guide.

Orange means the part being added

Each detail page isolates the new hardware in orange. The assembled view also uses orange for those same parts; earlier parts are neutral gray. Read the parts list, perform the action, and pass the check before moving on.

Use all three views

BEFORE shows the starting point. PARTS ADDED shows only the new components in their modeled relative positions. AFTER shows where they belong. The center image is a parts-location view, not an exploded insertion path; image sizes are not a measuring scale.

Left and right are separate builds

Follow the page for the correct side. Do not copy a belt-bearing height from one side to the other. The rear Z arm has its own pages because its body and cable attachment differ from the side arms.

Hardware identification still requires review

Quantities on detail pages count the selected CAD occurrences. Plain metric labels are reproduced where available. A supplier ID marked size pending must be resolved against the actual kit before installation. This draft is not yet a beginner-ready release.

Understand a screw label

Example: 2 x M3 x 12 countersunk screw means two screws, M3 metric thread, 12 mm nominal length, and a countersunk head. Head shape matters: do not exchange a cap-head screw and a countersunk screw just because their thread and length match.

Keep one step in one tray

Collect only the hardware listed on the current detail page. Count it before fitting. Put unused parts back in the labeled bag rather than carrying loose screws from one operation to the next.

Start screws by hand

Hold the parts together without forcing them. Start the thread gently and make sure the screw turns freely before tightening with a driver. If it stops early, check alignment, screw length and thread compatibility.

Stop at an unresolved detail

A catalog size identifies a screw; it does not prove its engagement in this design. If a step calls for verification or the pictured hardware does not match your kit, resolve the difference before tightening or closing the assembly.

2 Frame and captive panels

Build the left frame, add crossmembers, install the rear Z support and partition, then close the electronics bay. Install hardware that becomes inaccessible before closing a joint.

3-5 XY motion and belts

Prepare the beam and rails; fit the XY joints, rear drives and front idlers. Verify the two belt planes and the final belt clamp before tensioning.

6 Z system and bed

Prepare the three arms, install the screw/nut assemblies and rails, seat the bed on its supports, and check cable-chain travel.

7-9 Electronics layout and enclosure

Dry-fit the electronics, prepare the harness plan, then fit skirts, vents, panels and doors. Follow the release checklist before powering the machine.

Tools

Metric hex and Torx drivers matching the supplied screws; machinist square; straightedge; calipers; tape measure; labelled parts trays; and suitable rail cleaning/lubrication supplies. Use an insert tool only where an insert and its specification are confirmed.

Prepare the parts

Remove burrs and support material without enlarging functional holes. Check countersinks, tapped holes, pockets and bearing seats. Keep rail carriages on their rails or use the rail supplier's transfer block.

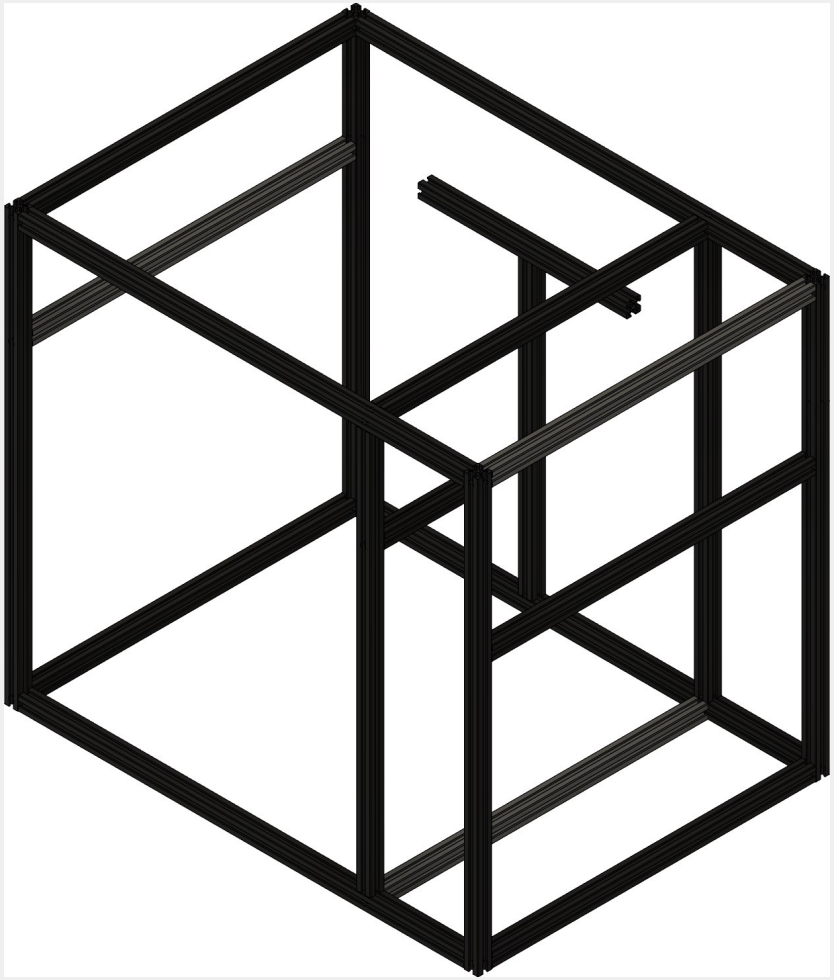
Sort the hardware

Separate frame fasteners, rail fasteners, drive hardware, Z-arm hardware and panel hardware. The model contains supplier part numbers as well as plain size labels. Resolve supplier numbers against the actual kit before assembly.

Printed-part settings

Material, wall count, infill, orientation and insert installation temperature are not established by this model. Use the approved Hybrid print specification for load-bearing and heated-chamber parts; do not copy Hypernova slicer settings automatically.

FRAME - IDENTIFY THE STRUCTURE



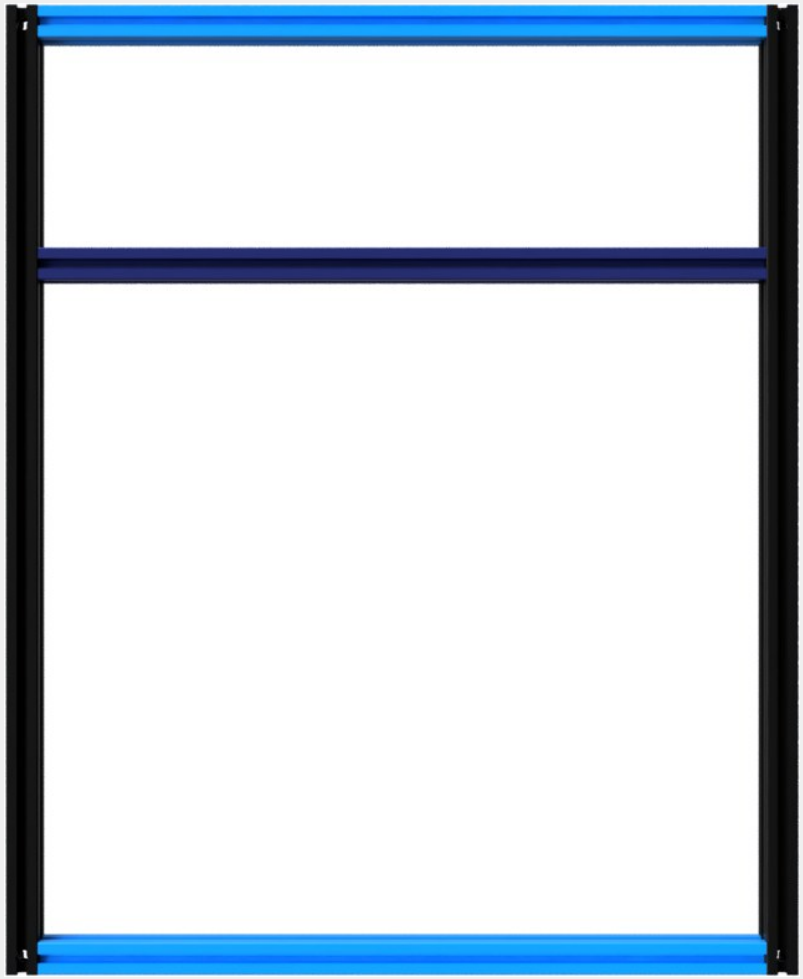
Identify the main print chamber and the narrower electronics bay. Keep the rear Z-rail support and the two Y-rail supports separate from the outer frame members.

MODEL INVENTORY
21 extrusion occurrences in the frame group.
This is not a verified cut list.

CHECK: Match the machining and access holes as well as the length. Several CAD names and measured lengths differ.

Current Hybrid assembly

BUILD THE LEFT FRAME



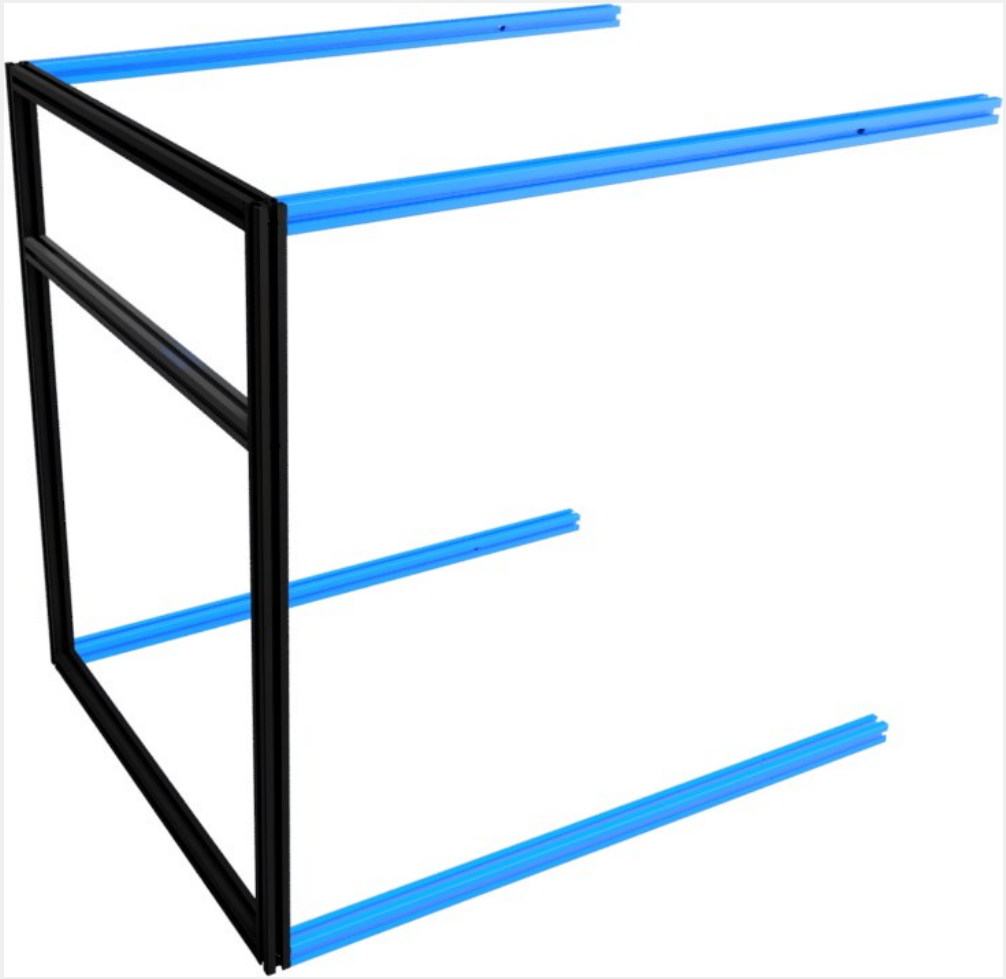
Lay out the two uprights and the three connecting extrusions as shown. Assemble the joints loosely on a flat surface. Check the Y-rail support orientation before adding the next members.

Two uprights
Three connecting extrusions

CHECK: The side frame is square and lies flat. Do not lock in a twist.

Existing Hybrid frame guide - step 1

ADD FOUR HORIZONTAL EXTRUSIONS



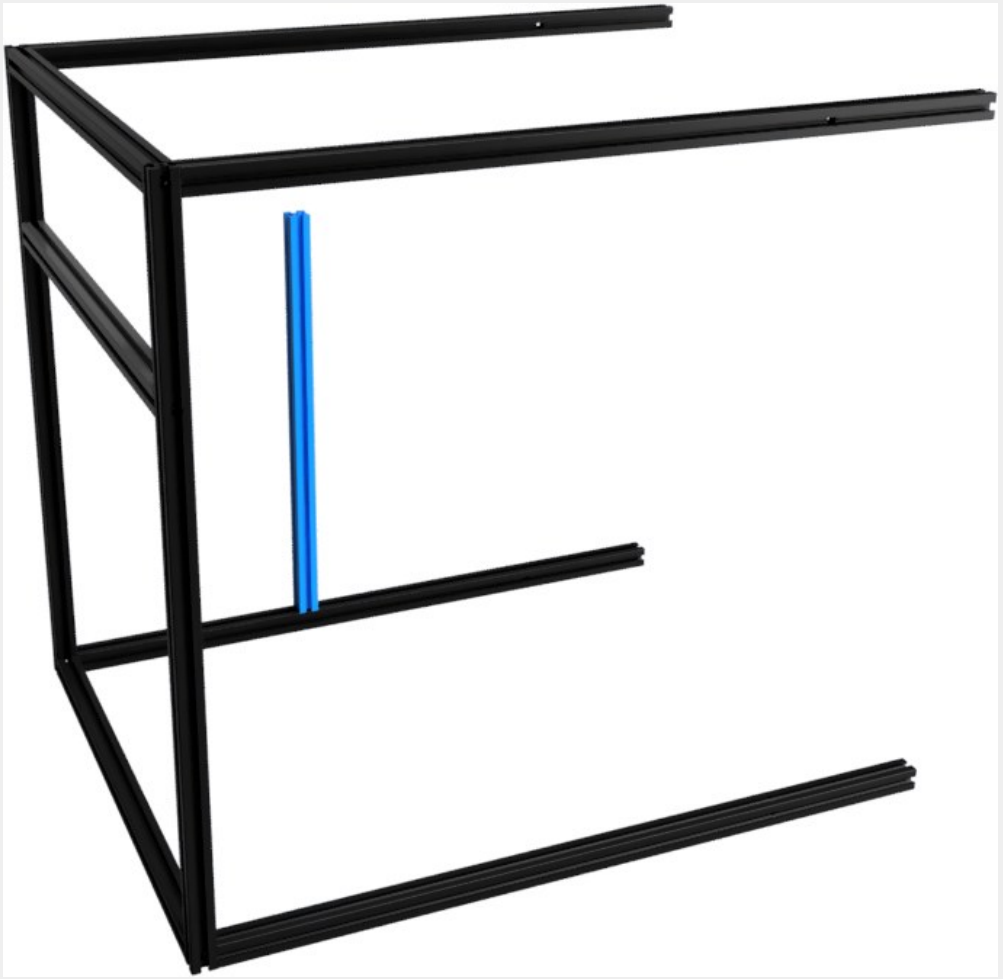
Attach the four highlighted crossmembers to the left frame. Support the free ends while starting the fasteners. Keep every joint adjustable.

Four horizontal extrusions
Matching frame fasteners

CHECK: All four members point toward the electronics side and their access holes remain reachable.

Existing Hybrid frame guide - step 2

INSTALL THE REAR Z-RAIL EXTRUSION



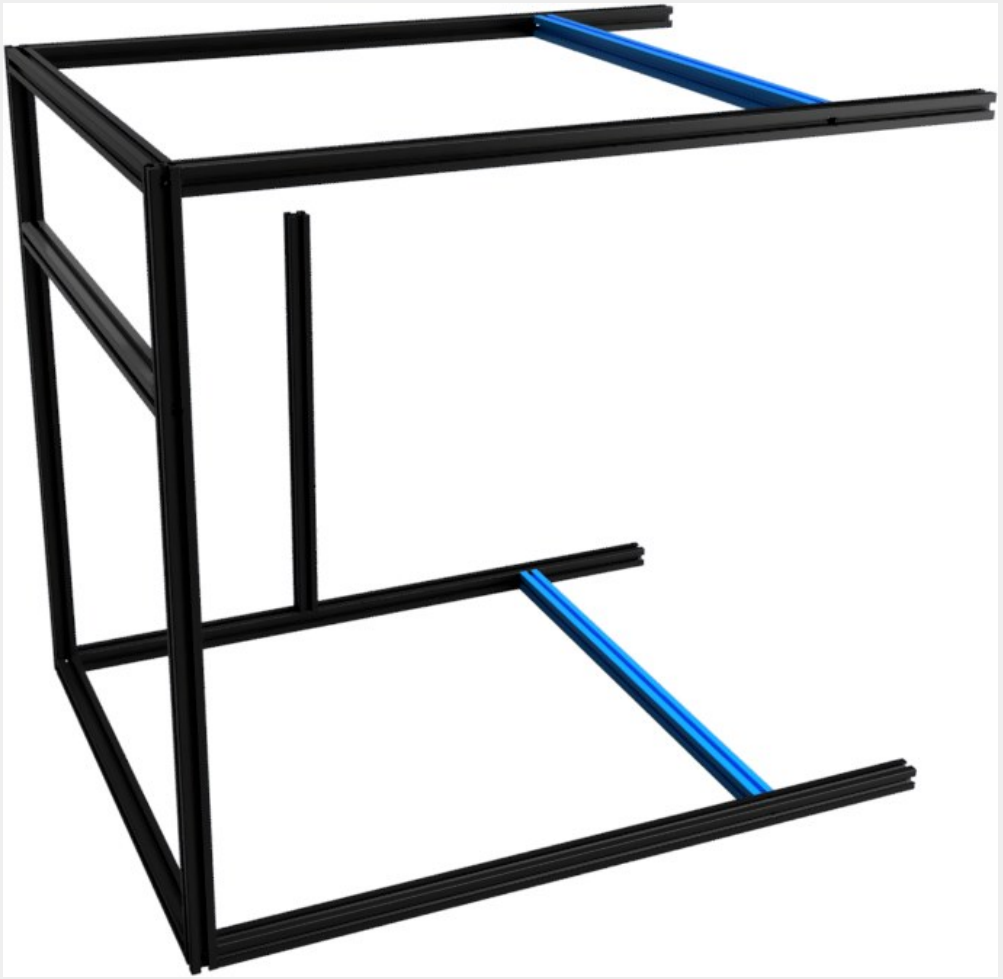
Fit the rear vertical Z support in the orientation shown. Check its attachment to the lower frame and its relationship to the rear upper support.

Rear Z-support extrusion
Associated joints

CHECK: The rail-mounting face points into the print chamber.

Existing Hybrid frame guide - step 3

SLIDE IN UPPER AND LOWER MID-FRAME MEMBERS



Slide the highlighted upper and lower members into position. Start their fasteners without fully tightening. These members define the boundary of the print chamber.

Upper and lower partition members

CHECK: The partition members remain parallel and square to the side frame.

Existing Hybrid frame guide - step 4

PREINSTALL THE RIGHT Y-RAIL SUPPORT

Add the upright and support members shown in the reference view. Establish the right Y-rail mounting position before enclosing this side.

Right Y-support frame members

CHECK: The two Y mounting surfaces are aligned; use the confirmed drawing for their final height.

Existing Hybrid frame guide - step 5

INSTALL THE INNER PANEL

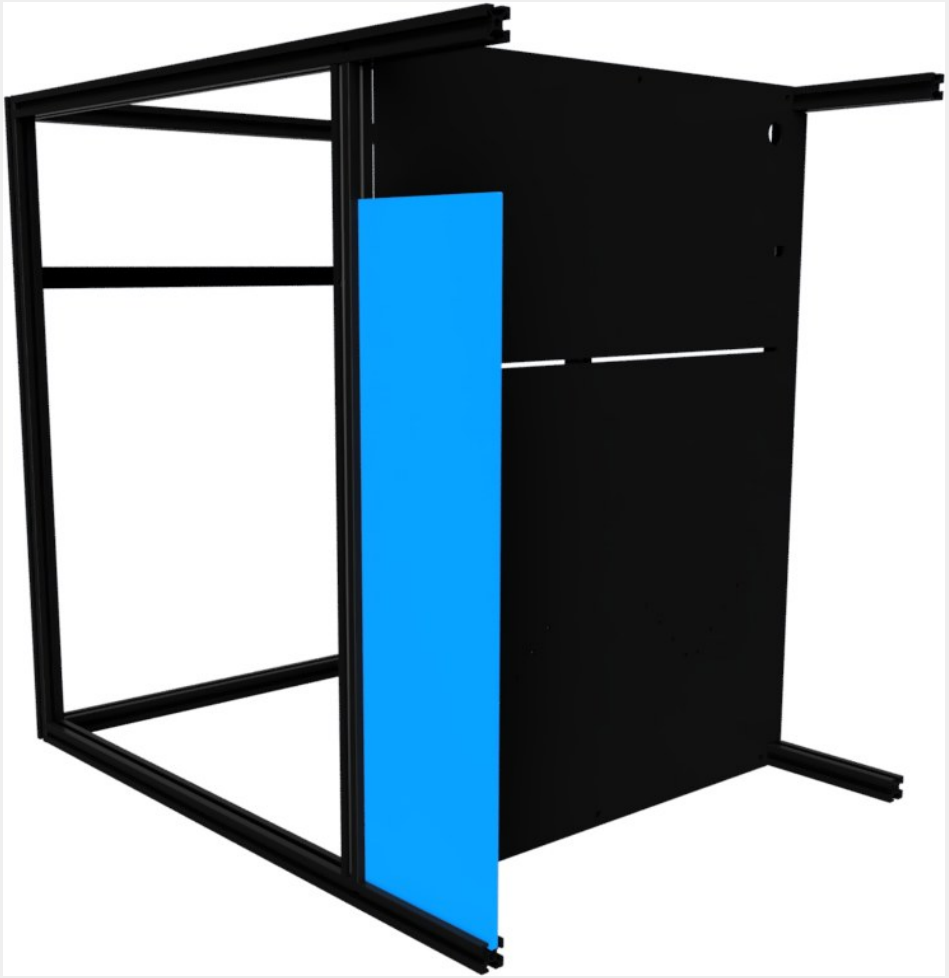
Preinstall the panel roll-in T-nuts, then fit the inner panel. Start all panel fasteners loosely so the panel can settle without distorting the frame.

Inner panel
Panel T-nuts and fasteners

CHECK: The panel seats without force and does not hide hardware still needed for the frame.

Existing Hybrid frame guide - step 6

INSTALL THE FRONT ELECTRONICS PANEL



Front electronics panel

Fit the narrow front electronics panel as shown. Check the display opening and the front/rear orientation before fastening.

CHECK: Edges align with the partition and the outer bay; the panel is not bowed.

Existing Hybrid frame guide - step 7

INSTALL THE CORNER EXTRUSIONS

Add the two highlighted outside uprights. Keep the corner joints loose until the remaining bay panels and members are fitted.

Two outer uprights

CHECK: The bay remains square and the mounting faces point in the correct direction.

Existing Hybrid frame guide - step 8

INSTALL THE LOWER BAY PANEL AND EXTRUSION

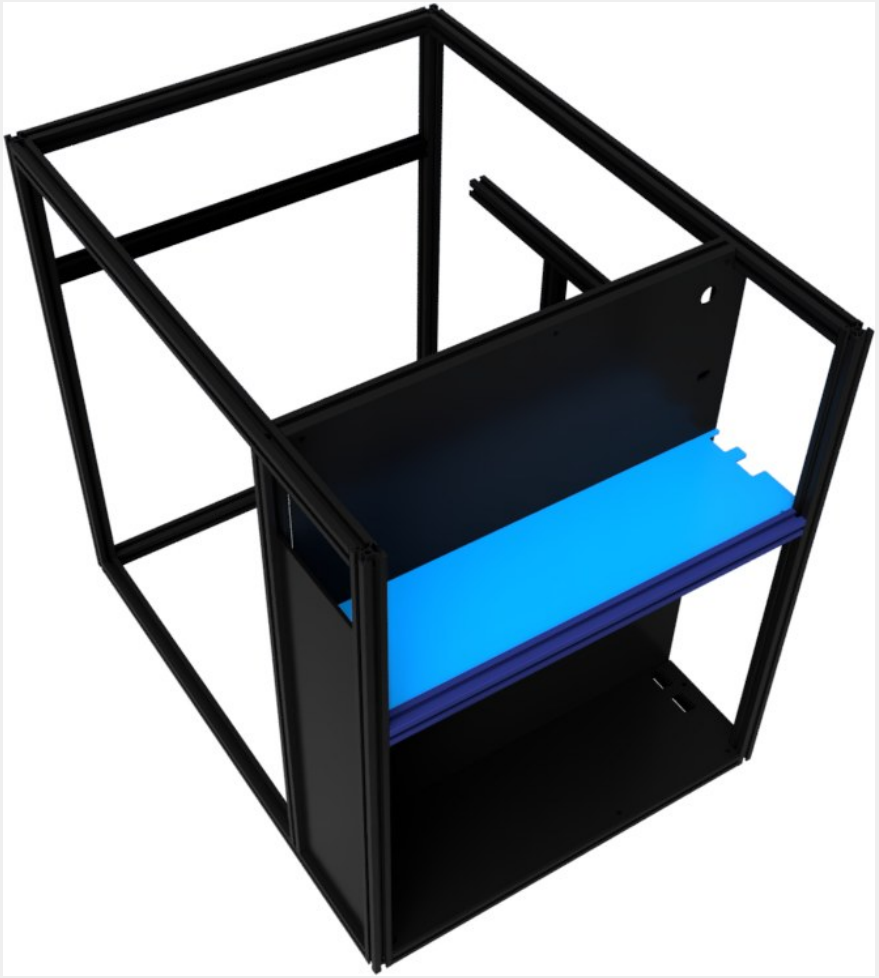
Preinstall its roll-in T-nuts. Fit the lower electronics panel and the highlighted bottom member before closing the final corner.

Lower electronics panel
Bottom bay extrusion

CHECK: The panel sits in the intended location without pinching its edge.

Existing Hybrid frame guide - step 9

ANGLE THE INNER BAY PANEL INTO POSITION



Angle the inner electronics panel into the bay and slide the supporting extrusion into place, following the illustrated order. Keep adjacent fasteners loose enough to allow placement.

Inner electronics panel
Supporting extrusion

CHECK: The panel clears the surrounding frame and sits flat after placement.

Existing Hybrid frame guide - step 10

INSTALL THE FINAL CORNER AND SQUARE THE FRAME

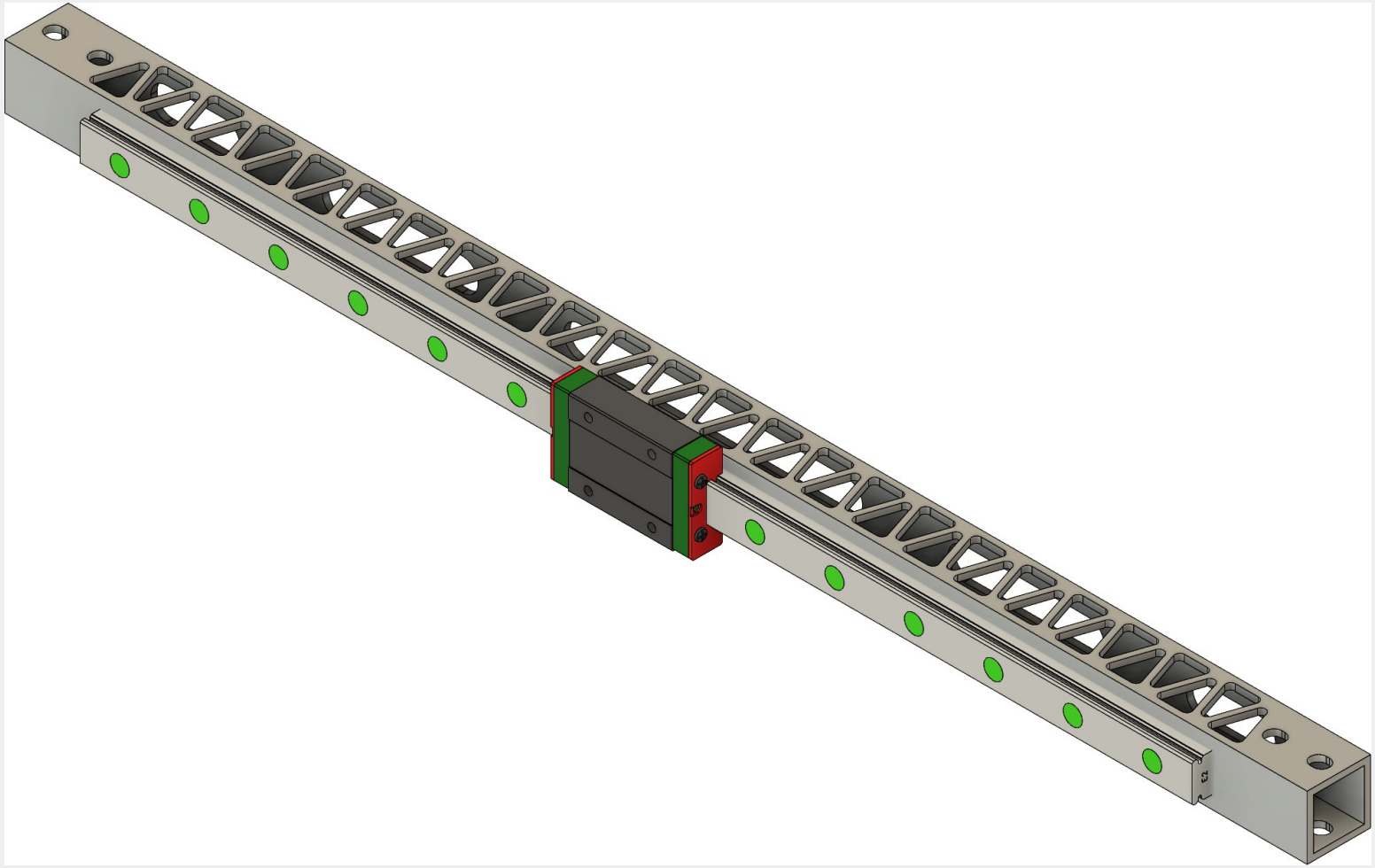
Fit the last highlighted member. Check the frame diagonals and the rail-support alignment, then tighten the joints progressively while checking squareness.

Final bay member
Frame-joint hardware

CHECK: The completed frame is square, stable and untwisted; captive panels are seated.

Existing Hybrid frame guide - step 11

PREPARE THE X BEAM AND RAIL



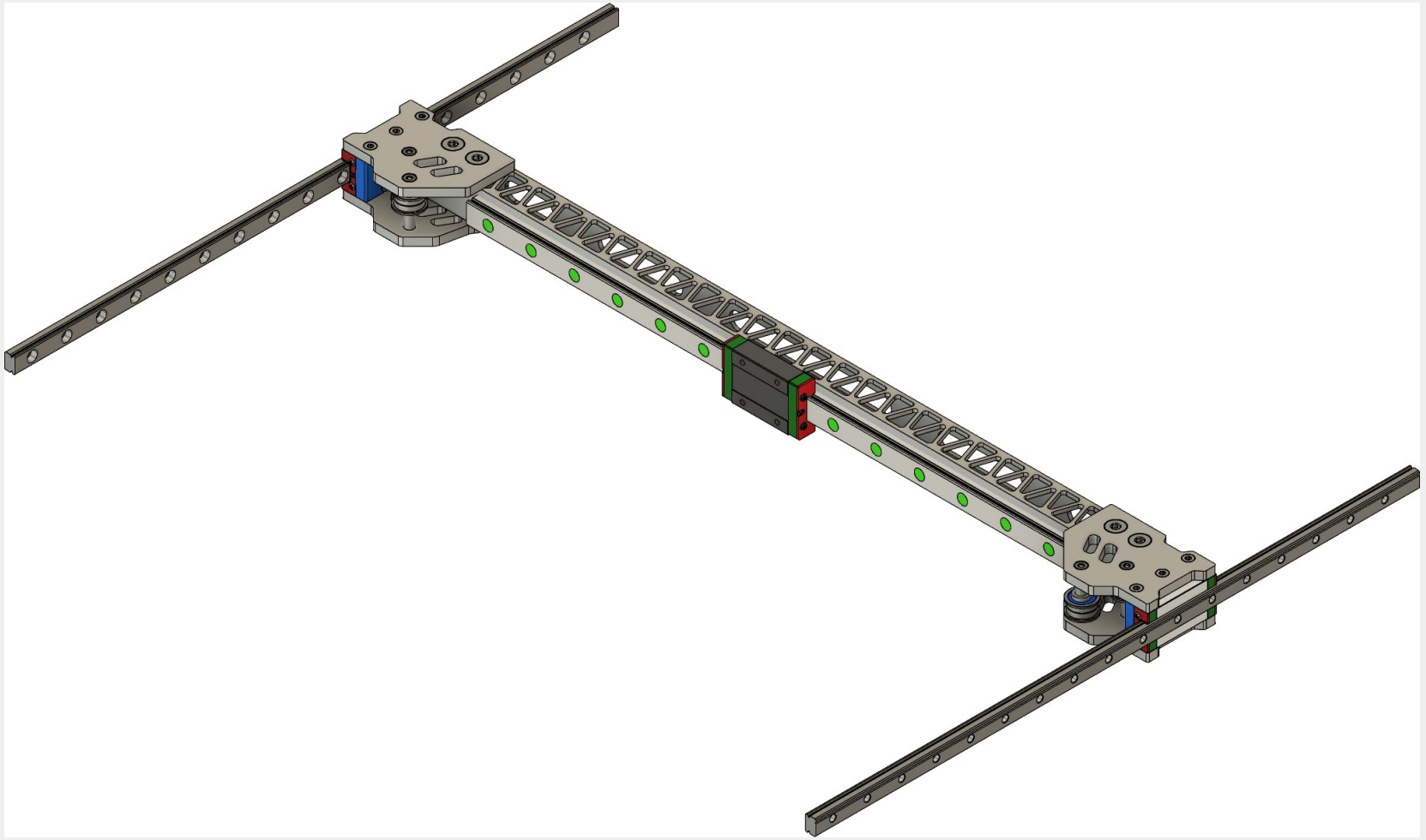
Match the MGN12 beam, rail and carriage shown. Clean the mounting face, start the rail fasteners, and align the rail progressively. Move the carriage along the rail to check for tight spots.

CAD GROUP
Lightweight_X_mgn12h
Rail and carriage models included

CHECK: The carriage moves smoothly. Confirm actual rail length and screw engagement; names in this group are not a cut specification.

Current Hybrid assembly

ASSEMBLE THE XY JOINTS



Identify the left and right joint plates and carriage blocks. Assemble each side in the modeled orientation and attach the ends of the beam. Preserve the upper and lower belt clearances.

CAD ITEMS
Left/right joint plates
Carriage blocks
28 mm pin labels in idler group

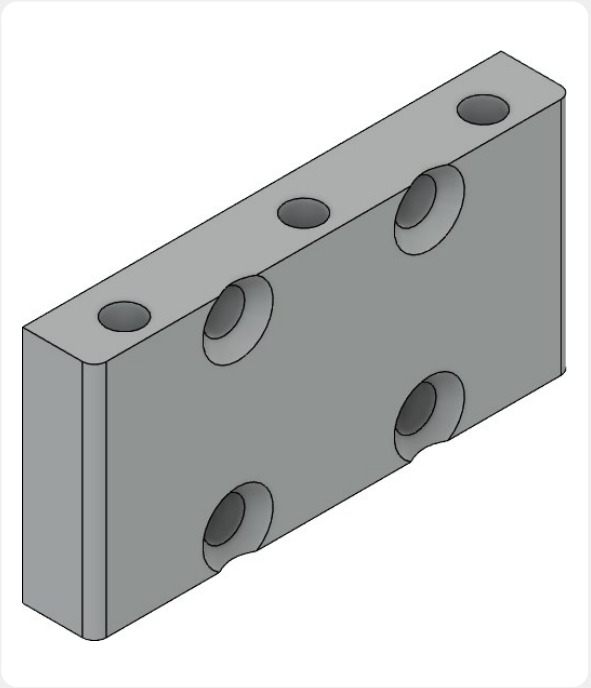
CHECK: The beam is square to the Y motion and both ends move together without racking.

Current Hybrid assembly

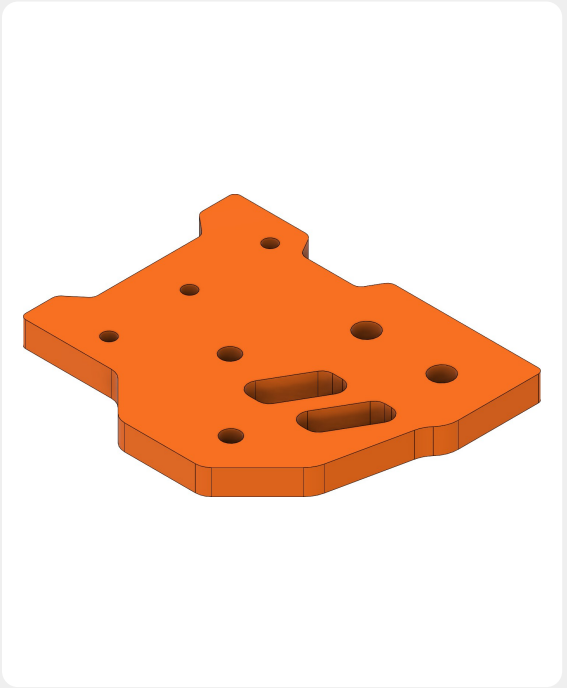
POSITION THE LOWER JOINT PLATE

LEFT XY JOINT | DETAIL 1

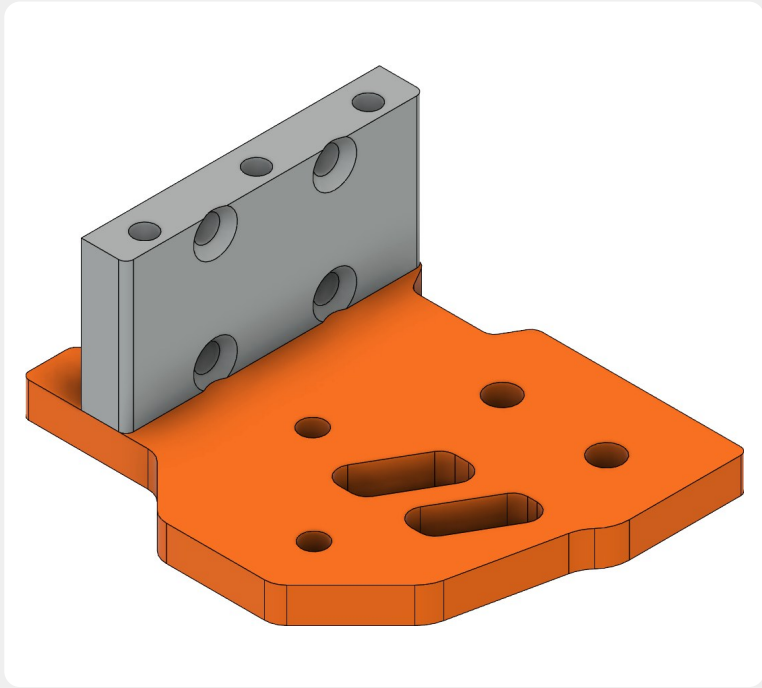
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match the lower plate to the carriage block. Face the countersunk holes toward their screw heads. Keep the beam mounting holes accessible.

PARTS IN THIS CAD STEP

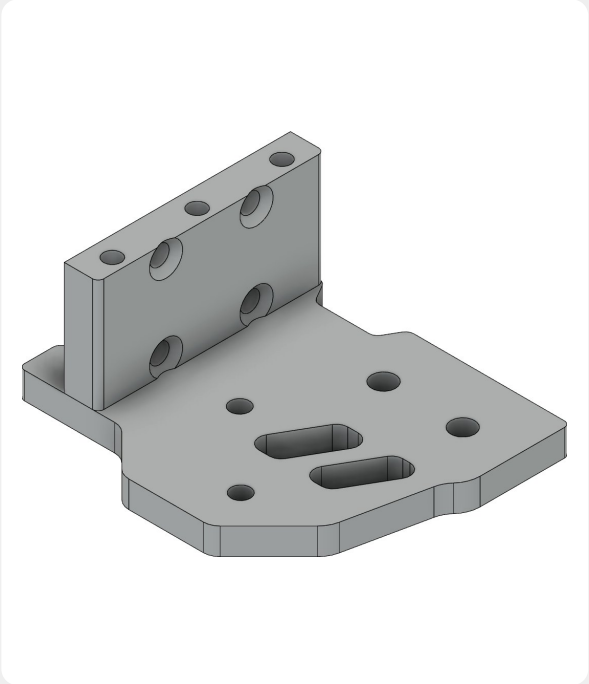
1 x XY Plate Left Bottom

CHECK BEFORE CONTINUING: The plate is the correct hand; the beam attachment points face inward.

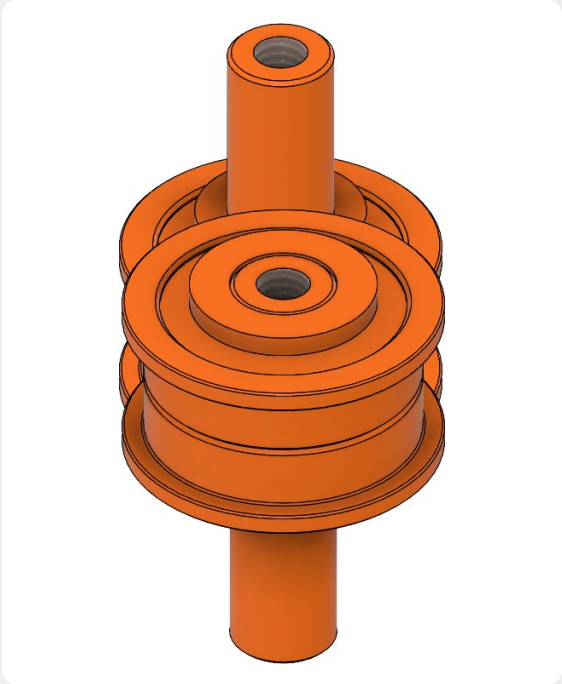
FIT THE PINS, WASHERS AND BEARINGS

LEFT XY JOINT | DETAIL 2

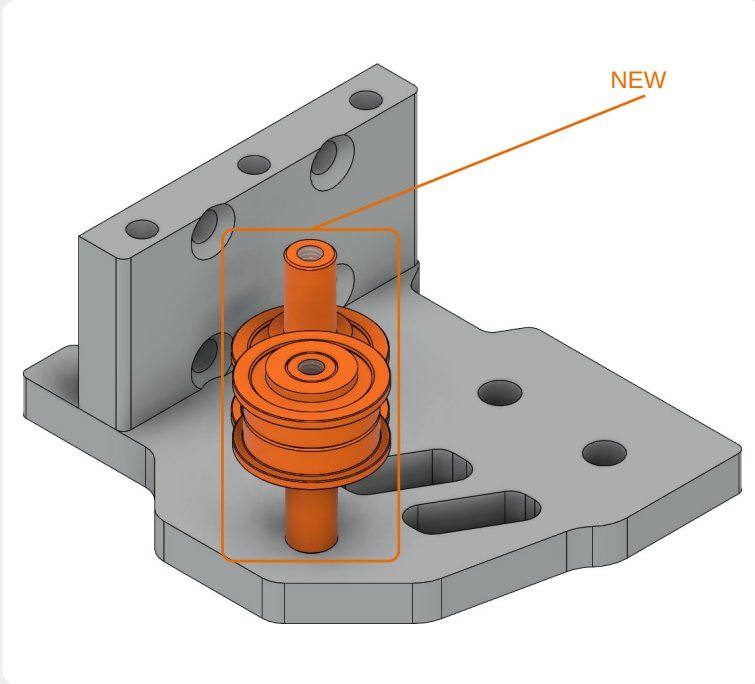
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Build the two idler positions using the pins, washers and bearing pairs shown. Compare the upper and lower belt planes before closing the joint.

PARTS IN THIS CAD STEP

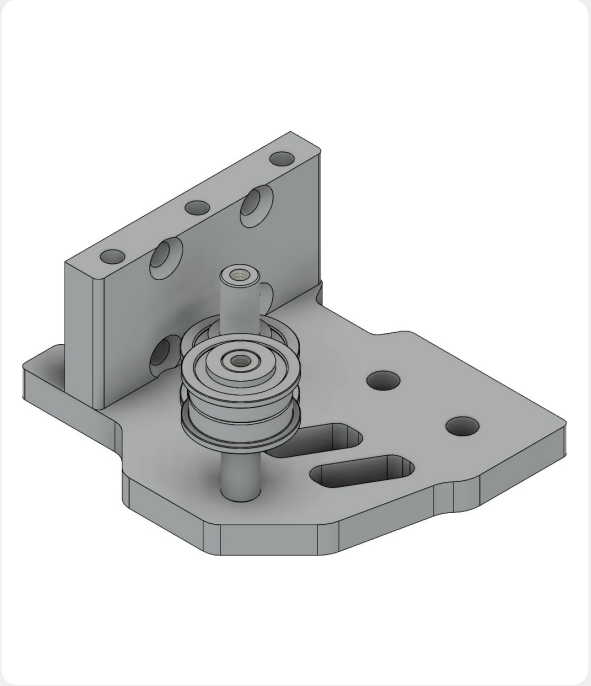
2 x 28 mm pin
2 x M5 washer
4 x F695 flanged bearing

CHECK BEFORE CONTINUING: Each bearing pair rotates freely; washers remain at their illustrated ends.

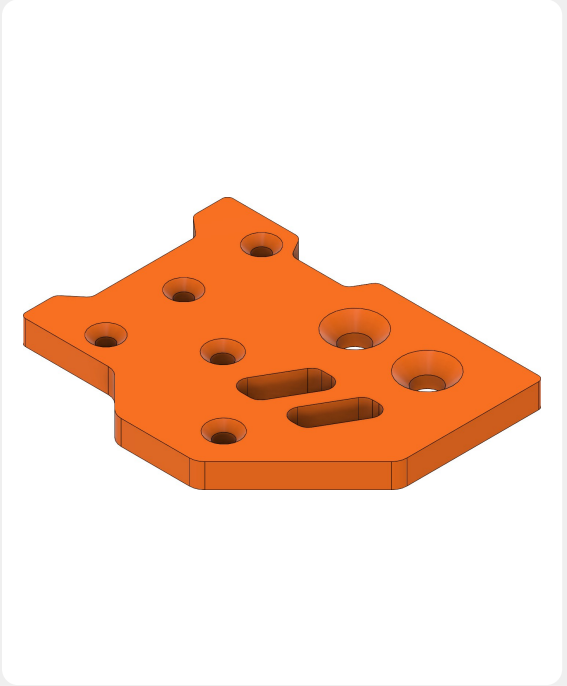
ADD THE UPPER PLATE

LEFT XY JOINT | DETAIL 3

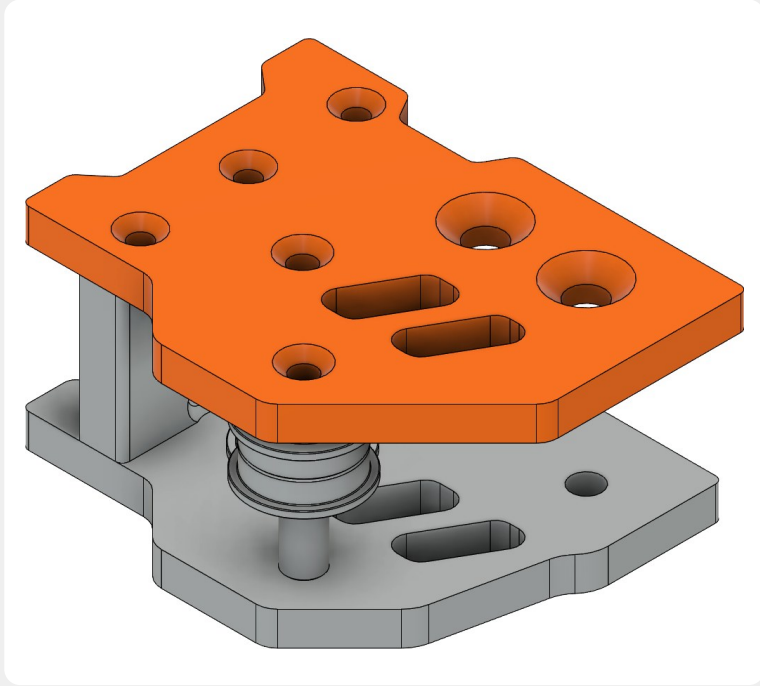
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the upper plate over the two pins and onto the carriage block. Keep all pin ends aligned while lowering the plate.

PARTS IN THIS CAD STEP

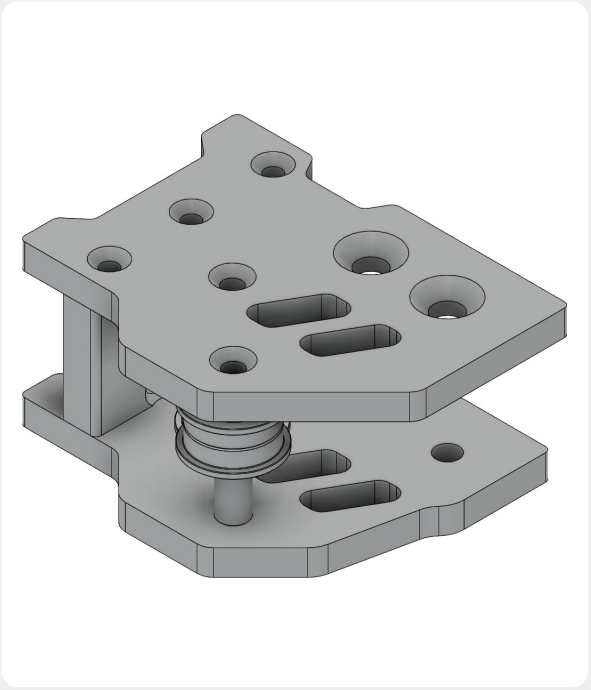
1 x XY Plate Left Bottom(Mirror)

CHECK BEFORE CONTINUING: The two plates sit parallel without force.

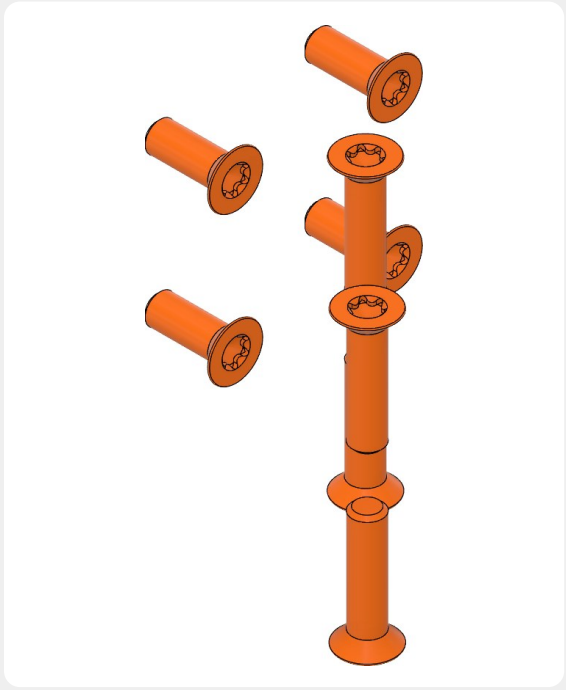
FIT THE JOINT RETAINING SCREWS

LEFT XY JOINT | DETAIL 4

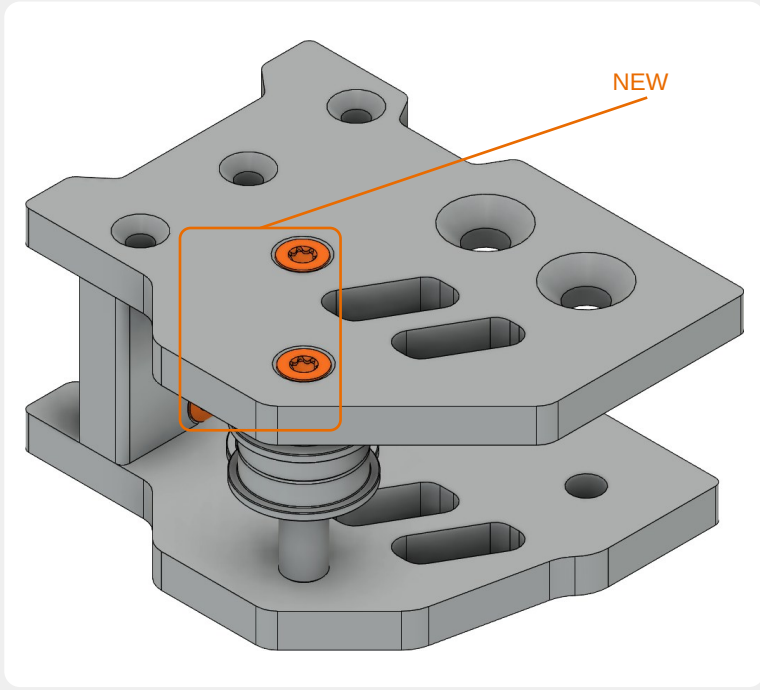
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use the four M3 x 8 and four M3 x 12 Torx flat-head screws in the locations shown. Keep the two lengths separate. Start all screws before tightening the plate pair.

PARTS IN THIS CAD STEP

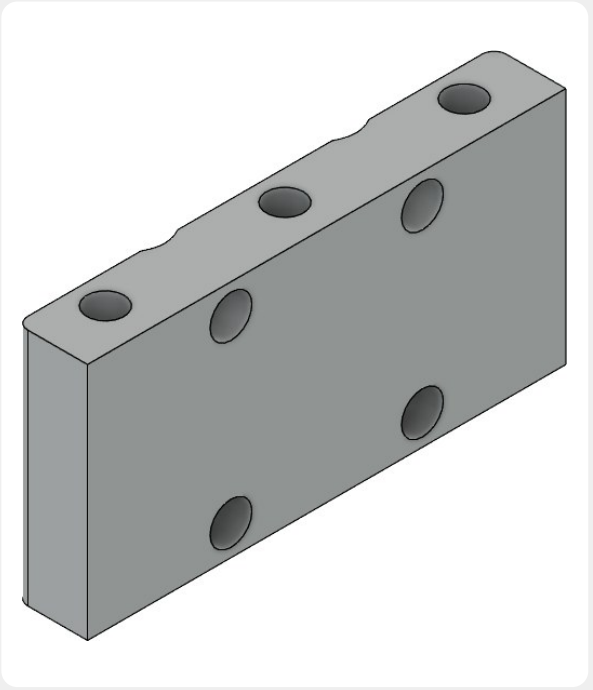
4 x M3 x 8 Torx flat-head [90236A120]
4 x M3 x 12 Torx flat-head [90236A122]

CHECK BEFORE CONTINUING: Both idler stacks turn freely. Confirm screw engagement; fit the beam/block mounting screws when attaching the beam.

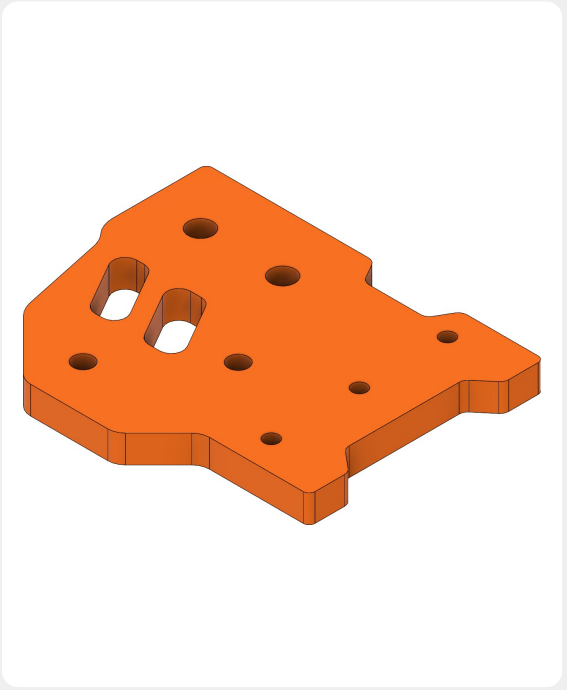
POSITION THE LOWER JOINT PLATE

RIGHT XY JOINT | DETAIL 1

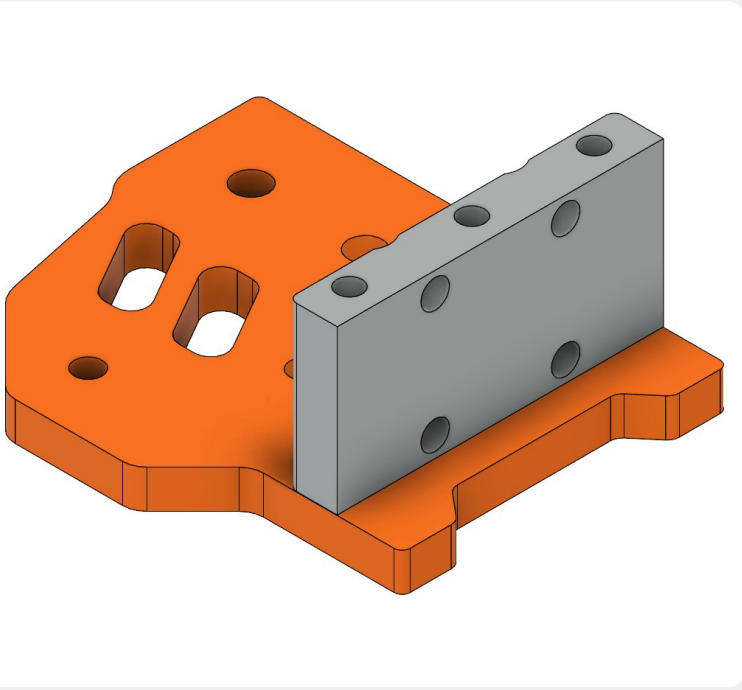
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match the lower plate to the carriage block. Face the countersunk holes toward their screw heads. Keep the beam mounting holes accessible.

PARTS IN THIS CAD STEP

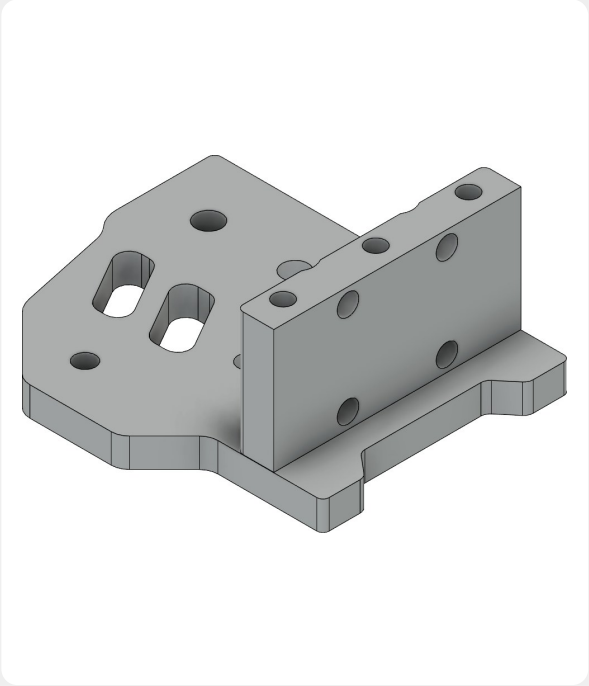
1 x XY Plate Left Bottom(Mirror)

CHECK BEFORE CONTINUING: The plate is the correct hand; the beam attachment points face inward.

FIT THE PINS, WASHERS AND BEARINGS

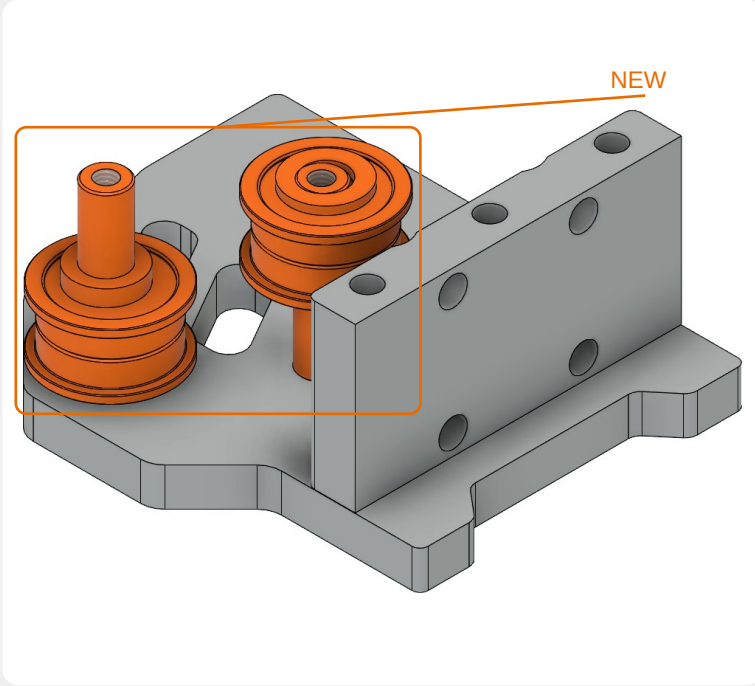
RIGHT XY JOINT | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Build the two idler positions using the pins, washers and bearing pairs shown. Compare the upper and lower belt planes before closing the joint.

PARTS IN THIS CAD STEP

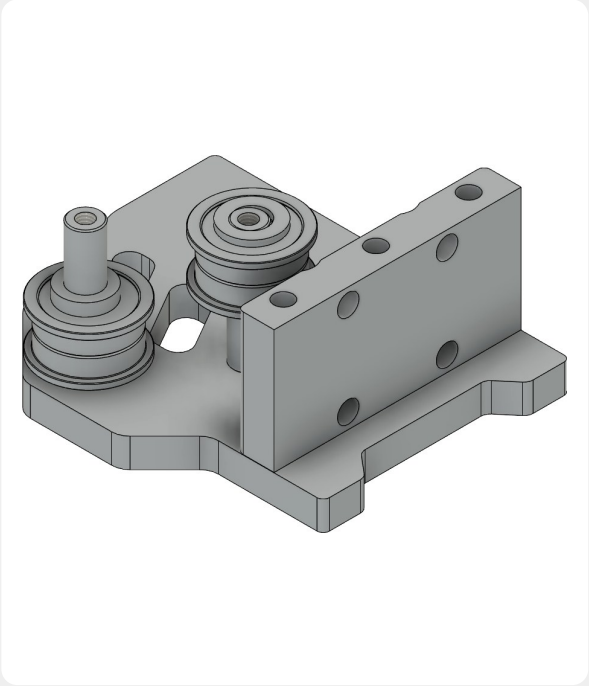
2 x 28 mm pin
2 x M5 washer
4 x F695 flanged bearing

CHECK BEFORE CONTINUING: Each bearing pair rotates freely; washers remain at their illustrated ends.

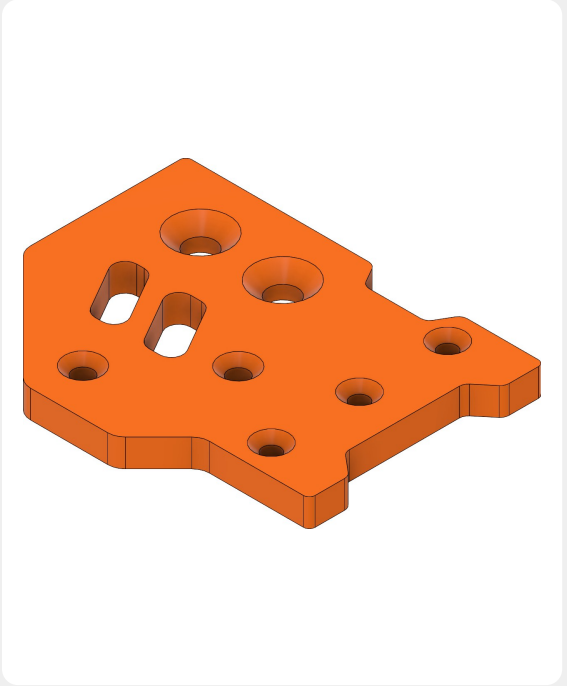
ADD THE UPPER PLATE

RIGHT XY JOINT | DETAIL 3

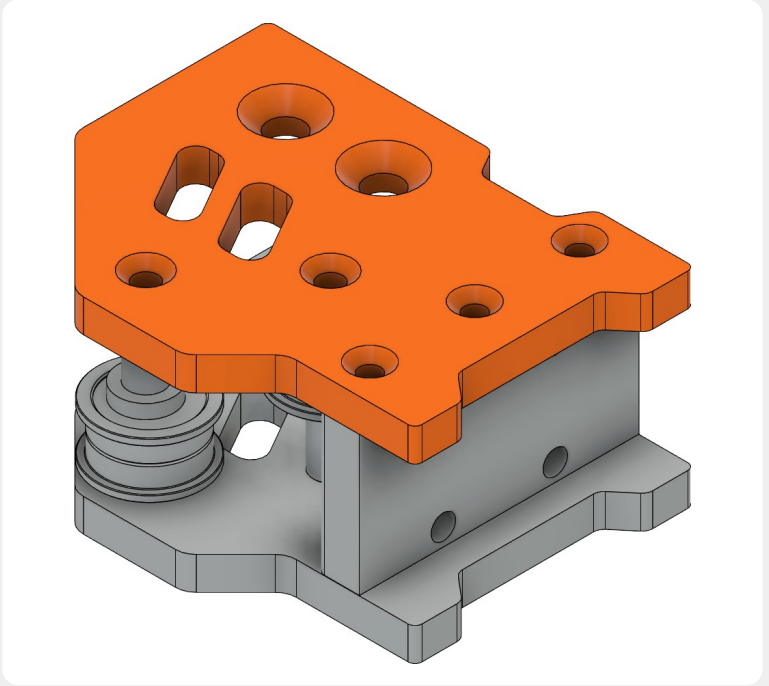
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the upper plate over the two pins and onto the carriage block. Keep all pin ends aligned while lowering the plate.

PARTS IN THIS CAD STEP

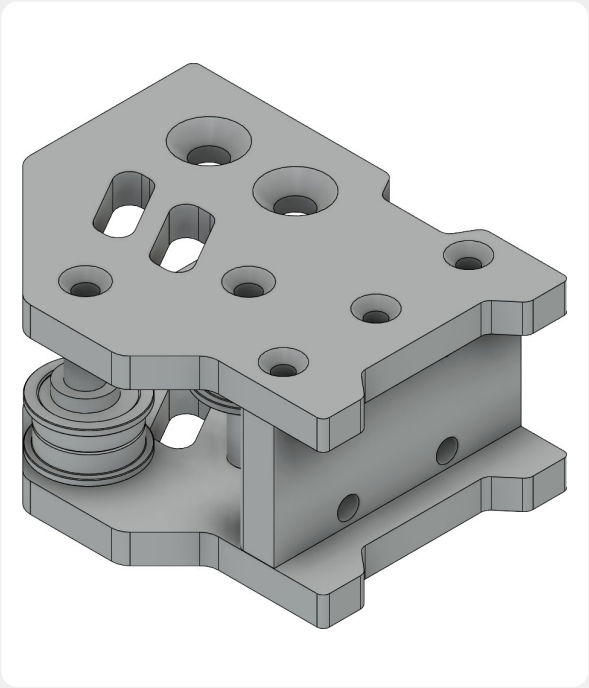
1 x XY Plate Left Bottom(Mirror)(Mirror)

CHECK BEFORE CONTINUING: The two plates sit parallel without force.

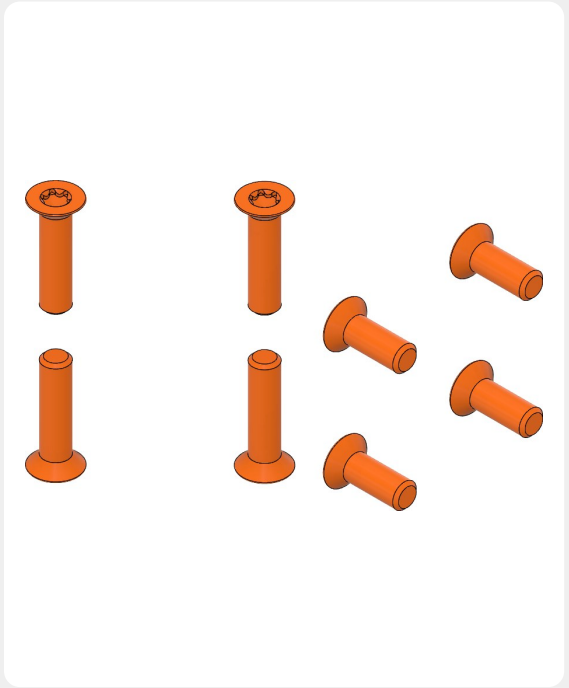
FIT THE JOINT RETAINING SCREWS

RIGHT XY JOINT | DETAIL 4

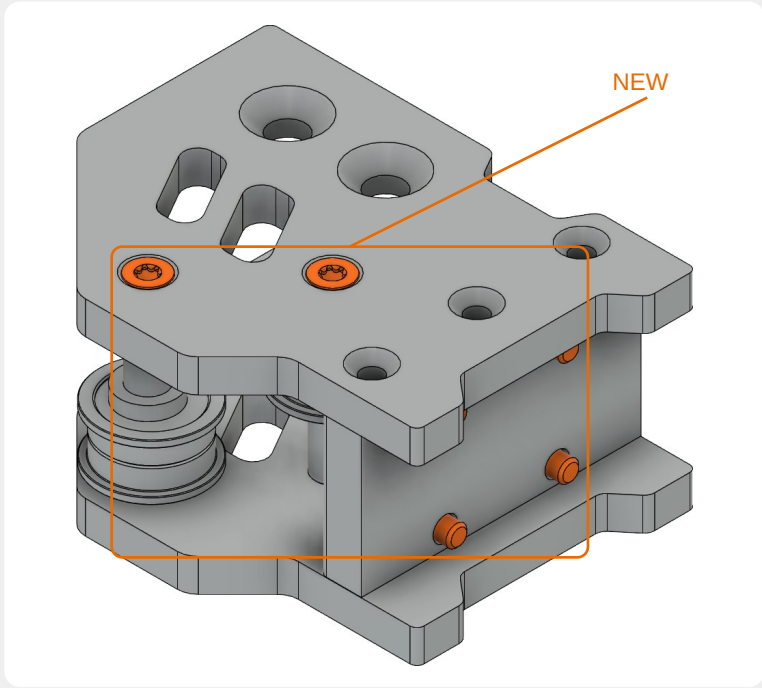
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

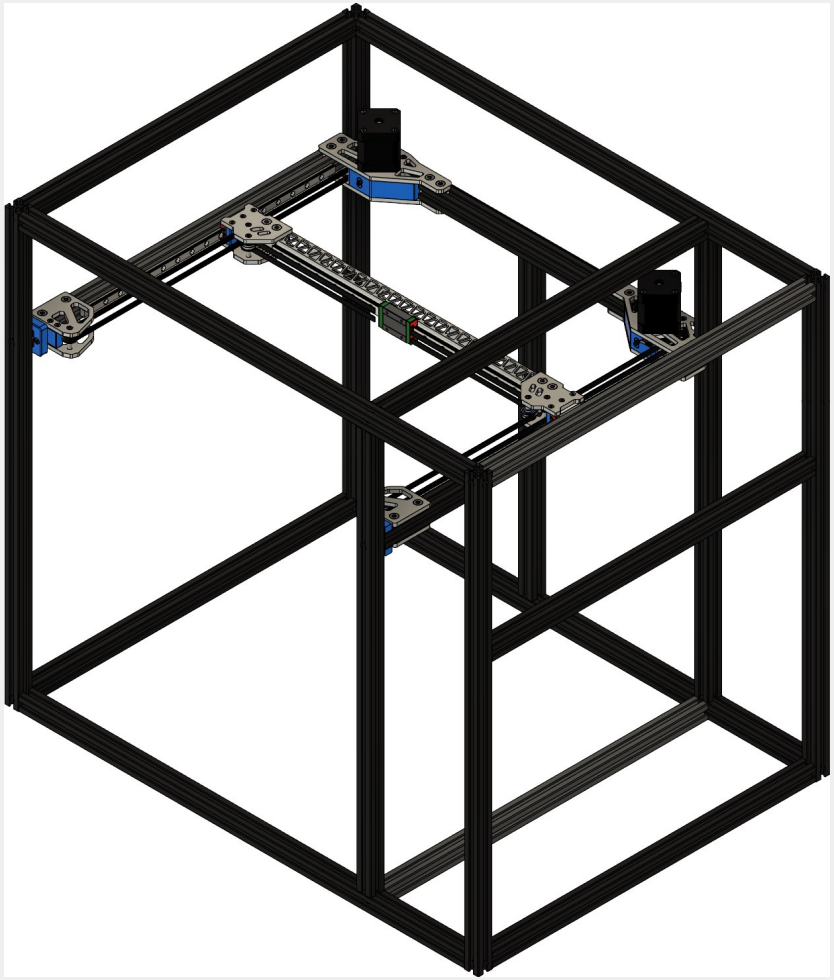
Use the four M3 x 8 and four M3 x 12 Torx flat-head screws in the locations shown. Keep the two lengths separate. Start all screws before tightening the plate pair.

PARTS IN THIS CAD STEP

4 x M3 x 8 Torx flat-head [90236A120]
4 x M3 x 12 Torx flat-head [90236A122]

CHECK BEFORE CONTINUING: Both idler stacks turn freely. Confirm screw engagement; fit the beam/block mounting screws when attaching the beam.

FIT XY MOTION TO THE FRAME



Fit the Y rails to the prepared supports. Move the gantry by hand through its intended range while aligning the second rail. Tighten progressively and repeat the travel check.

INSTALLATION CHECK
Parallel Y rails
Square X beam
Free carriage movement

CHECK: Travel remains smooth after tightening. Drive and idler assemblies shown here are located in the following steps.

Current Hybrid assembly

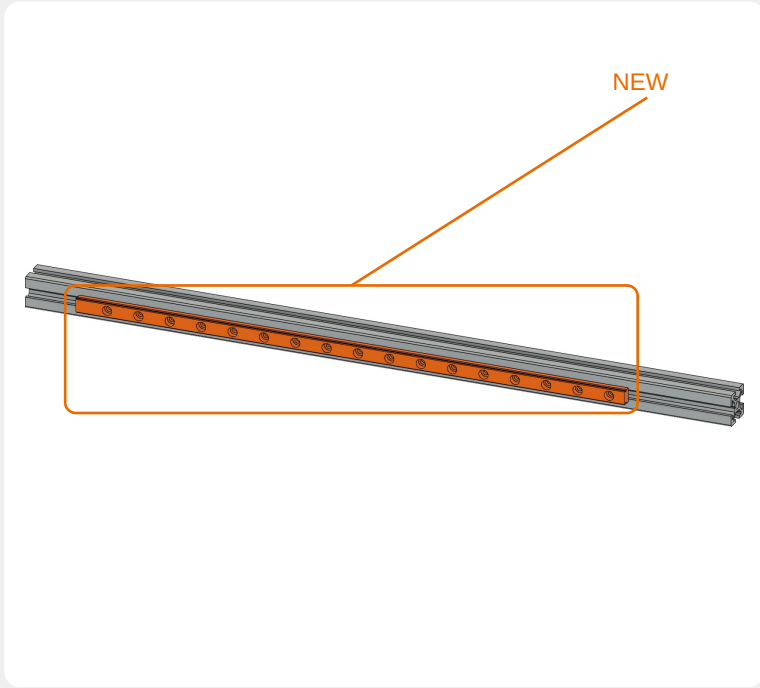
POSITION THE RAIL ON ITS EXTRUSION

RIGHT Y RAIL | DETAIL 1

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use the rail for this side and keep its carriage on the rail. Align the mounting holes with the extrusion slot. Match the rail orientation shown; leave room to adjust its position before tightening.

PARTS IN THIS CAD STEP

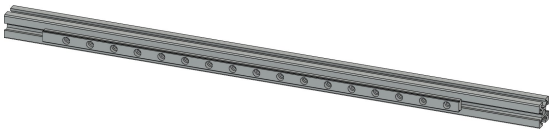
1 x 9 mm Wide x 350 mm Long 440 Stainless Steel Rail galaxy 350

CHECK BEFORE CONTINUING: The rail sits flat on the mounting face. Keep the carriage from sliding off either end.

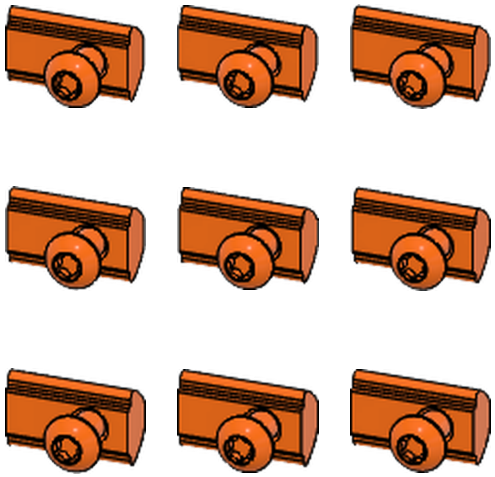
INSTALL THE NINE RAIL FASTENERS

RIGHT Y RAIL | DETAIL 2

1 BEFORE

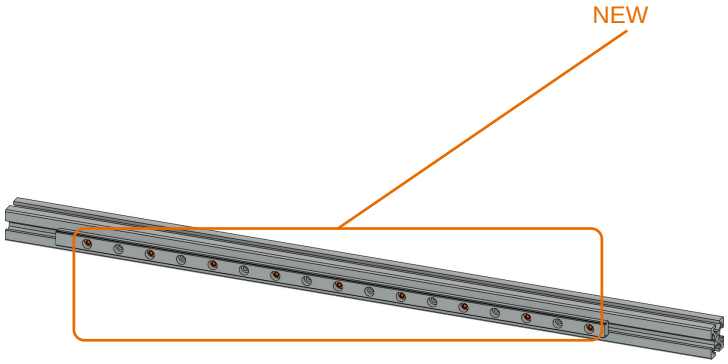


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align nine M3 roll-in T-nuts in the slot with the rail holes. Start all nine M3 x 8 Torx button-head screws loosely. Align the rail, then tighten progressively while checking smooth carriage travel.

PARTS IN THIS CAD STEP

9 x M3 x 8 Torx button-head [90991A113]
9 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: Every screw engages a T-nut. Slide the carriage through its usable travel; stop and realign if it binds. CAD nut centers are not installation measurements.

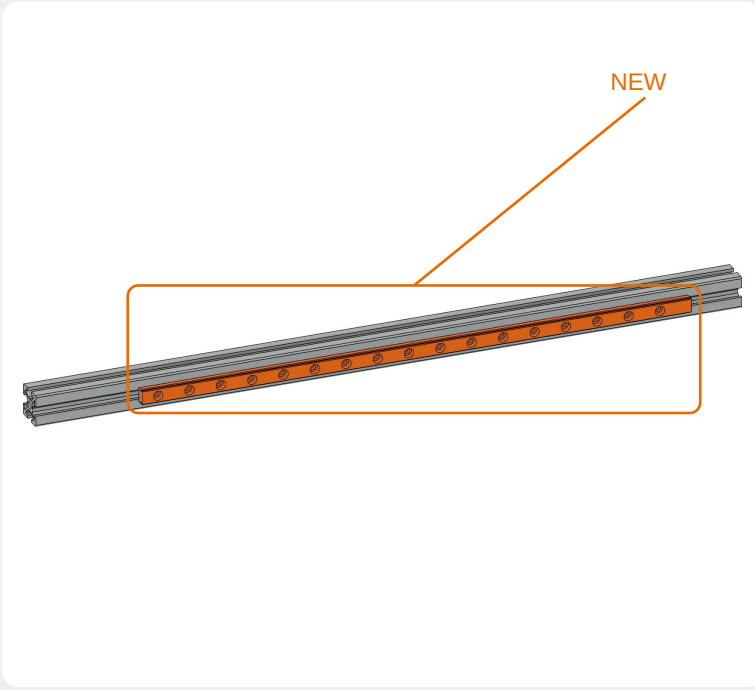
POSITION THE RAIL ON ITS EXTRUSION

LEFT Y RAIL | DETAIL 1

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use the rail for this side and keep its carriage on the rail. Align the mounting holes with the extrusion slot. Match the rail orientation shown; leave room to adjust its position before tightening.

PARTS IN THIS CAD STEP

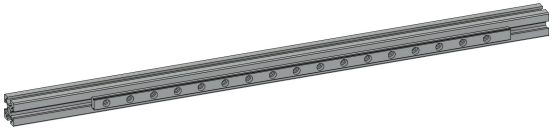
1 x 9 mm Wide x 350 mm Long 440 Stainless Steel Rail galaxy 350

CHECK BEFORE CONTINUING: The rail sits flat on the mounting face. Keep the carriage from sliding off either end.

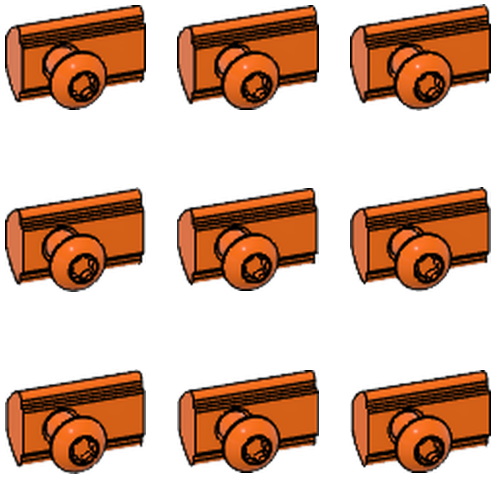
INSTALL THE NINE RAIL FASTENERS

LEFT Y RAIL | DETAIL 2

1 BEFORE

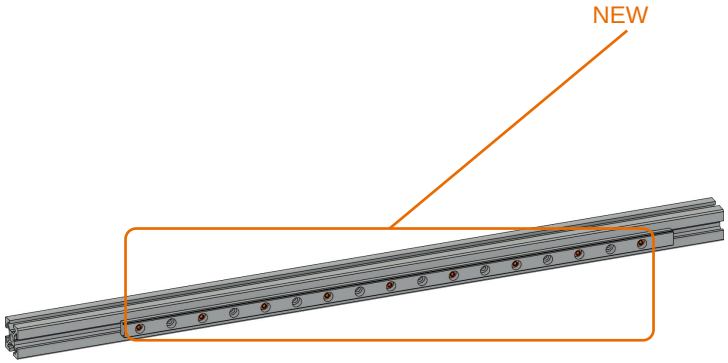


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

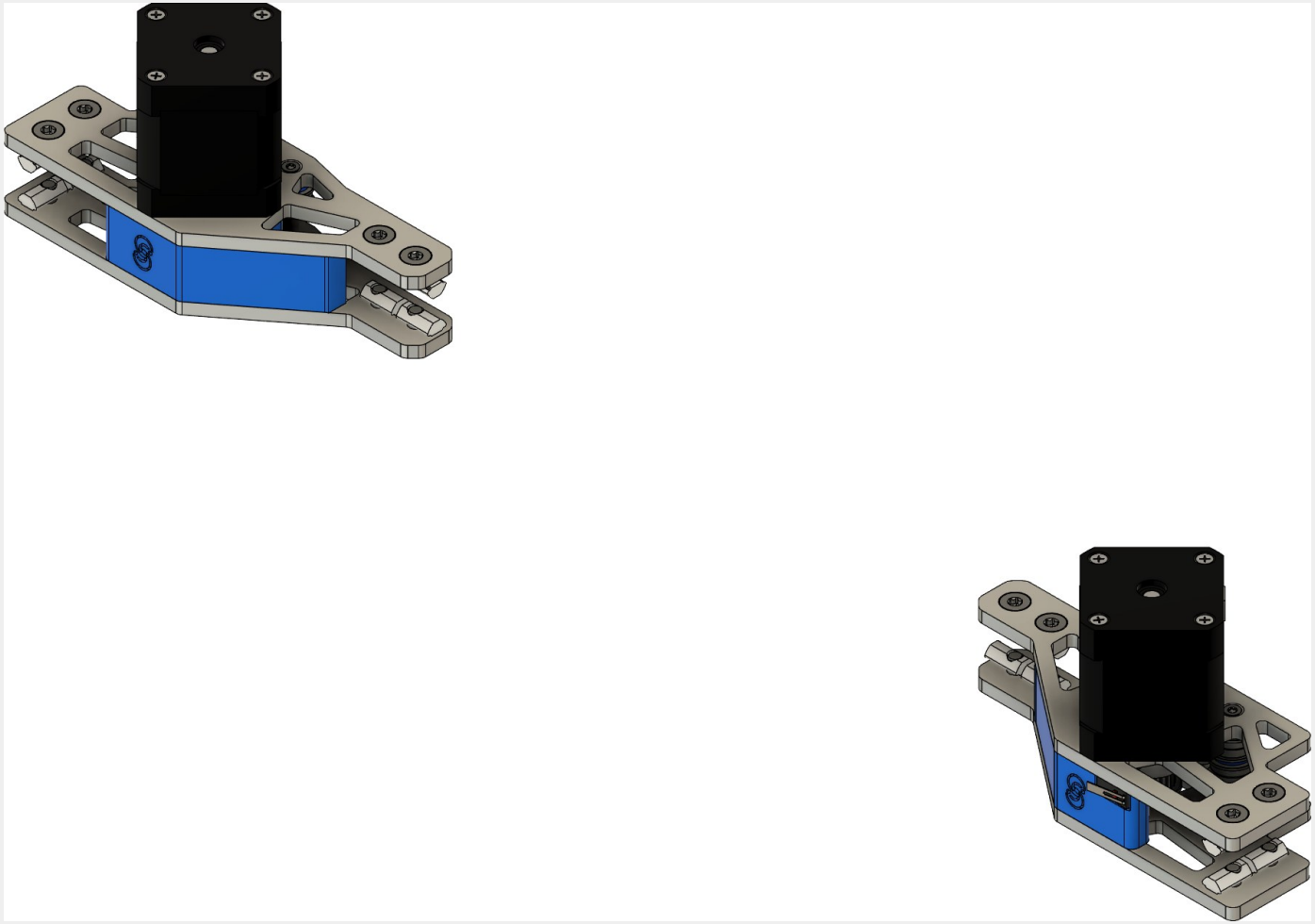
Align nine M3 roll-in T-nuts in the slot with the rail holes. Start all nine M3 x 8 Torx button-head screws loosely. Align the rail, then tighten progressively while checking smooth carriage travel.

PARTS IN THIS CAD STEP

9 x M3 x 8 Torx button-head [90991A113]
9 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: Every screw engages a T-nut. Slide the carriage through its usable travel; stop and realign if it binds. CAD nut centers are not installation measurements.

ASSEMBLE THE REAR MOTOR DRIVES



Keep each motor with its matching plate pair and stiffener. Fit the pins, spacers and bearings in the modeled stack. Align each pulley with its belt plane before securing it.

CAD ITEMS
Two NEMA17 motor occurrences
Two plate pairs
20 mm pin/spacer labels

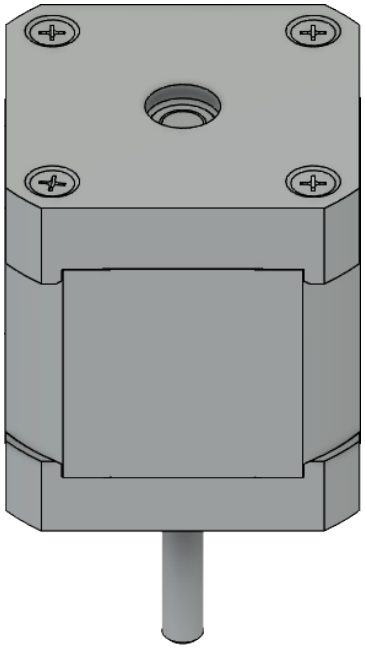
CHECK: Each idler rotates freely and each pulley is secure. Confirm pulley specification and all fastener lengths from the kit.

Current Hybrid assembly

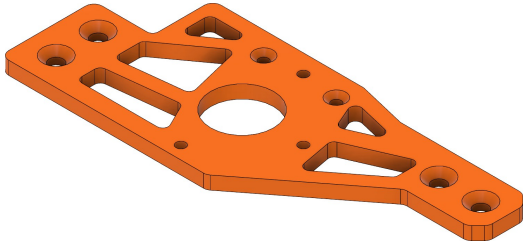
POSITION THE MOTOR MOUNTING PLATE

LEFT REAR DRIVE | DETAIL 1

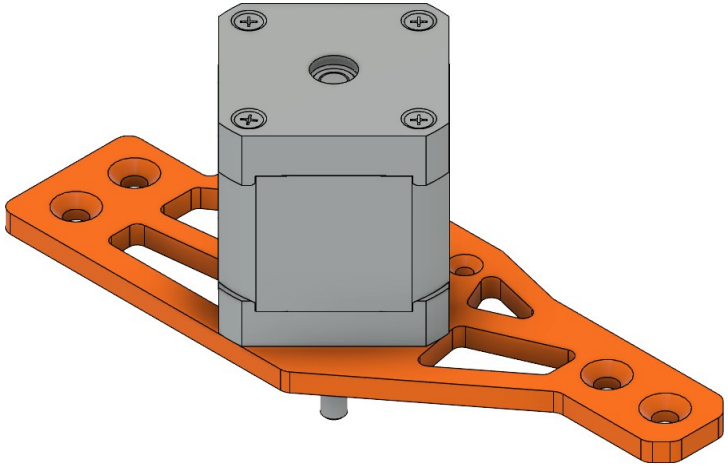
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the motor-side plate against the motor mounting face. Match the motor shaft opening and screw holes. Note: the CAD name says Bottom even though this plate is above the idlers in the assembled printer.

PARTS IN THIS CAD STEP

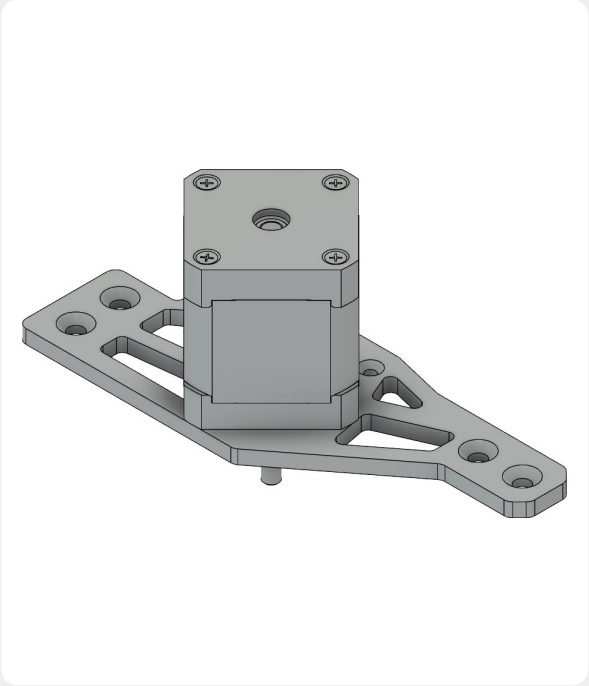
1 x B Motor Plate Bottom(Mirror)

CHECK BEFORE CONTINUING: Motor shaft is centered in the opening; cable exit remains accessible.

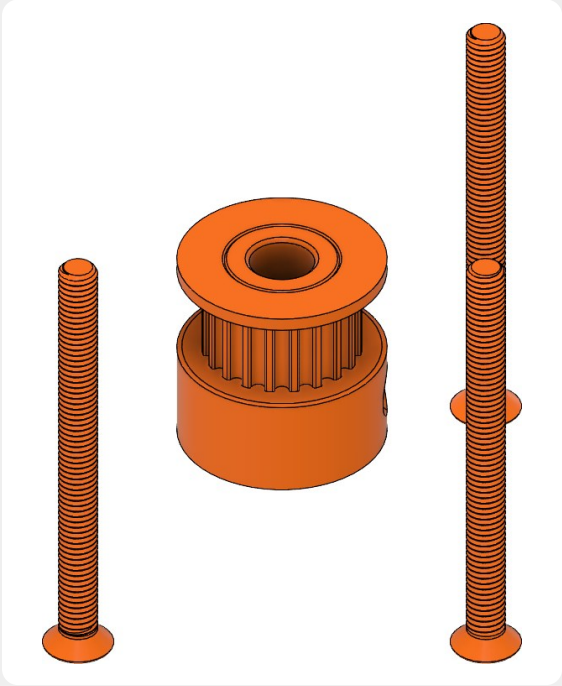
FIT THE MOTOR SCREWS AND PULLEY

LEFT REAR DRIVE | DETAIL 2

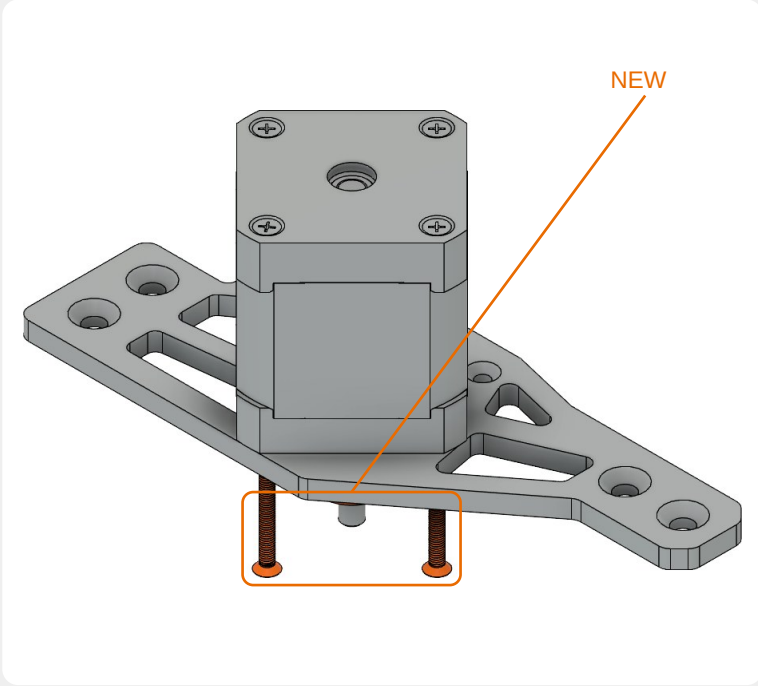
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the three M3 x 35 Torx flat-head screws at the highlighted motor mounting points. Slide the pulley onto the shaft in the illustrated orientation. Leave its final height adjustment until belt alignment.

PARTS IN THIS CAD STEP

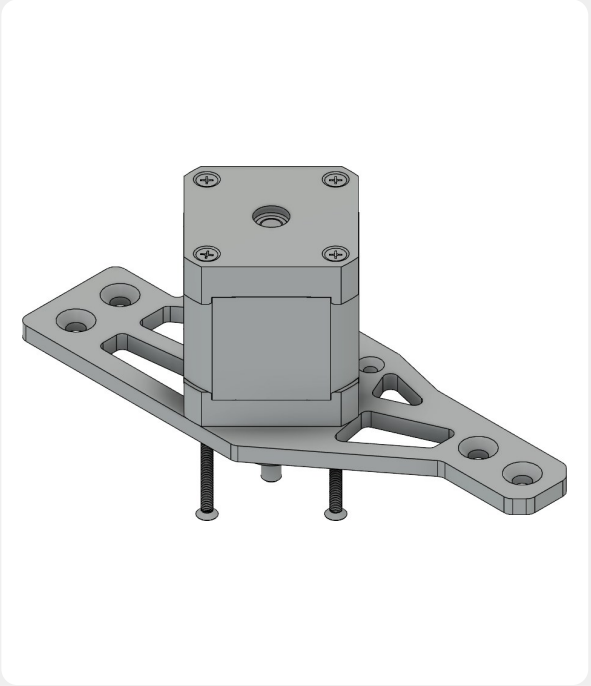
3 x M3 x 35 Torx flat-head [90236A127]
1 x GT2 20T 6MM pulley (CAD label)

CHECK BEFORE CONTINUING: Pulley does not rub the plate. Verify the actual shaft bore and grub screws before securing it.

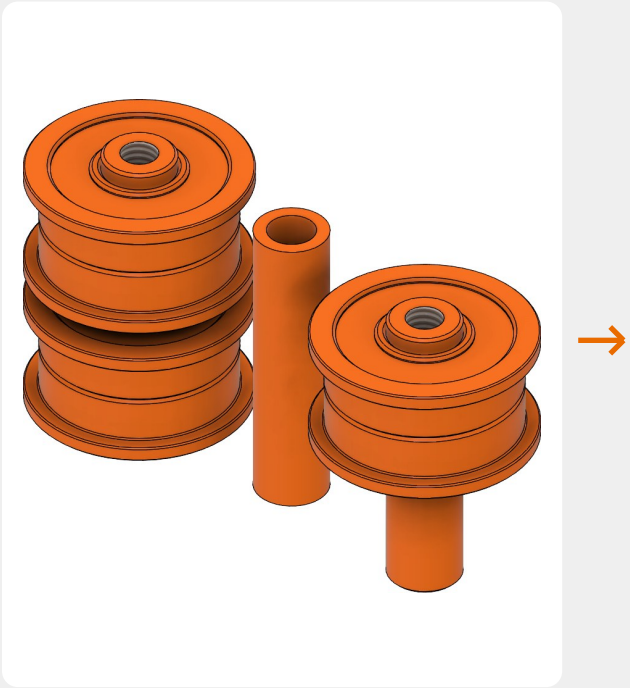
FIT THE SPACERS AND IDLER STACKS

LEFT REAR DRIVE | DETAIL 3

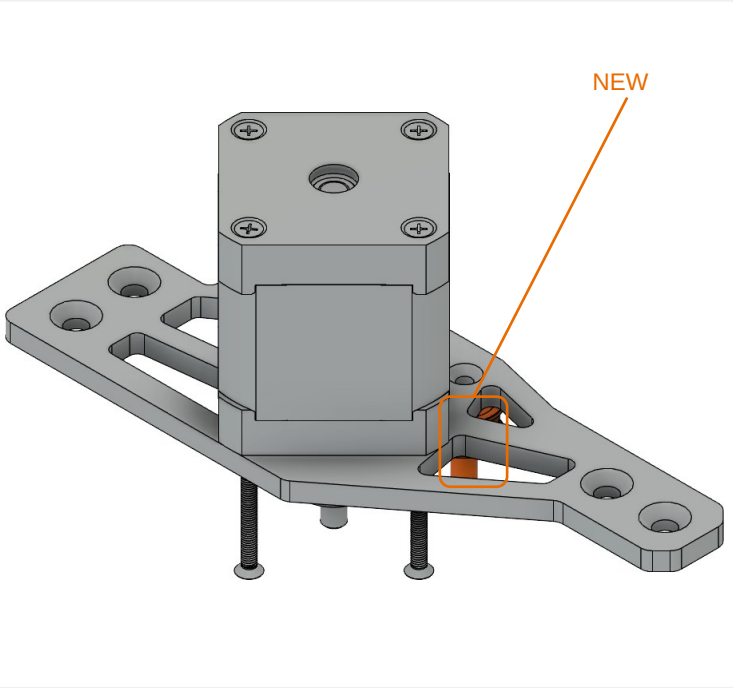
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the spacer and two idler pins with the bearing groups shown. Compare each pin separately: the bearing quantities and heights are not the same on both pins.

PARTS IN THIS CAD STEP

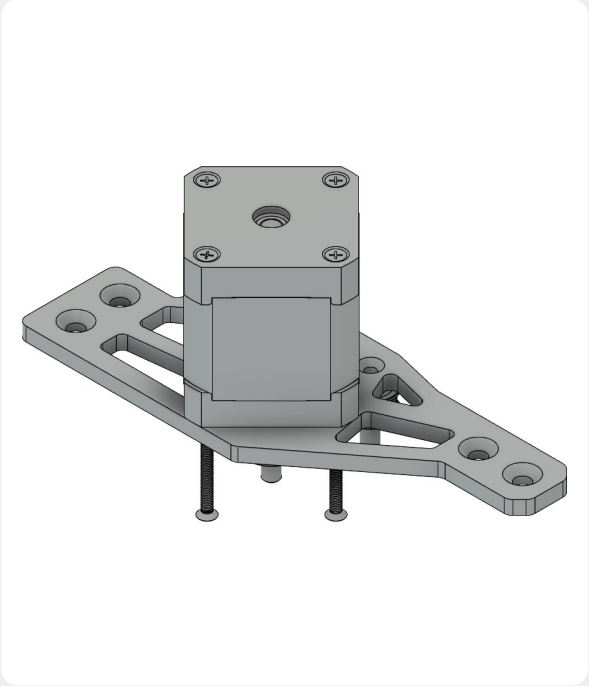
1 x 20 mm spacer
2 x 20 mm idler pin
6 x F695 flanged bearing

CHECK BEFORE CONTINUING: Every bearing turns freely and each flange faces the modeled direction.

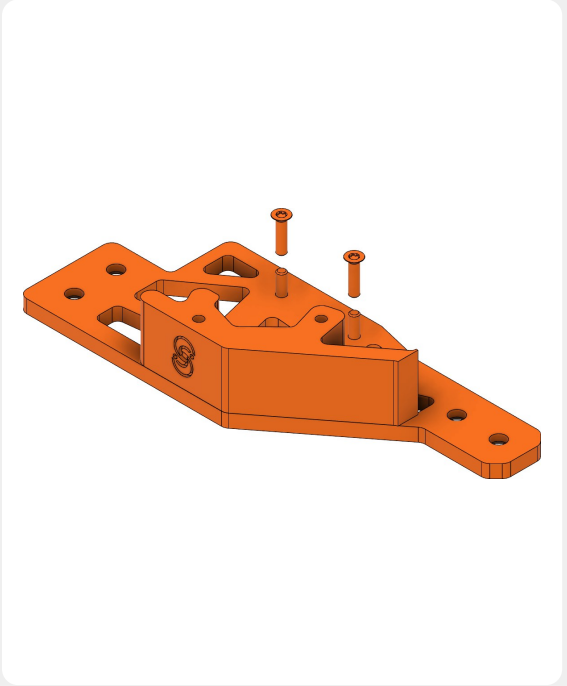
CLOSE THE DRIVE PLATES

LEFT REAR DRIVE | DETAIL 4

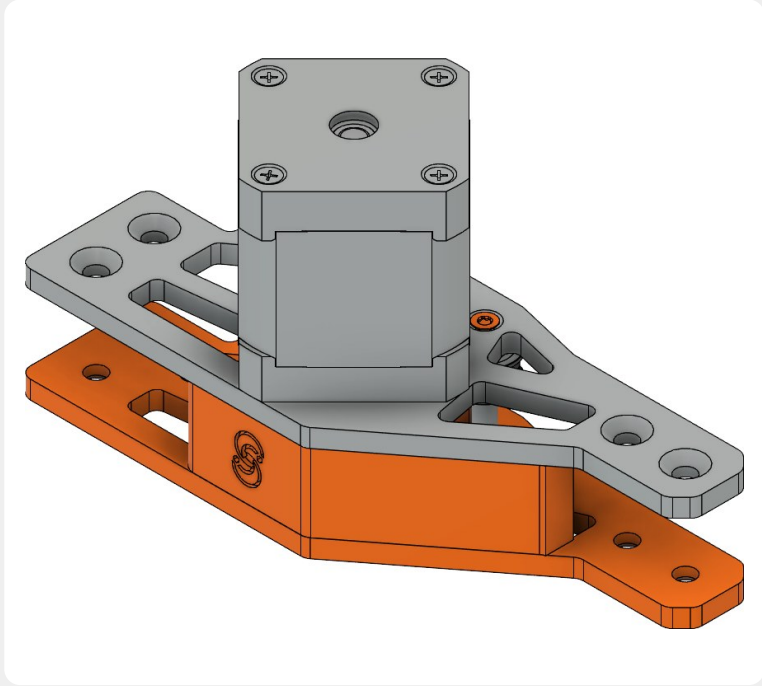
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the remaining plate and stiffener around the pins. Start the four M3 x 12 Torx flat-head retaining screws. Align the pin ends before tightening; do not use the screws to force the plates together.

PARTS IN THIS CAD STEP

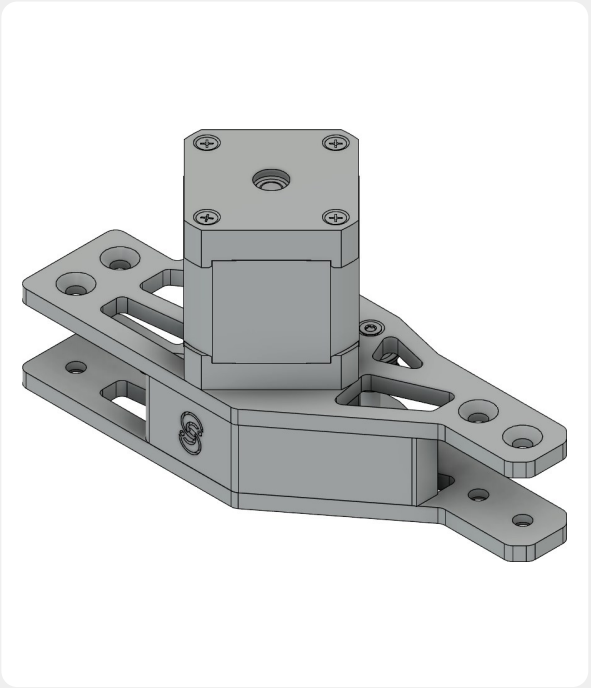
1 x B Motor Plate Top(Mirror)
1 x Back Motor Siffener left(Mirror)
4 x M3 x 12 Torx flat-head [90236A122]

CHECK BEFORE CONTINUING: Plates are parallel; rotating surfaces are not clamped. The right switch installation needs its own confirmed hardware detail.

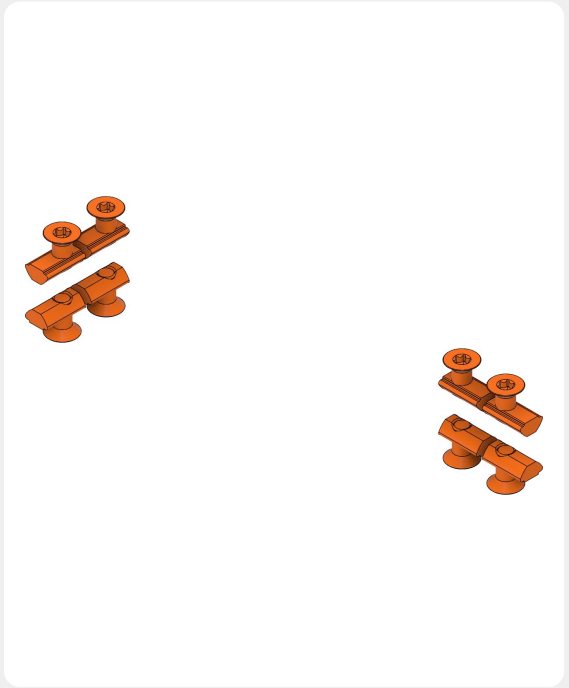
PREPARE THE FRAME FASTENERS

LEFT REAR DRIVE | DETAIL 5

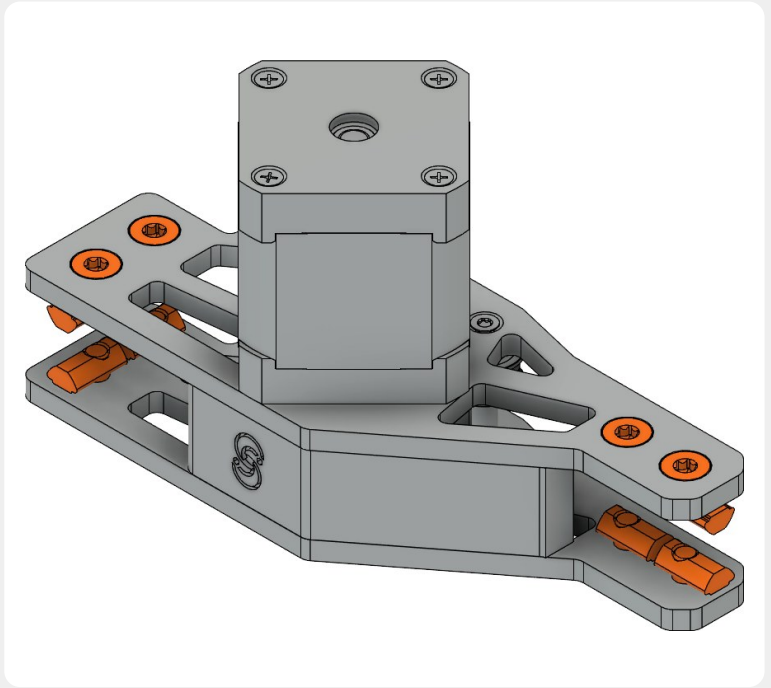
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Pair the frame screws and M5 T-nuts at the highlighted attachment points. Leave them loose for fitting and squaring the drive assembly on the frame.

PARTS IN THIS CAD STEP

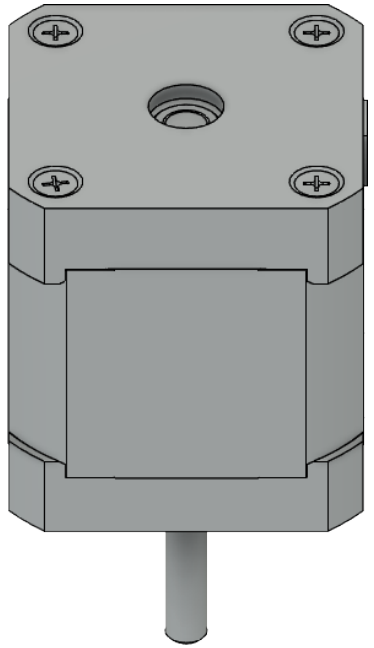
8 x M5 roll-in T-nut
8 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: All mounting points are accessible and nuts engage the extrusion slots before final tightening.

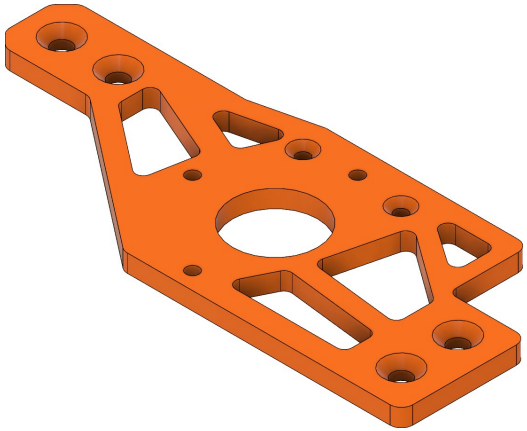
POSITION THE MOTOR MOUNTING PLATE

RIGHT REAR DRIVE | DETAIL 1

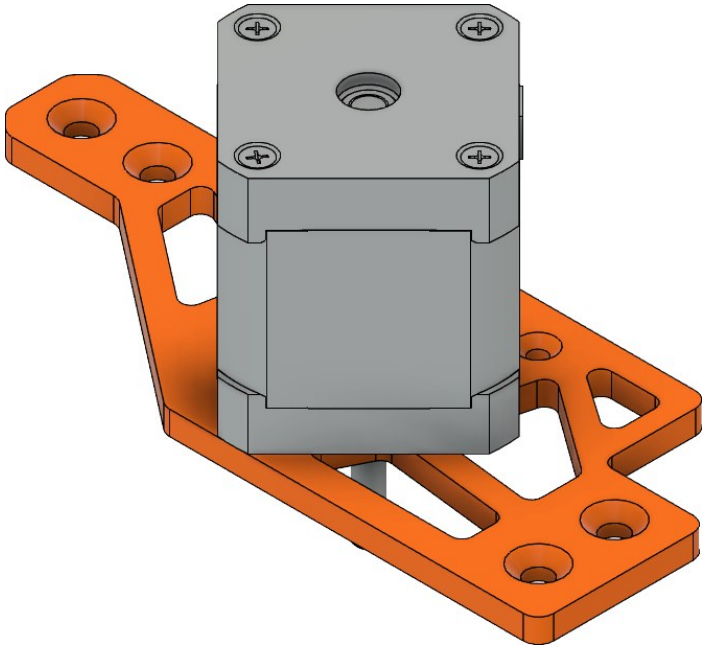
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the motor-side plate against the motor mounting face. Match the motor shaft opening and screw holes. Note: the CAD name says Bottom even though this plate is above the idlers in the assembled printer.

PARTS IN THIS CAD STEP

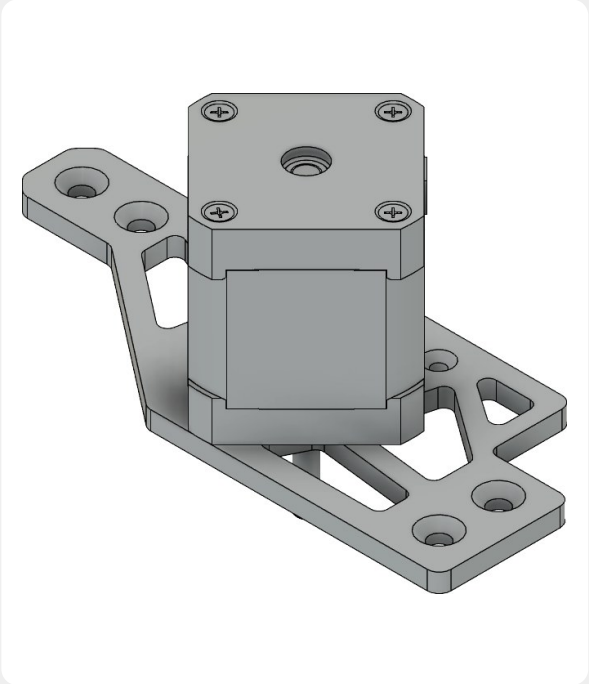
1 x B Motor Plate Bottom

CHECK BEFORE CONTINUING: Motor shaft is centered in the opening; cable exit remains accessible.

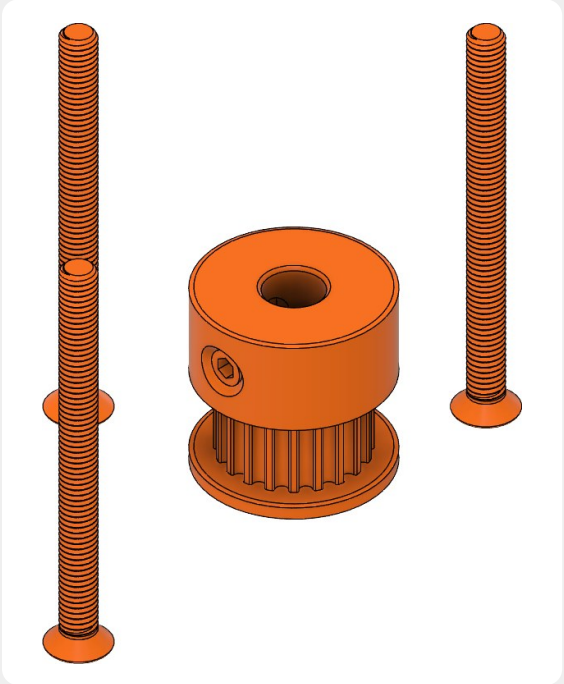
FIT THE MOTOR SCREWS AND PULLEY

RIGHT REAR DRIVE | DETAIL 2

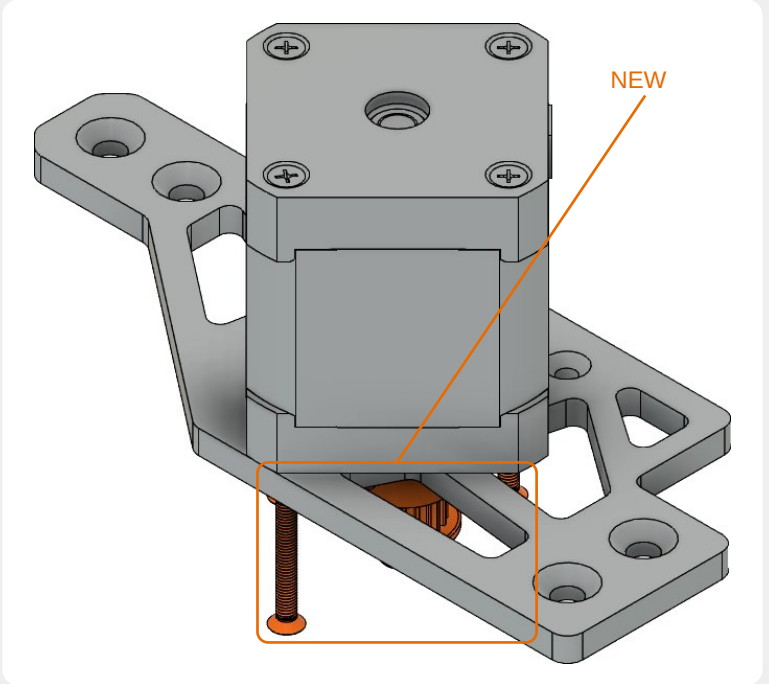
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the three M3 x 35 Torx flat-head screws at the highlighted motor mounting points. Slide the pulley onto the shaft in the illustrated orientation. Leave its final height adjustment until belt alignment.

PARTS IN THIS CAD STEP

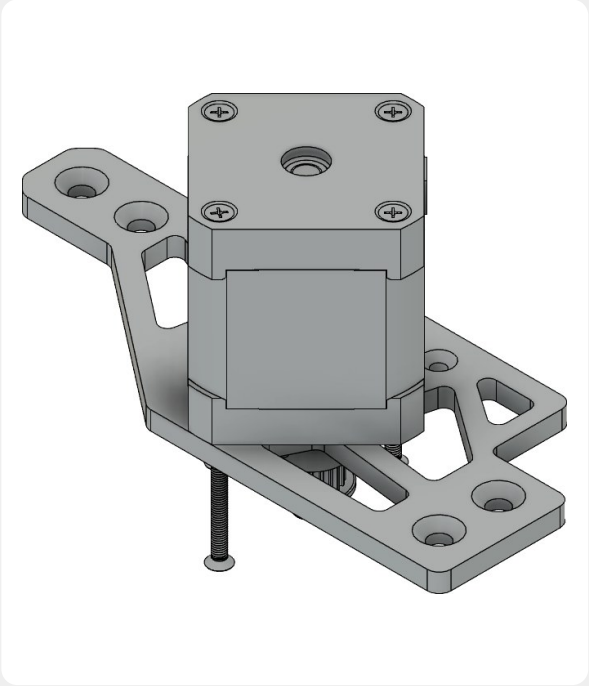
3 x M3 x 35 Torx flat-head [90236A127]
1 x GT2 20T 6MM pulley (CAD label)

CHECK BEFORE CONTINUING: Pulley does not rub the plate. Verify the actual shaft bore and grub screws before securing it.

FIT THE SPACERS AND IDLER STACKS

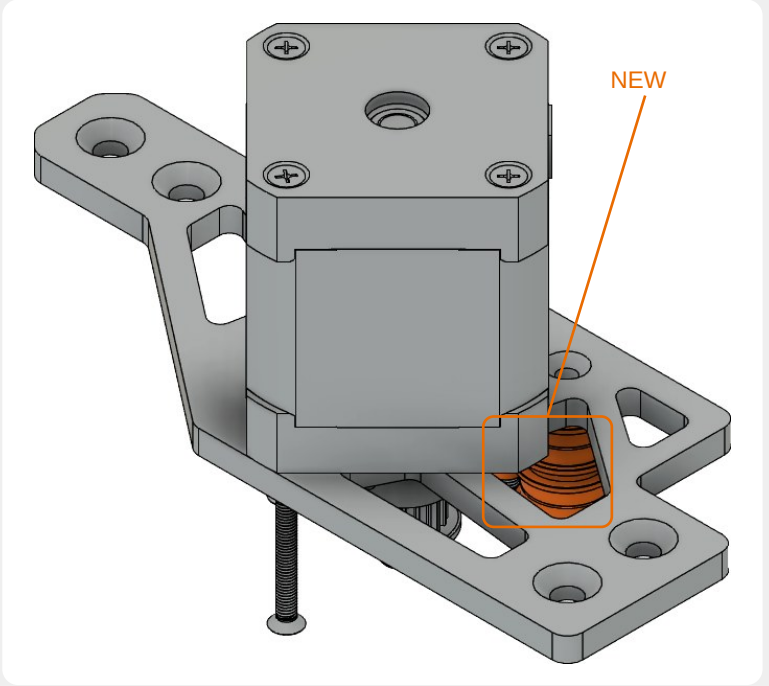
RIGHT REAR DRIVE | DETAIL 3

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the spacer and two idler pins with the bearing groups shown. Compare each pin separately: the bearing quantities and heights are not the same on both pins.

PARTS IN THIS CAD STEP

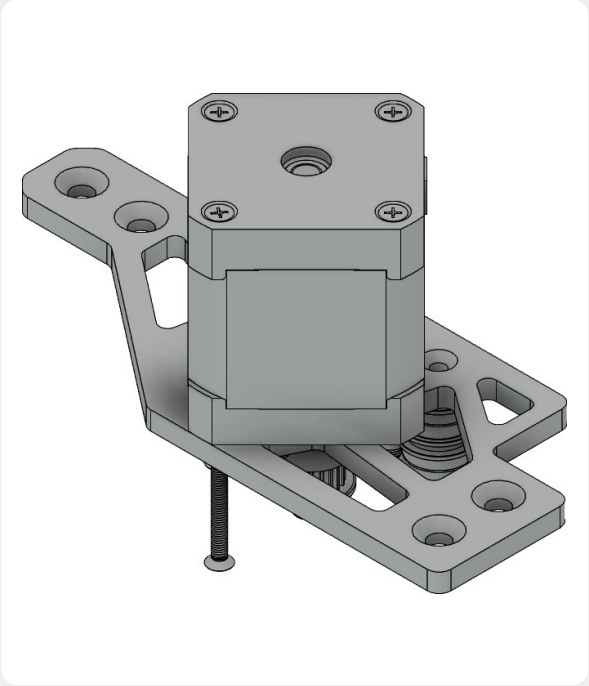
1 x 20 mm spacer
2 x 20 mm idler pin
6 x F695 flanged bearing

CHECK BEFORE CONTINUING: Every bearing turns freely and each flange faces the modeled direction.

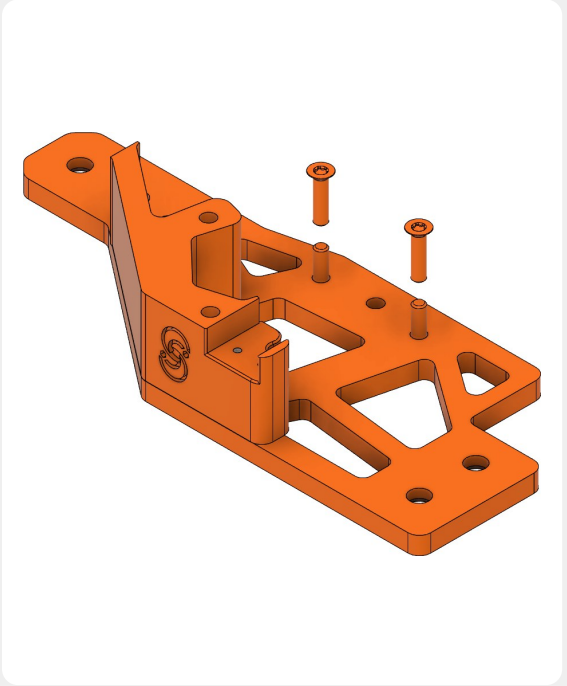
CLOSE THE DRIVE PLATES

RIGHT REAR DRIVE | DETAIL 4

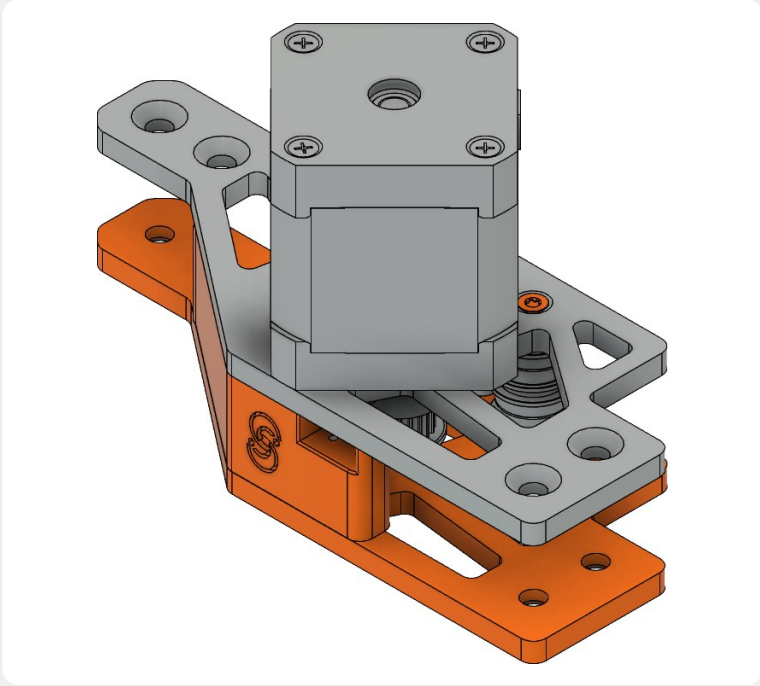
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the remaining plate and stiffener around the pins. Start the four M3 x 12 Torx flat-head retaining screws. Align the pin ends before tightening; do not use the screws to force the plates together.

PARTS IN THIS CAD STEP

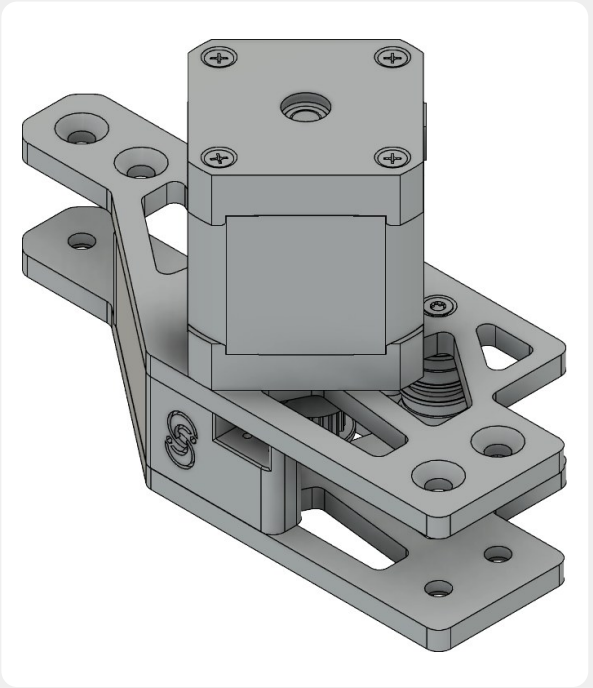
1 x B Motor Plate Top
1 x Back Motor Siffener Right
4 x M3 x 12 Torx flat-head [90236A122]

CHECK BEFORE CONTINUING: Plates are parallel; rotating surfaces are not clamped. The right switch installation needs its own confirmed hardware detail.

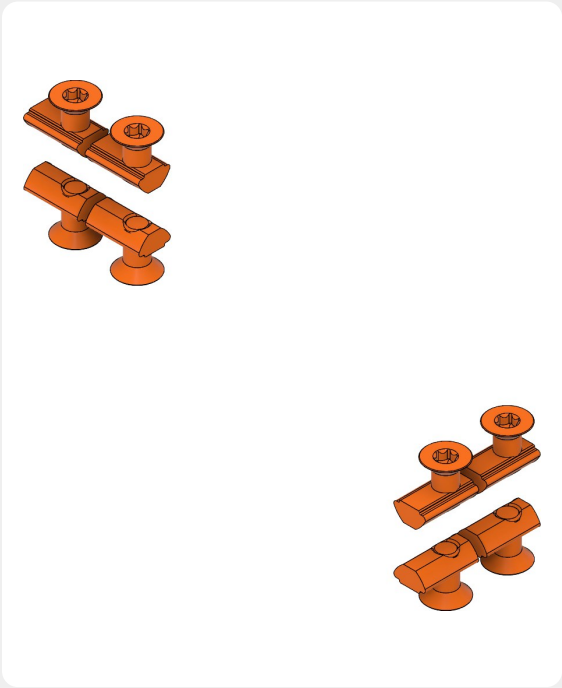
PREPARE THE FRAME FASTENERS

RIGHT REAR DRIVE | DETAIL 5

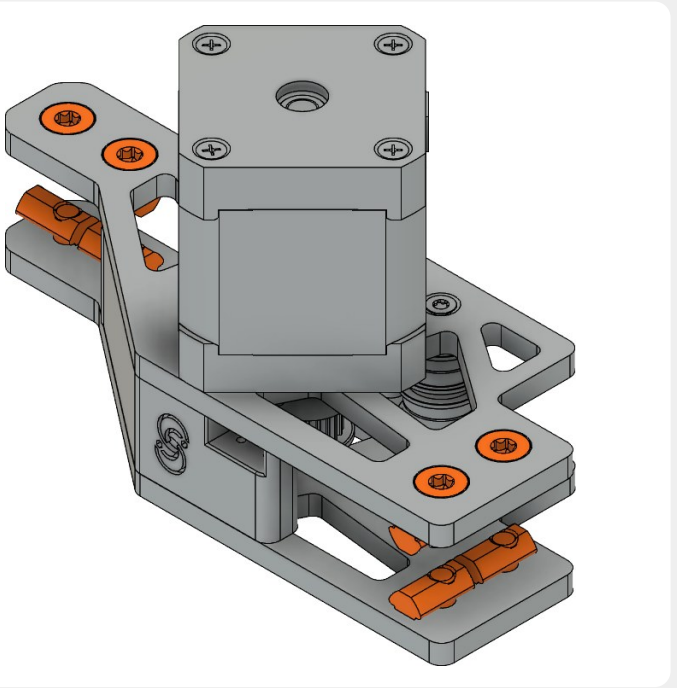
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

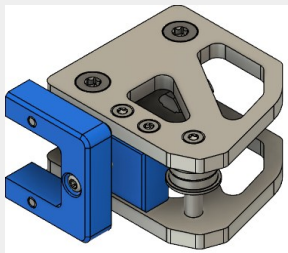
Pair the frame screws and M5 T-nuts at the highlighted attachment points. Leave them loose for fitting and squaring the drive assembly on the frame.

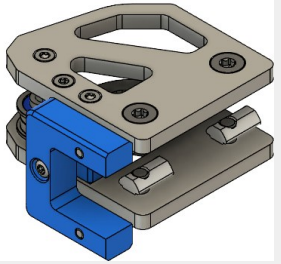
PARTS IN THIS CAD STEP

8 x M5 roll-in T-nut
8 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: All mounting points are accessible and nuts engage the extrusion slots before final tightening.

ASSEMBLE THE FRONT IDLERS





Build the two mirrored idler assemblies. Fit the tension blocks and adjusters in the orientation shown. Begin with useful adjustment travel available in both directions.

CAD ITEMS
Two plate pairs
Left/right tension blocks
Two 20 mm idler pin occurrences

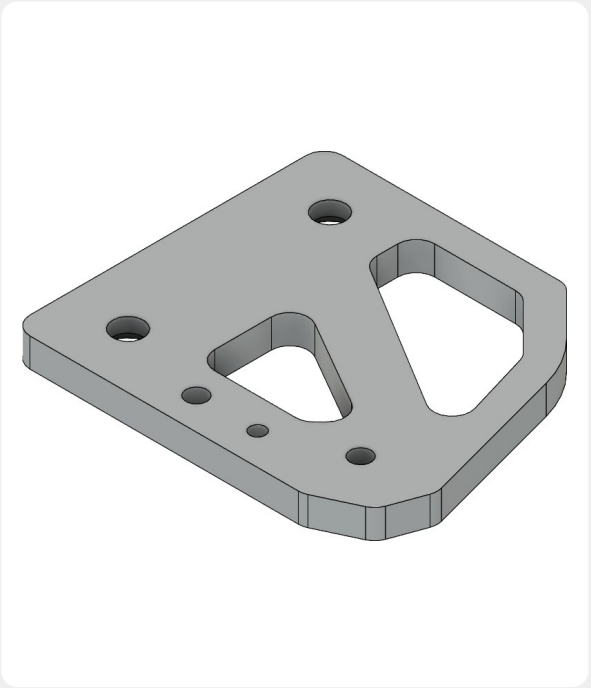
CHECK: The tension blocks move freely and bearings rotate without the plates clamping the rotating surfaces.

Current Hybrid assembly

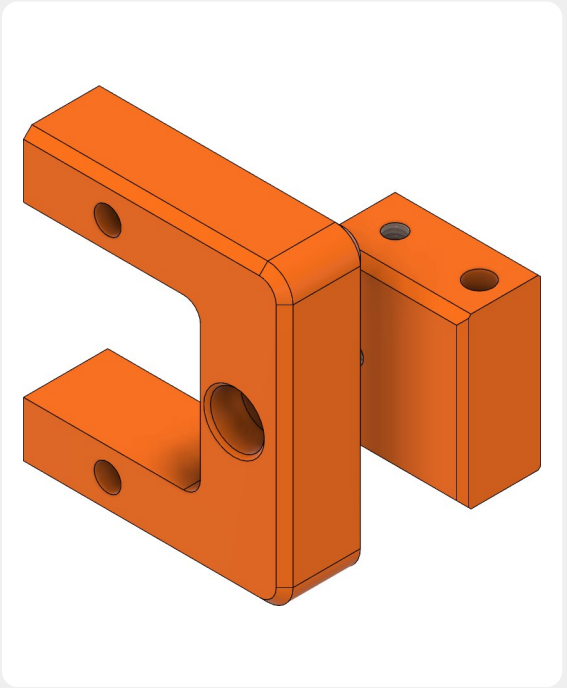
FIT THE TENSION BLOCK

LEFT FRONT IDLER | DETAIL 1

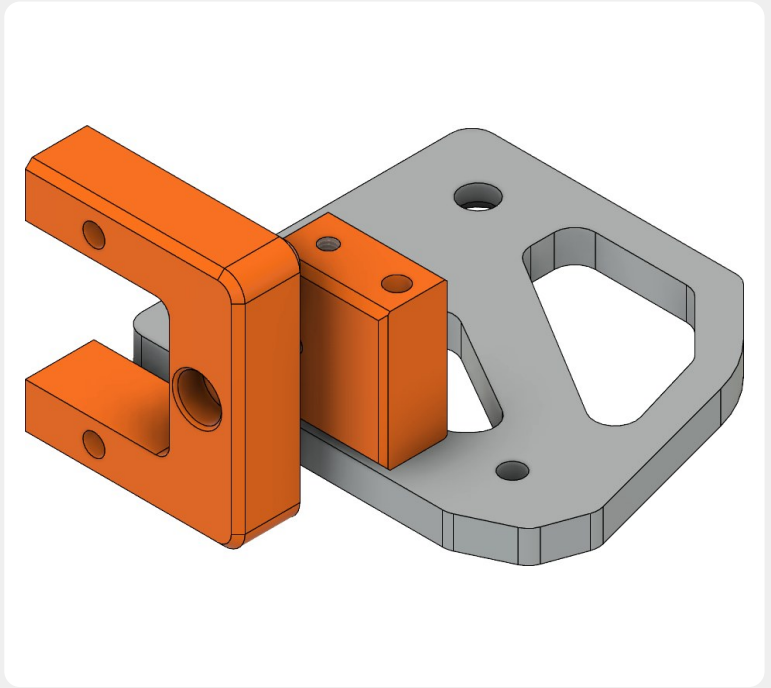
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the tension block and its matching adjuster support on the lower plate. Match the holes and the open side shown. Keep the block free to slide.

PARTS IN THIS CAD STEP

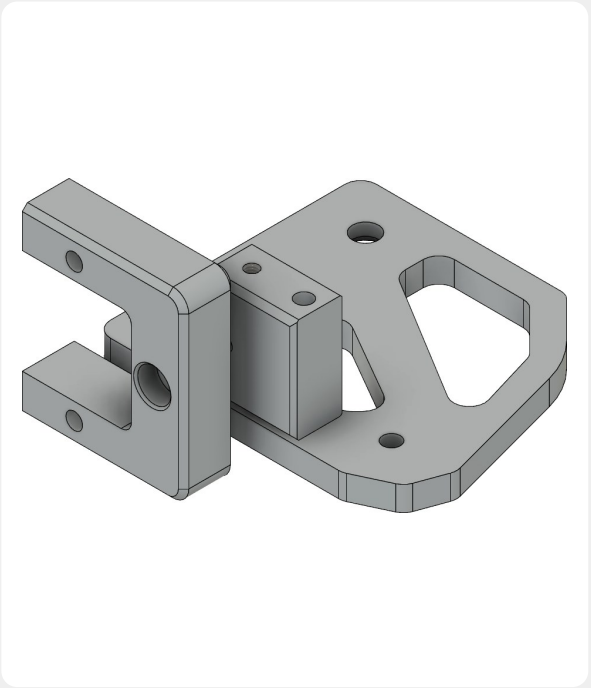
1 x Idler Tension Block Left
1 x Front idler Tensor Left

CHECK BEFORE CONTINUING: The block is the correct handed part and moves without scraping.

BUILD THE IDLER BEARING STACK

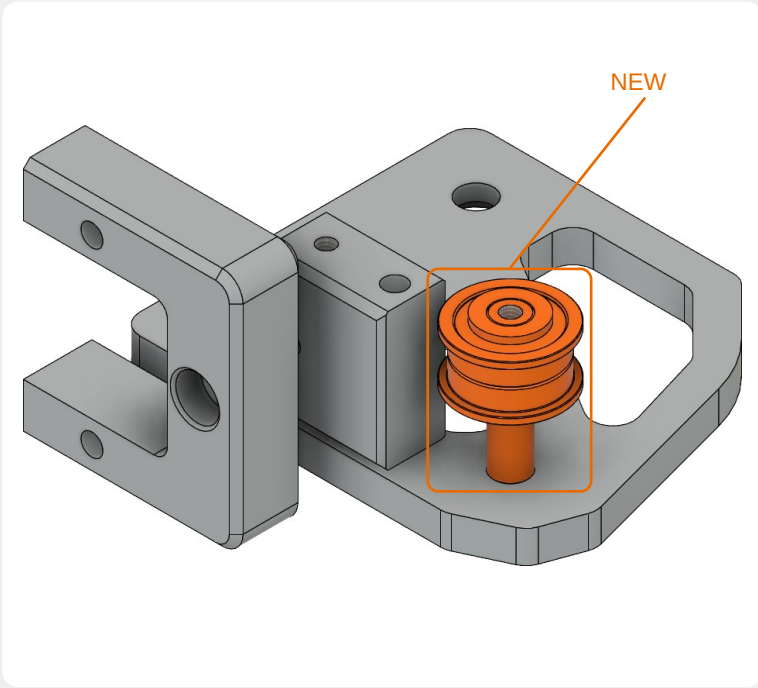
LEFT FRONT IDLER | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the pin, washer and two F695 bearings at the idler location. Use the parts-only view to match the washer end and the two bearing flanges.

PARTS IN THIS CAD STEP

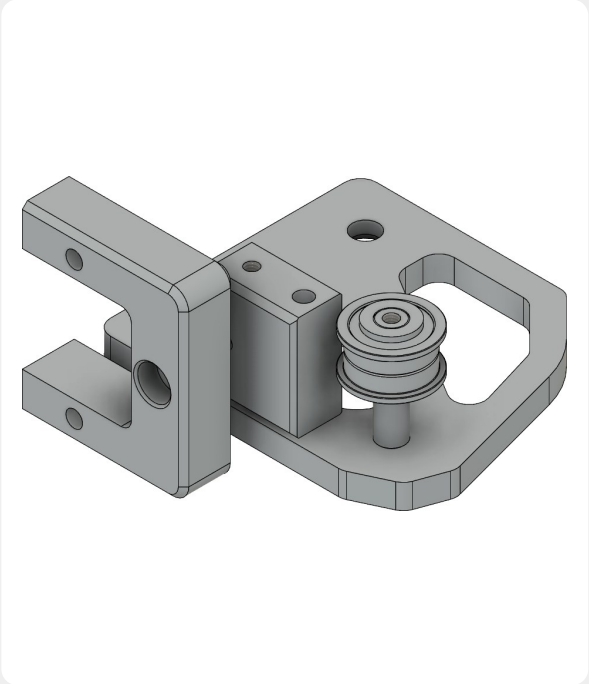
1 x 20 mm idler pin
1 x M5 washer
2 x F695 flanged bearing

CHECK BEFORE CONTINUING: Both bearings rotate freely. The left and right stacks occupy different belt heights.

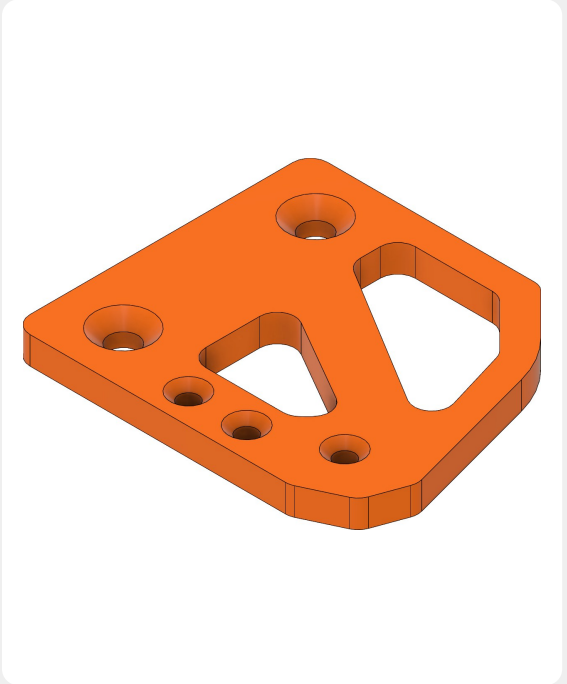
CLOSE THE PLATE PAIR

LEFT FRONT IDLER | DETAIL 3

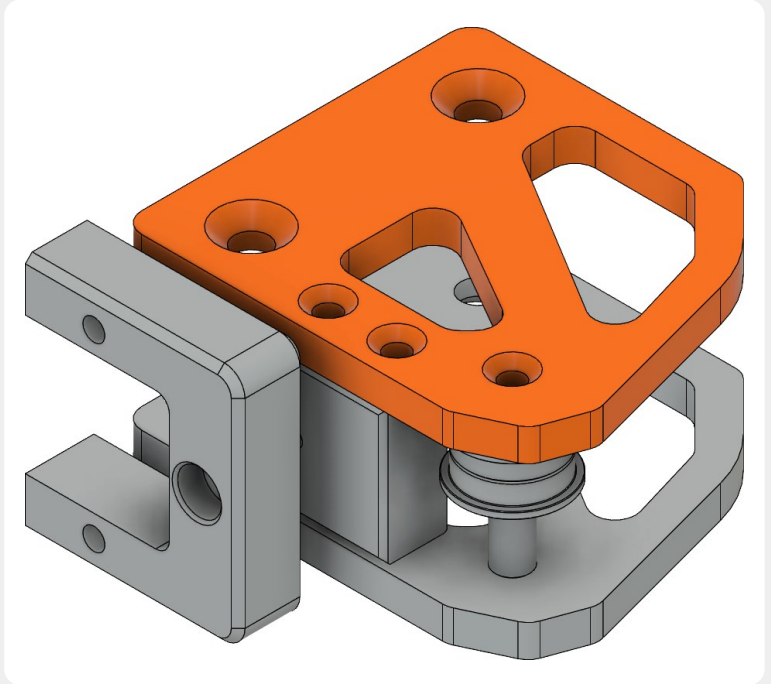
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Lower the upper plate over the pin and supports. Keep the pin ends aligned with their holes; do not pull a misaligned plate into place using screws.

PARTS IN THIS CAD STEP

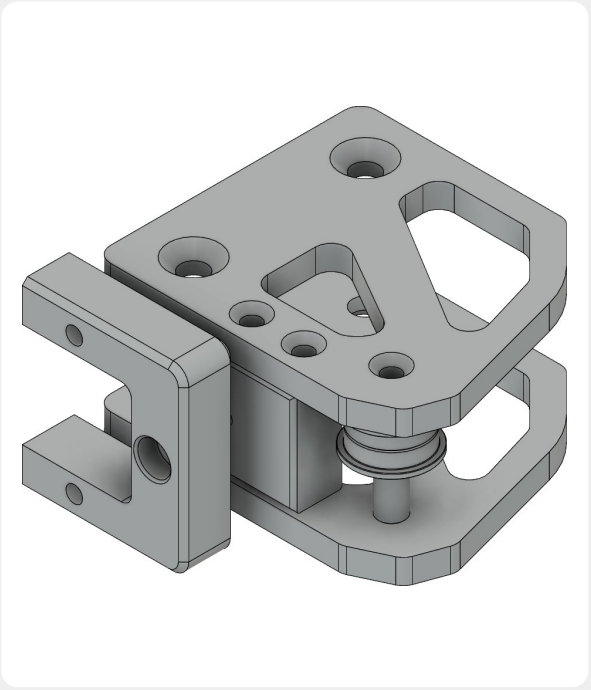
1 x Front idler Left Top

CHECK BEFORE CONTINUING: The plates sit parallel without squeezing the rotating bearing surfaces.

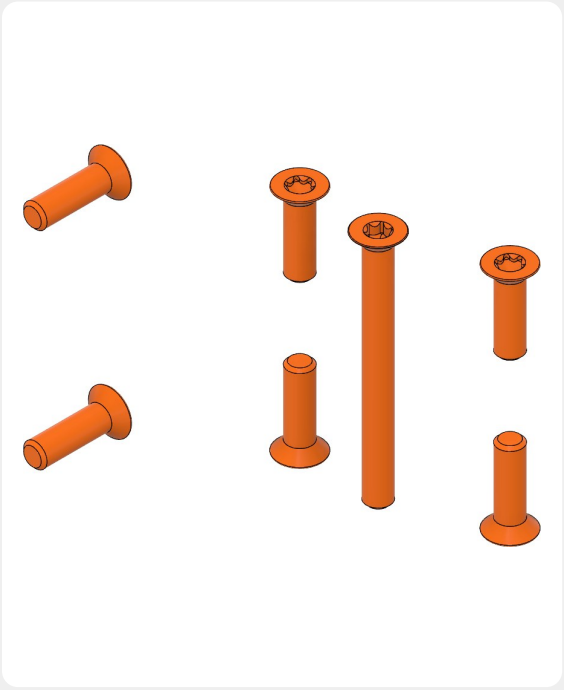
FIT THE RETAINING SCREWS

LEFT FRONT IDLER | DETAIL 4

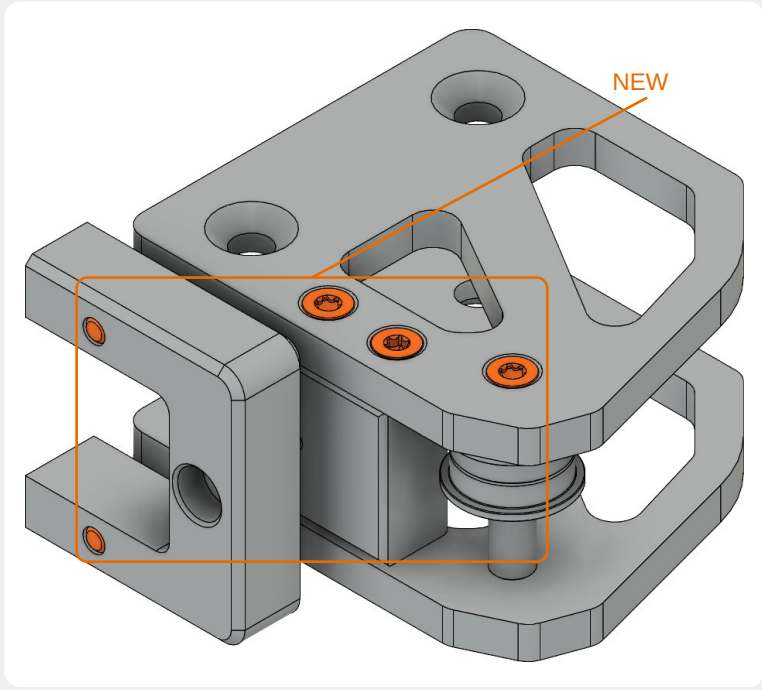
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use six M3 x 10 Torx flat-head screws and one M3 x 30 Torx flat-head screw at the highlighted locations. Start every thread before tightening. Keep the sliding tension block free to move.

PARTS IN THIS CAD STEP

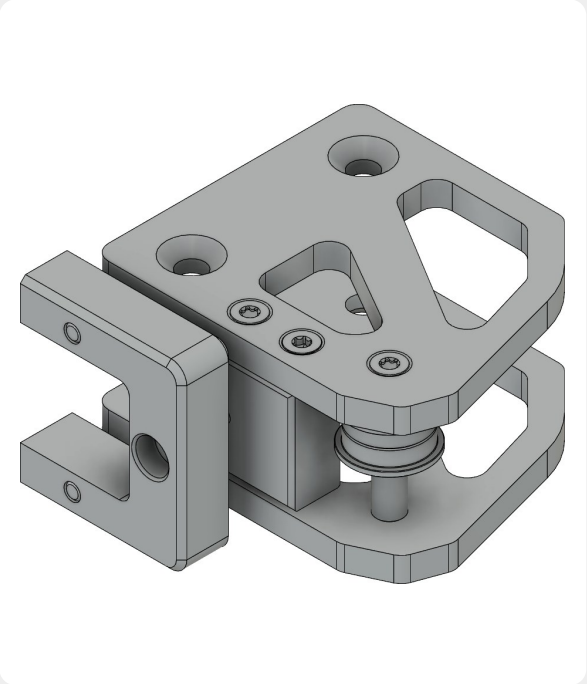
6 x M3 x 10 Torx flat-head [90236A121]
1 x M3 x 30 Torx flat-head [92703A457]

CHECK BEFORE CONTINUING: Match the kit hardware to the listed sizes and check thread engagement. The tension block must still slide and the idler must spin.

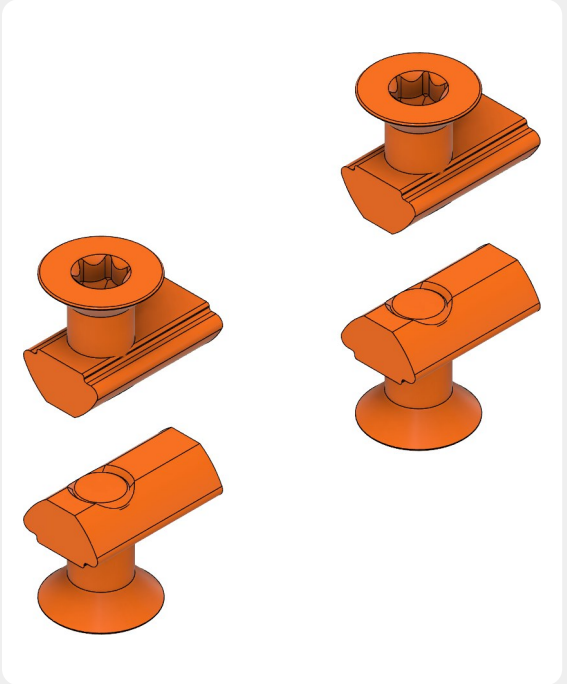
PREPARE FRAME MOUNTING HARDWARE

LEFT FRONT IDLER | DETAIL 5

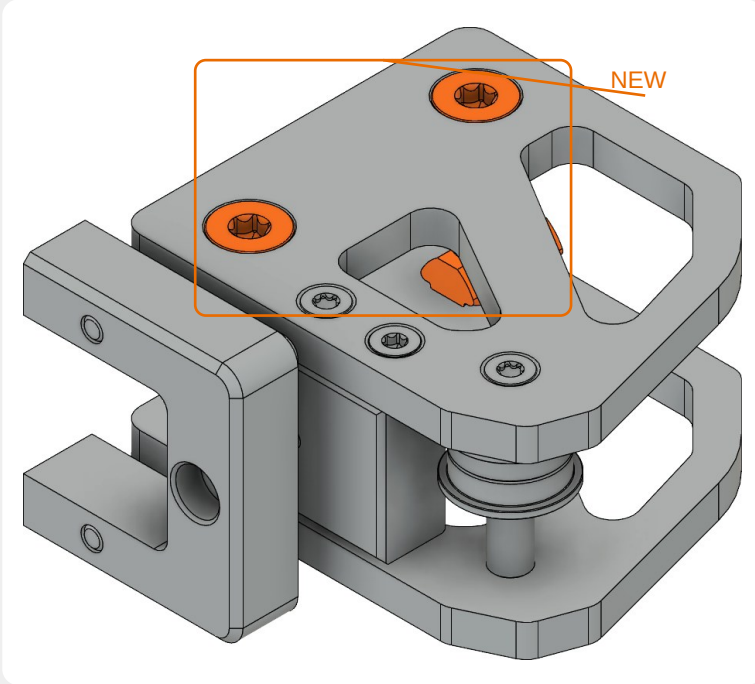
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Pair the frame screws with the M5 roll-in T-nuts at the highlighted slots. Start the nuts only loosely before placing the assembly on the frame.

PARTS IN THIS CAD STEP

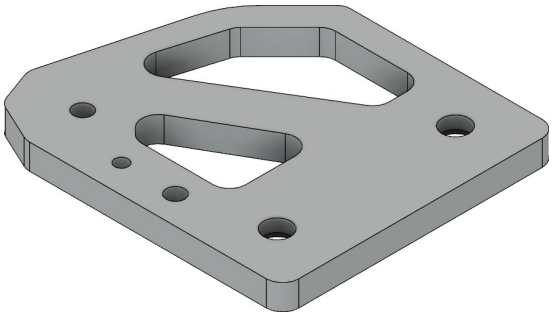
4 x M5 roll-in T-nut
4 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: Each T-nut engages the slot correctly. Tension-adjuster screw specification and the overlapping left CAD entries need review.

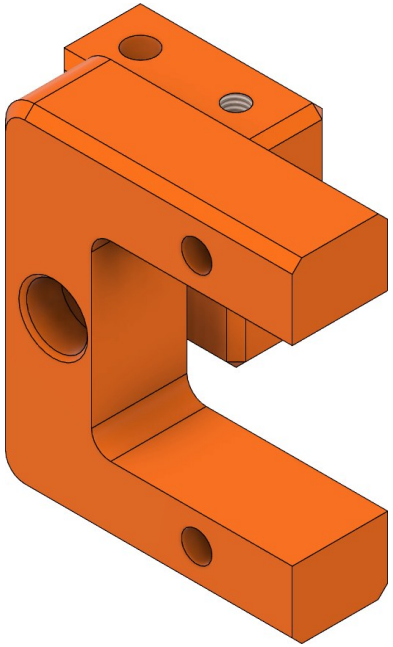
FIT THE TENSION BLOCK

RIGHT FRONT IDLER | DETAIL 1

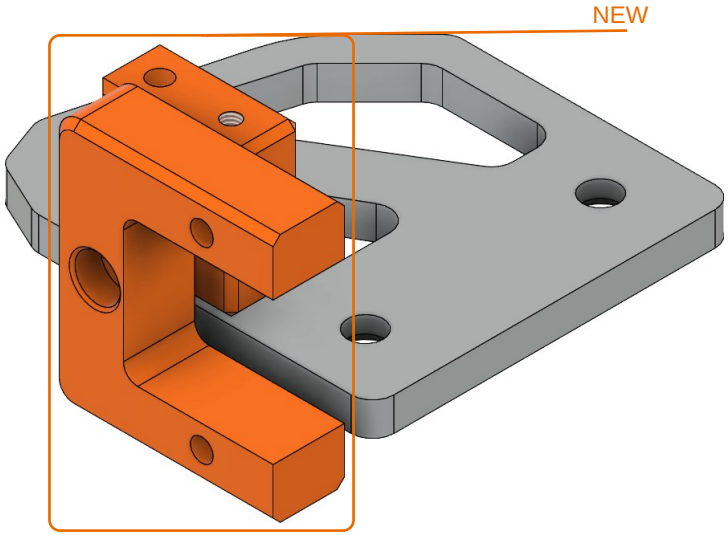
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the tension block and its matching adjuster support on the lower plate. Match the holes and the open side shown. Keep the block free to slide.

PARTS IN THIS CAD STEP

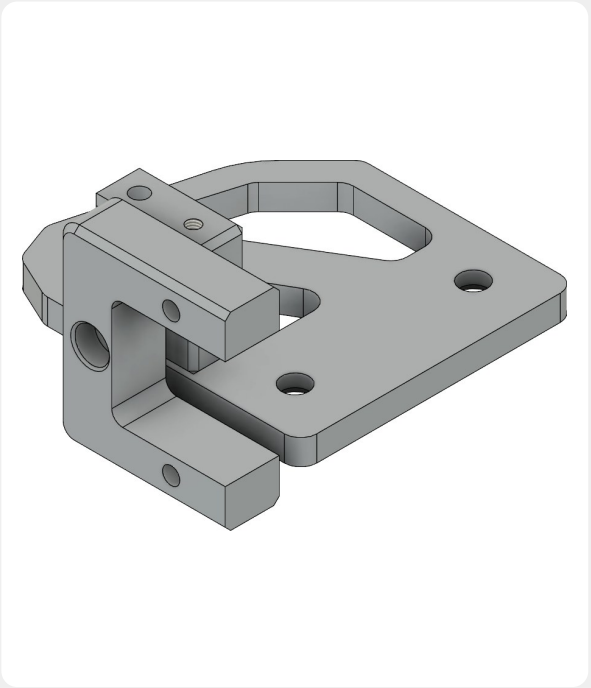
1 x Idler Tension Block Right
1 x Front idler Tensor Left(Mirror)

CHECK BEFORE CONTINUING: The block is the correct handed part and moves without scraping.

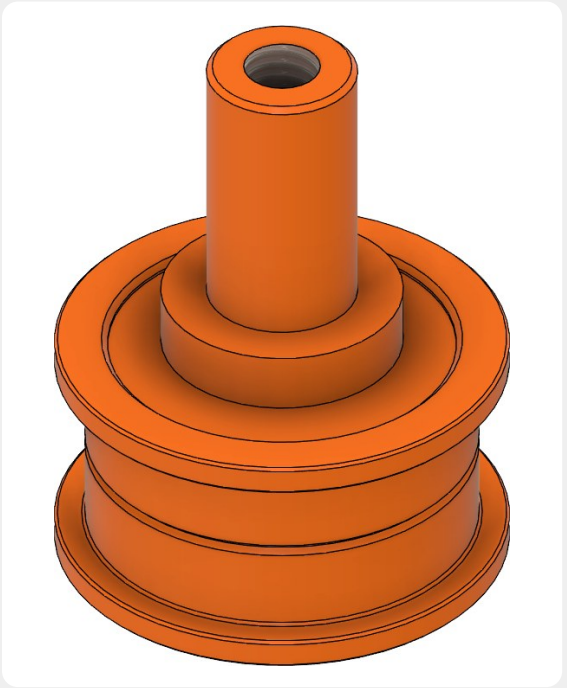
BUILD THE IDLER BEARING STACK

RIGHT FRONT IDLER | DETAIL 2

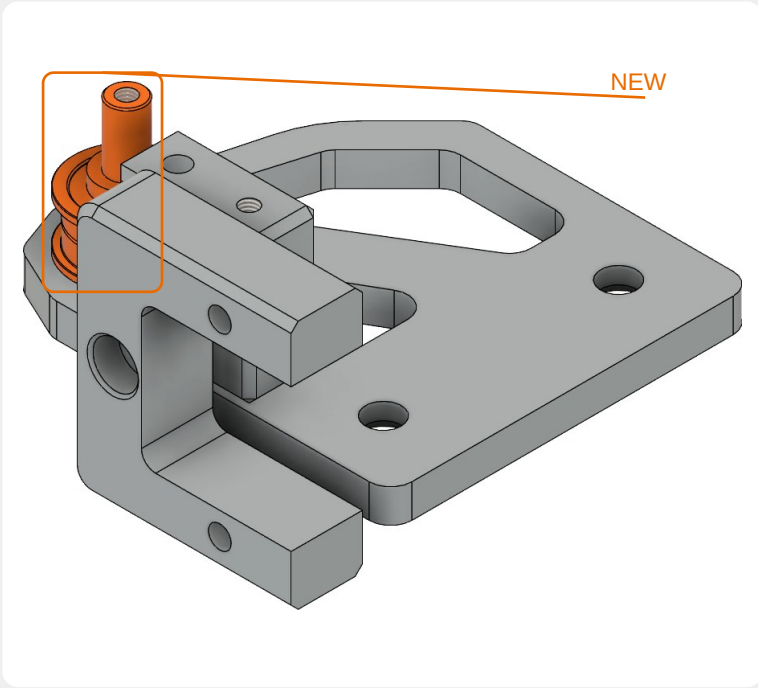
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the pin, washer and two F695 bearings at the idler location. Use the parts-only view to match the washer end and the two bearing flanges.

PARTS IN THIS CAD STEP

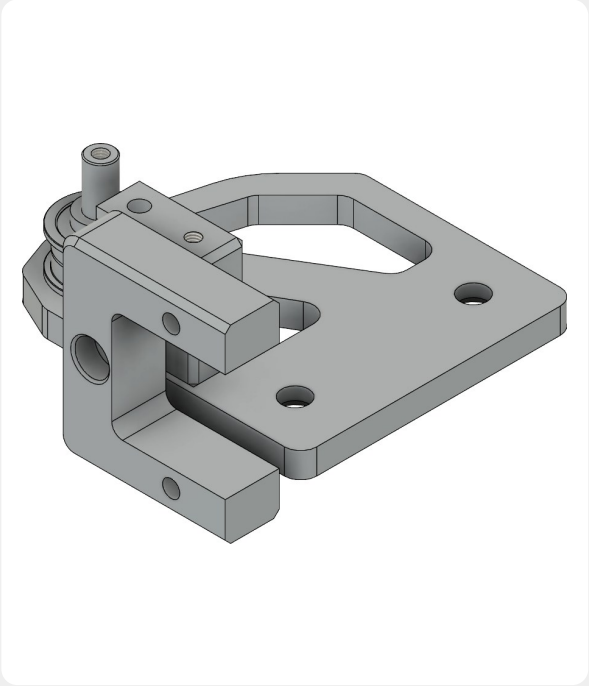
1 x 20 mm idler pin
1 x M5 washer
2 x F695 flanged bearing

CHECK BEFORE CONTINUING: Both bearings rotate freely. The left and right stacks occupy different belt heights.

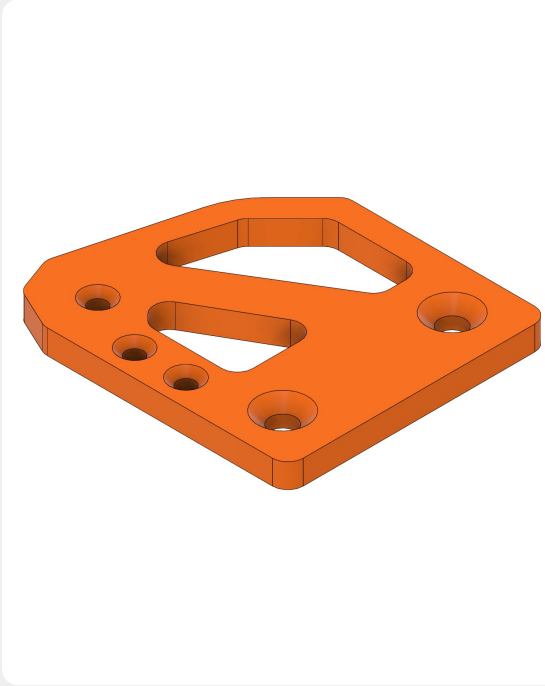
CLOSE THE PLATE PAIR

RIGHT FRONT IDLER | DETAIL 3

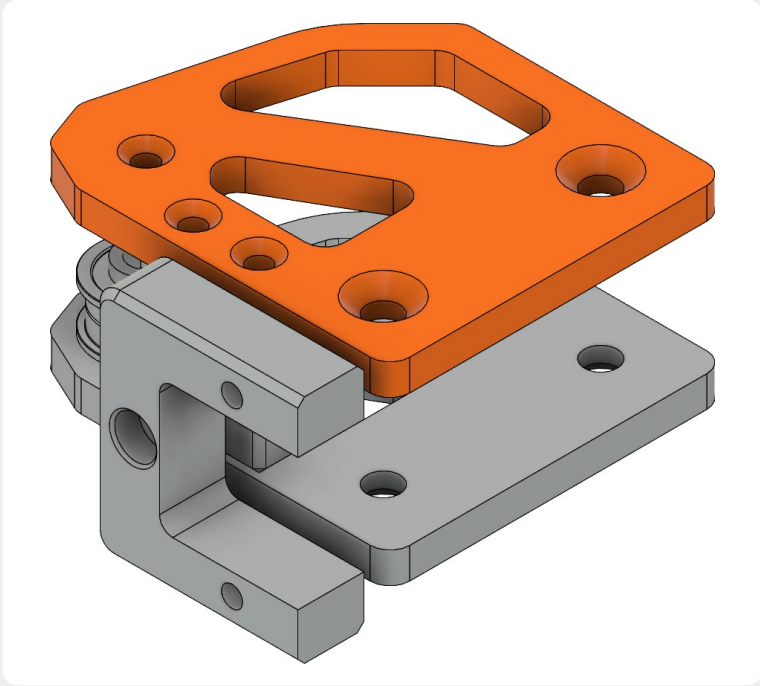
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Lower the upper plate over the pin and supports. Keep the pin ends aligned with their holes; do not pull a misaligned plate into place using screws.

PARTS IN THIS CAD STEP

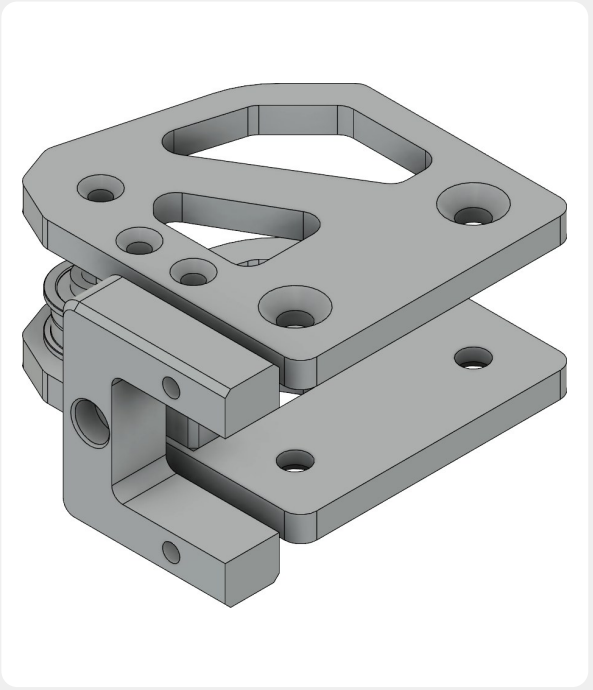
1 x Front idler Left Top(Mirror)

CHECK BEFORE CONTINUING: The plates sit parallel without squeezing the rotating bearing surfaces.

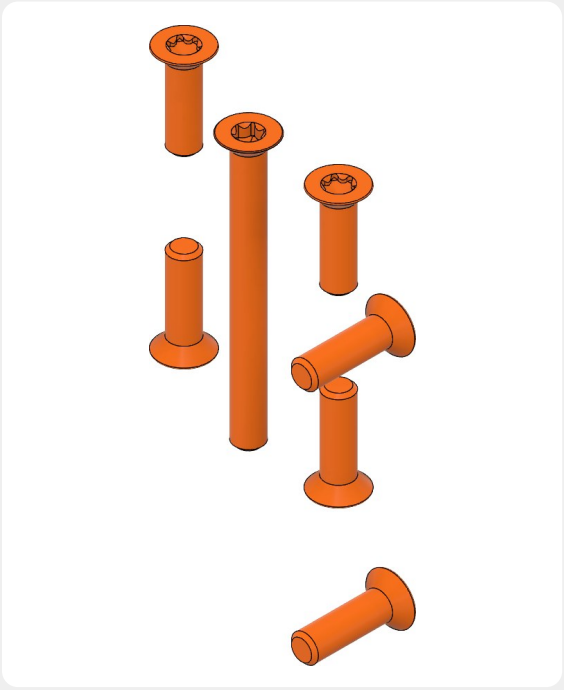
FIT THE RETAINING SCREWS

RIGHT FRONT IDLER | DETAIL 4

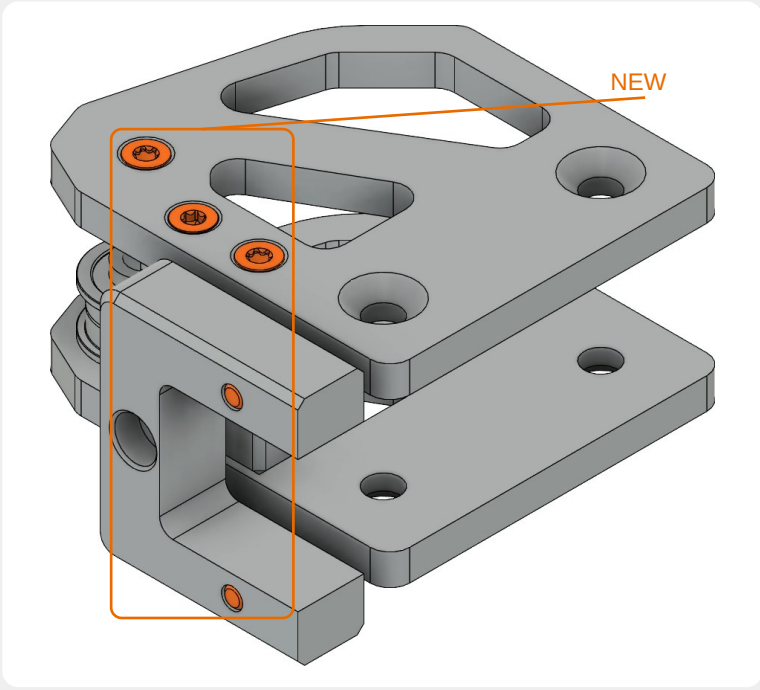
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use six M3 x 10 Torx flat-head screws and one M3 x 30 Torx flat-head screw at the highlighted locations. Start every thread before tightening. Keep the sliding tension block free to move.

PARTS IN THIS CAD STEP

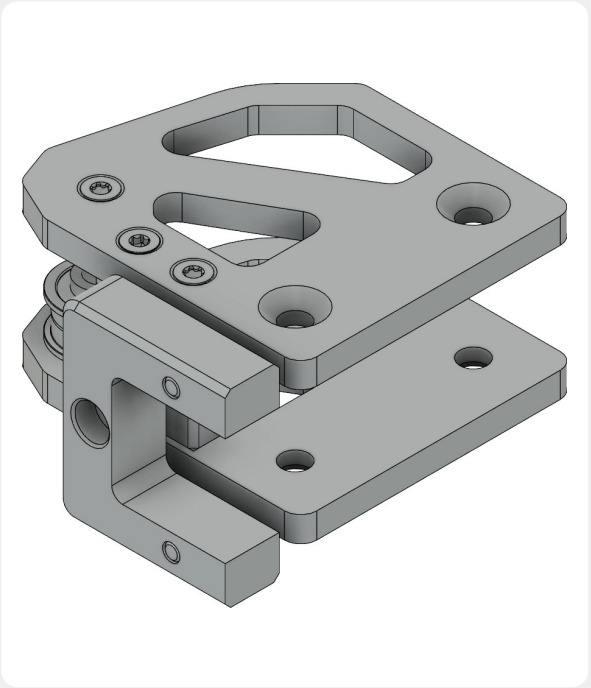
6 x M3 x 10 Torx flat-head [90236A121]
1 x M3 x 30 Torx flat-head [92703A457]

CHECK BEFORE CONTINUING: Match the kit hardware to the listed sizes and check thread engagement. The tension block must still slide and the idler must spin.

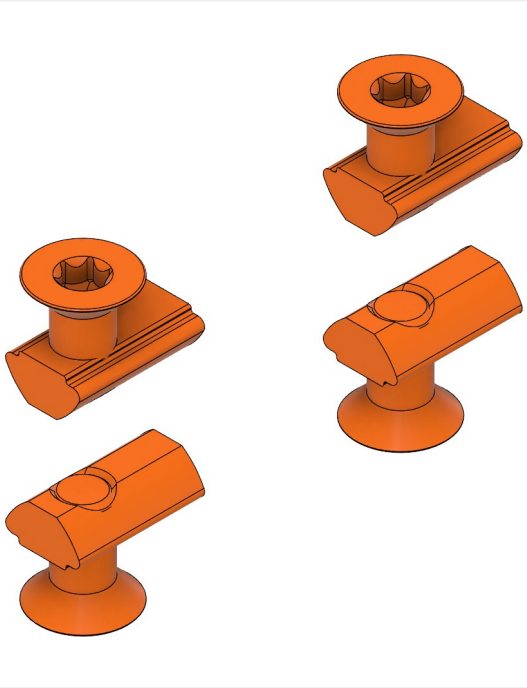
PREPARE FRAME MOUNTING HARDWARE

RIGHT FRONT IDLER | DETAIL 5

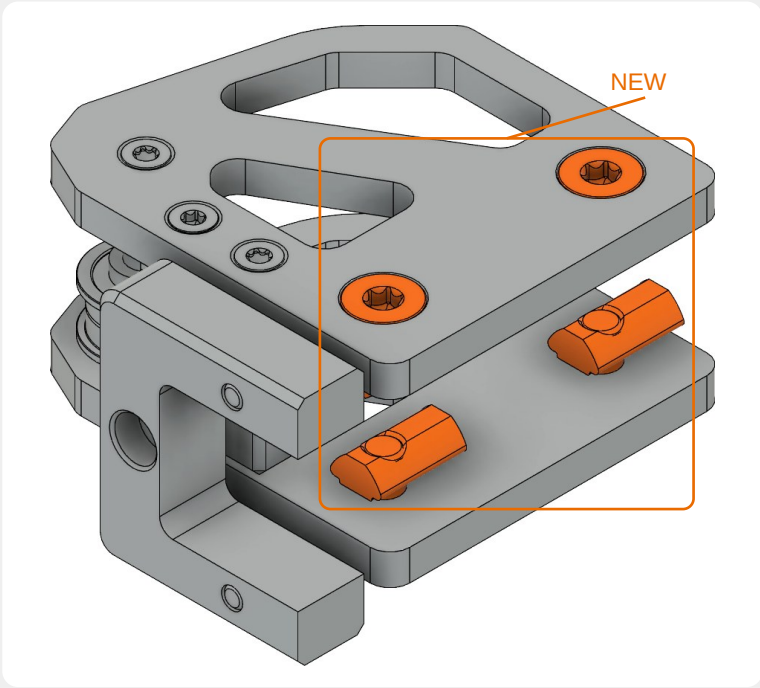
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

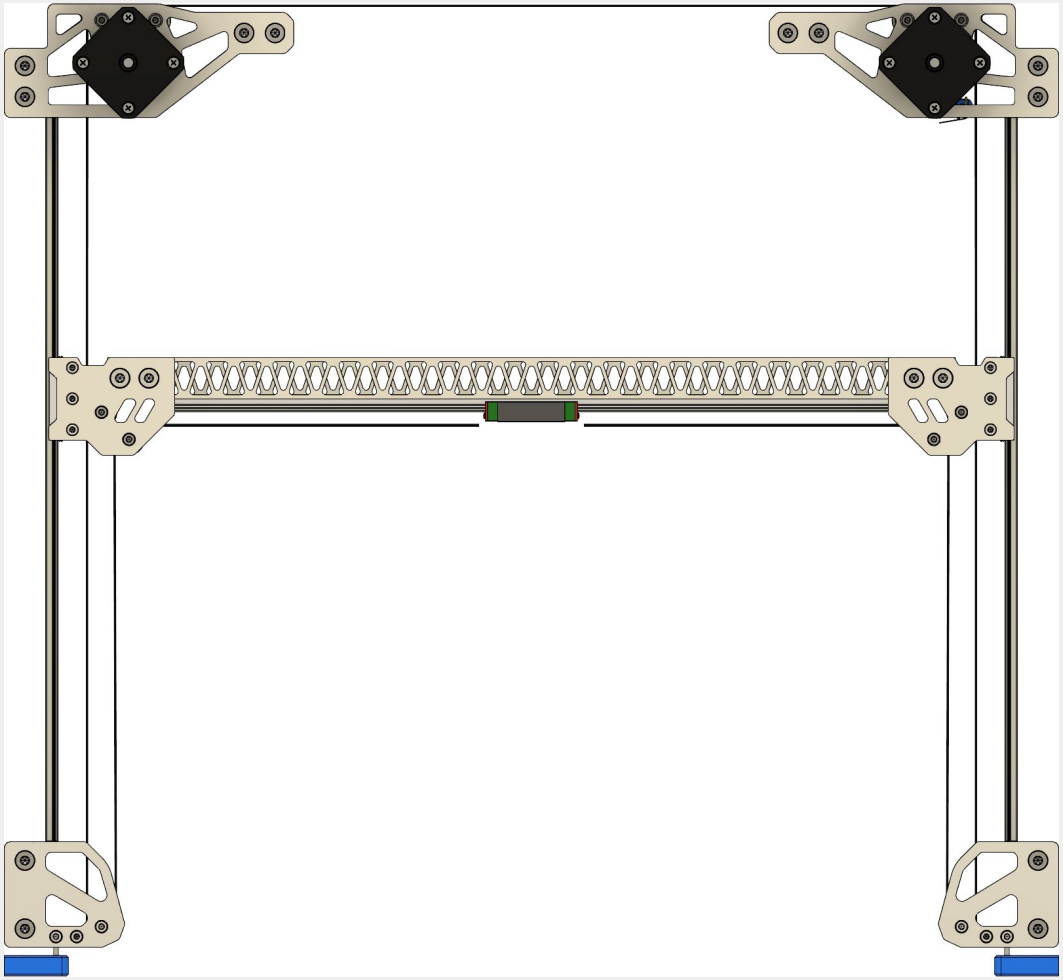
Pair the frame screws with the M5 roll-in T-nuts at the highlighted slots. Start the nuts only loosely before placing the assembly on the frame.

PARTS IN THIS CAD STEP

4 x M5 roll-in T-nut
4 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: Each T-nut engages the slot correctly. Tension-adjuster screw specification and the overlapping left CAD entries need review.

IDENTIFY THE TWO BELT PATHS



Use this top view to identify the rear drives, front idlers and XY joints. Trace each belt through its own plane before fastening its ends. Check the belt path from both above and below.

MODEL REFERENCE
Two belt bodies are present.
This view does not establish the final toolhead clamp.

CHECK: Resolve the toolhead and belt termination first. Do not cut belts to length or set final tension from this overview alone.

Current Hybrid assembly

Confirm the complete route

Record a separate route for each belt, including toothed/smooth contact faces, upper/lower plane and end clamp. The exported view confirms the general layout, but the incomplete toolhead prevents a verified termination instruction.

Install without twist

Keep each belt in its intended plane. Move the gantry slowly through the entire usable area and look for flange rubbing, crossed spans and contact with fasteners or printed parts.

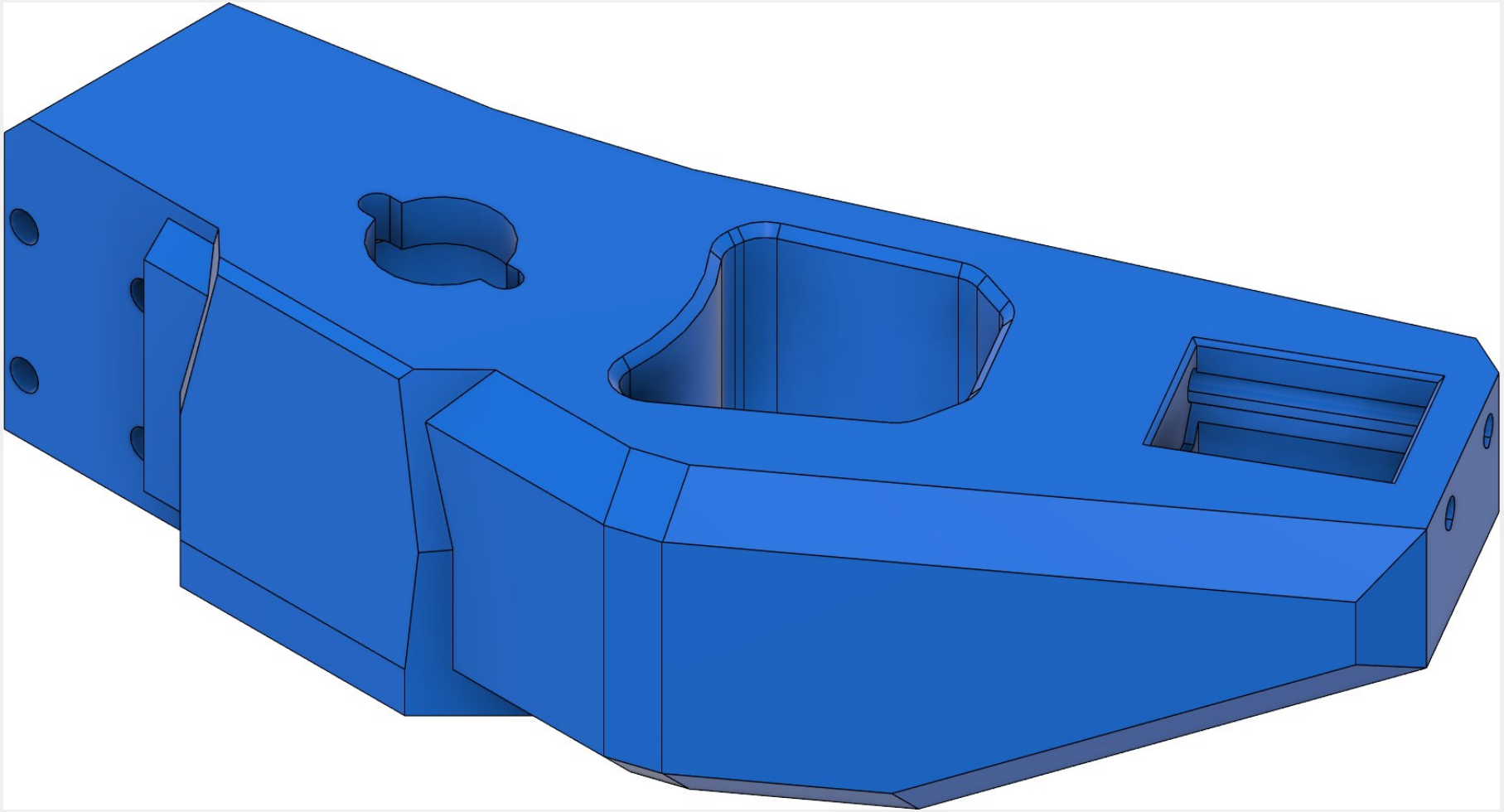
Tension and alignment

Use the designer-approved tension method once specified. Adjust incrementally, recheck gantry squareness and retain tensioner travel. Do not use an arbitrary frequency or tension value.

Pass condition

Both belts track consistently through full XY travel, fasteners remain accessible, clamps hold the belt ends, and the carriage does not bind. Final belt length, clamp hardware and target tension remain release items.

PREPARE THE LEFT Z ARM



Identify the left arm and inspect the rail mounting face, leadscrew opening and bed-support pocket. Clear loose debris and check the hardware fit before assembling the arm.

CAD GROUP
Z-Axis Arm Assembly Left v2
Use the matching printed revision.

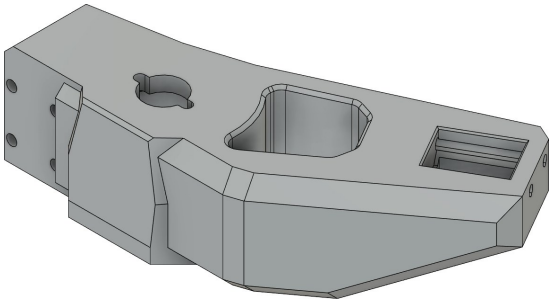
CHECK: Pockets and mounting faces are clean. Do not enlarge pin seats or tapped features to force a fit.

Current Hybrid assembly

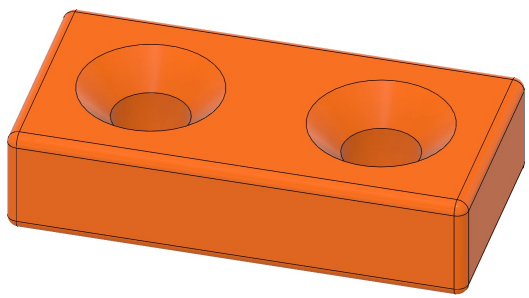
SEAT THE MAGNET

LEFT Z ARM | DETAIL 1

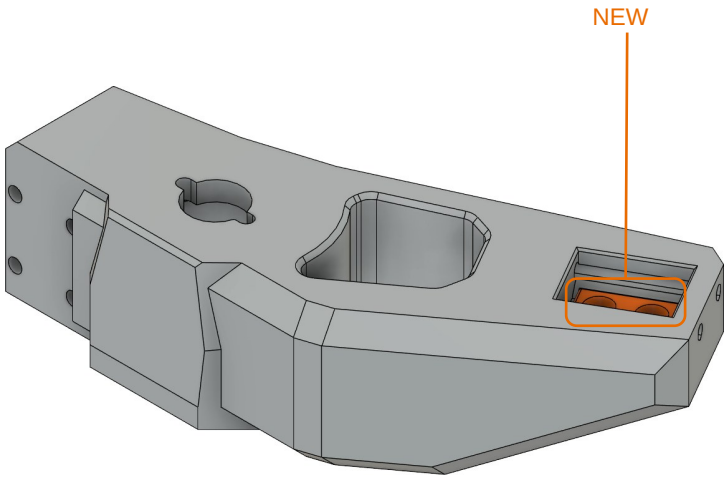
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the magnet in the rectangular pocket. Face its two countersunk holes toward the screw heads. Align both holes before adding screws.

PARTS IN THIS CAD STEP

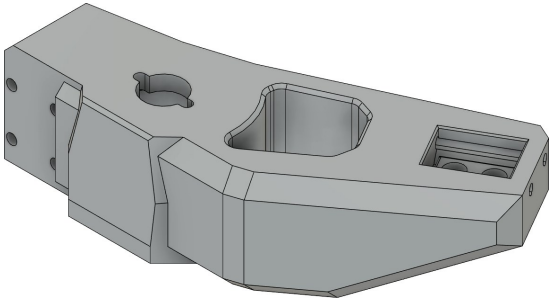
1 x 20 x 10 x 5 mm magnet

CHECK BEFORE CONTINUING: The magnet sits flat; both holes line up with the arm.

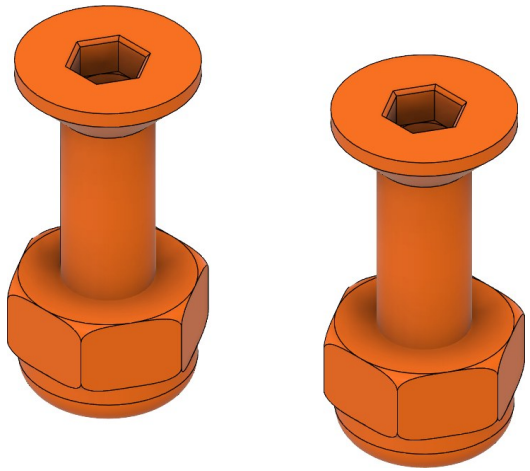
SECURE THE MAGNET

LEFT Z ARM | DETAIL 2

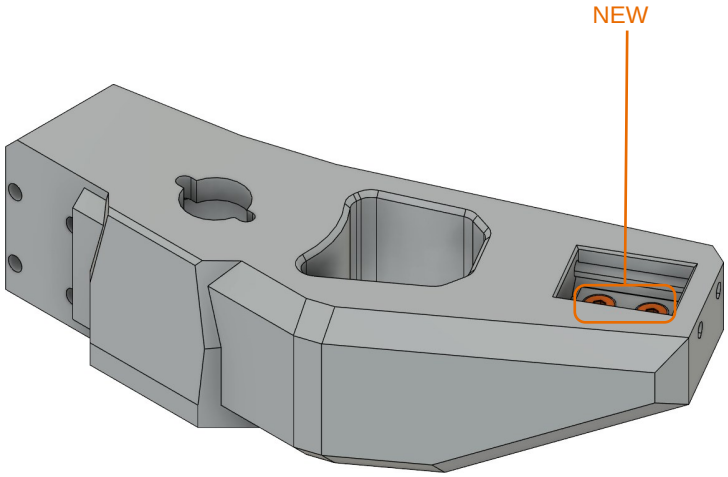
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the two M3 locking nuts in their matching seats. Pass the two M3 x 12 countersunk screws through the magnet and start both threads. Tighten gently while holding the nuts.

PARTS IN THIS CAD STEP

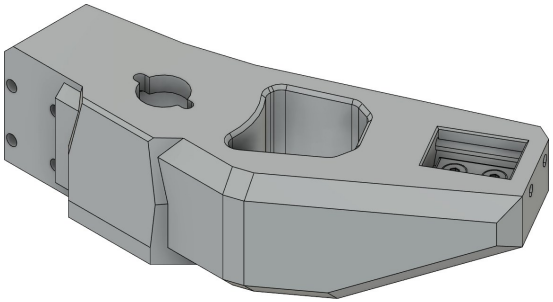
2 x M3 x 12 countersunk screw
2 x M3 locking nut

CHECK BEFORE CONTINUING: Both screw heads sit in the countersinks. The magnet is held without rocking or cracking.

INSTALL THE TWO SUPPORT PINS

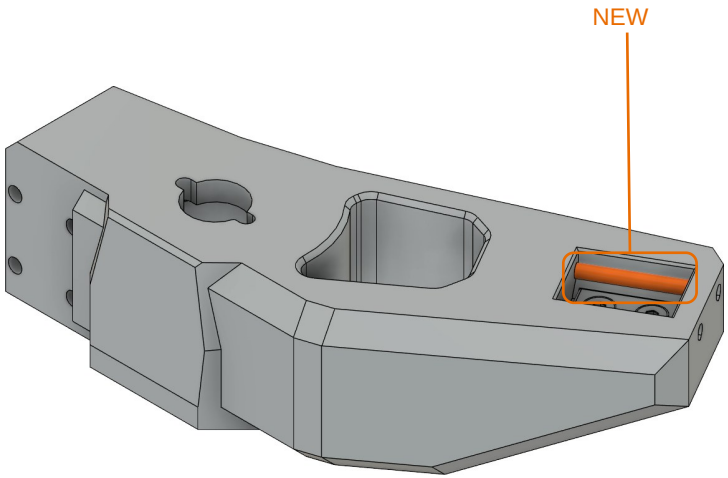
LEFT Z ARM | DETAIL 3

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the two 3 x 35 mm dowel pins into the parallel seats beside the magnet. Compare their position with the orange parts in the finished view.

PARTS IN THIS CAD STEP

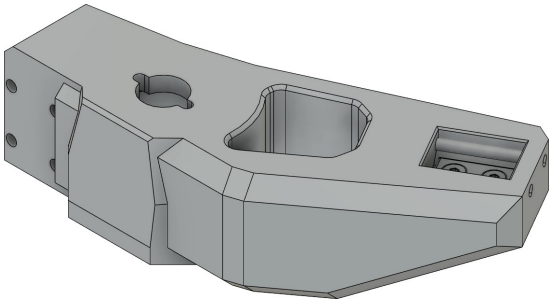
2 x 3 x 35 mm dowel pin

CHECK BEFORE CONTINUING: Both pins are seated, parallel and clean. Do not hammer a pin into a blocked or undersized seat.

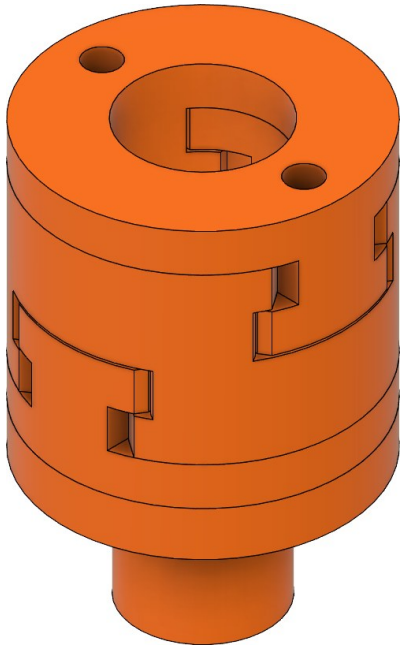
POSITION THE NUT AND DECOUPLER

LEFT Z ARM | DETAIL 4

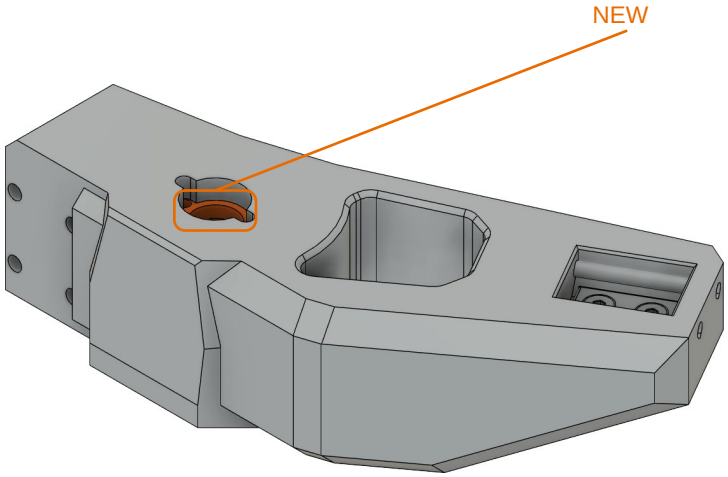
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the POM leadscrew nut and decoupler at the leadscrew opening. Match their vertical order and mounting-hole orientation to the separate parts view.

PARTS IN THIS CAD STEP

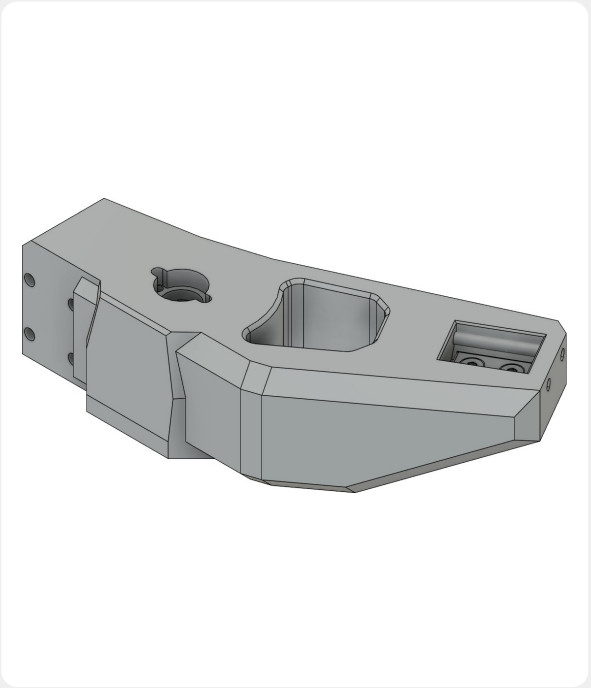
1 x TR8*4 POM leadscrew nut
1 x Lead screw decoupler

CHECK BEFORE CONTINUING: The leadscrew opening is aligned. The floating interface is not trapped by the arm.

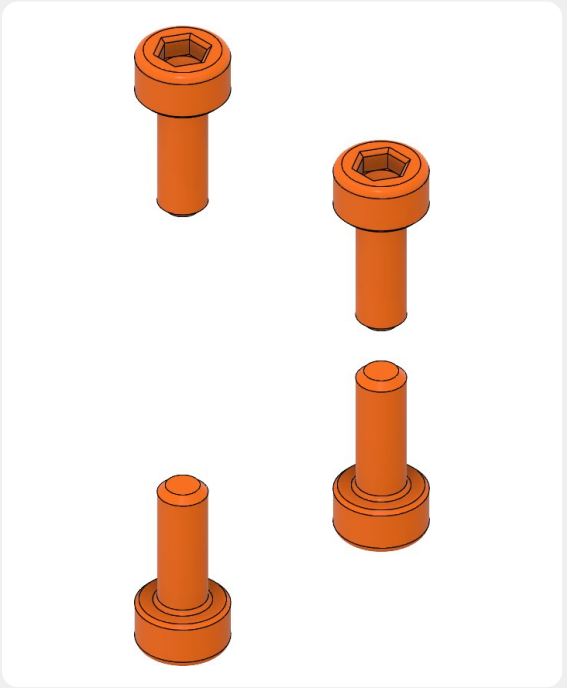
ADD THE FOUR M3 X 8 SCREWS

LEFT Z ARM | DETAIL 5

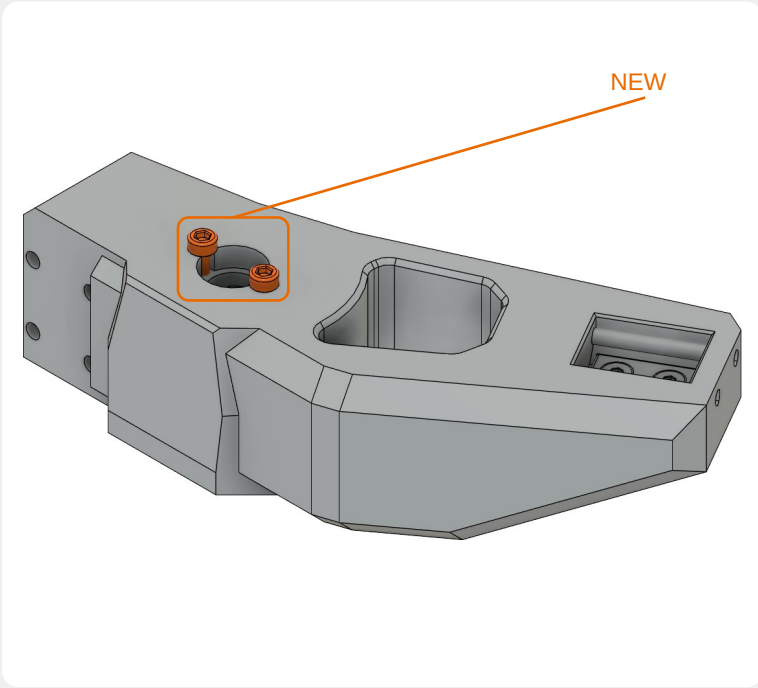
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start all four M3 x 8 cap-head screws at the highlighted nut/decoupler locations. Engage the threads by hand before using the driver; tighten progressively.

PARTS IN THIS CAD STEP

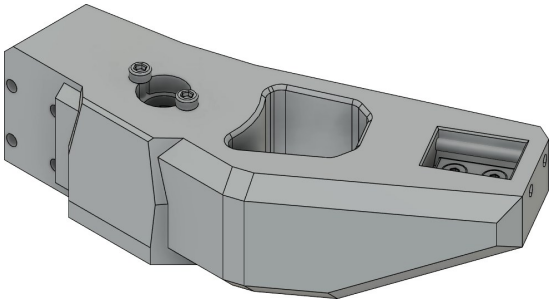
4 x M3 x 8 cap-head screw

CHECK BEFORE CONTINUING: All four heads are seated and the decoupler can perform its intended floating movement.

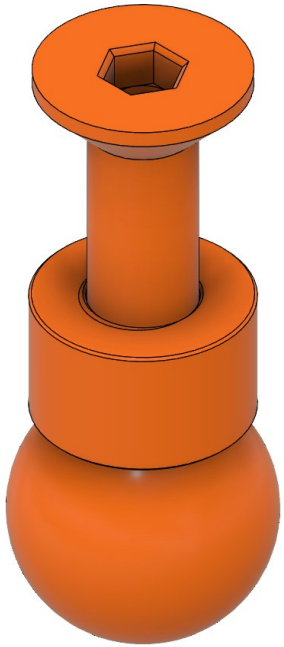
IDENTIFY THE BED CONTACT HARDWARE

LEFT Z ARM | DETAIL 6

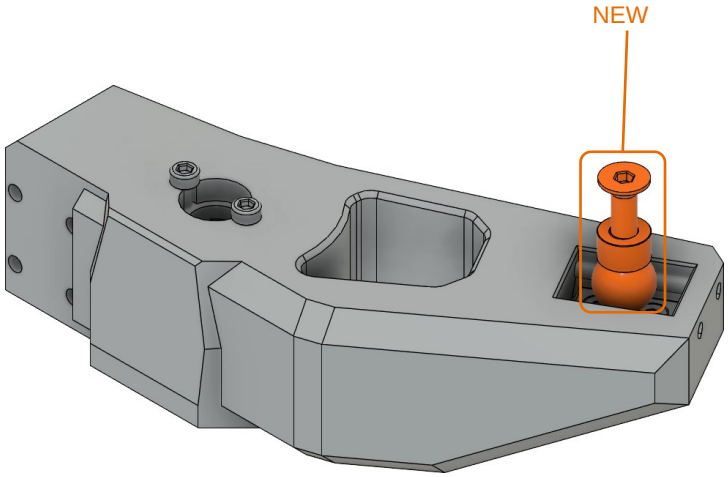
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

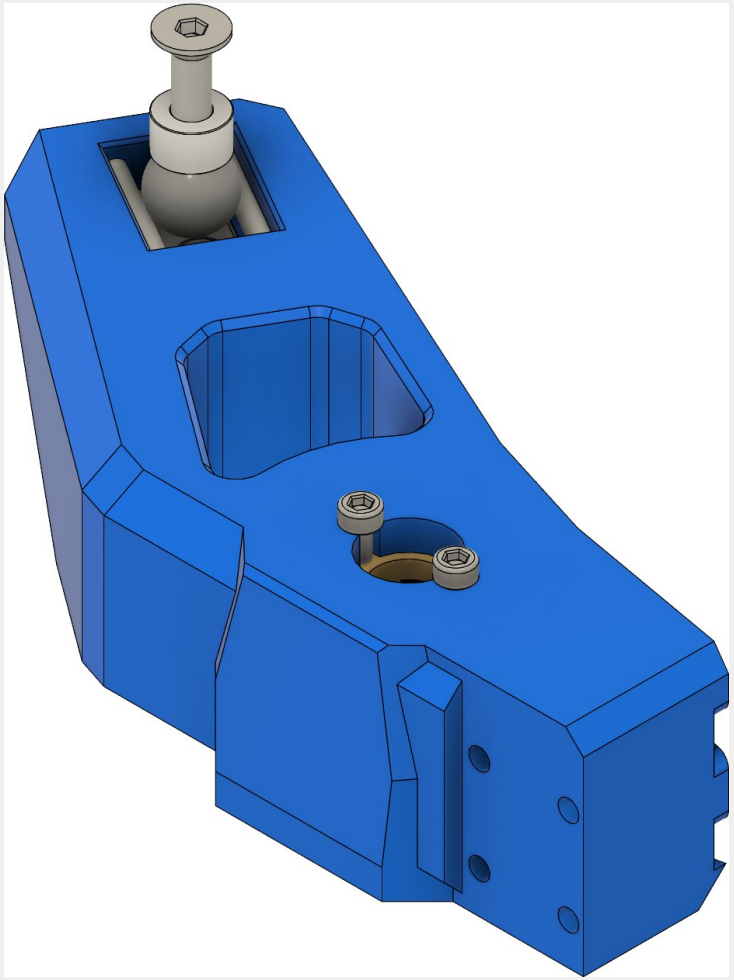
Set aside the 12 mm M5-threaded ball, 6 mm spacer and M5 x 16 screw for the bed-support joint. The view shows their installed relationship; fit them with the bed mounting stack during bed installation.

PARTS IN THIS CAD STEP

1 x 12 mm ball - M5 thread
1 x 6 mm spacer
1 x M5 x 16 countersunk screw

CHECK BEFORE CONTINUING: Do not fasten this stack without the bed. Verify thread engagement through the actual bed and spacer.

ASSEMBLE THE MIRRORED RIGHT ARM



Build the right-hand arm using the mirrored component. Repeat the magnet, pins, nut, decoupler and support hardware arrangement, checking the direction of every mounting face.

CAD GROUP
Z-Axis Arm Assembly Left v2(Mirror)
Hardware is mirrored with the arm.

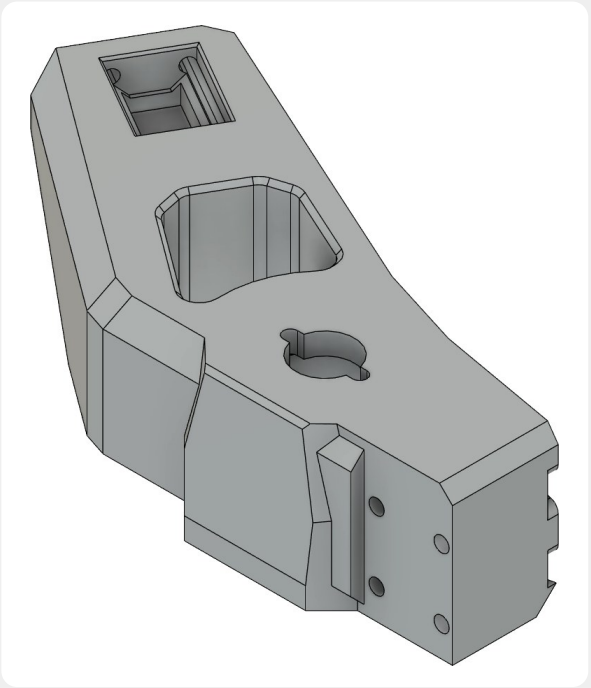
CHECK: The arm points inward from the right rail and its bed-support contact faces upward.

Current Hybrid assembly

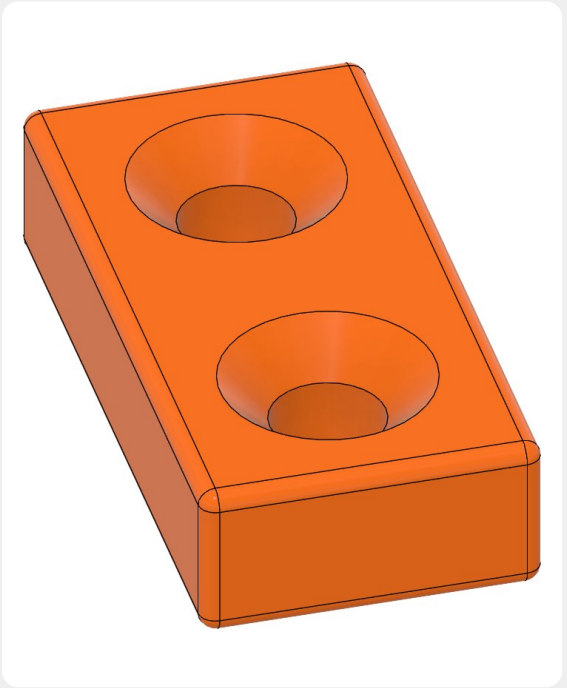
SEAT THE MAGNET

RIGHT Z ARM | DETAIL 1

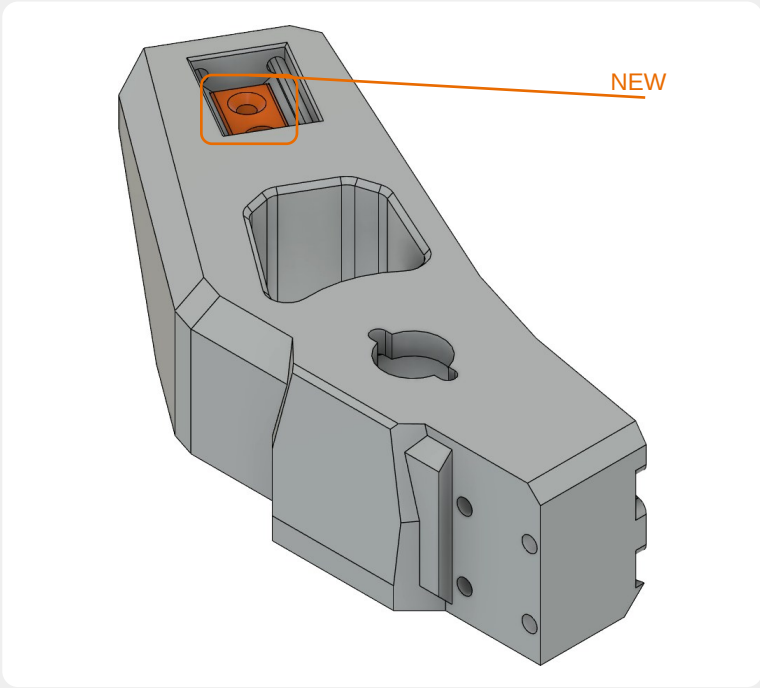
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the magnet in the rectangular pocket. Face its two countersunk holes toward the screw heads. Align both holes before adding screws.

PARTS IN THIS CAD STEP

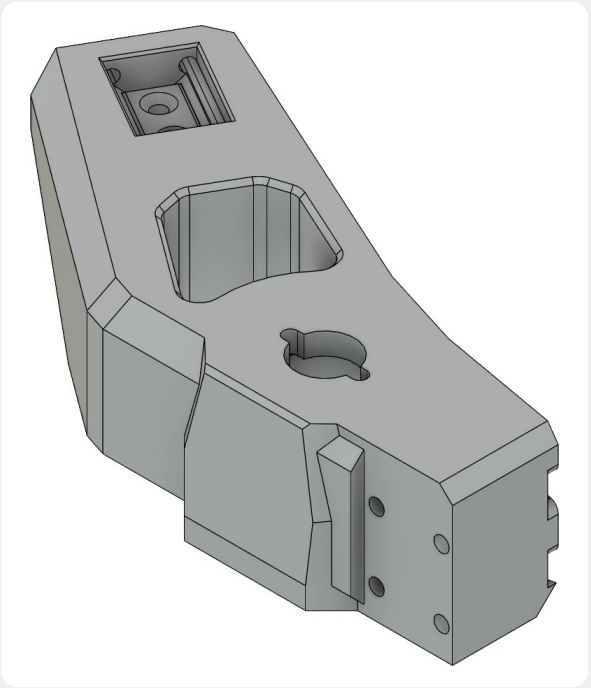
1 x 20 x 10 x 5 mm magnet

CHECK BEFORE CONTINUING: The magnet sits flat; both holes line up with the arm.

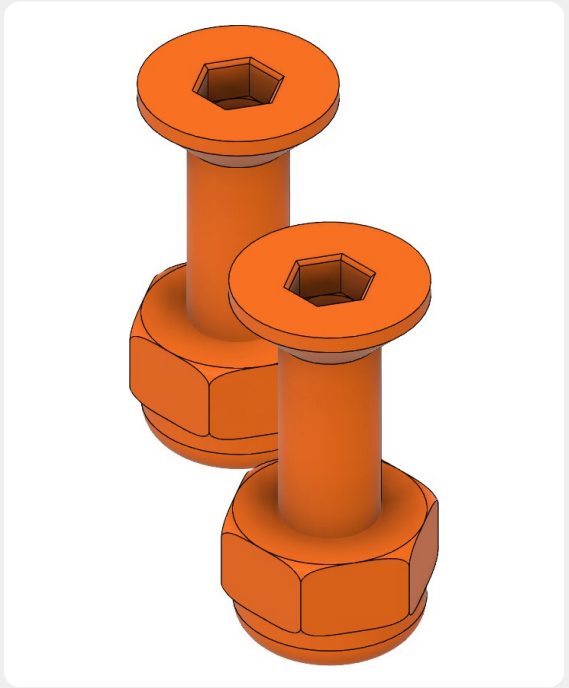
SECURE THE MAGNET

RIGHT Z ARM | DETAIL 2

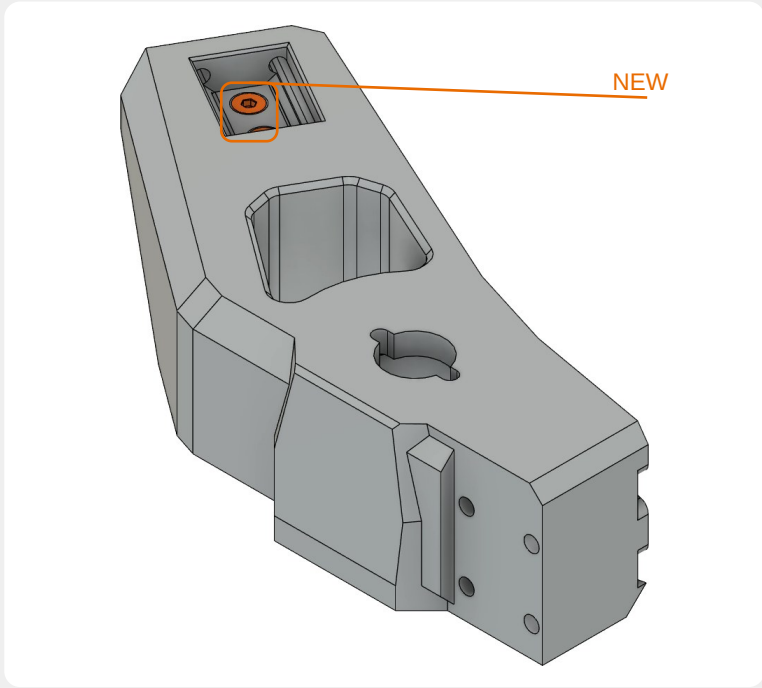
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the two M3 locking nuts in their matching seats. Pass the two M3 x 12 countersunk screws through the magnet and start both threads. Tighten gently while holding the nuts.

PARTS IN THIS CAD STEP

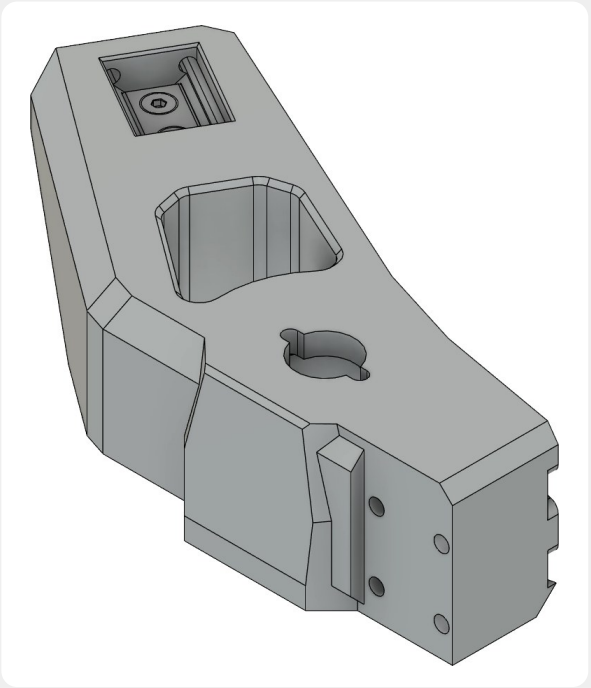
2 x M3 x 12 countersunk screw
2 x M3 locking nut

CHECK BEFORE CONTINUING: Both screw heads sit in the countersinks. The magnet is held without rocking or cracking.

INSTALL THE TWO SUPPORT PINS

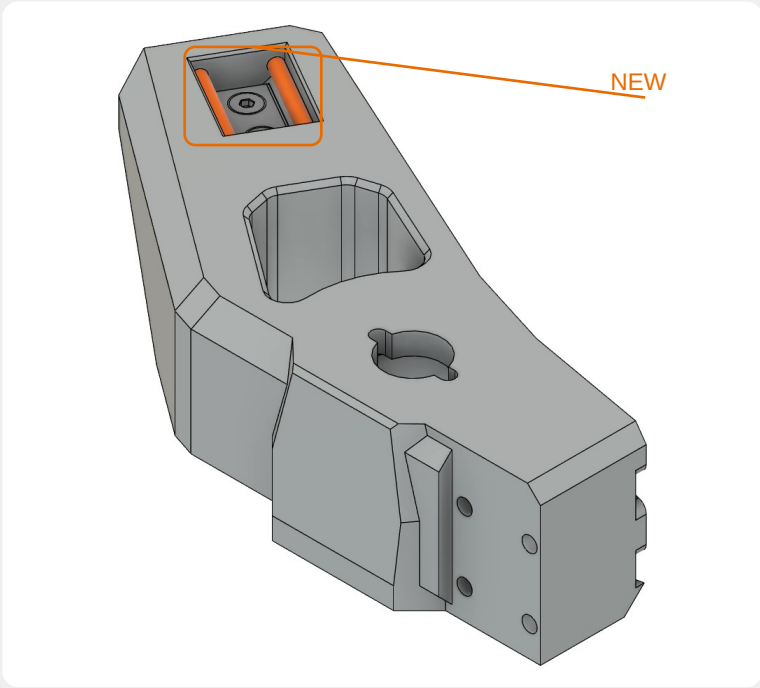
RIGHT Z ARM | DETAIL 3

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the two 3 x 35 mm dowel pins into the parallel seats beside the magnet. Compare their position with the orange parts in the finished view.

PARTS IN THIS CAD STEP

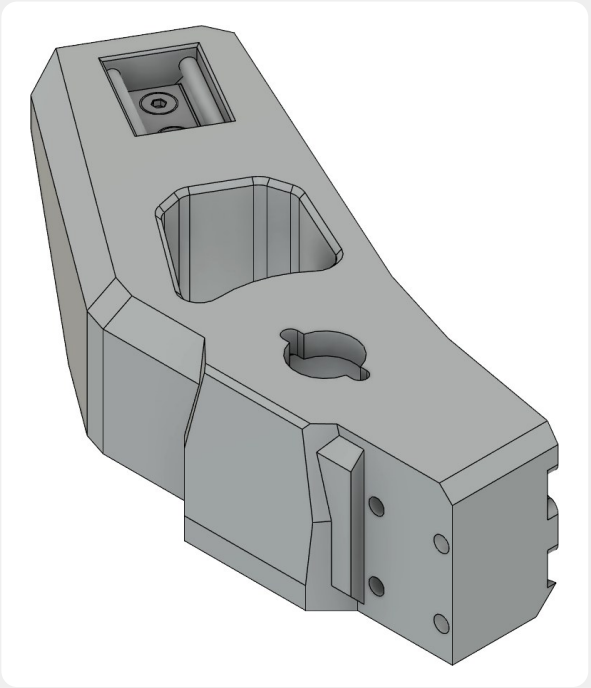
2 x 3 x 35 mm dowel pin

CHECK BEFORE CONTINUING: Both pins are seated, parallel and clean. Do not hammer a pin into a blocked or undersized seat.

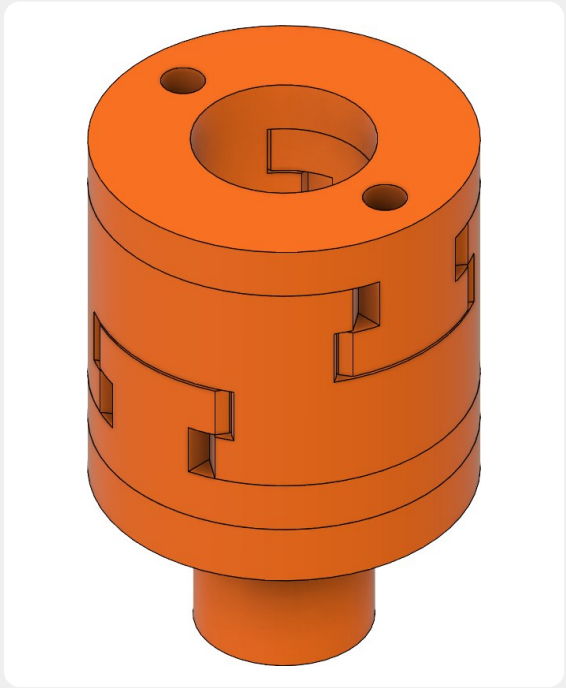
POSITION THE NUT AND DECOUPLER

RIGHT Z ARM | DETAIL 4

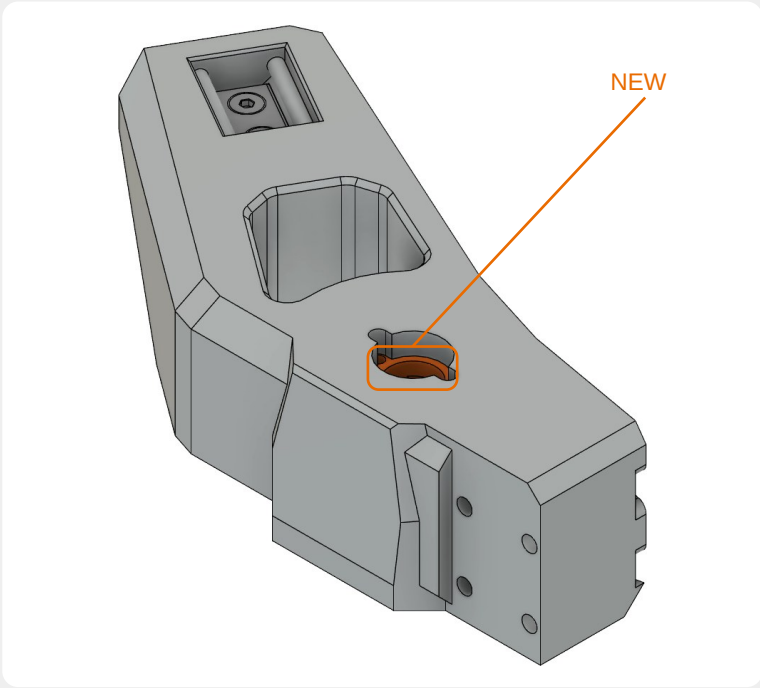
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the POM leadscrew nut and decoupler at the leadscrew opening. Match their vertical order and mounting-hole orientation to the separate parts view.

PARTS IN THIS CAD STEP

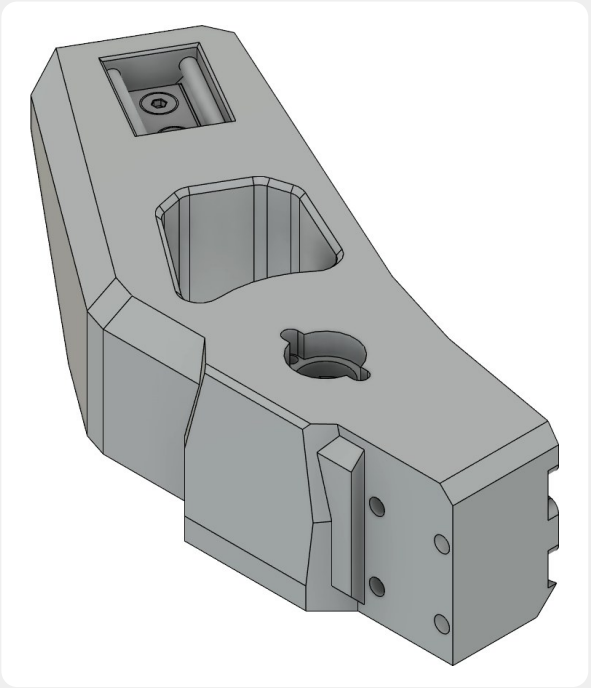
1 x TR8*4 POM leadscrew nut
1 x Lead screw decoupler

CHECK BEFORE CONTINUING: The leadscrew opening is aligned. The floating interface is not trapped by the arm.

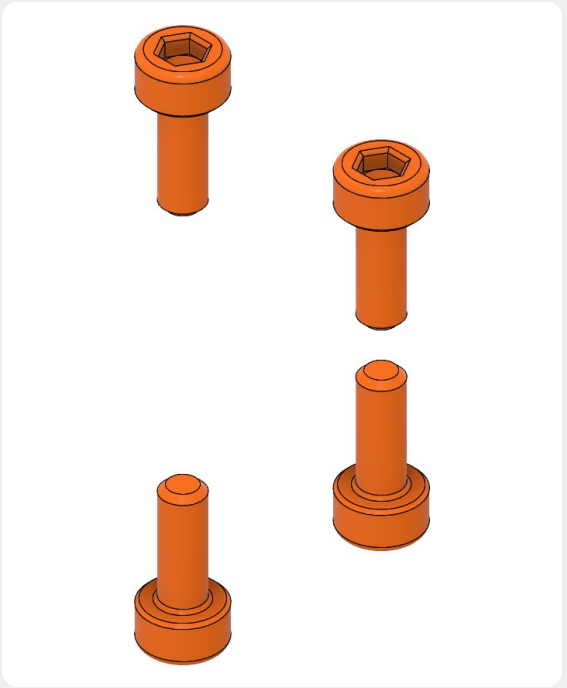
ADD THE FOUR M3 X 8 SCREWS

RIGHT Z ARM | DETAIL 5

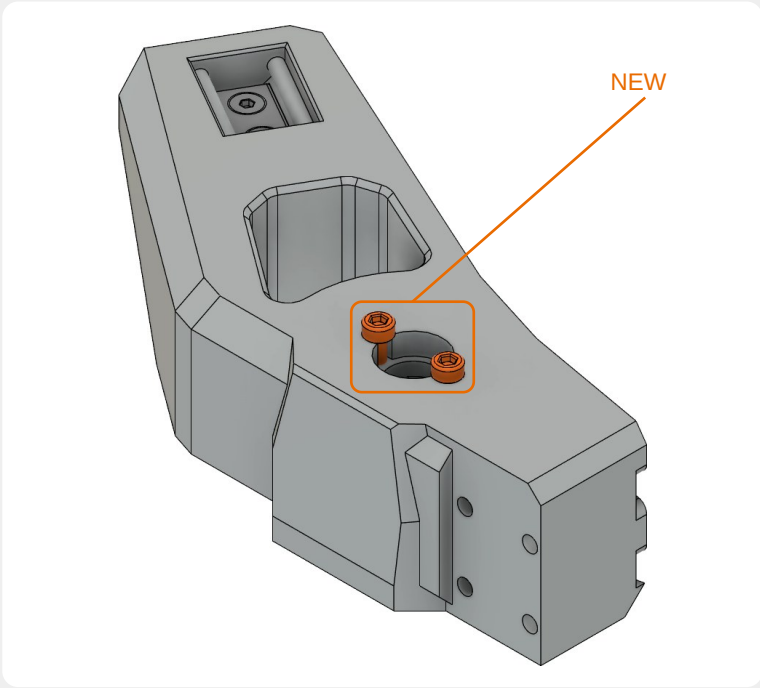
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start all four M3 x 8 cap-head screws at the highlighted nut/decoupler locations. Engage the threads by hand before using the driver; tighten progressively.

PARTS IN THIS CAD STEP

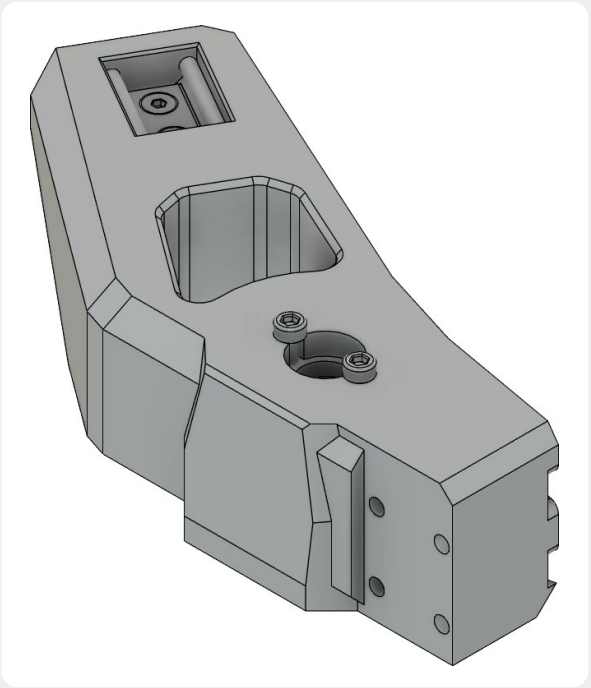
4 x M3 x 8 cap-head screw

CHECK BEFORE CONTINUING: All four heads are seated and the decoupler can perform its intended floating movement.

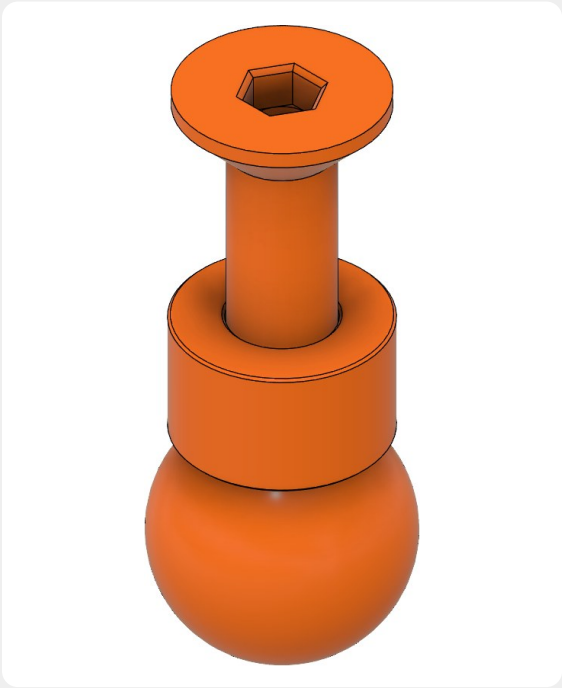
IDENTIFY THE BED CONTACT HARDWARE

RIGHT Z ARM | DETAIL 6

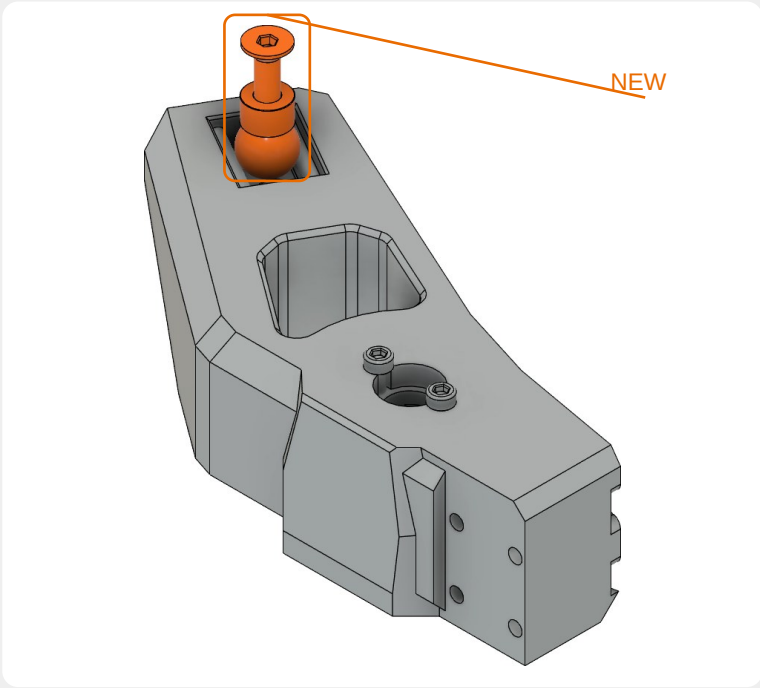
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

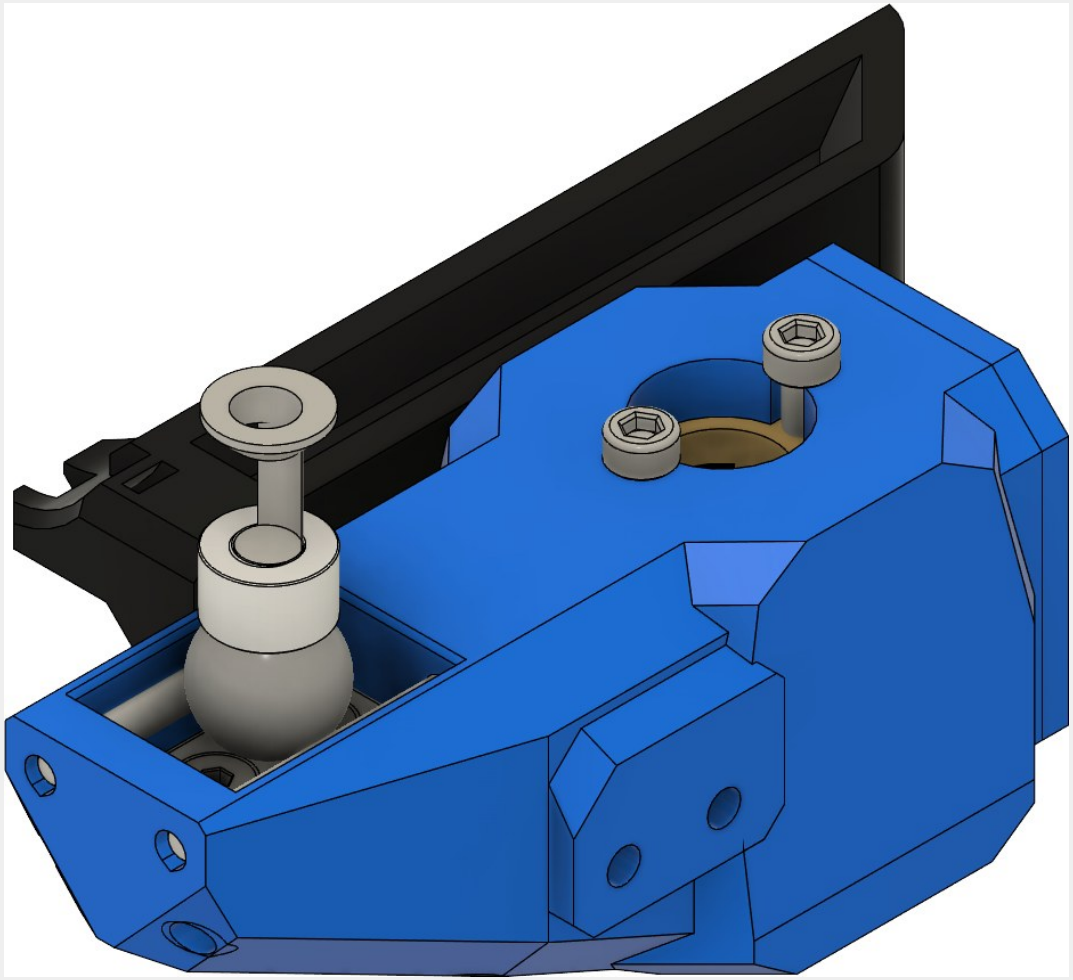
Set aside the 12 mm M5-threaded ball, 6 mm spacer and M5 x 16 screw for the bed-support joint. The view shows their installed relationship; fit them with the bed mounting stack during bed installation.

PARTS IN THIS CAD STEP

1 x 12 mm ball - M5 thread
1 x 6 mm spacer
1 x M5 x 16 countersunk screw

CHECK BEFORE CONTINUING: Do not fasten this stack without the bed. Verify thread engagement through the actual bed and spacer.

ASSEMBLE THE REAR Z ARM



Use the dedicated rear arm and carriage interface. Fit its nut, decoupler and bed-support hardware in the positions shown. Check the rear cable-chain connection before mounting the arm.

CAD GROUP
Z-Axis Arm Assembly Back v3
Rear interface differs from the side arms.

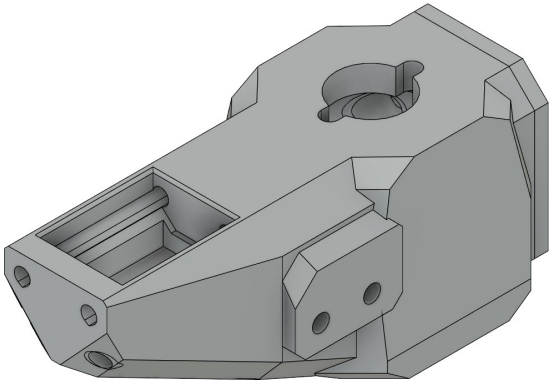
CHECK: The rear carriage mount and chain attachment remain accessible. Do not substitute a side arm.

Current Hybrid assembly

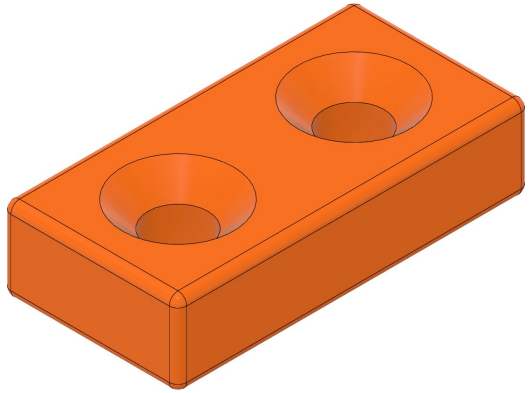
SEAT THE MAGNET

REAR Z ARM | DETAIL 1

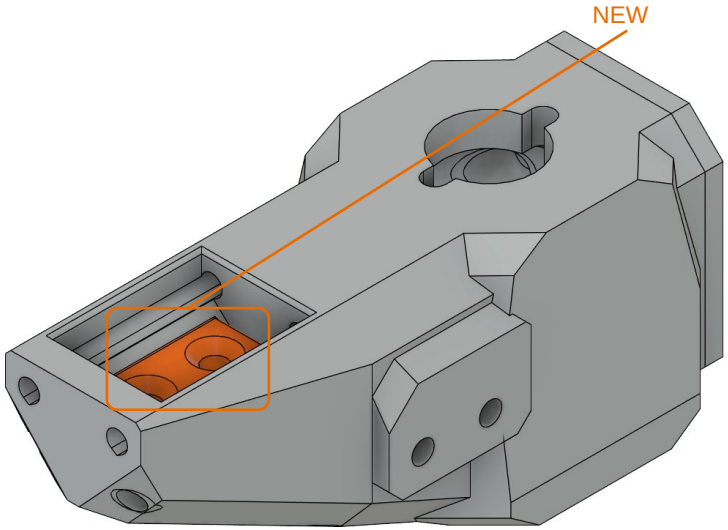
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the magnet in the rectangular pocket. Face its two countersunk holes toward the screw heads. Align both holes before adding screws.

PARTS IN THIS CAD STEP

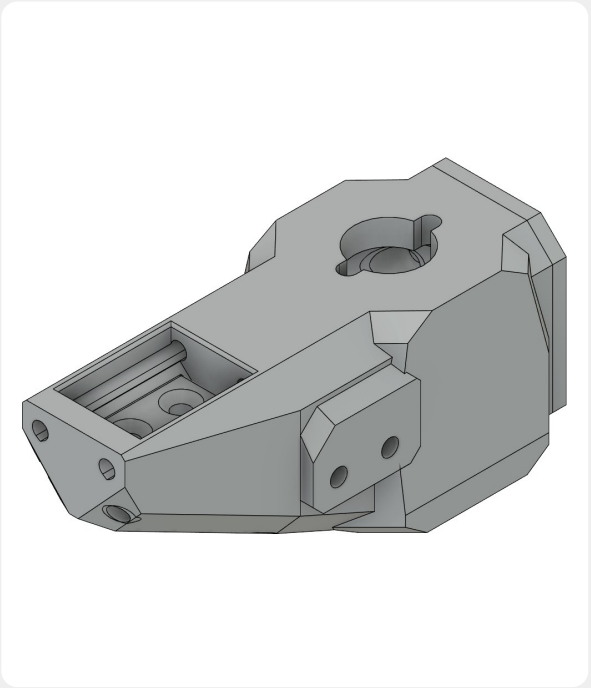
1 x 20 x 10 x 5 mm magnet

CHECK BEFORE CONTINUING: The magnet sits flat; both holes line up with the arm.

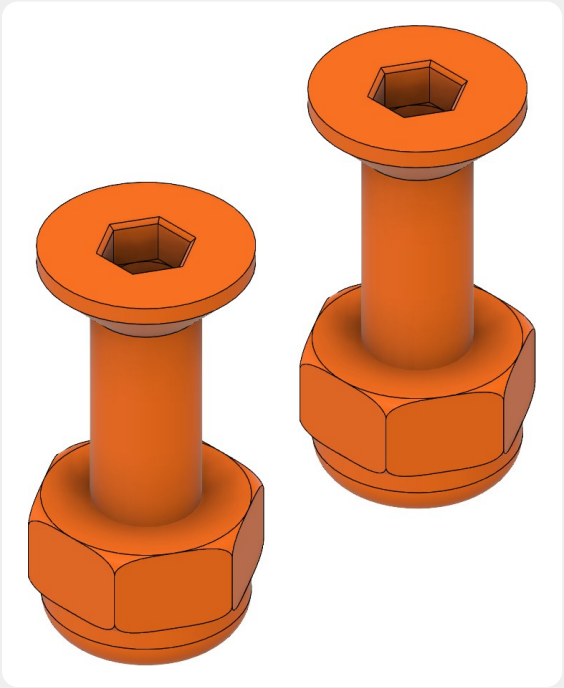
SECURE THE MAGNET

REAR Z ARM | DETAIL 2

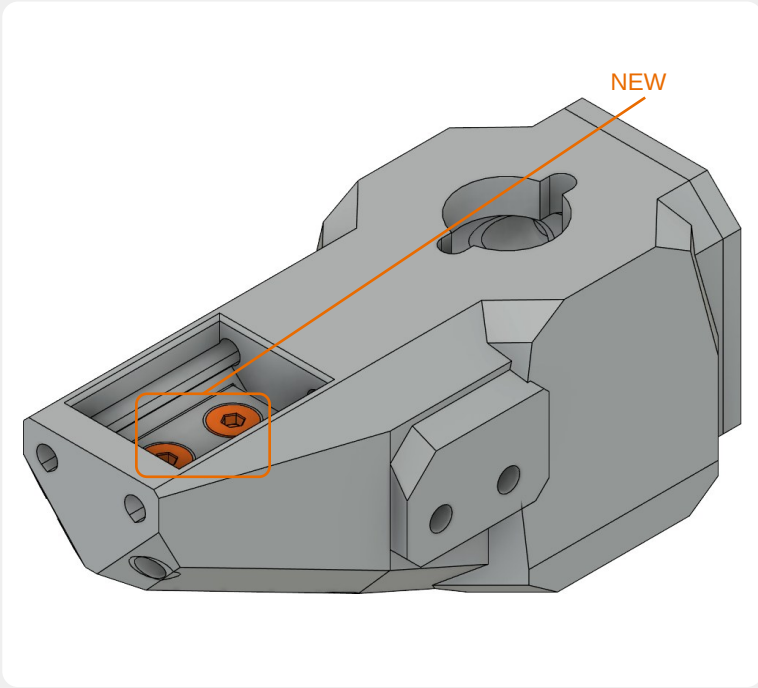
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the two M3 locking nuts in their matching seats. Pass the two M3 x 12 countersunk screws through the magnet and start both threads. Tighten gently while holding the nuts.

PARTS IN THIS CAD STEP

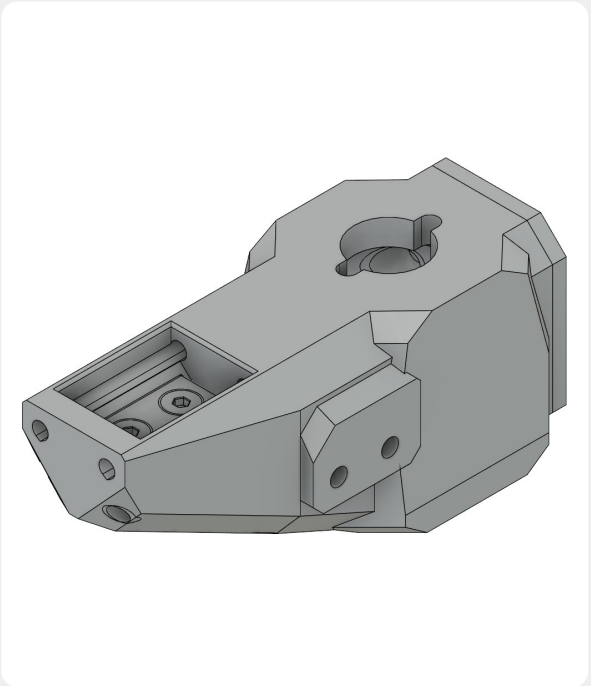
2 x M3 x 12 countersunk screw
2 x M3 locking nut

CHECK BEFORE CONTINUING: Both screw heads sit in the countersinks. The magnet is held without rocking or cracking.

INSTALL THE TWO SUPPORT PINS

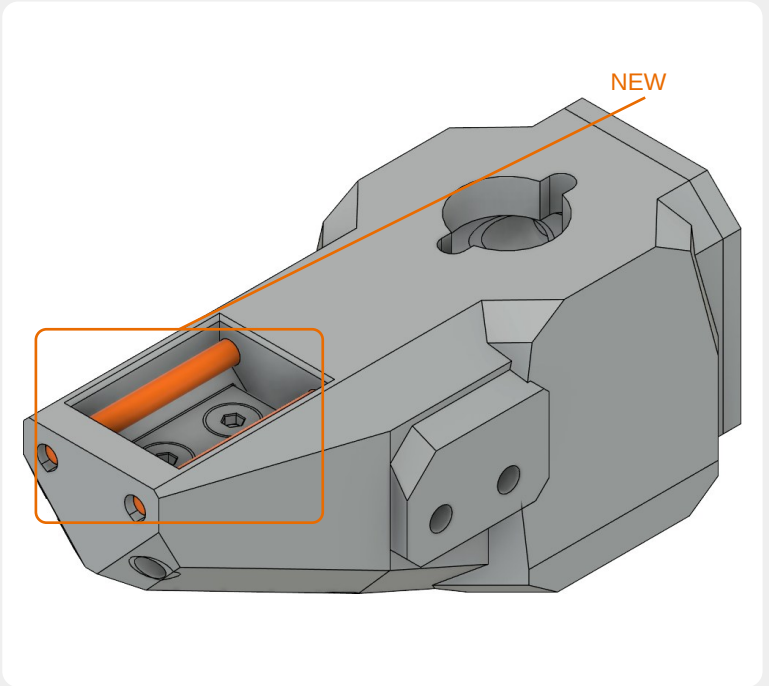
REAR Z ARM | DETAIL 3

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the two 3 x 35 mm dowel pins into the parallel seats beside the magnet. Compare their position with the orange parts in the finished view.

PARTS IN THIS CAD STEP

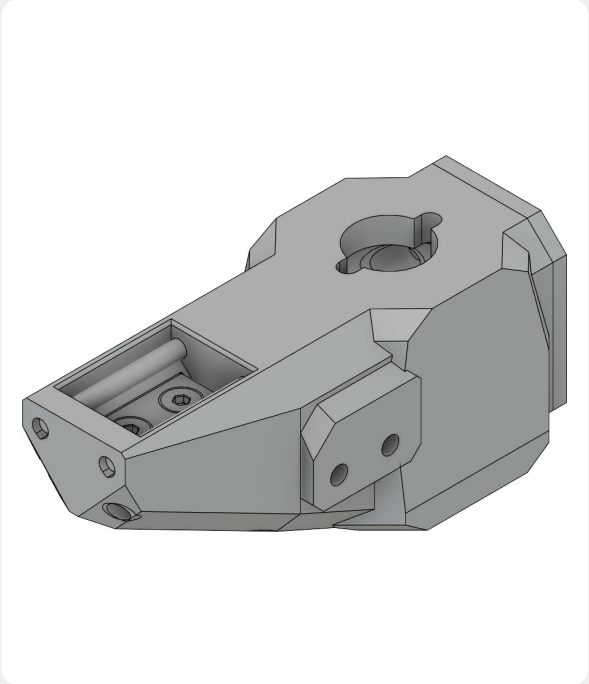
2 x 3 x 35 mm dowel pin

CHECK BEFORE CONTINUING: Both pins are seated, parallel and clean. Do not hammer a pin into a blocked or undersized seat.

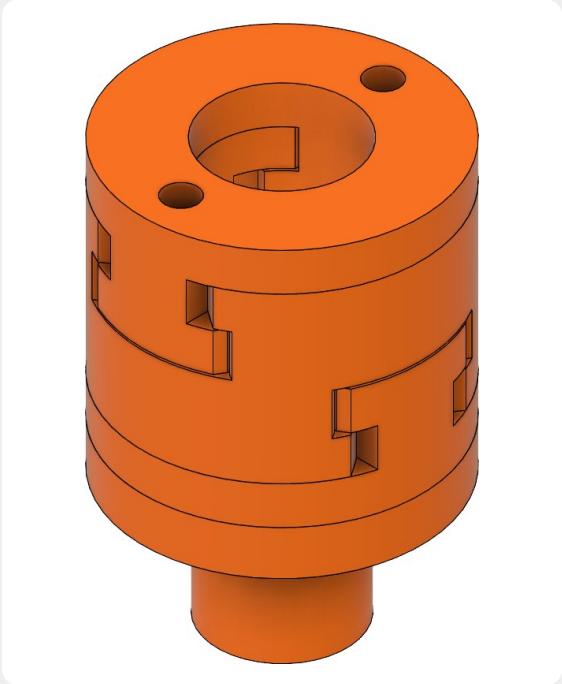
POSITION THE NUT AND DECOUPLER

REAR Z ARM | DETAIL 4

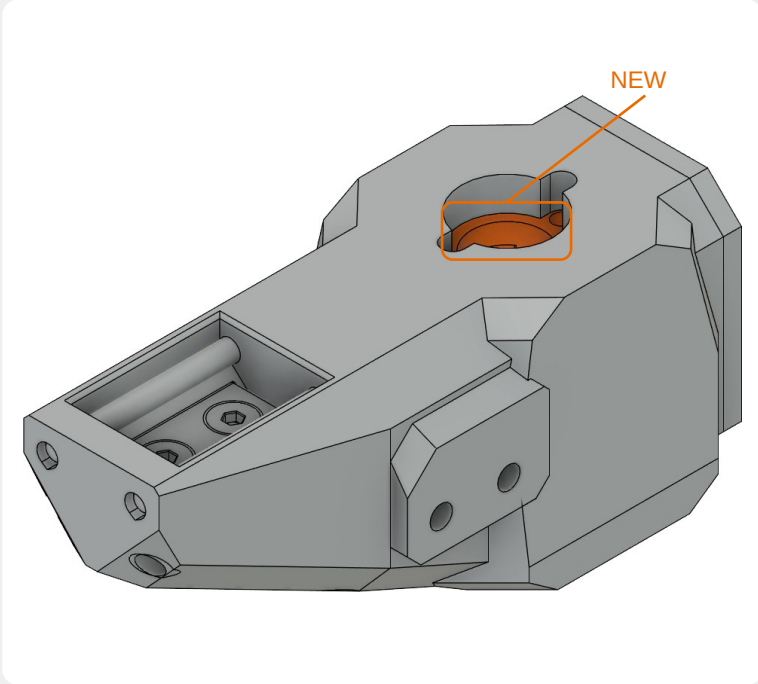
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the POM leadscrew nut and decoupler at the leadscrew opening. Match their vertical order and mounting-hole orientation to the separate parts view.

PARTS IN THIS CAD STEP

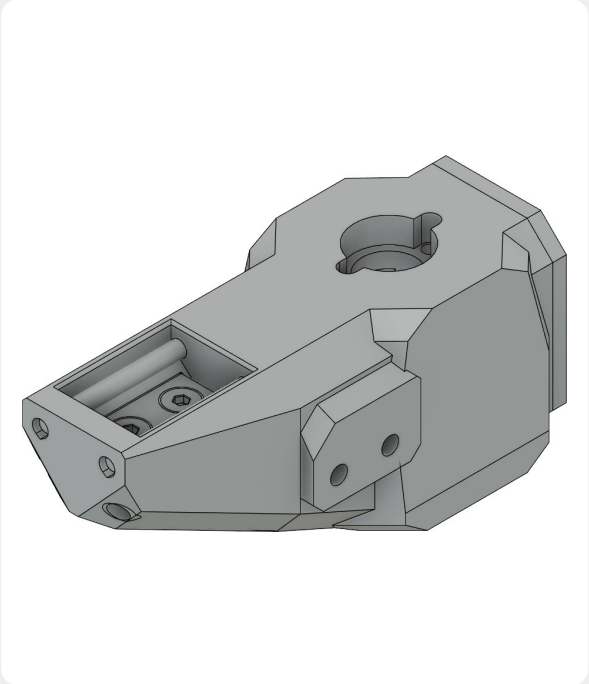
1 x TR8*4 POM leadscrew nut
1 x Lead screw decoupler

CHECK BEFORE CONTINUING: The leadscrew opening is aligned. The floating interface is not trapped by the arm.

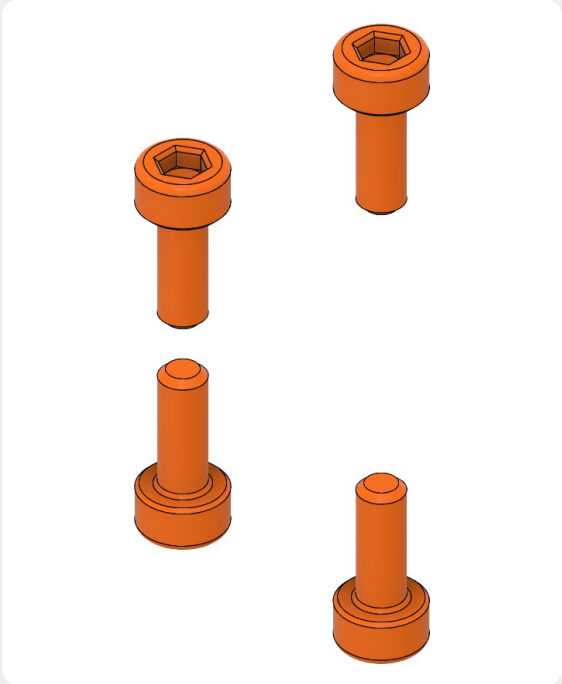
ADD THE FOUR M3 X 8 SCREWS

REAR Z ARM | DETAIL 5

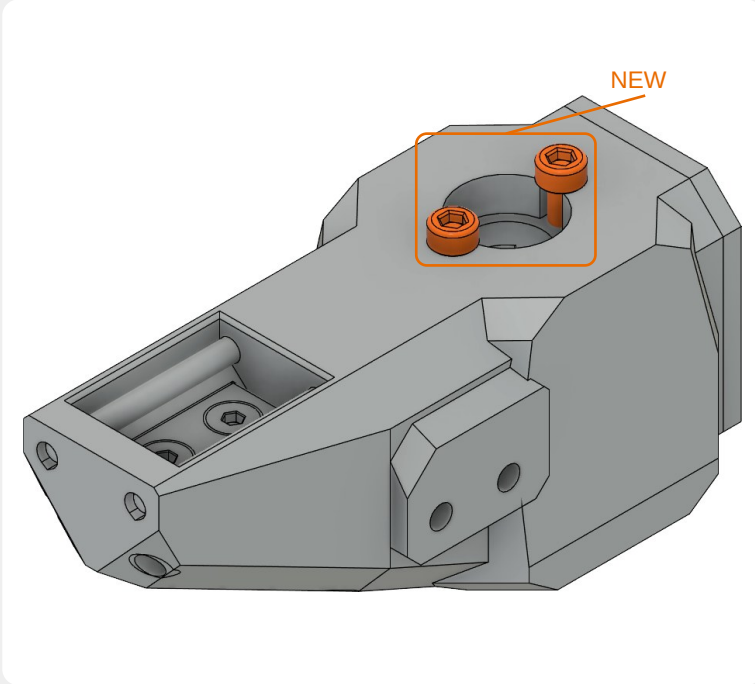
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start all four M3 x 8 cap-head screws at the highlighted nut/decoupler locations. Engage the threads by hand before using the driver; tighten progressively.

PARTS IN THIS CAD STEP

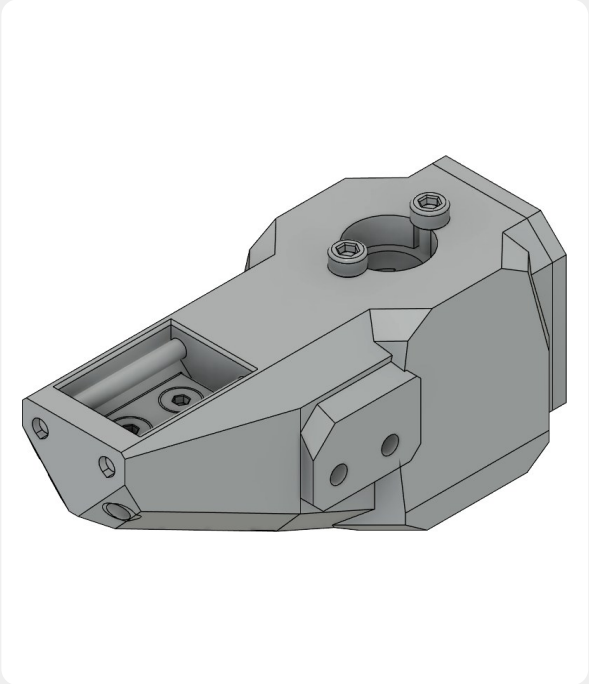
4 x M3 x 8 cap-head screw

CHECK BEFORE CONTINUING: All four heads are seated and the decoupler can perform its intended floating movement.

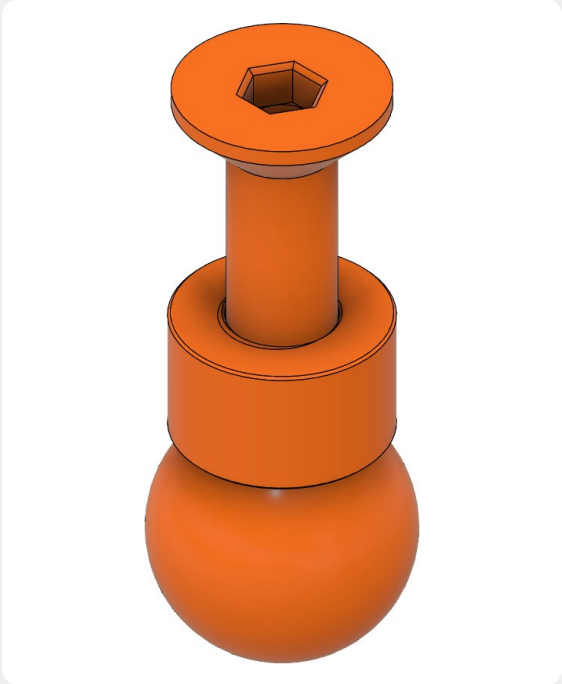
IDENTIFY THE BED CONTACT HARDWARE

REAR Z ARM | DETAIL 6

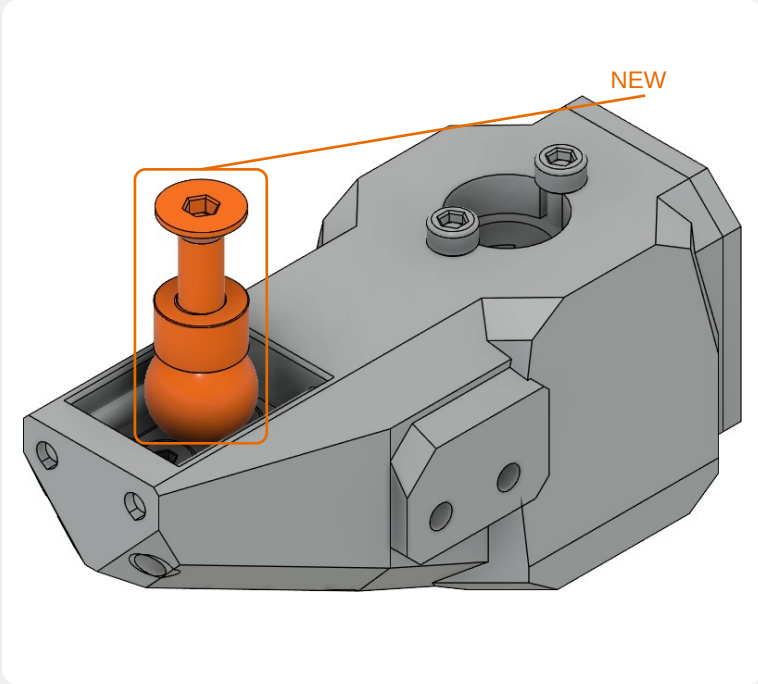
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Set aside the 12 mm M5-threaded ball, 6 mm spacer and M5 x 16 screw for the bed-support joint. The view shows their installed relationship; fit them with the bed mounting stack during bed installation.

PARTS IN THIS CAD STEP

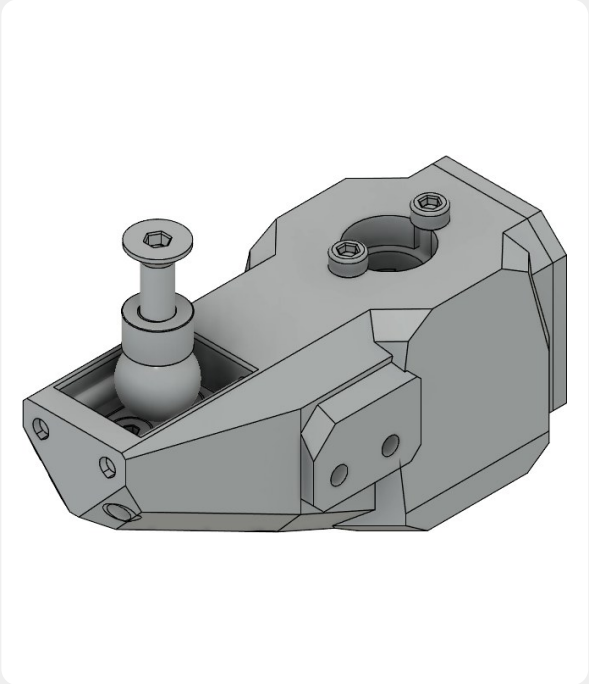
1 x 12 mm ball - M5 thread
1 x 6 mm spacer
1 x M5 x 16 countersunk screw

CHECK BEFORE CONTINUING: Do not fasten this stack without the bed. Verify thread engagement through the actual bed and spacer.

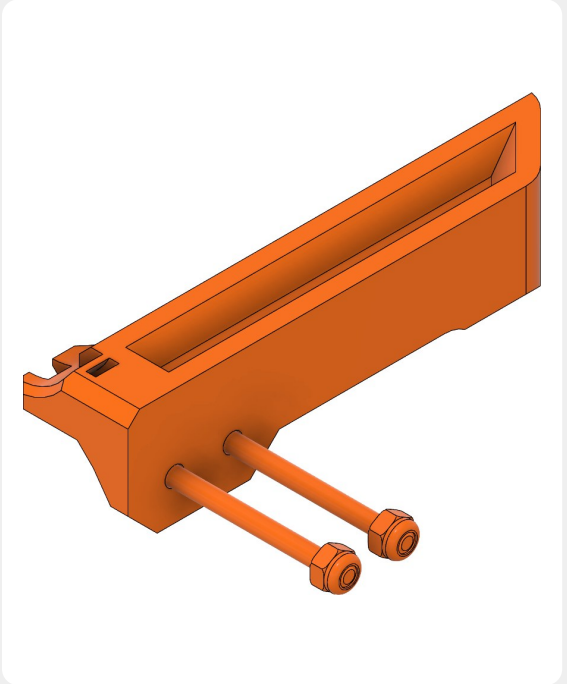
FIT THE REAR CABLE RELIEF

REAR Z ARM | DETAIL 7

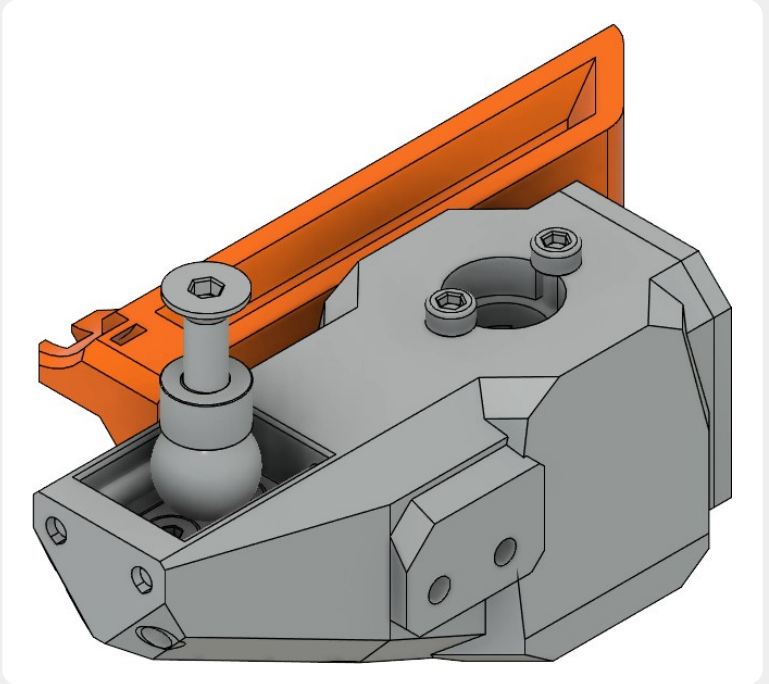
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position the rear cable relief against the rear arm. Start the two M3 x 35 screws and matching locking nuts. Leave cable clamping until the correct harness is fitted.

PARTS IN THIS CAD STEP

1 x bed cable relief
2 x M3 x 35 cap-head screw
2 x M3 locking nut

CHECK BEFORE CONTINUING: The relief is secure without crushing the future harness. Confirm the remaining rear mounting screws against the kit.

INSTALL THE Z RAILS AND DRIVES

Install the left, right and rear motor plates and rail assemblies at their matching frame locations. Align each leadscrew with its nut and guide rail before attaching the arms.

CAD ITEMS
Three motor plates
Three motor occurrences
Three 400mm_tr8 occurrences

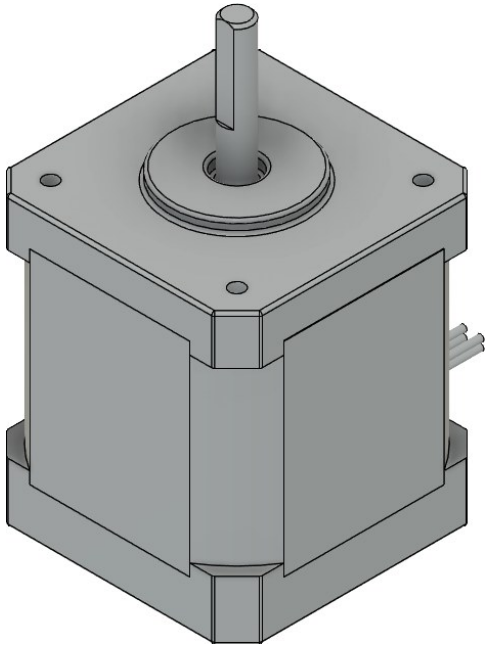
CHECK: Each arm travels without side-loading the leadscrew. Confirm rail offsets and travel stops from approved dimensions.

Current Hybrid assembly

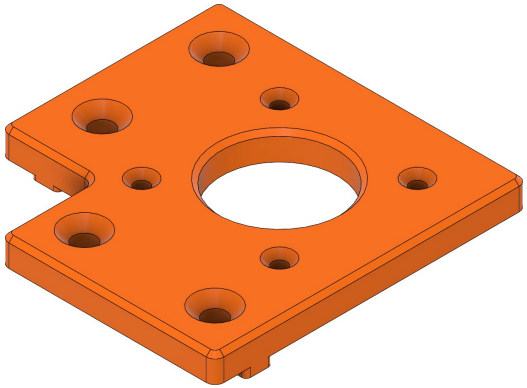
ALIGN THE MOTOR PLATE

LEFT Z MOTOR MOUNT | DETAIL 1

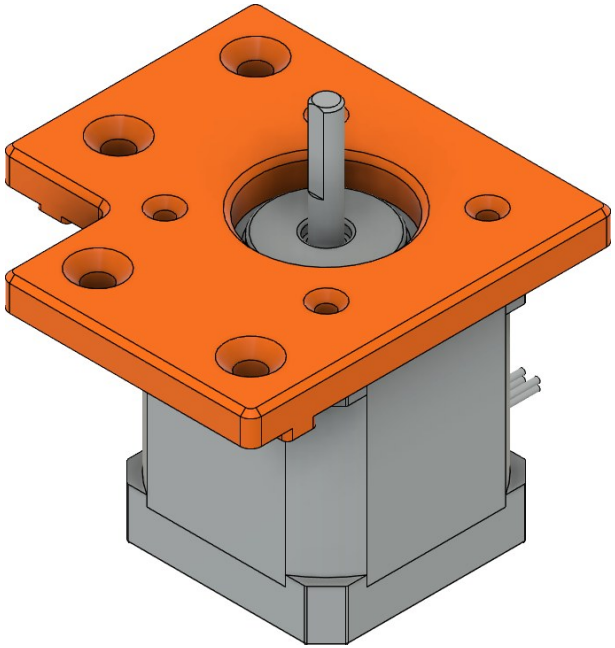
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position the matching Z motor plate over the motor face. Match the shaft opening and four mounting holes. Keep the lead exit accessible and the plate seated flat.

PARTS IN THIS CAD STEP

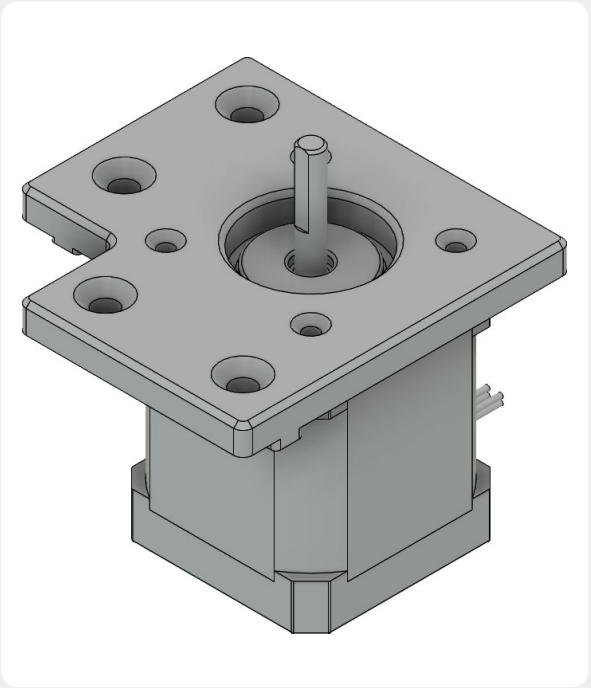
1 x Z Motor Plate Left

CHECK BEFORE CONTINUING: The plate is the correct left, right or rear part. The shaft passes through the opening without rubbing.

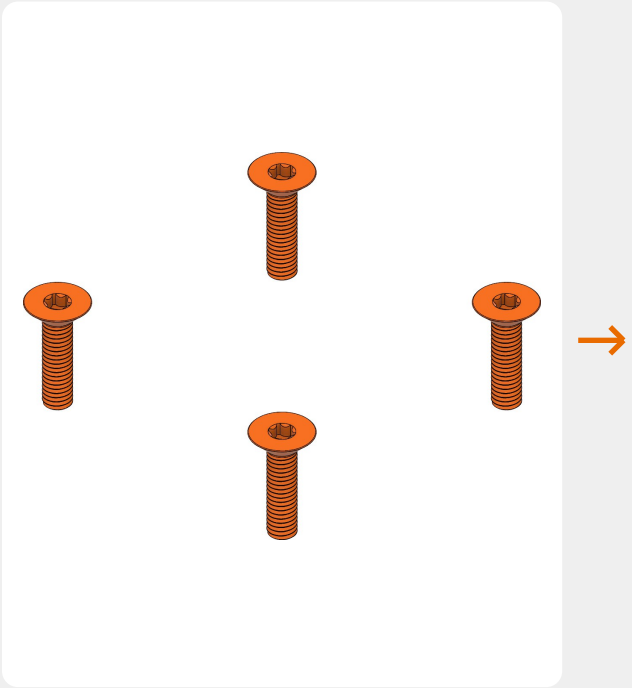
INSTALL THE FOUR M3 X 12 MOTOR SCREWS

LEFT Z MOTOR MOUNT | DETAIL 2

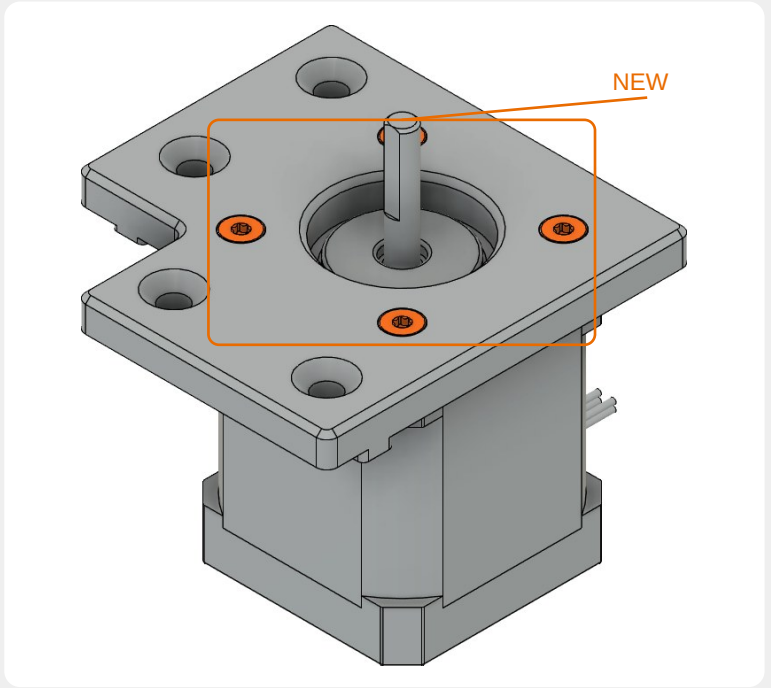
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the four M3 x 12 Torx flat-head screws through the motor plate into the motor. Engage all four threads by hand, then tighten progressively with the plate seated flat.

PARTS IN THIS CAD STEP

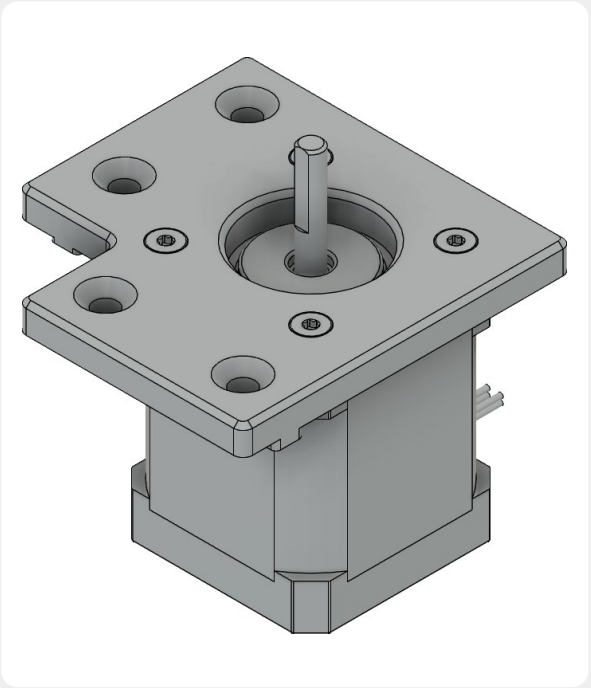
4 x M3 x 12 Torx flat-head [92703A151]

CHECK BEFORE CONTINUING: All four heads seat in the plate recesses. Verify engagement on the actual motor and confirm that no screw bottoms before the plate is clamped.

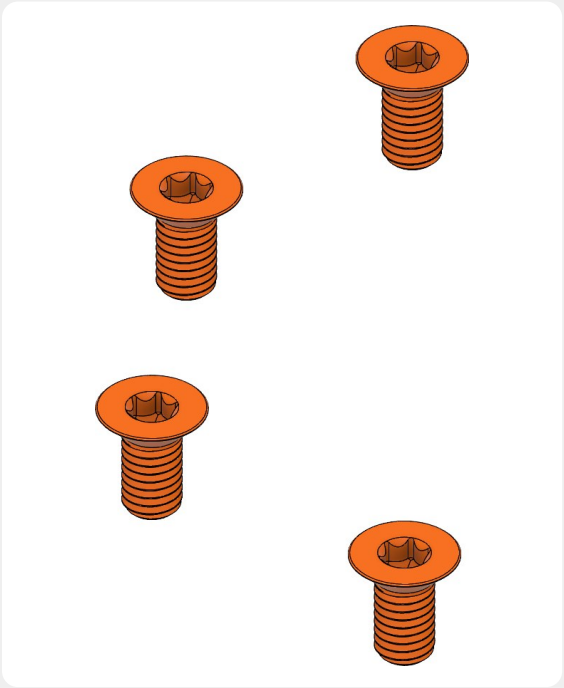
PREPARE THE FRAME SCREWS

LEFT Z MOTOR MOUNT | DETAIL 3

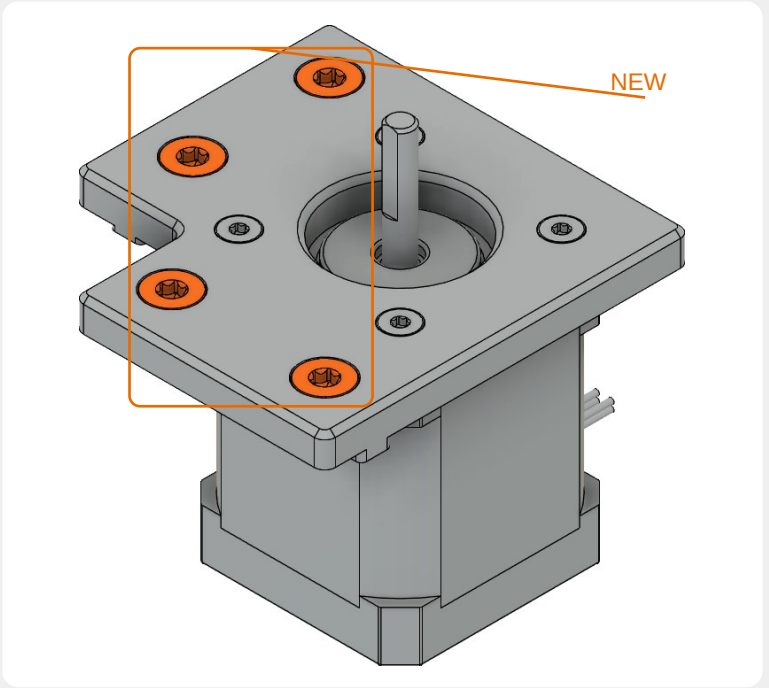
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the M5 x 10 Torx flat-head frame screws around the motor plate. Align the plate with its frame mounting locations before engaging the matching frame threads or reviewed slot hardware.

PARTS IN THIS CAD STEP

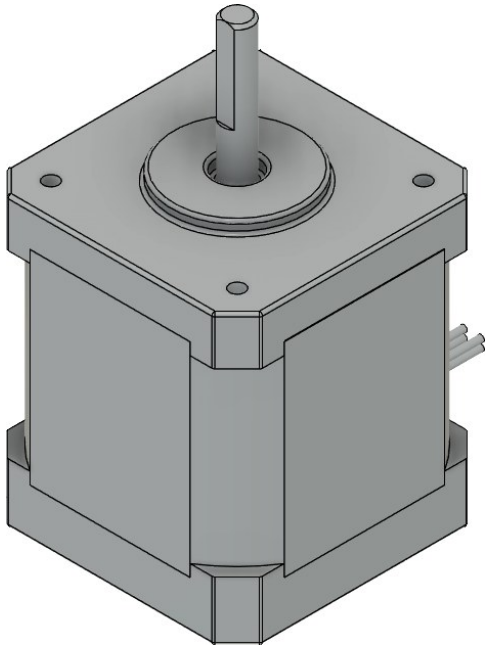
4 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: All mounting points align without bending the plate. Verify the mating thread or T-nut arrangement before tightening.

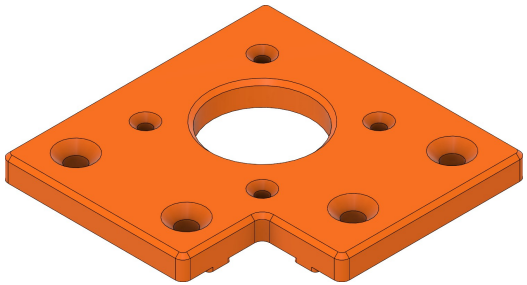
ALIGN THE MOTOR PLATE

RIGHT Z MOTOR MOUNT | DETAIL 1

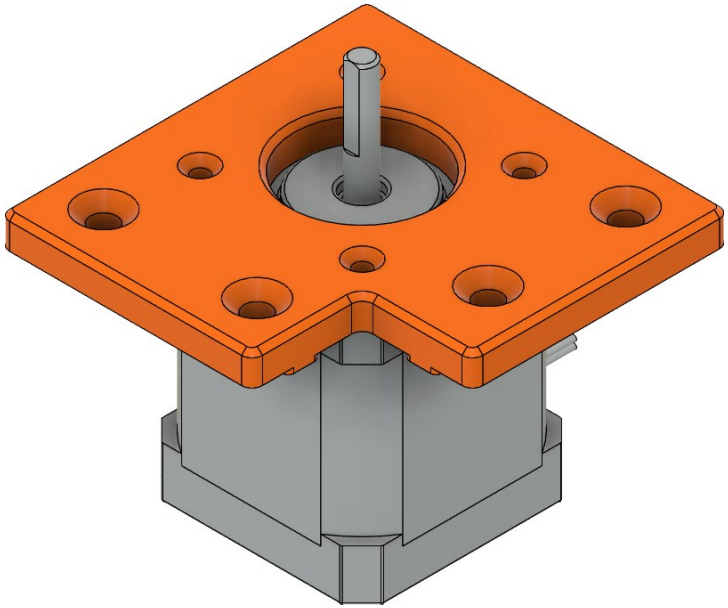
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position the matching Z motor plate over the motor face. Match the shaft opening and four mounting holes. Keep the lead exit accessible and the plate seated flat.

PARTS IN THIS CAD STEP

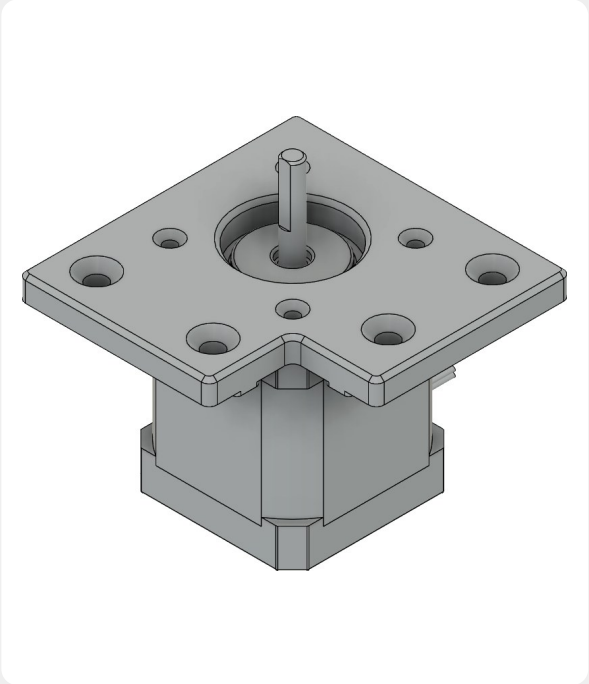
1 x Z Motor Plate Right

CHECK BEFORE CONTINUING: The plate is the correct left, right or rear part. The shaft passes through the opening without rubbing.

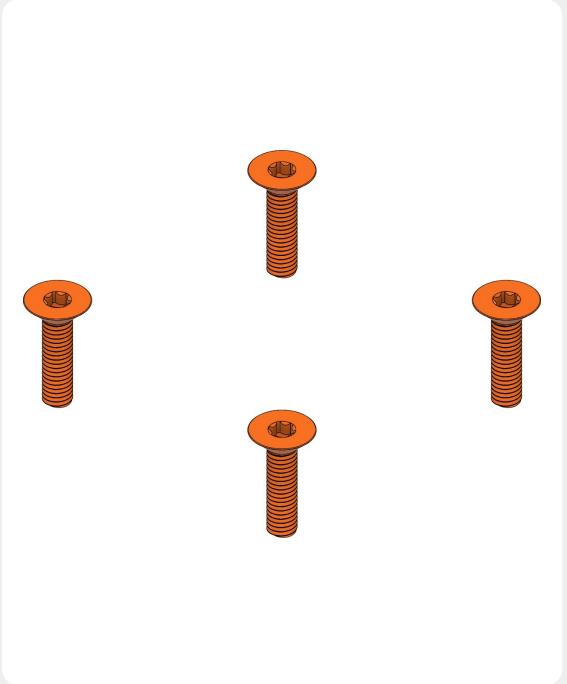
INSTALL THE FOUR M3 X 12 MOTOR SCREWS

RIGHT Z MOTOR MOUNT | DETAIL 2

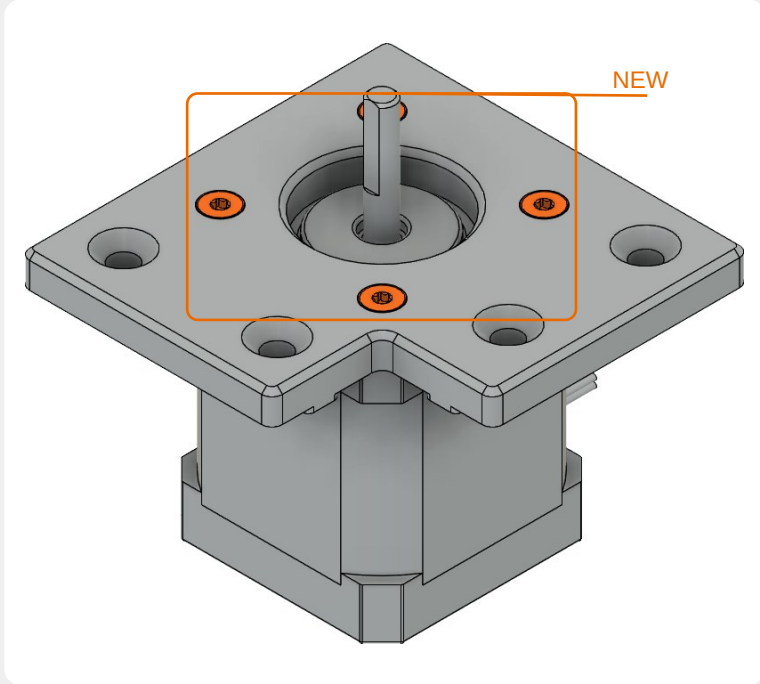
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the four M3 x 12 Torx flat-head screws through the motor plate into the motor. Engage all four threads by hand, then tighten progressively with the plate seated flat.

PARTS IN THIS CAD STEP

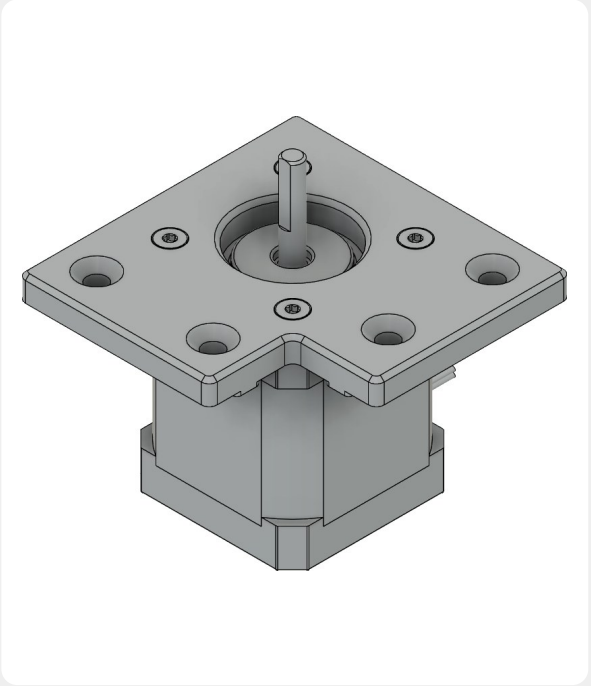
4 x M3 x 12 Torx flat-head [92703A151]

CHECK BEFORE CONTINUING: All four heads seat in the plate recesses. Verify engagement on the actual motor and confirm that no screw bottoms before the plate is clamped.

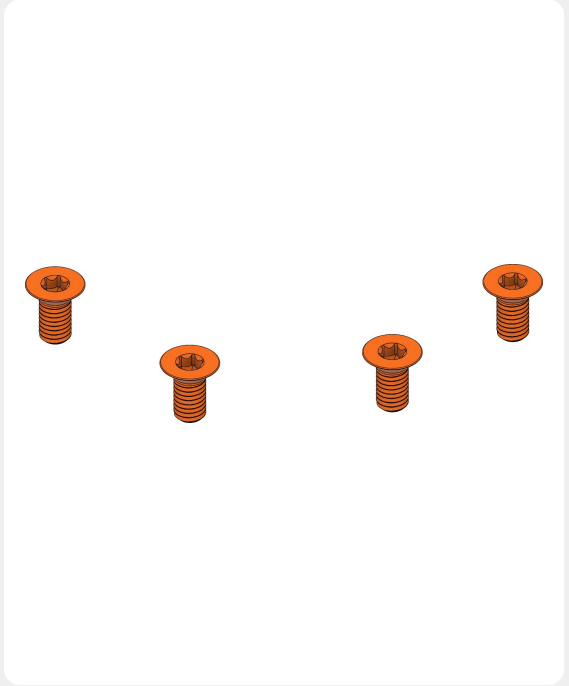
PREPARE THE FRAME SCREWS

RIGHT Z MOTOR MOUNT | DETAIL 3

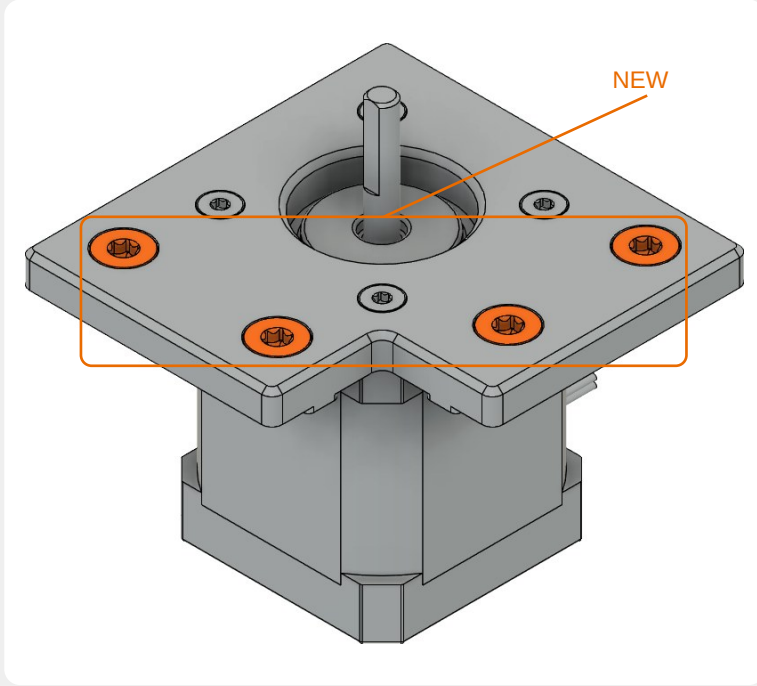
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the M5 x 10 Torx flat-head frame screws around the motor plate. Align the plate with its frame mounting locations before engaging the matching frame threads or reviewed slot hardware.

PARTS IN THIS CAD STEP

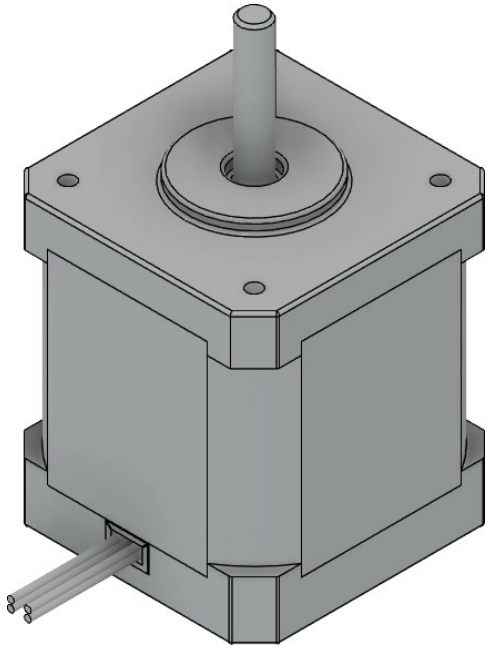
4 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: All mounting points align without bending the plate. Verify the mating thread or T-nut arrangement before tightening.

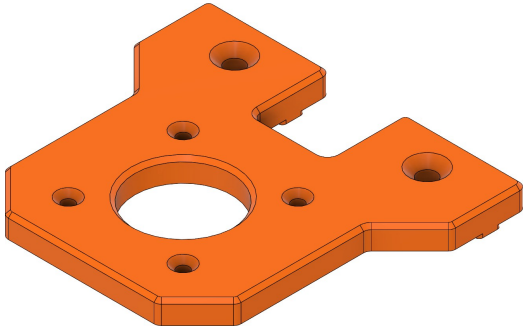
ALIGN THE MOTOR PLATE

REAR Z MOTOR MOUNT | DETAIL 1

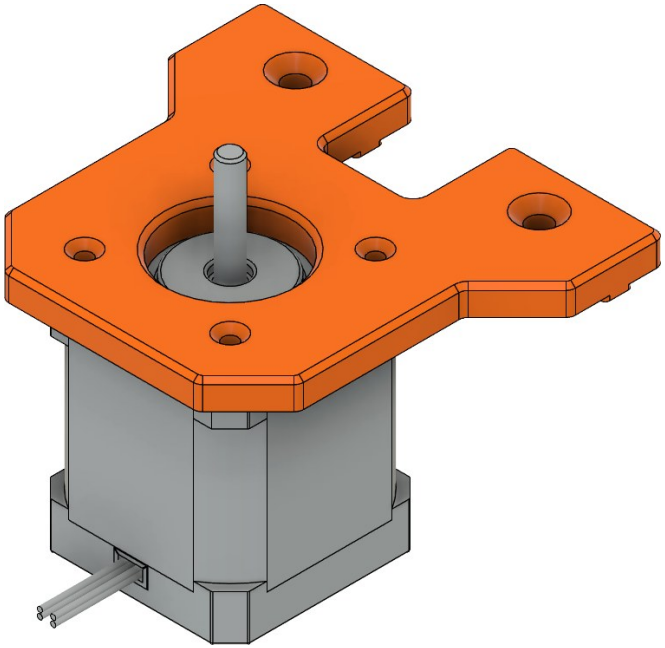
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position the matching Z motor plate over the motor face. Match the shaft opening and four mounting holes. Keep the lead exit accessible and the plate seated flat.

PARTS IN THIS CAD STEP

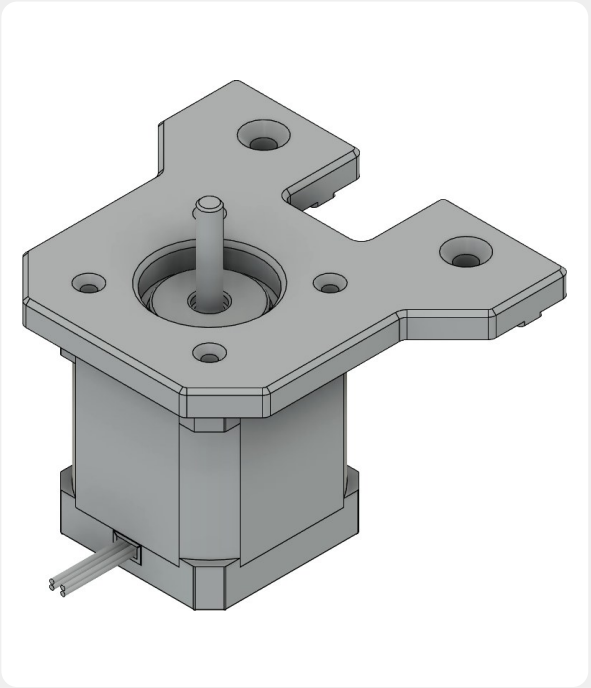
1 x Z Motor Plate Rear

CHECK BEFORE CONTINUING: The plate is the correct left, right or rear part. The shaft passes through the opening without rubbing.

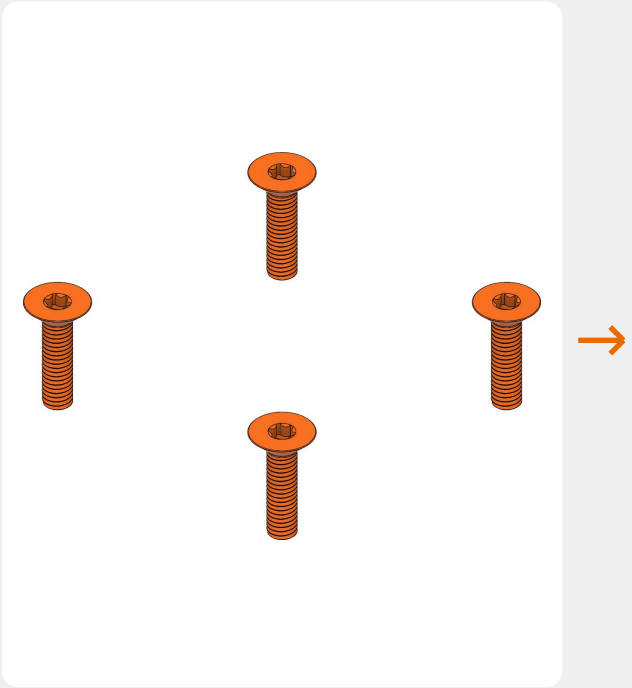
INSTALL THE FOUR M3 X 12 MOTOR SCREWS

REAR Z MOTOR MOUNT | DETAIL 2

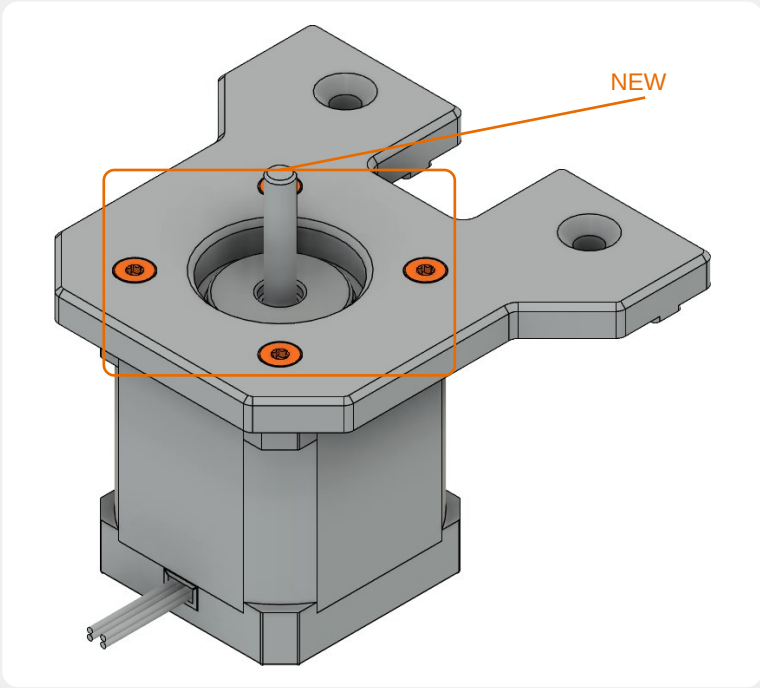
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the four M3 x 12 Torx flat-head screws through the motor plate into the motor. Engage all four threads by hand, then tighten progressively with the plate seated flat.

PARTS IN THIS CAD STEP

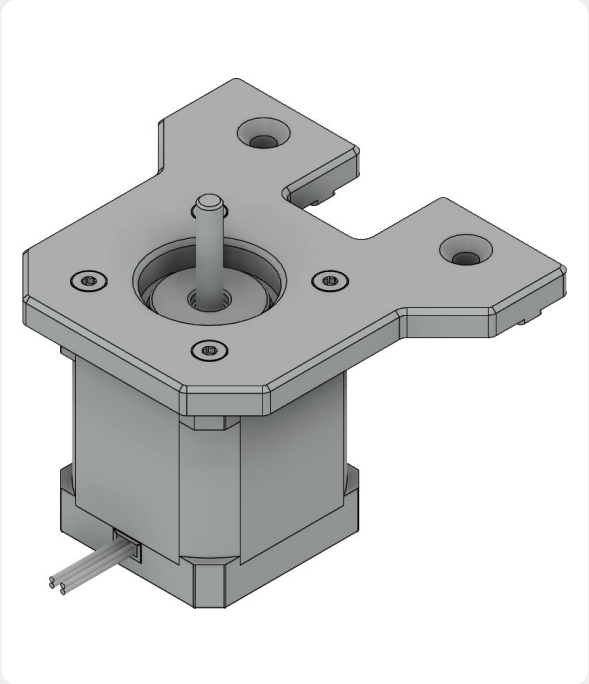
4 x M3 x 12 Torx flat-head [92703A151]

CHECK BEFORE CONTINUING: All four heads seat in the plate recesses. Verify engagement on the actual motor and confirm that no screw bottoms before the plate is clamped.

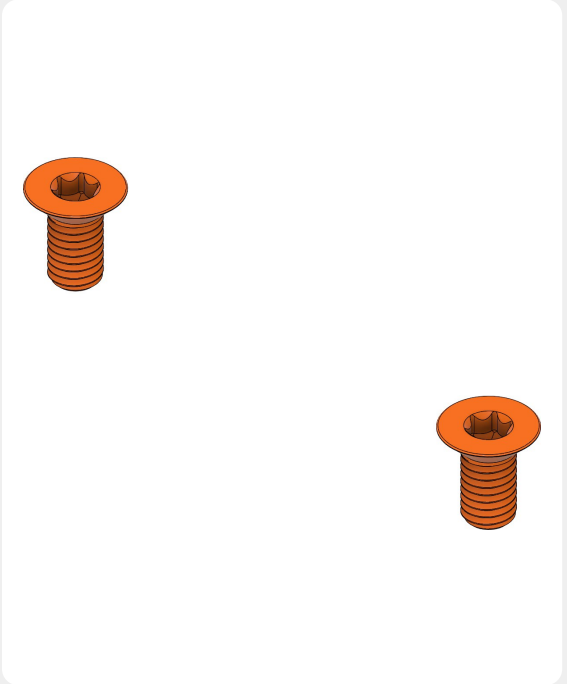
PREPARE THE FRAME SCREWS

REAR Z MOTOR MOUNT | DETAIL 3

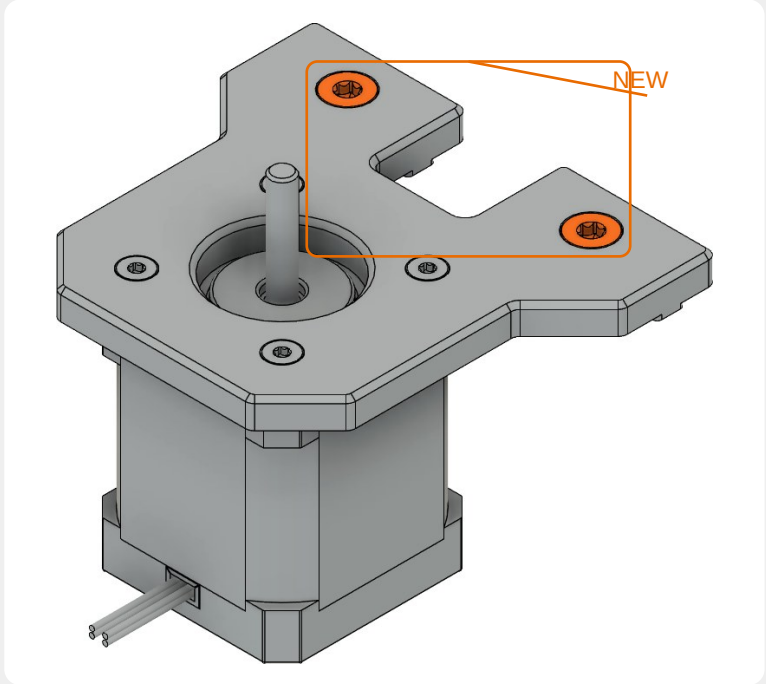
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the M5 x 10 Torx flat-head frame screws around the motor plate. Align the plate with its frame mounting locations before engaging the matching frame threads or reviewed slot hardware.

PARTS IN THIS CAD STEP

2 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: All mounting points align without bending the plate. Verify the mating thread or T-nut arrangement before tightening.

ORIENT THE REAR UPPER BRACE

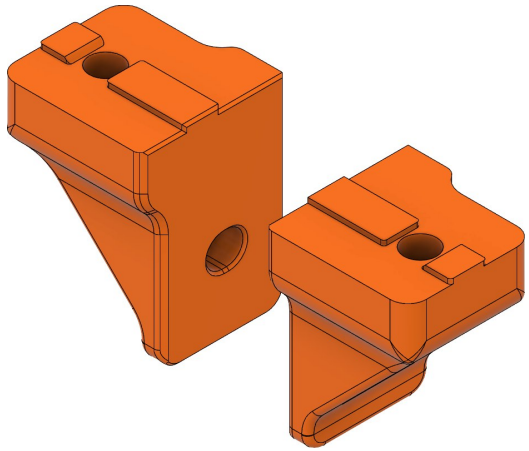
REAR UPPER BRACE | DETAIL 1

1 BEFORE

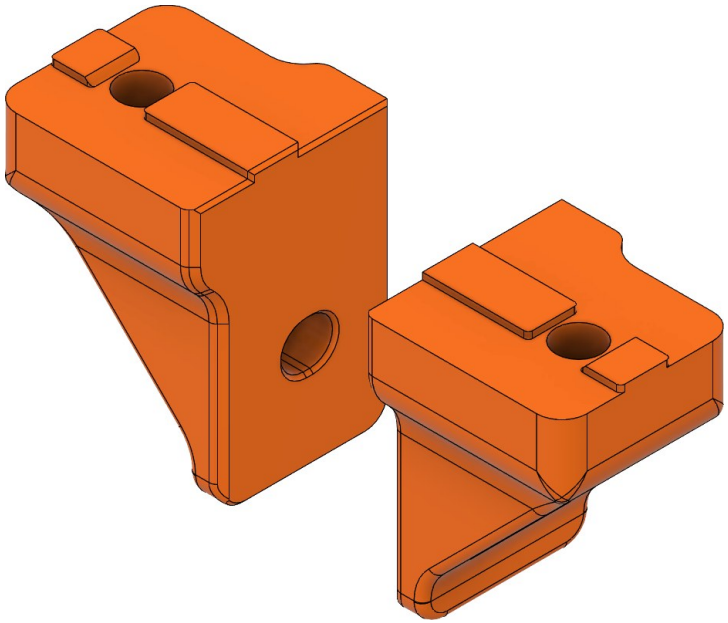
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the brace at the upper rear vertical extrusion, matching the orientation shown. Align the four mounting holes before inserting hardware.

PARTS IN THIS CAD STEP

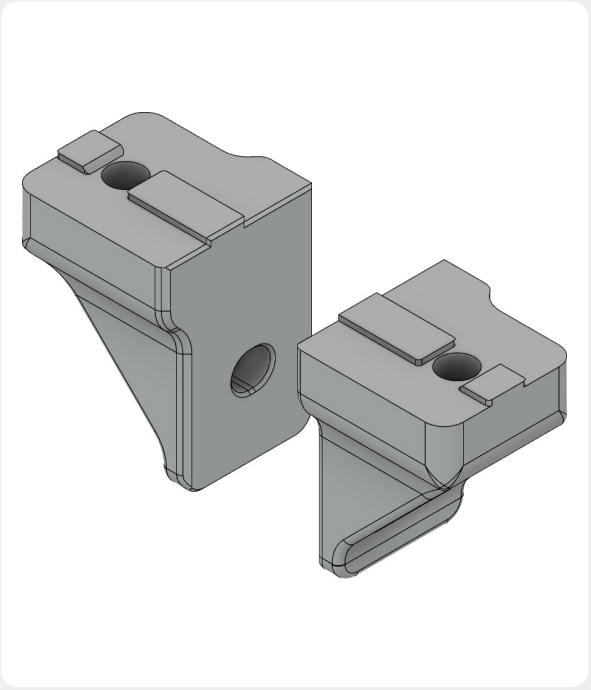
1 x Rear VertExt Upper Brace

CHECK BEFORE CONTINUING: The brace is centered on the rear upright; no nearby moving part is trapped.

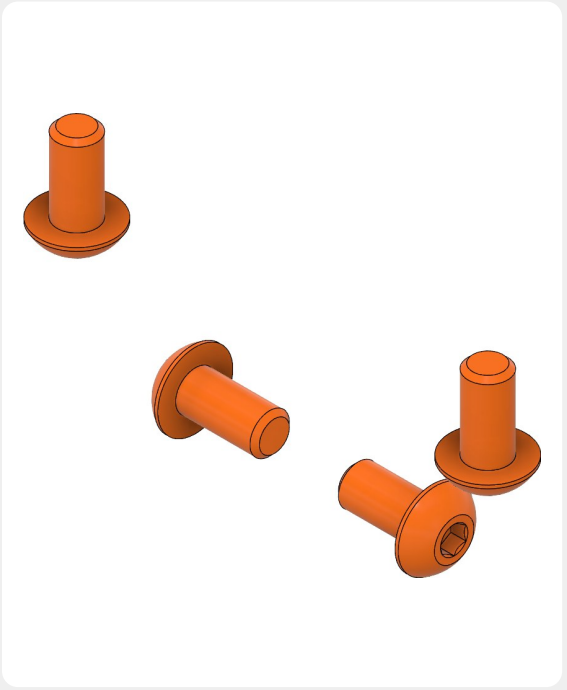
INSTALL THE FOUR BRACE SCREWS

REAR UPPER BRACE | DETAIL 2

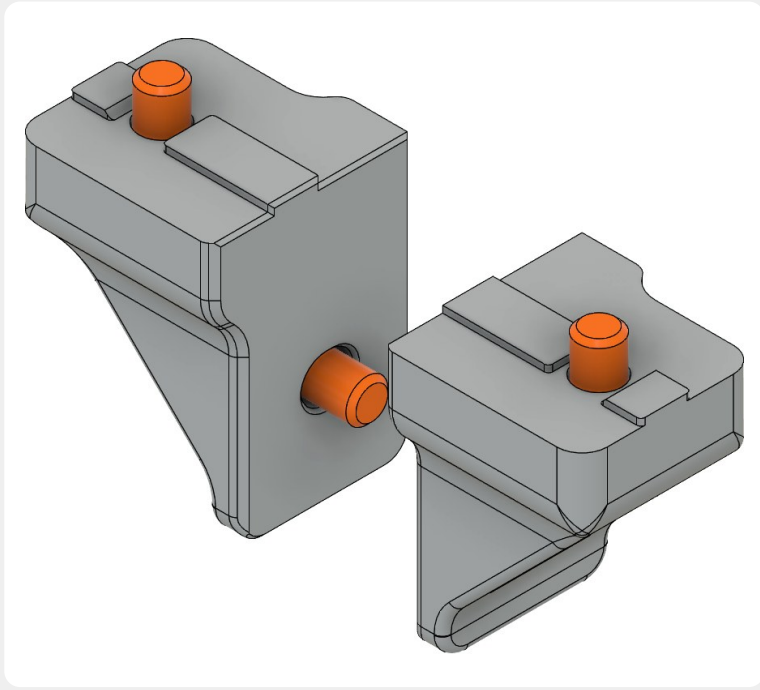
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the four M5 x 10 button-head screws in the highlighted holes. Bring the brace evenly against the extrusion, then tighten progressively.

PARTS IN THIS CAD STEP

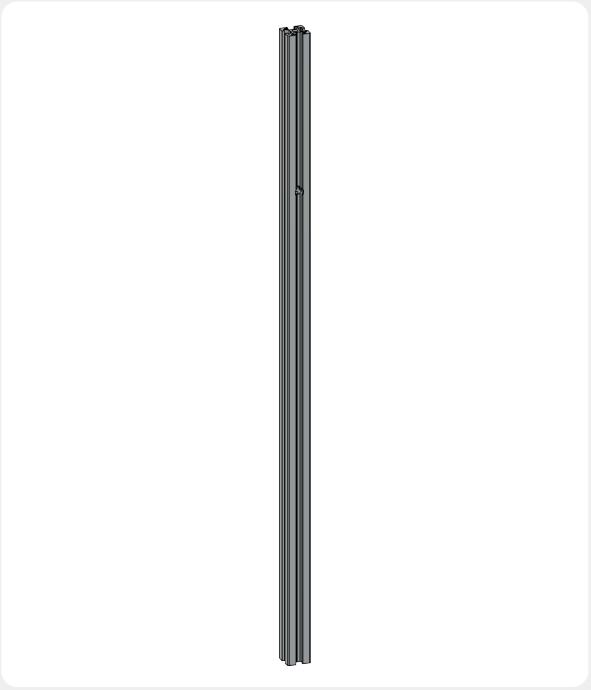
4 x M5x10 BHCS

CHECK BEFORE CONTINUING: The brace sits flush and remains square to the upright.

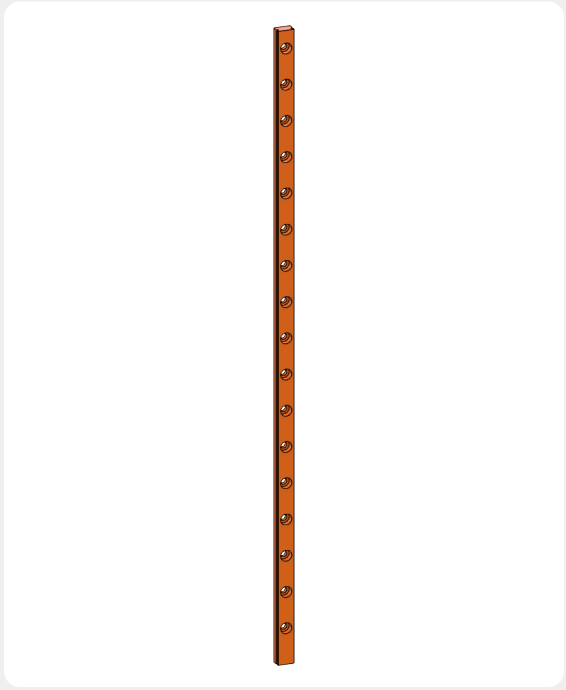
POSITION THE RAIL ON ITS EXTRUSION

RIGHT Z RAIL | DETAIL 1

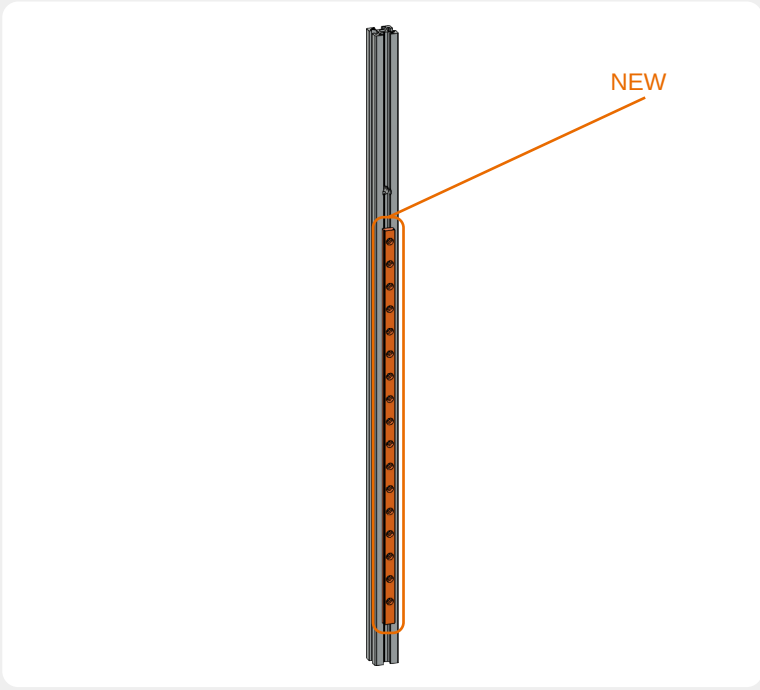
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use the rail for this side and keep its carriage on the rail. Align the mounting holes with the extrusion slot. Match the rail orientation shown; leave room to adjust its position before tightening.

PARTS IN THIS CAD STEP

1 x 9 mm Wide x 350 mm Long 440 Stainless Steel Rail galaxy 350

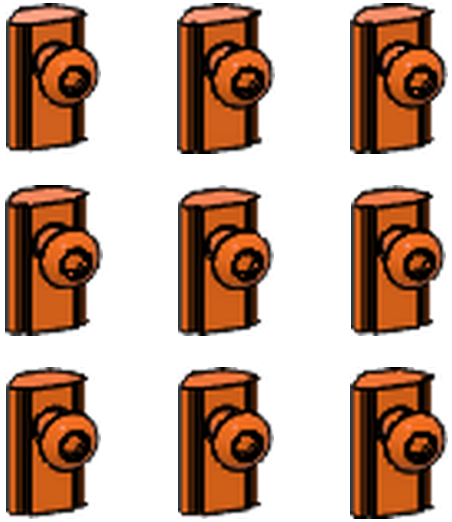
CHECK BEFORE CONTINUING: The rail sits flat on the mounting face. Keep the carriage from sliding off either end.

INSTALL THE NINE RAIL FASTENERS

RIGHT Z RAIL | DETAIL 2

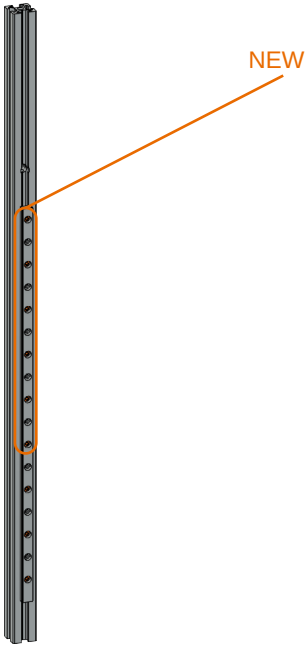
1 BEFORE

2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align nine M3 roll-in T-nuts in the slot with the rail holes. Start all nine M3 x 8 Torx button-head screws loosely. Align the rail, then tighten progressively while checking smooth carriage travel.

PARTS IN THIS CAD STEP

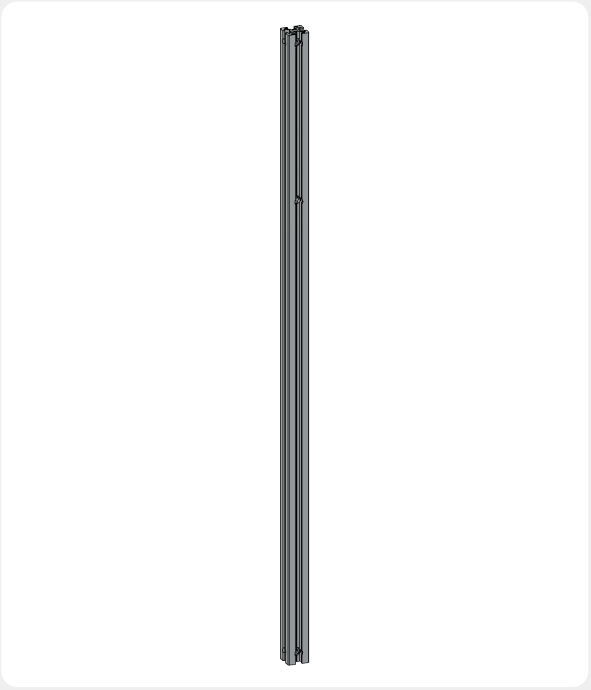
9 x M3 x 8 Torx button-head [90991A113]
9 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: Every screw engages a T-nut. Slide the carriage through its usable travel; stop and realign if it binds. CAD nut centers are not installation measurements.

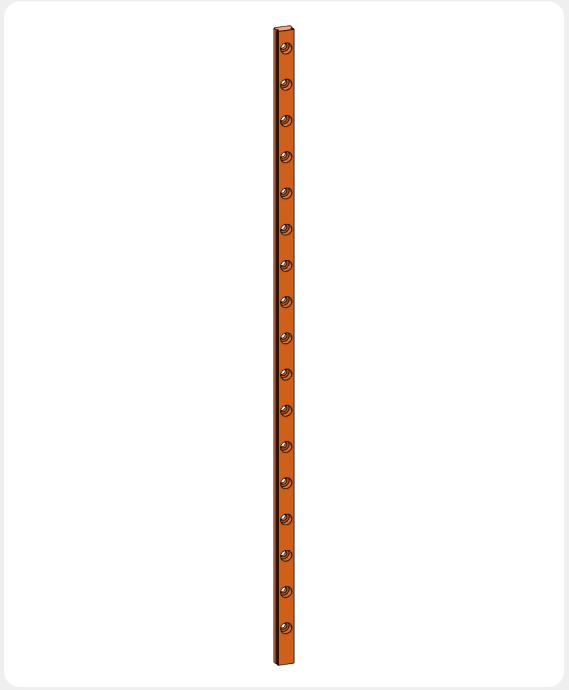
POSITION THE RAIL ON ITS EXTRUSION

LEFT Z RAIL | DETAIL 1

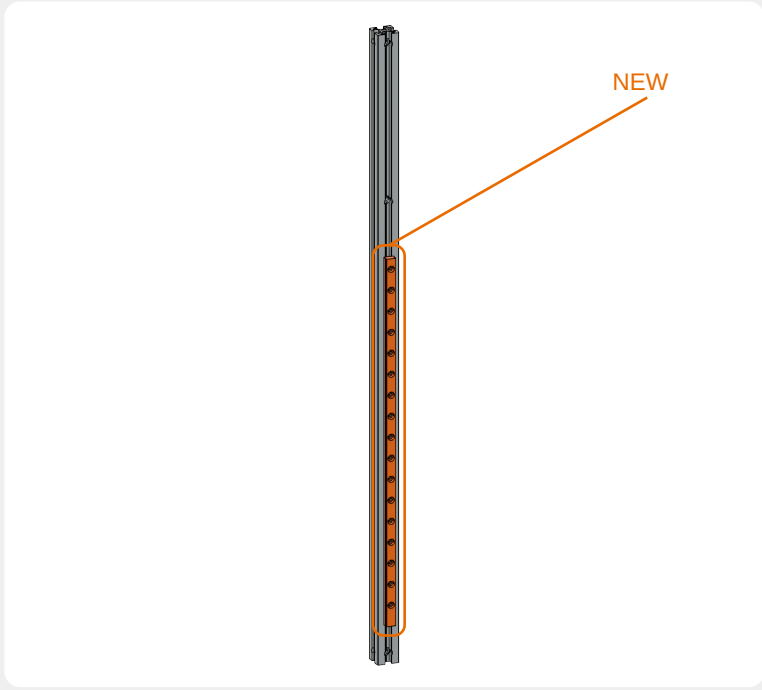
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Use the rail for this side and keep its carriage on the rail. Align the mounting holes with the extrusion slot. Match the rail orientation shown; leave room to adjust its position before tightening.

PARTS IN THIS CAD STEP

1 x 9 mm Wide x 350 mm Long 440 Stainless Steel Rail galaxy 350

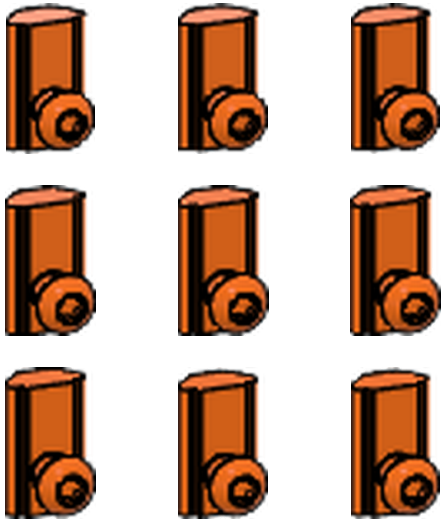
CHECK BEFORE CONTINUING: The rail sits flat on the mounting face. Keep the carriage from sliding off either end.

INSTALL THE NINE RAIL FASTENERS

LEFT Z RAIL | DETAIL 2

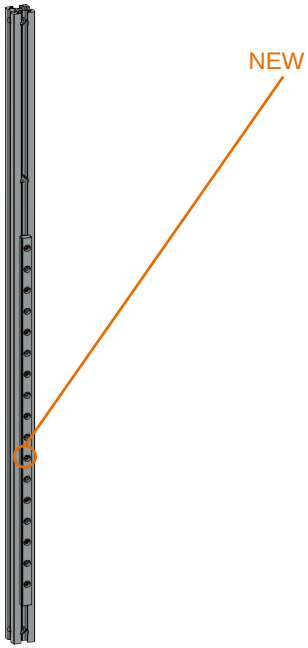
1 BEFORE

2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

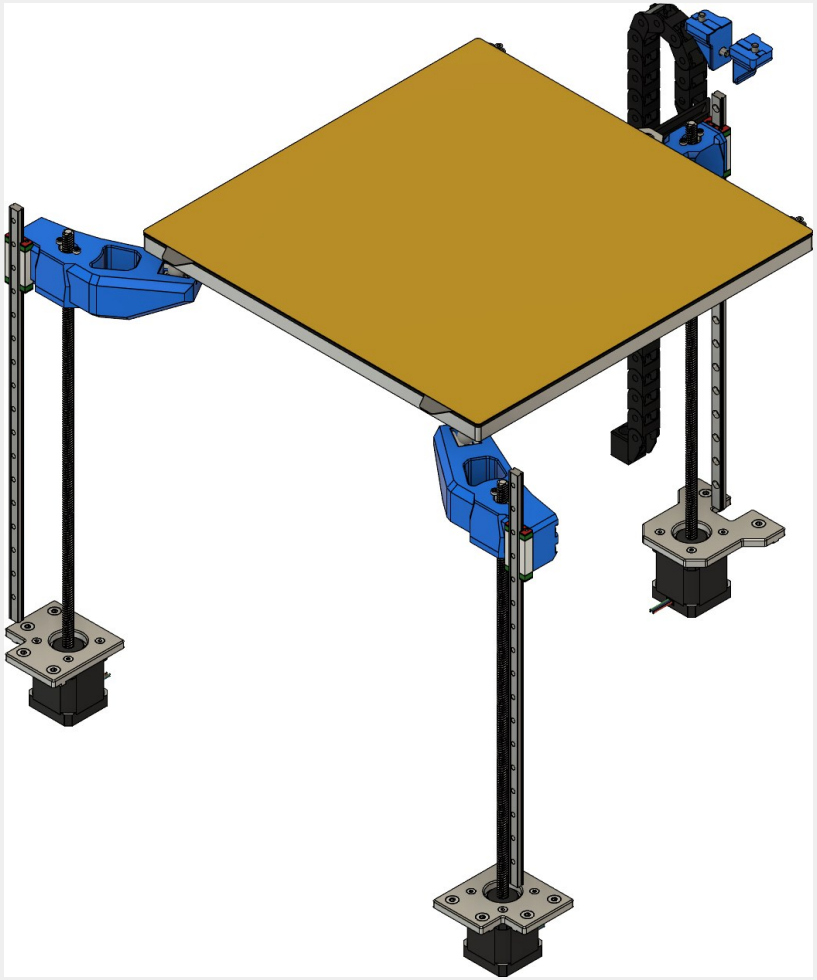
Align nine M3 roll-in T-nuts in the slot with the rail holes. Start all nine M3 x 8 Torx button-head screws loosely. Align the rail, then tighten progressively while checking smooth carriage travel.

PARTS IN THIS CAD STEP

9 x M3 x 8 Torx button-head [90991A113]
9 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: Every screw engages a T-nut. Slide the carriage through its usable travel; stop and realign if it binds. CAD nut centers are not installation measurements.

SEAT THE BED ON ITS THREE SUPPORTS



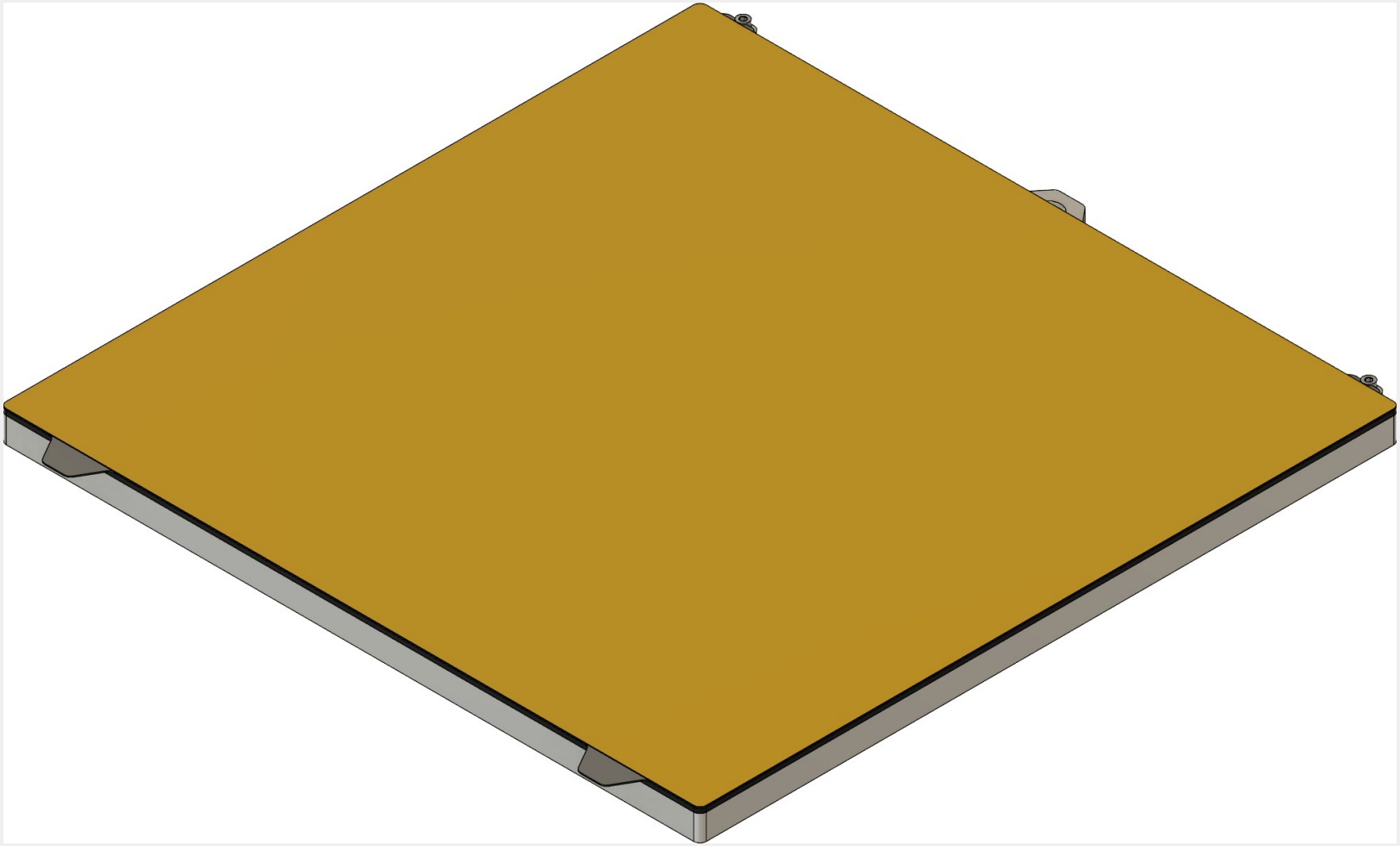
Support the bed while bringing all three arms to a common level. Place the bed on the three modeled support points and check that each contact seats without forcing the other two.

CAD BED LABEL
330 mm x 330 mm aluminum plate
Verify physical dimensions and hole pattern.

CHECK: All three supports are engaged. Check underside clearance and retention before moving the bed.

Current Hybrid assembly

CHECK THE BED SURFACE STACK



Identify the aluminum plate, magnetic layer, spring steel and PEI surface in the model. Match the installed kit to its own heater and surface instructions before attaching any adhesive layers.

CAD LAYERS
Aluminum plate
Magnet
Spring steel
PEI

CHECK: Heater attachment, electrical protection and thermal-sensor placement need their own verified instructions.

Current Hybrid assembly

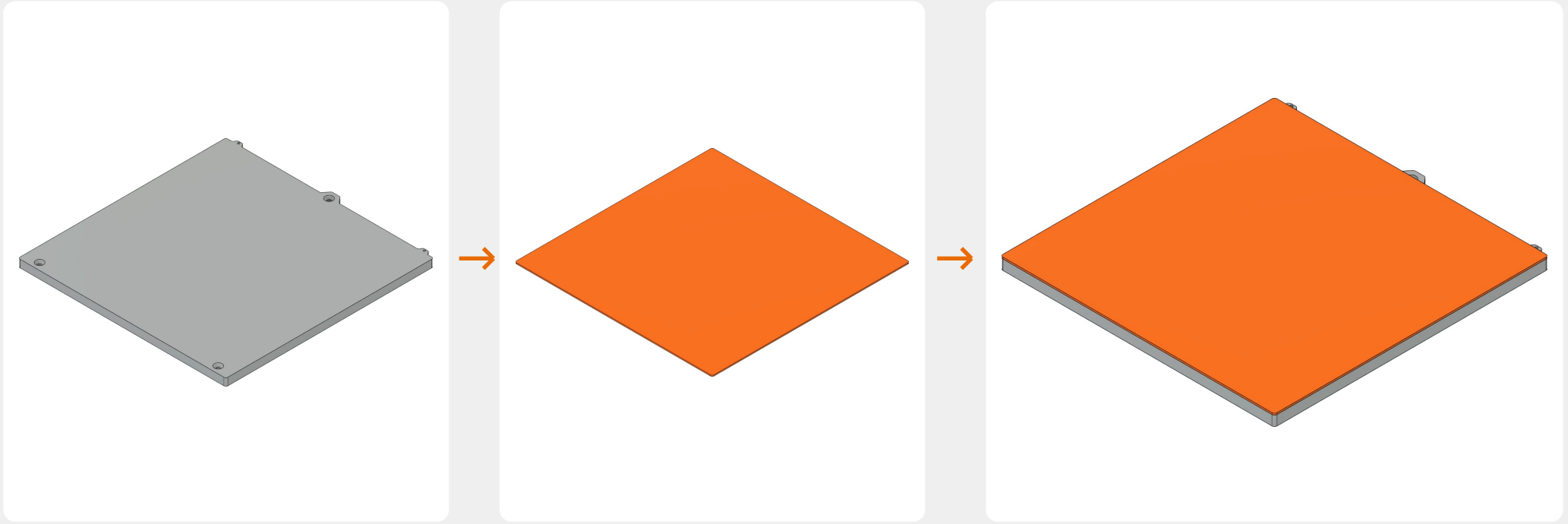
POSITION THE MAGNETIC LAYER

BUILD PLATE LAYERS | DETAIL 1

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Clean the plate and position the magnetic layer to match the outline shown. Check alignment before committing any adhesive backing. Follow the magnetic sheet supplier's application instructions.

PARTS IN THIS CAD STEP

1 x magnet

CHECK BEFORE CONTINUING: The sheet lies flat without trapped debris, bubbles or overhanging edges.

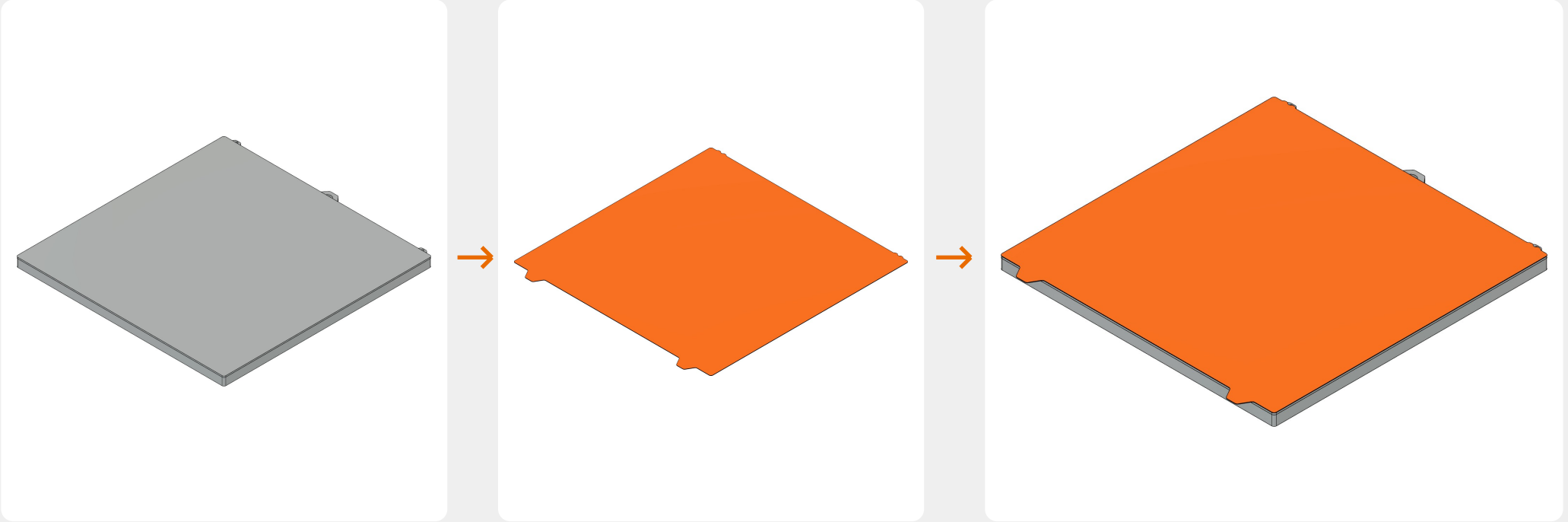
PLACE THE SPRING-STEEL SHEET

BUILD PLATE LAYERS | DETAIL 2

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Lower the spring-steel sheet onto the magnetic layer. Keep its edges aligned with the bed and keep fingers clear as the magnet draws it down.

PARTS IN THIS CAD STEP

1 x spring steel

CHECK BEFORE CONTINUING: The sheet is fully supported and can be lifted using its intended removal edge.

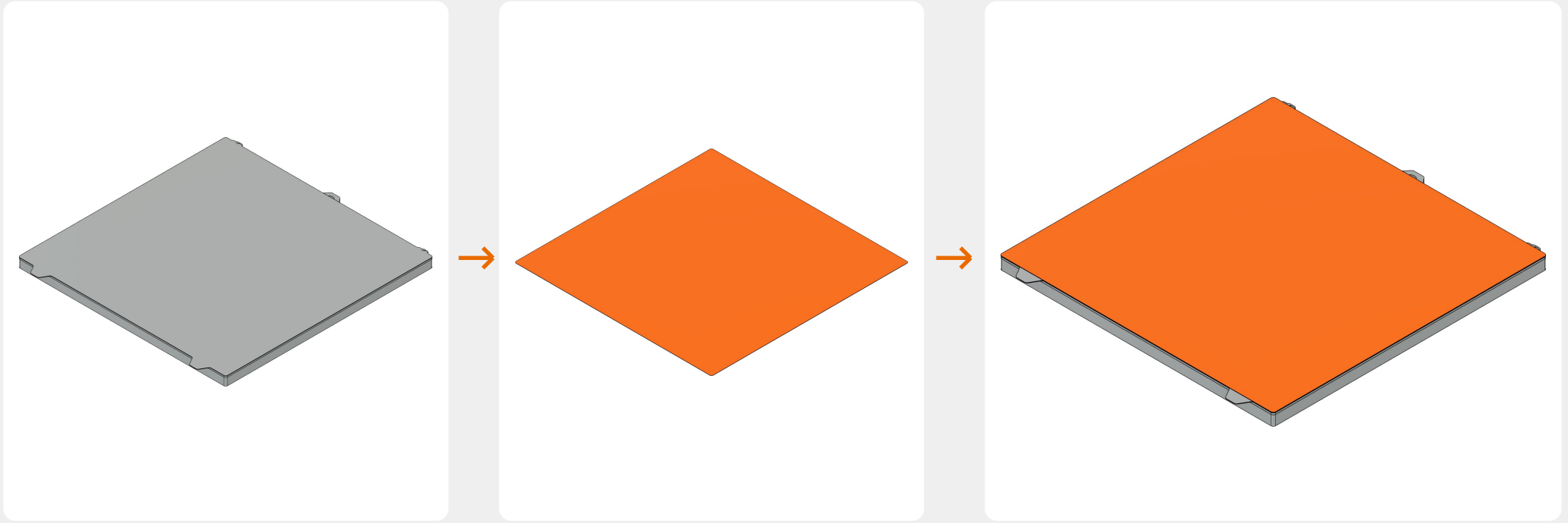
CHECK THE PEI PRINTING SURFACE

BUILD PLATE LAYERS | DETAIL 3

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

The CAD shows PEI as a separate top layer. If your spring steel is already PEI-coated, use that finished sheet. For a separate adhesive PEI film, follow its supplier's application instructions.

PARTS IN THIS CAD STEP

1 x pei

CHECK BEFORE CONTINUING: The printing surface is clean and flat. Do not add a second PEI layer to an already coated sheet.

ROUTE THE REAR CABLE CHAIN

Fit the fixed and moving ends of the rear chain to their modeled mounts. Plan a service loop at both ends and test the chain with the bed supported through its intended travel.

CAD GROUP
10x11 Chain X
Rear chain mount
Rear arm attachment

CHECK: The chain bends freely and clears the rear support. Final harness length and bend radius depend on the actual cable.

Current Hybrid assembly

ORIENT THE LOWER CHAIN MOUNT

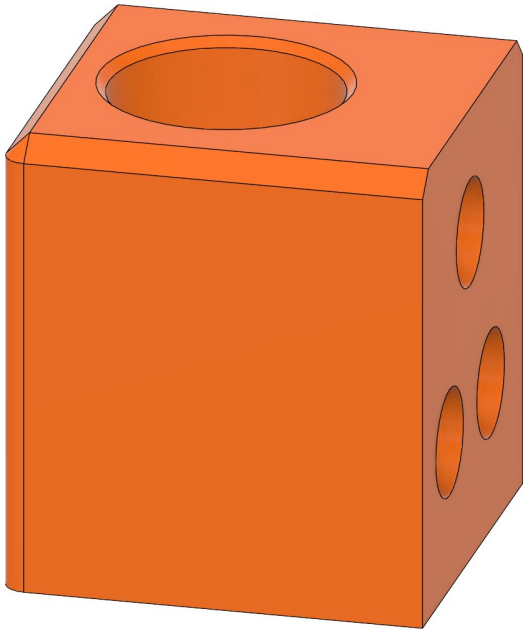
BED CABLE CHAIN | DETAIL 1

1 BEFORE

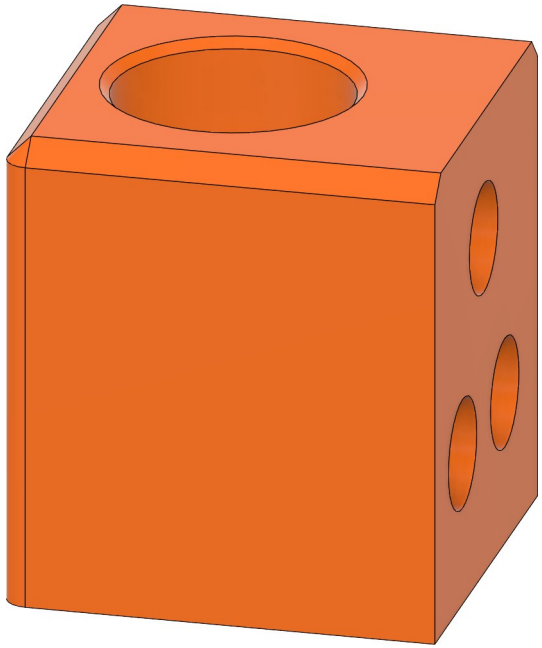
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the lower chain mount at the rear frame, as shown in the overview. Orient its end-link seat to match this view. The CAD export does not identify its mounting fasteners; confirm the fixing hardware before fastening it.

PARTS IN THIS CAD STEP

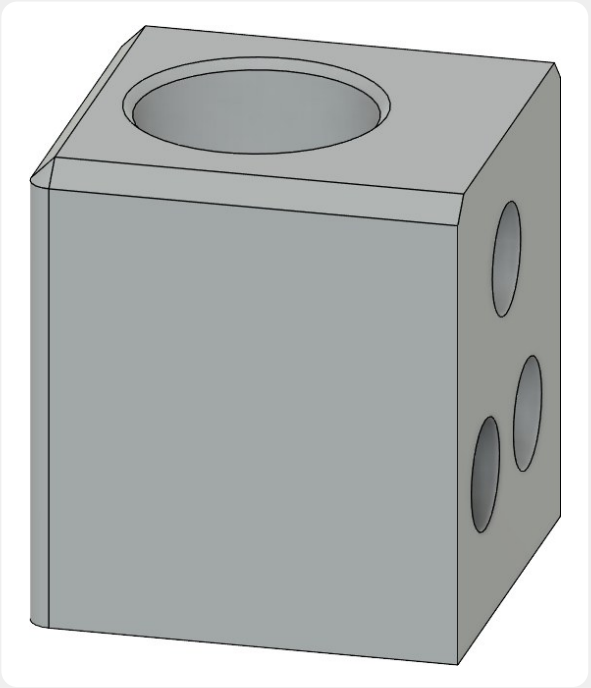
1 x chian mount

CHECK BEFORE CONTINUING: The mount orientation is established. Mounting fastener size and quantity remain to be confirmed.

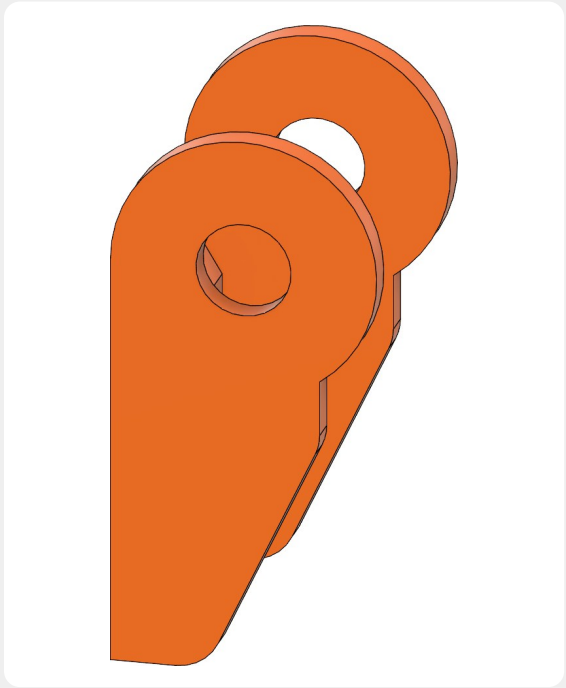
POSITION THE LOWER END FITTING

BED CABLE CHAIN | DETAIL 2

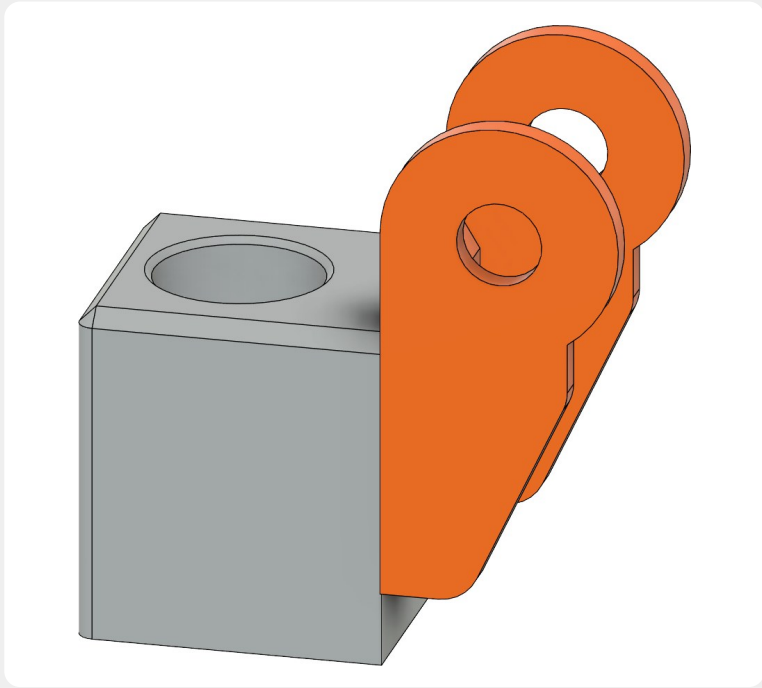
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the lower end fitting to the mount in the orientation shown. Align the fixing holes. End-fixing screws are not identified in this CAD group; confirm the actual end-fitting hardware before securing it.

PARTS IN THIS CAD STEP

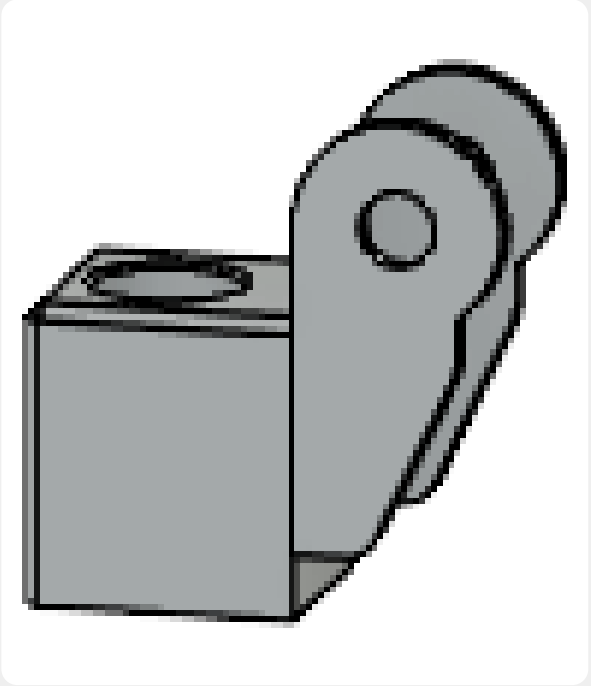
1 x 10x11 Unfixed End

CHECK BEFORE CONTINUING: The end fitting points into the vertical chain run and its pivot area remains clear.

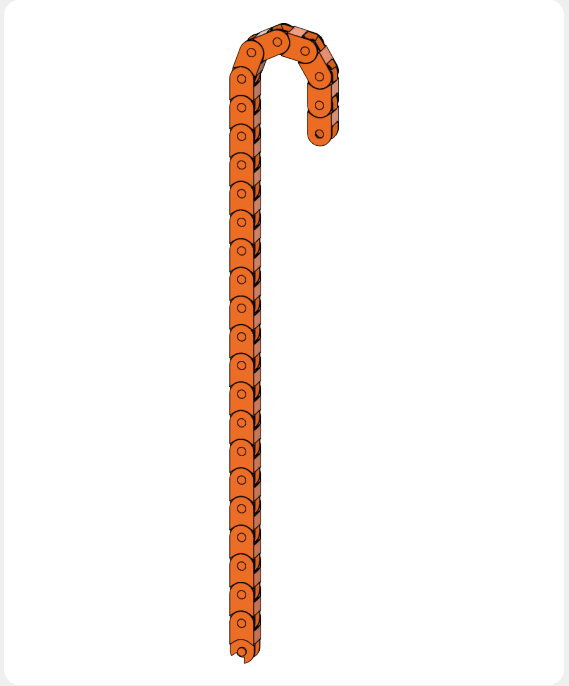
CONNECT AND CURVE THE CHAIN LINKS

BED CABLE CHAIN | DETAIL 3

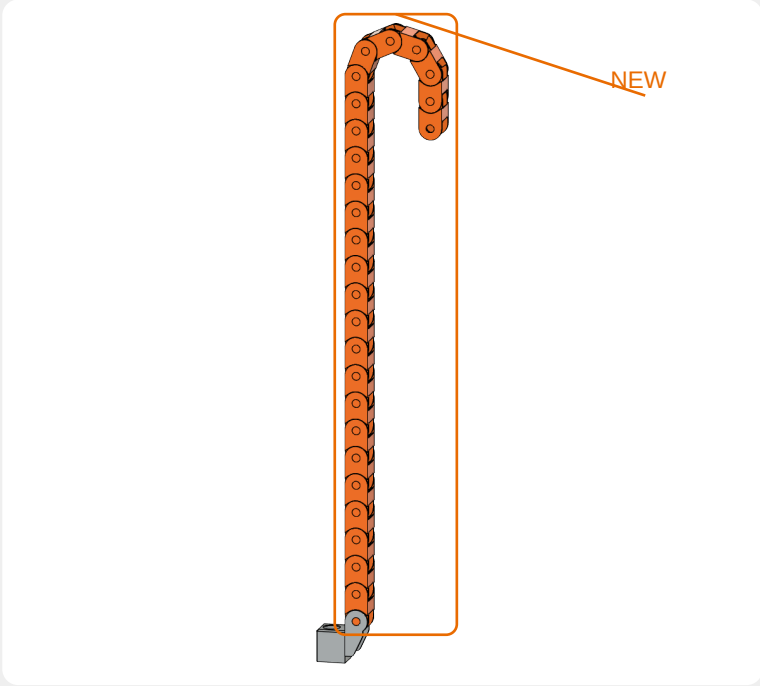
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Connect the chain links to the lower end fitting and form the return bend shown. The CAD contains 26 links; confirm the required length with the actual travel. Use the chain supplier's method for joining links and opening crossbars.

PARTS IN THIS CAD STEP

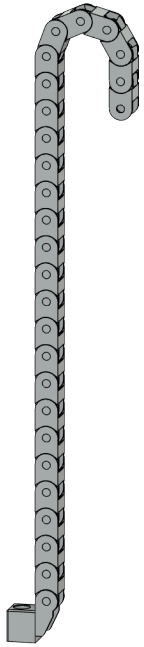
26 x 10x11 Chain Link

CHECK BEFORE CONTINUING: Links pivot freely in their intended direction. Do not force a link to bend backward or use the drawing as a cable-length measurement.

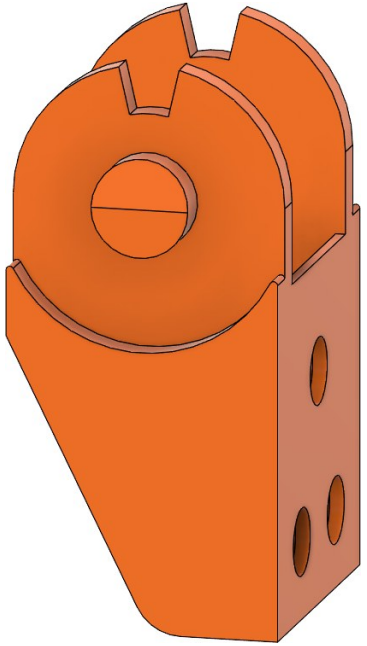
POSITION THE UPPER END FITTING

BED CABLE CHAIN | DETAIL 4

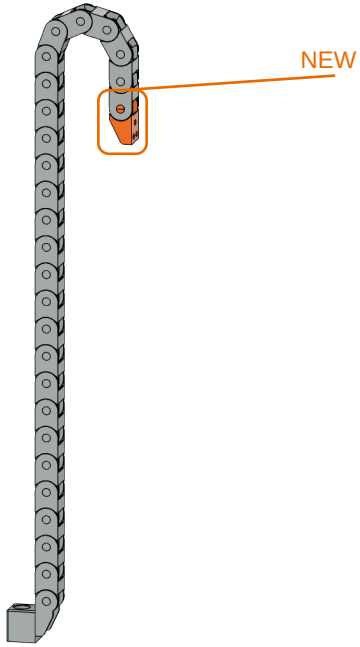
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

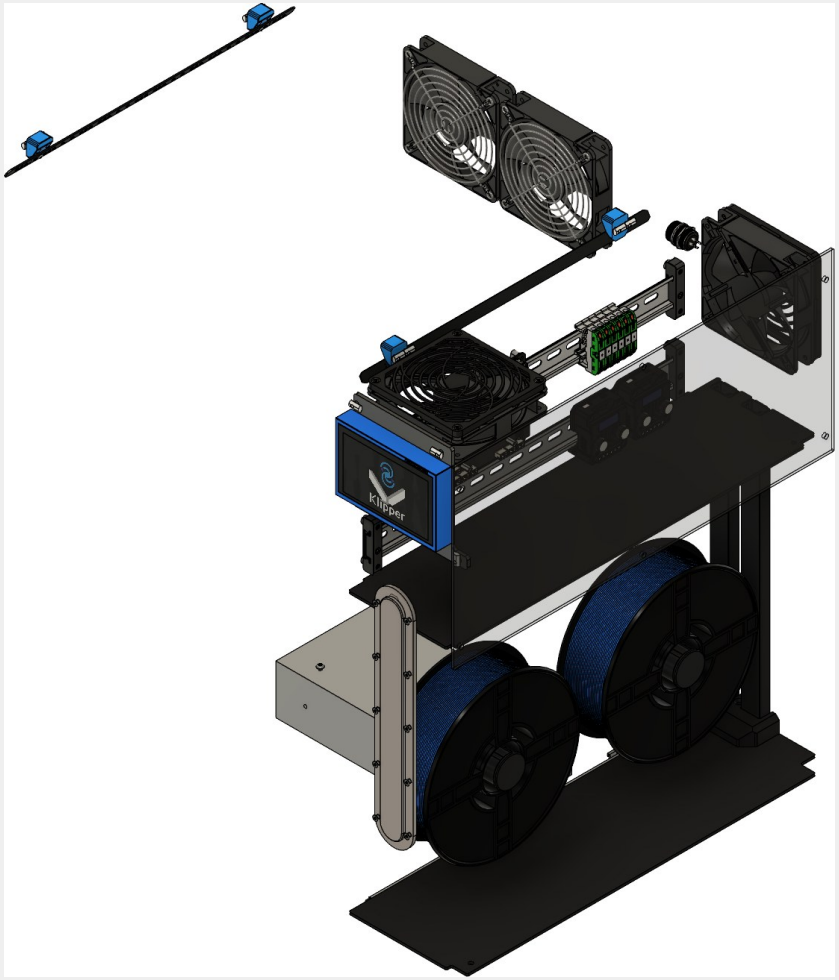
Join the upper end fitting to the chain and align it with the moving bed-side mounting point shown in the overview. Confirm its fasteners before attachment. Leave cable slack for full travel and provide strain relief at both ends.

PARTS IN THIS CAD STEP

1 x 10x11 Fixed End

CHECK BEFORE CONTINUING: Move the bed slowly through its intended travel before powering it. The chain must not pull tight, rub the frame or pinch cables. End fasteners remain unverified.

DRY-FIT THE ELECTRONICS LAYOUT



Use the CAD layout to identify the controller, power supply, DIN rails, terminals, display, relays, fans and wire channels. Dry-fit their mounting hardware while the bay is accessible.

MODELED LABELS
LDO Leviathan
BTT PI TFT50
XP Power X7
Confirm installed versions.

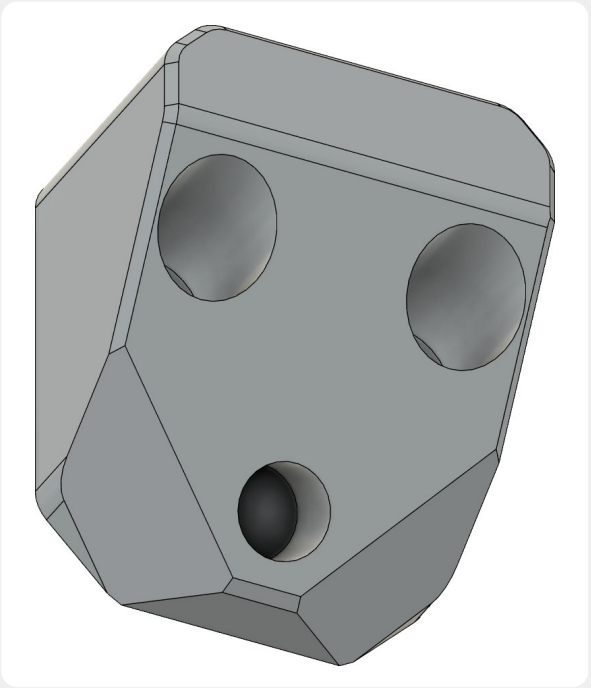
CHECK: Mounting access and ventilation remain clear. This illustration is a placement reference, not a wiring diagram.

Current Hybrid assembly

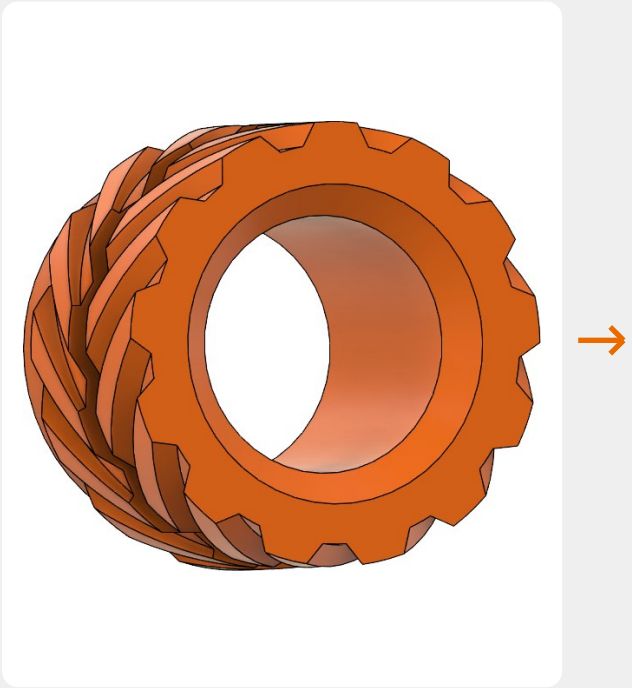
INSTALL THE M3 HEAT-SET INSERT

LEFT FRONT LIGHT BRACKET | DETAIL 1

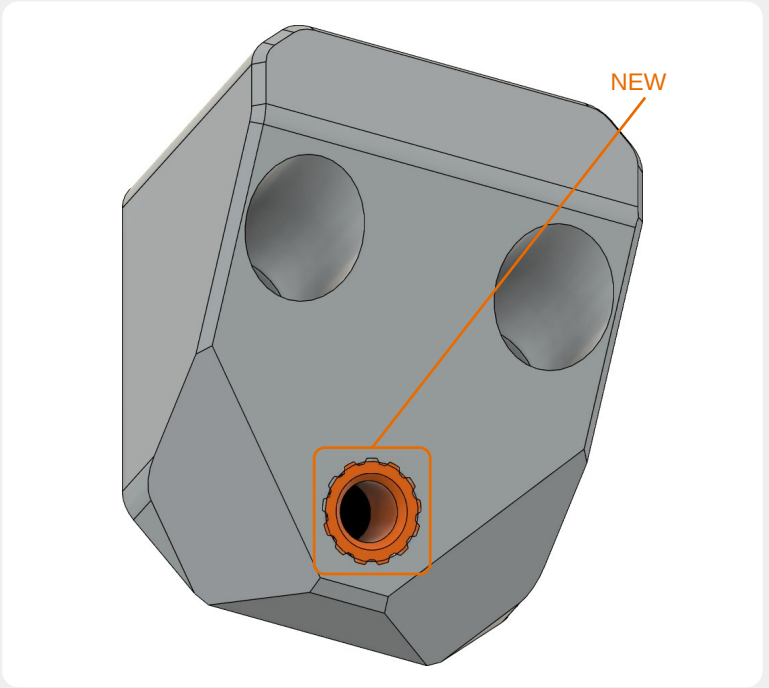
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the M3 heat-set insert in the highlighted pocket. Use an insert-setting tip and the temperature recommended for your printed material. Press straight and let it cool before fitting a screw.

PARTS IN THIS CAD STEP

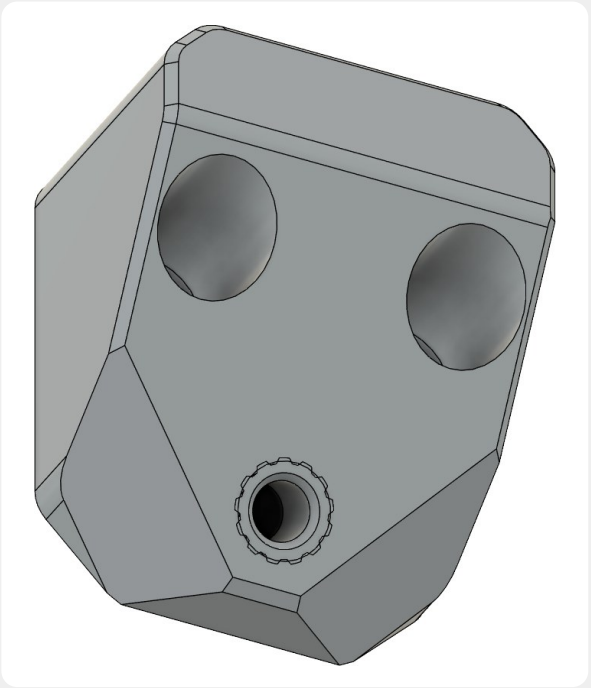
1 x M3 heat-set insert

CHECK BEFORE CONTINUING: The insert is straight and flush with its intended seat; the threaded opening remains clear.

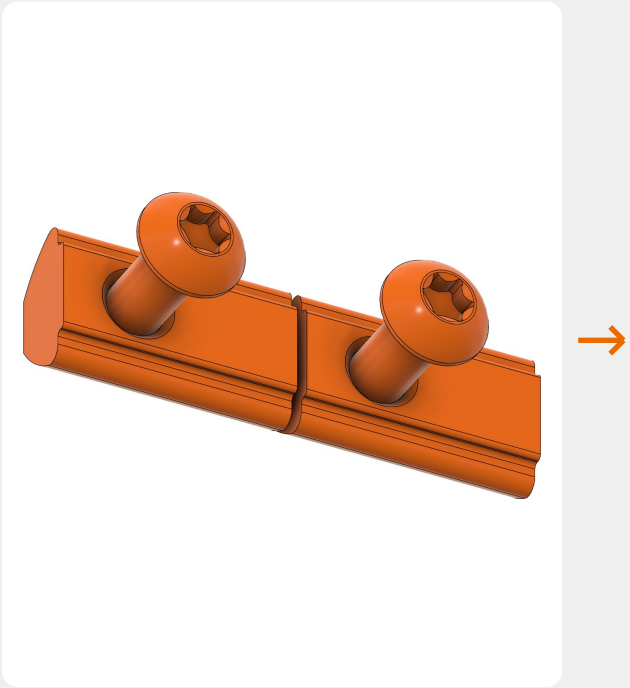
FIT THE FRAME MOUNTING HARDWARE

LEFT FRONT LIGHT BRACKET | DETAIL 2

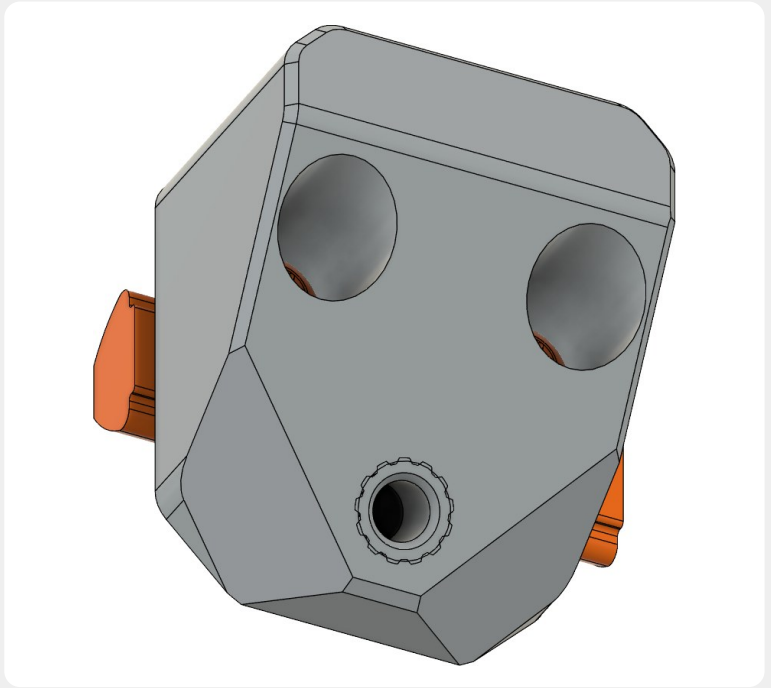
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit two M3 roll-in T-nuts into the frame slot. Position the bracket and start the two M3 x 8 Torx button-head screws through the deep access holes. Reach the recessed heads with a straight driver, then tighten evenly.

PARTS IN THIS CAD STEP

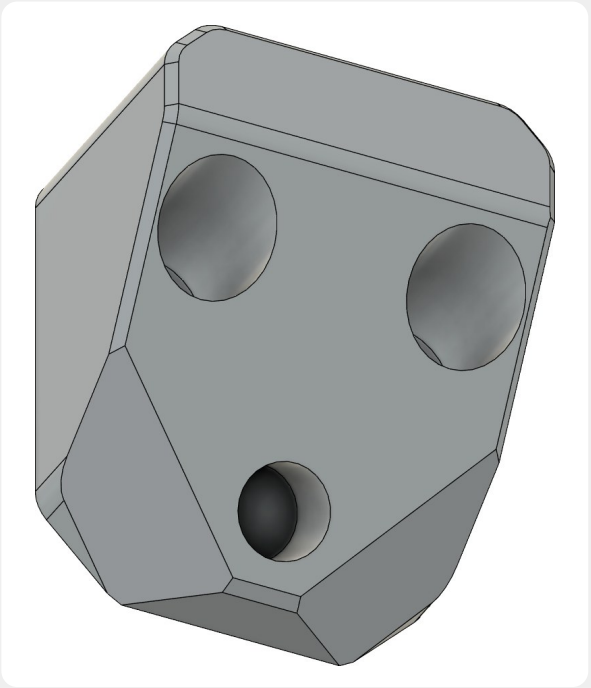
2 x M3 x 8 Torx button-head [90991A113]
2 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: The orange screw heads sit deep inside the two access holes. Both T-nuts engage the frame slot and the bracket is secure.

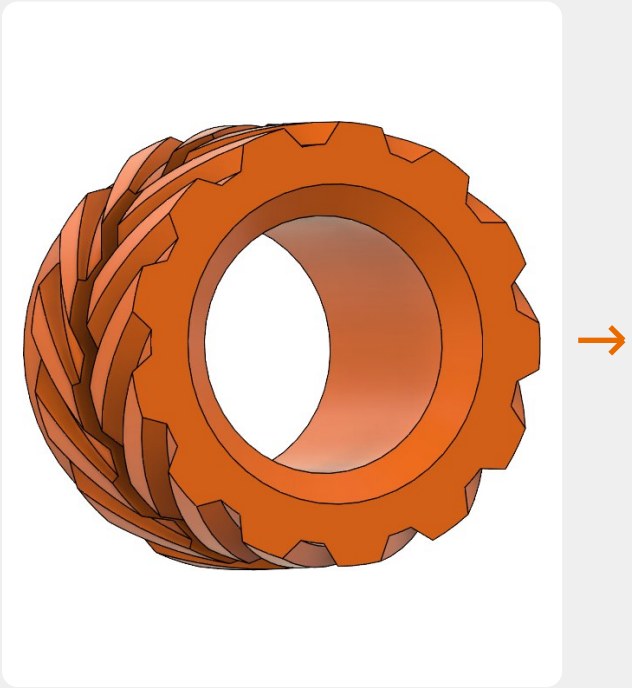
INSTALL THE M3 HEAT-SET INSERT

LEFT REAR LIGHT BRACKET | DETAIL 1

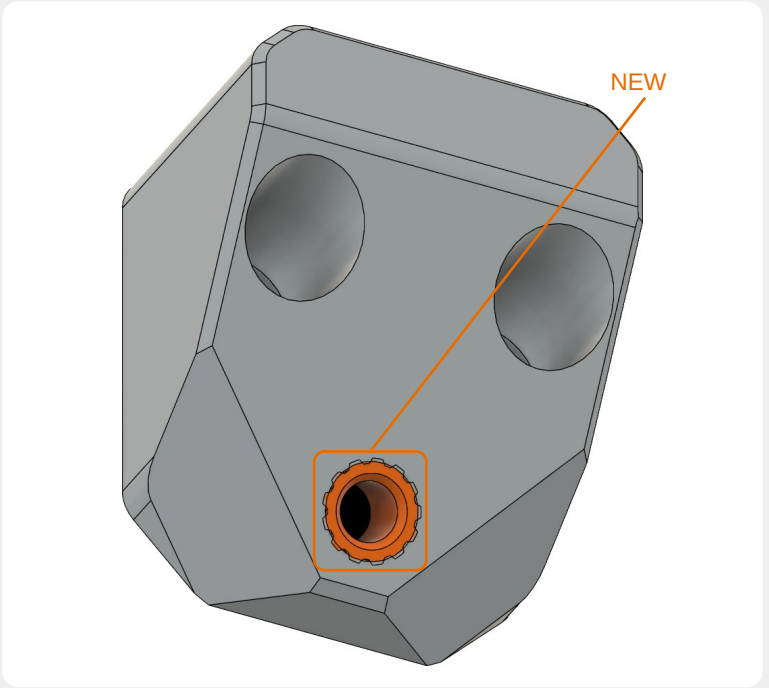
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the M3 heat-set insert in the highlighted pocket. Use an insert-setting tip and the temperature recommended for your printed material. Press straight and let it cool before fitting a screw.

PARTS IN THIS CAD STEP

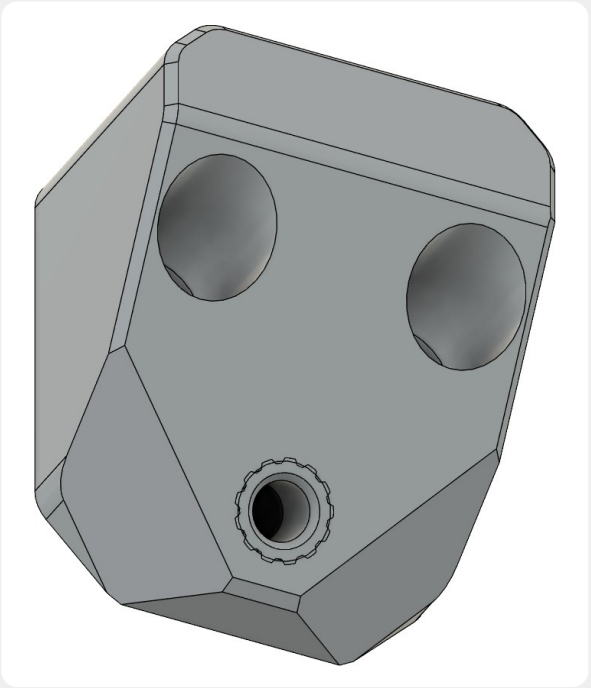
1 x M3 heat-set insert

CHECK BEFORE CONTINUING: The insert is straight and flush with its intended seat; the threaded opening remains clear.

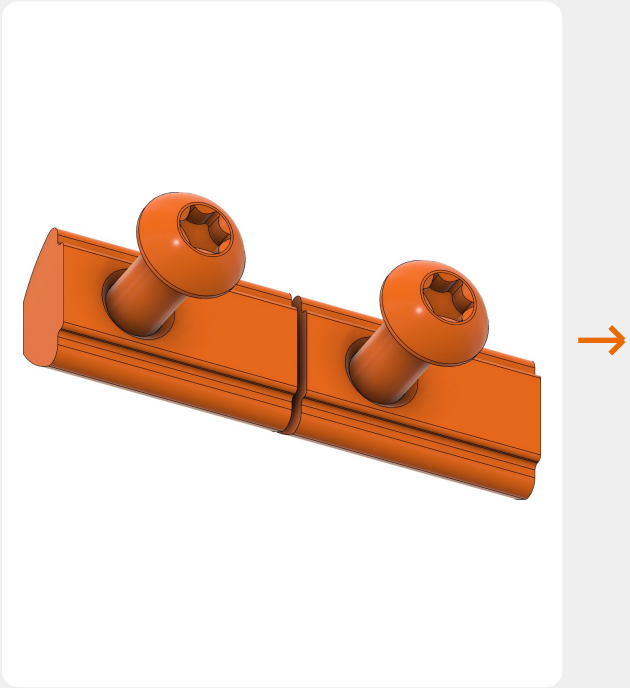
FIT THE FRAME MOUNTING HARDWARE

LEFT REAR LIGHT BRACKET | DETAIL 2

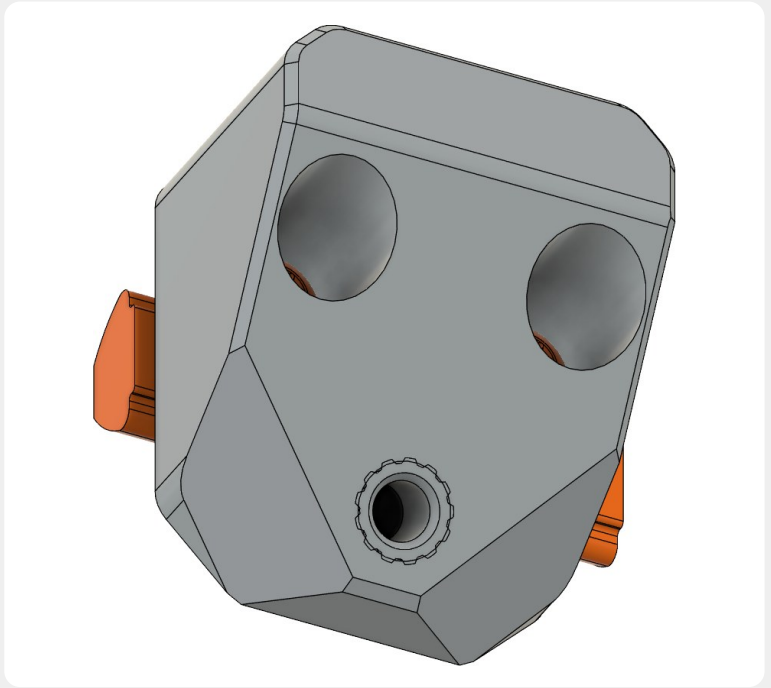
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit two M3 roll-in T-nuts into the frame slot. Position the bracket and start the two M3 x 8 Torx button-head screws through the deep access holes. Reach the recessed heads with a straight driver, then tighten evenly.

PARTS IN THIS CAD STEP

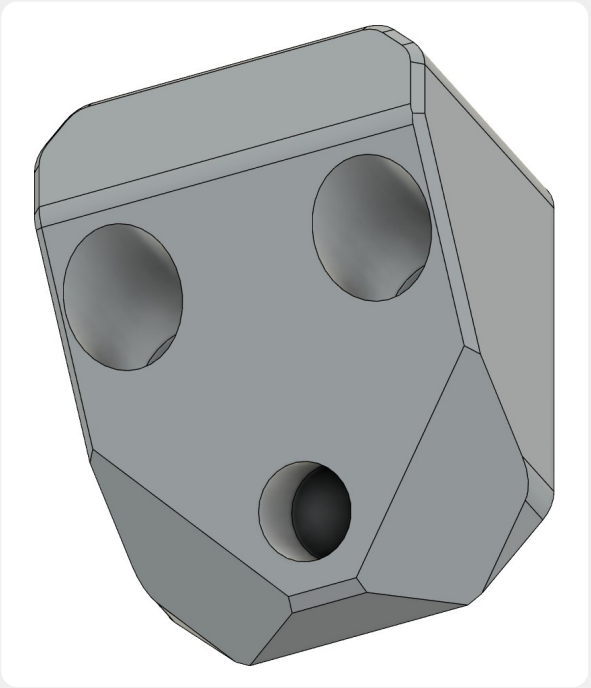
2 x M3 x 8 Torx button-head [90991A113]
2 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: The orange screw heads sit deep inside the two access holes. Both T-nuts engage the frame slot and the bracket is secure.

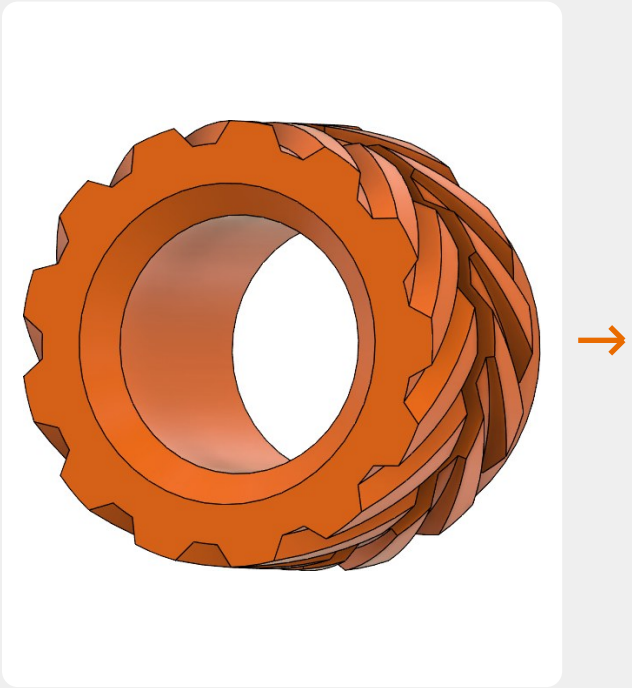
INSTALL THE M3 HEAT-SET INSERT

RIGHT FRONT LIGHT BRACKET | DETAIL 1

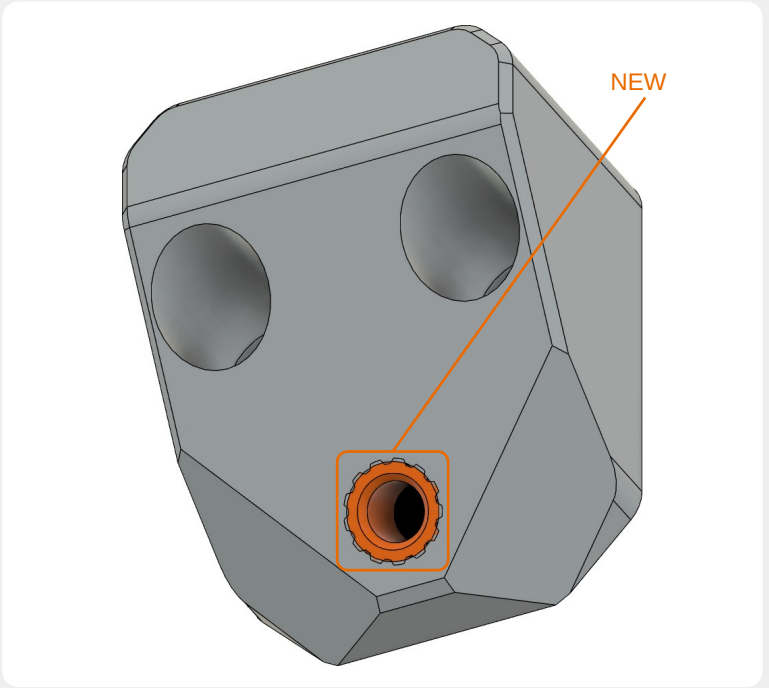
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the M3 heat-set insert in the highlighted pocket. Use an insert-setting tip and the temperature recommended for your printed material. Press straight and let it cool before fitting a screw.

PARTS IN THIS CAD STEP

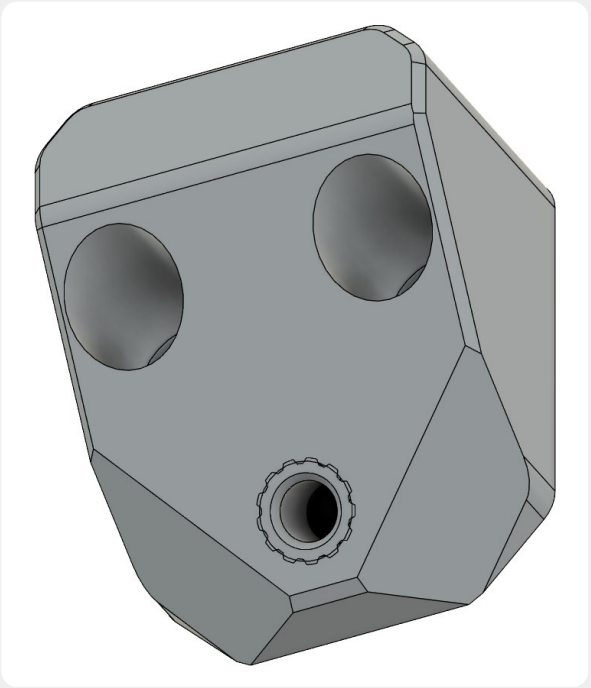
1 x M3 heat-set insert

CHECK BEFORE CONTINUING: The insert is straight and flush with its intended seat; the threaded opening remains clear.

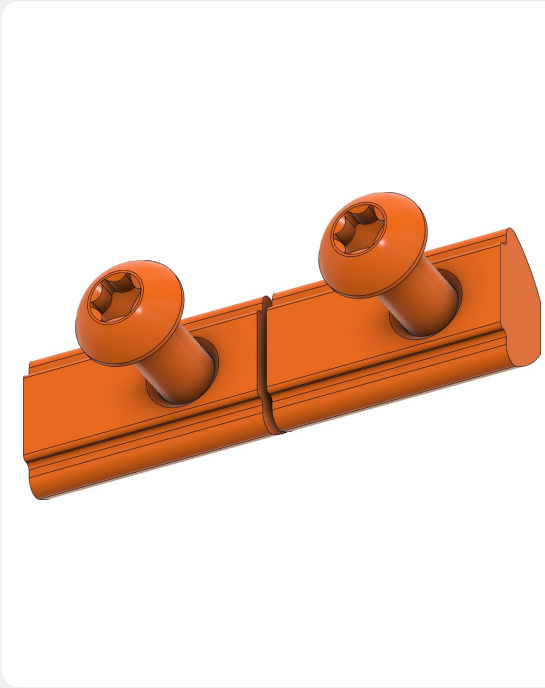
FIT THE FRAME MOUNTING HARDWARE

RIGHT FRONT LIGHT BRACKET | DETAIL 2

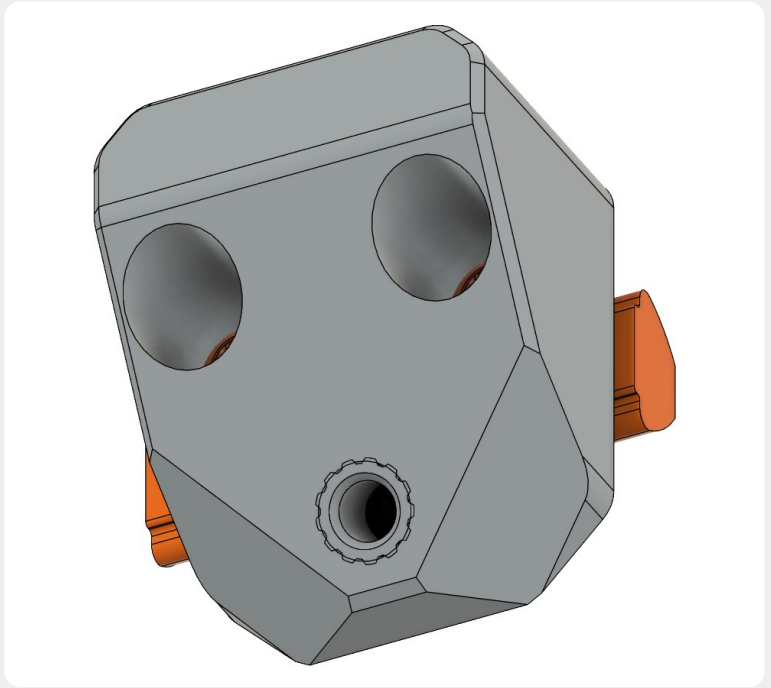
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit two M3 roll-in T-nuts into the frame slot. Position the bracket and start the two M3 x 8 Torx button-head screws through the deep access holes. Reach the recessed heads with a straight driver, then tighten evenly.

PARTS IN THIS CAD STEP

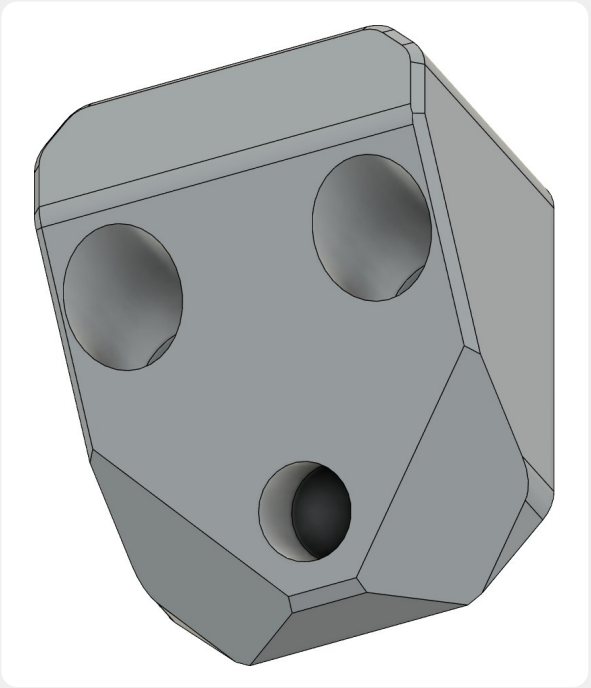
2 x M3 x 8 Torx button-head [90991A113]
2 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: The orange screw heads sit deep inside the two access holes. Both T-nuts engage the frame slot and the bracket is secure.

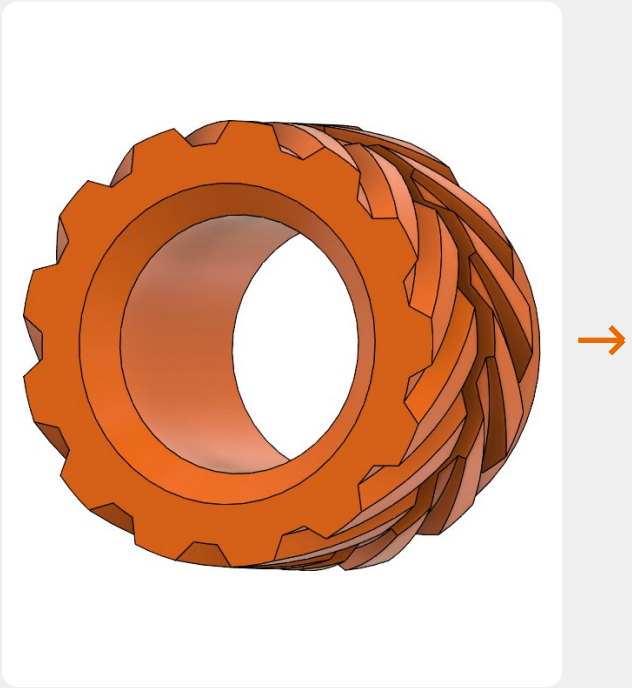
INSTALL THE M3 HEAT-SET INSERT

RIGHT REAR LIGHT BRACKET | DETAIL 1

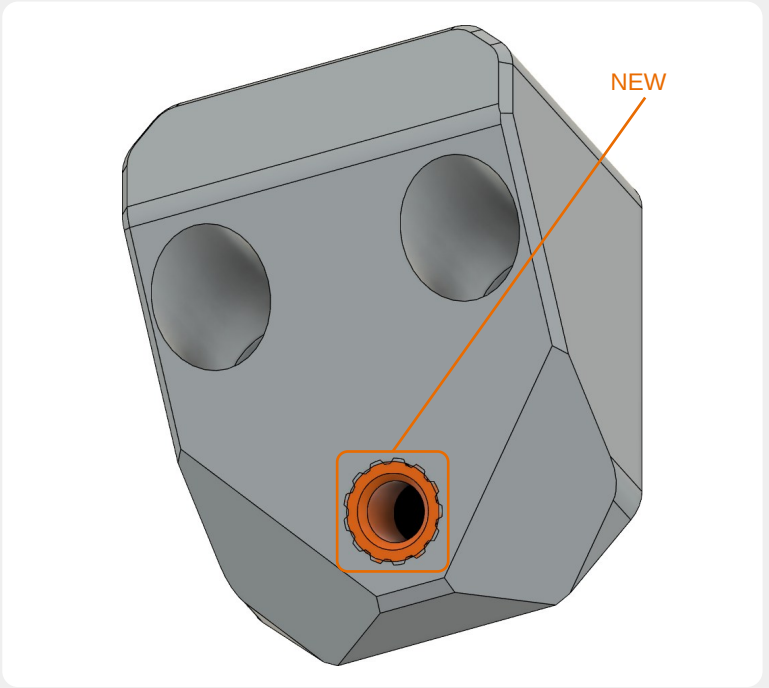
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the M3 heat-set insert in the highlighted pocket. Use an insert-setting tip and the temperature recommended for your printed material. Press straight and let it cool before fitting a screw.

PARTS IN THIS CAD STEP

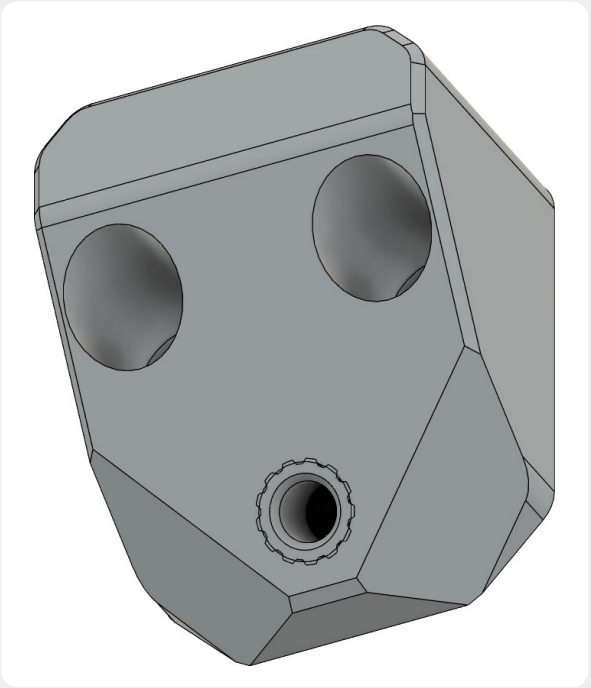
1 x M3 heat-set insert

CHECK BEFORE CONTINUING: The insert is straight and flush with its intended seat; the threaded opening remains clear.

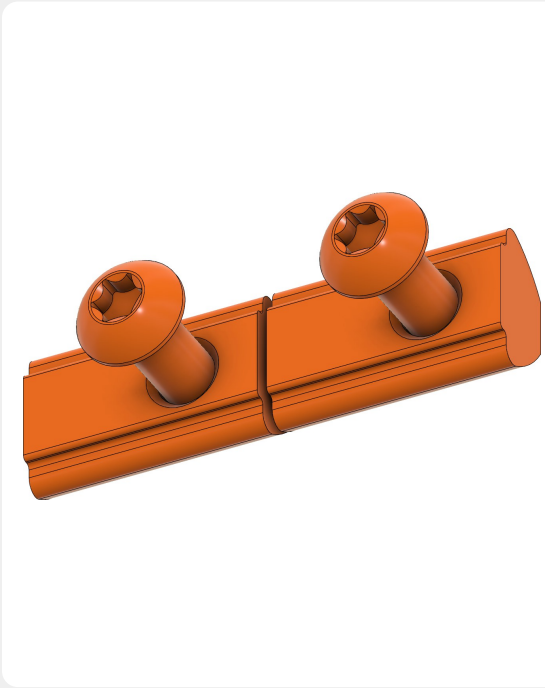
FIT THE FRAME MOUNTING HARDWARE

RIGHT REAR LIGHT BRACKET | DETAIL 2

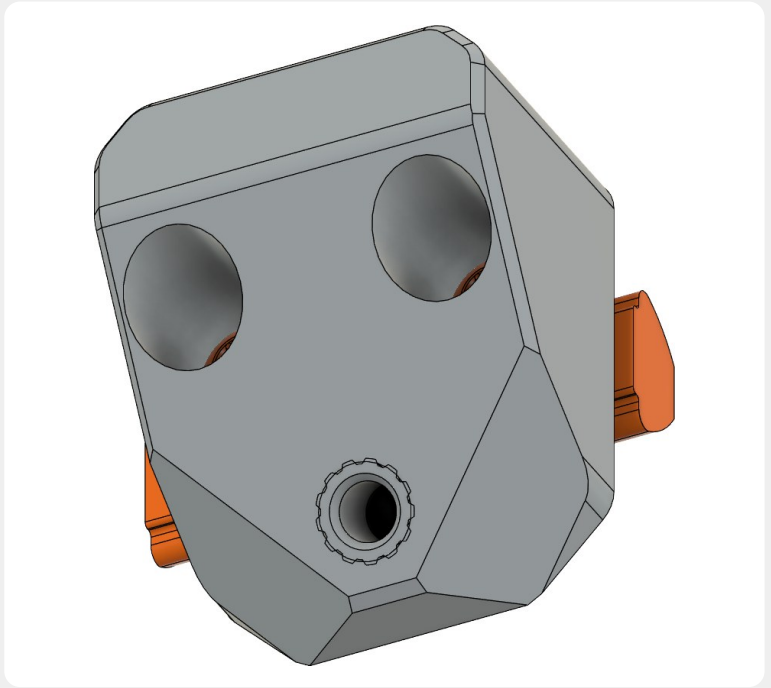
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit two M3 roll-in T-nuts into the frame slot. Position the bracket and start the two M3 x 8 Torx button-head screws through the deep access holes. Reach the recessed heads with a straight driver, then tighten evenly.

PARTS IN THIS CAD STEP

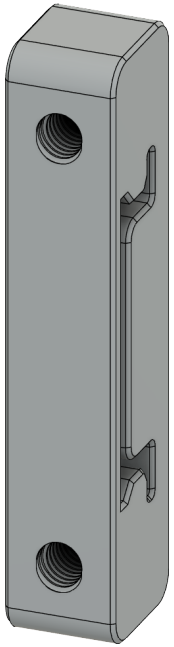
2 x M3 x 8 Torx button-head [90991A113]
2 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: The orange screw heads sit deep inside the two access holes. Both T-nuts engage the frame slot and the bracket is secure.

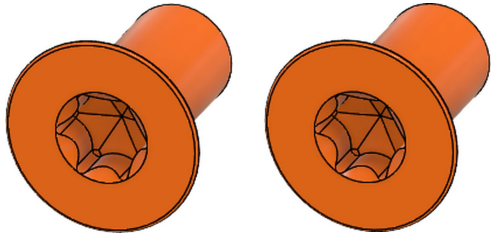
SECURE THE DIN-RAIL SUPPORT

UPPER FRONT DIN SUPPORT | DETAIL 1

1 BEFORE

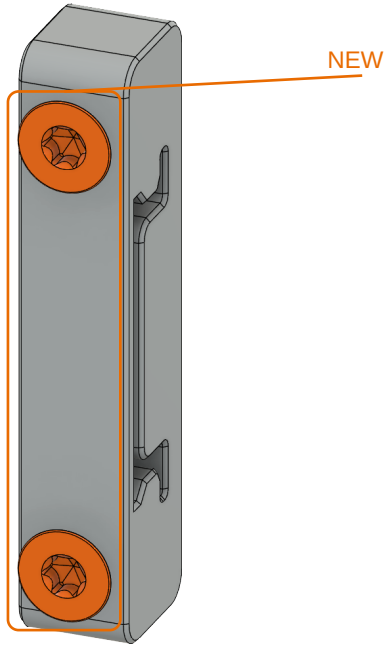


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position this support at its matching electronics-bay location. Start the two M5 x 10 Torx flat-head screws in the highlighted holes. Confirm the mating threads, then tighten evenly with the support seated flat.

PARTS IN THIS CAD STEP

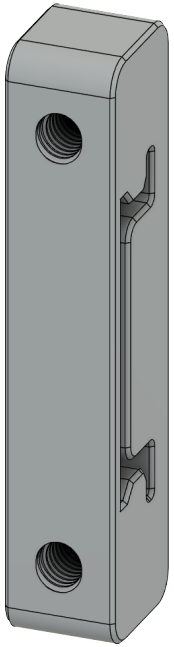
2 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: Both heads sit in their countersinks. The support is correctly oriented for the rail and remains square.

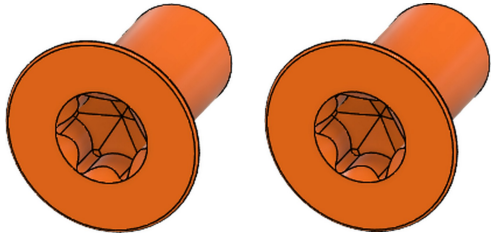
SECURE THE DIN-RAIL SUPPORT

UPPER REAR DIN SUPPORT | DETAIL 1

1 BEFORE

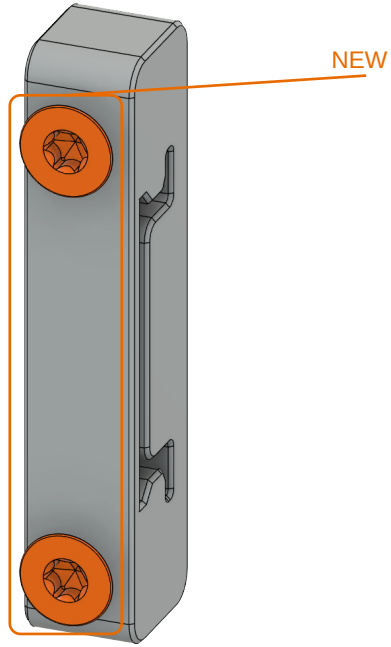


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position this support at its matching electronics-bay location. Start the two M5 x 10 Torx flat-head screws in the highlighted holes. Confirm the mating threads, then tighten evenly with the support seated flat.

PARTS IN THIS CAD STEP

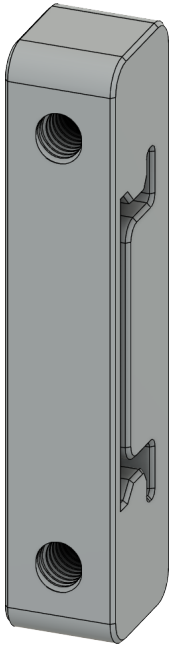
2 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: Both heads sit in their countersinks. The support is correctly oriented for the rail and remains square.

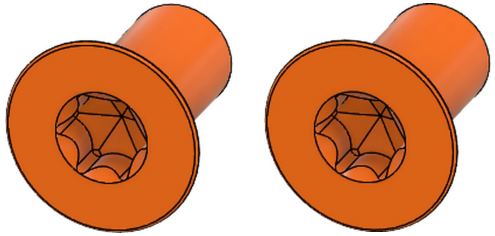
SECURE THE DIN-RAIL SUPPORT

LOWER FRONT DIN SUPPORT | DETAIL 1

1 BEFORE

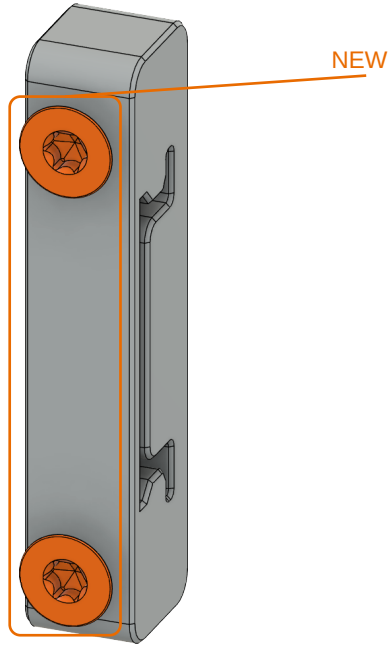


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position this support at its matching electronics-bay location. Start the two M5 x 10 Torx flat-head screws in the highlighted holes. Confirm the mating threads, then tighten evenly with the support seated flat.

PARTS IN THIS CAD STEP

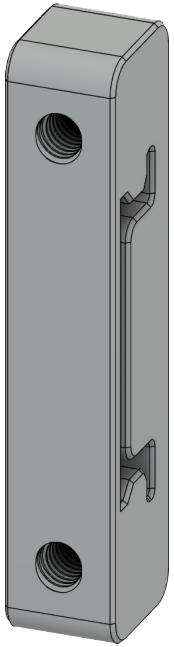
2 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: Both heads sit in their countersinks. The support is correctly oriented for the rail and remains square.

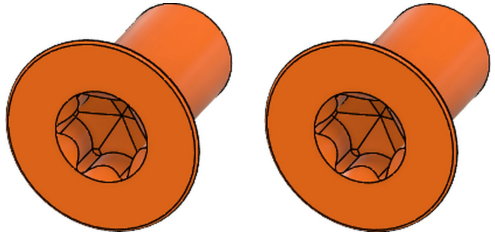
SECURE THE DIN-RAIL SUPPORT

LOWER REAR DIN SUPPORT | DETAIL 1

1 BEFORE

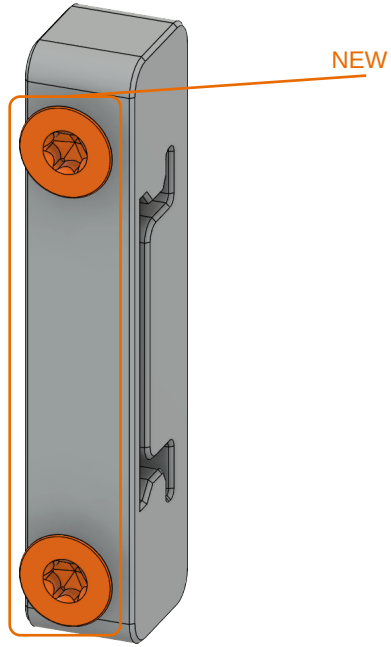


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Position this support at its matching electronics-bay location. Start the two M5 x 10 Torx flat-head screws in the highlighted holes. Confirm the mating threads, then tighten evenly with the support seated flat.

PARTS IN THIS CAD STEP

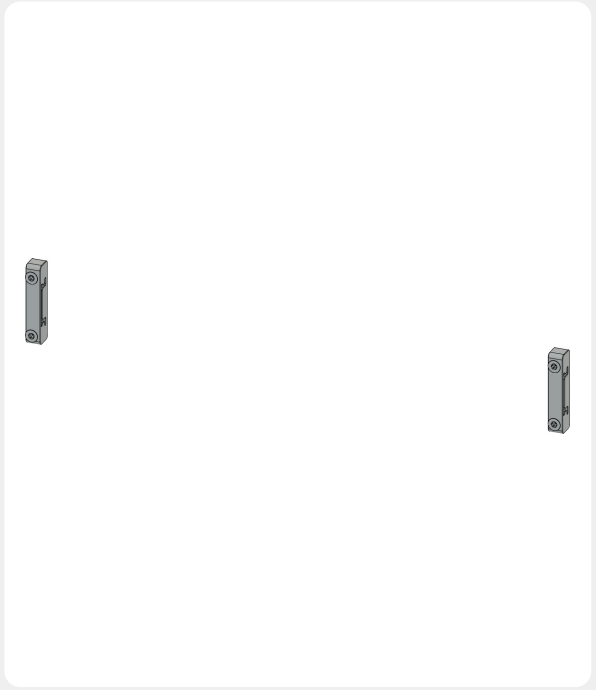
2 x M5 x 10 Torx flat-head [92703A161]

CHECK BEFORE CONTINUING: Both heads sit in their countersinks. The support is correctly oriented for the rail and remains square.

POSITION THE DIN RAIL

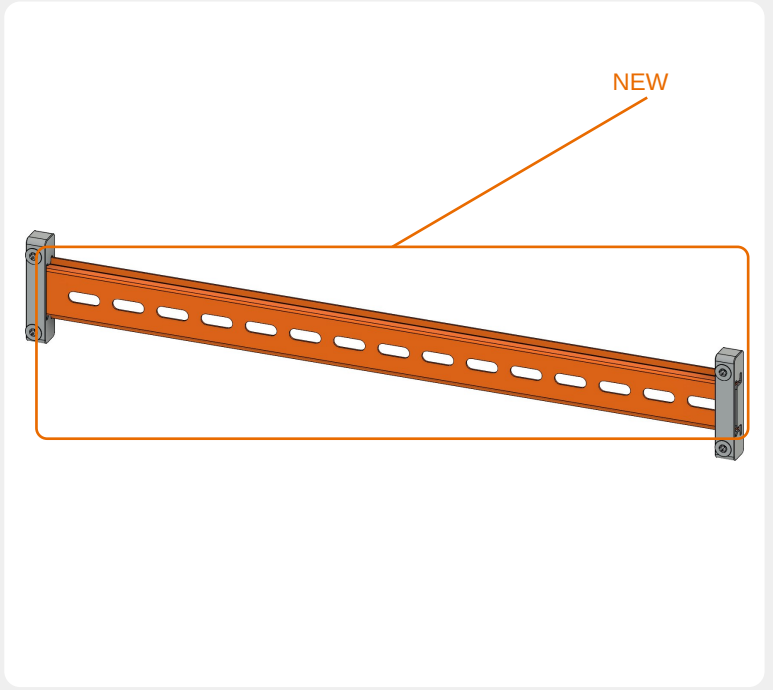
UPPER ELECTRONICS DIN RAIL | DETAIL 1

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Lay the DIN rail across the two matching supports. Align its end mounting slots with the threaded fixing points. Keep the rail oriented as shown so equipment clips can engage its edges.

PARTS IN THIS CAD STEP

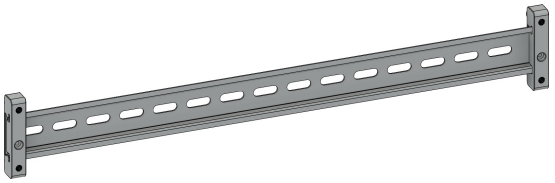
1 x Din Rail 17.6875 8961K93 Steel DIN 3 Rail 8961K93

CHECK BEFORE CONTINUING: Both ends sit on their supports without twisting. Leave the clips and equipment off until the rail is secured.

INSTALL THE TWO RAIL RETAINING SCREWS

UPPER ELECTRONICS DIN RAIL | DETAIL 2

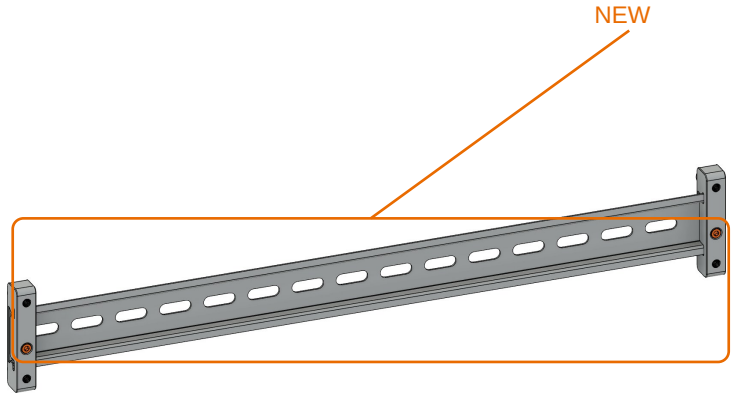
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start one M3 x 8 socket-head screw at each rail support. Align the rail, then tighten both ends progressively. Check that a loose equipment clip can engage the rail before loading it with electronics.

PARTS IN THIS CAD STEP

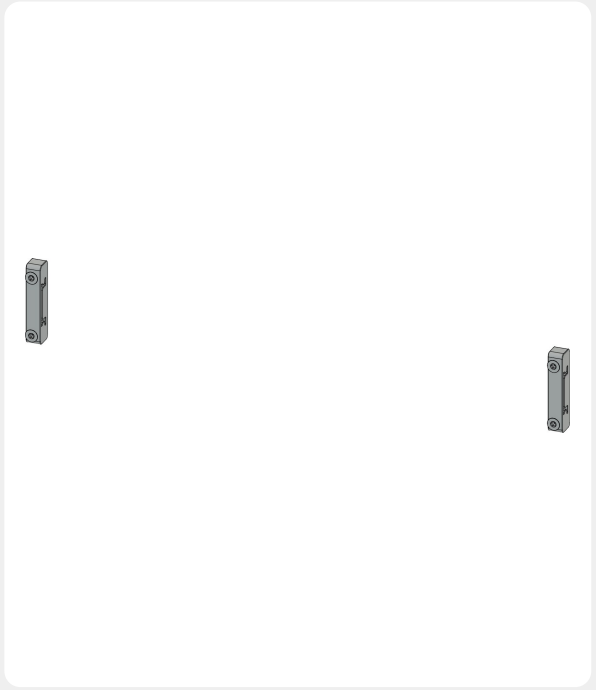
2 x M3x8 SHCS

CHECK BEFORE CONTINUING: Both rail ends are secure. The rail stays straight and the clip edges remain unobstructed.

POSITION THE DIN RAIL

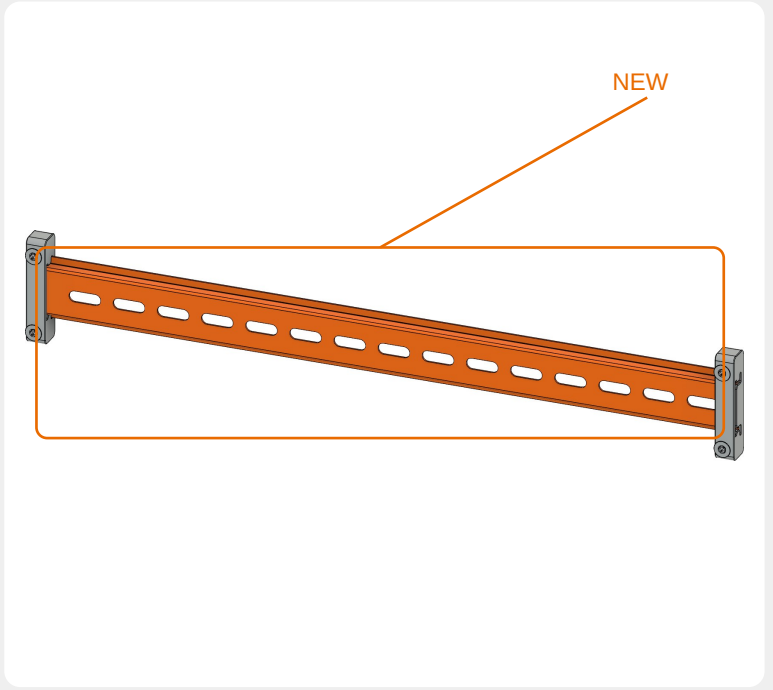
LOWER ELECTRONICS DIN RAIL | DETAIL 1

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Lay the DIN rail across the two matching supports. Align its end mounting slots with the threaded fixing points. Keep the rail oriented as shown so equipment clips can engage its edges.

PARTS IN THIS CAD STEP

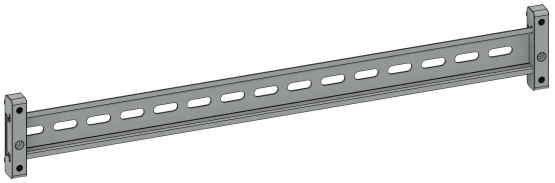
1 x Din Rail 17.6875 8961K93 Steel DIN 3 Rail 8961K93

CHECK BEFORE CONTINUING: Both ends sit on their supports without twisting. Leave the clips and equipment off until the rail is secured.

INSTALL THE TWO RAIL RETAINING SCREWS

LOWER ELECTRONICS DIN RAIL | DETAIL 2

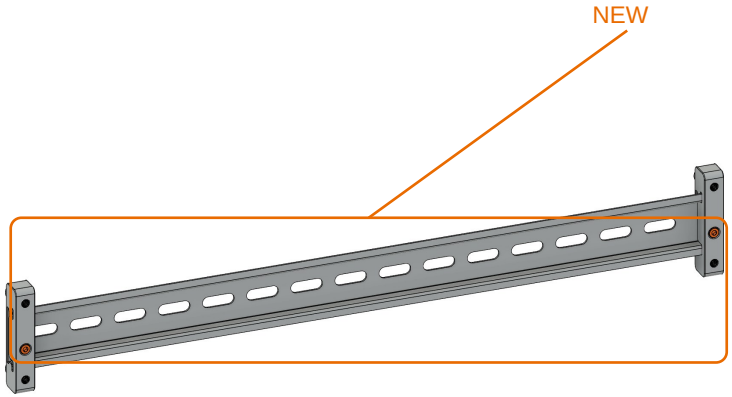
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start one M3 x 8 socket-head screw at each rail support. Align the rail, then tighten both ends progressively. Check that a loose equipment clip can engage the rail before loading it with electronics.

PARTS IN THIS CAD STEP

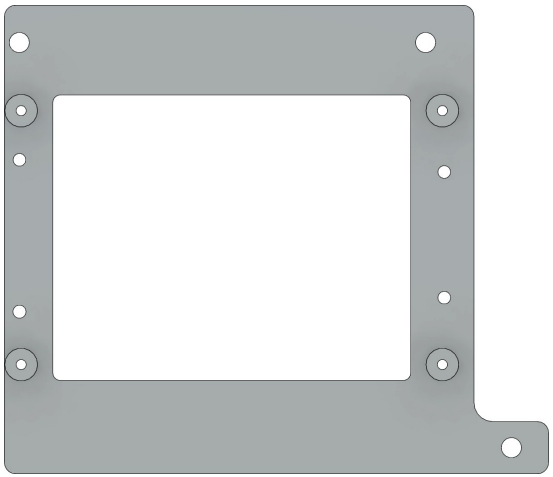
2 x M3x8 SHCS

CHECK BEFORE CONTINUING: Both rail ends are secure. The rail stays straight and the clip edges remain unobstructed.

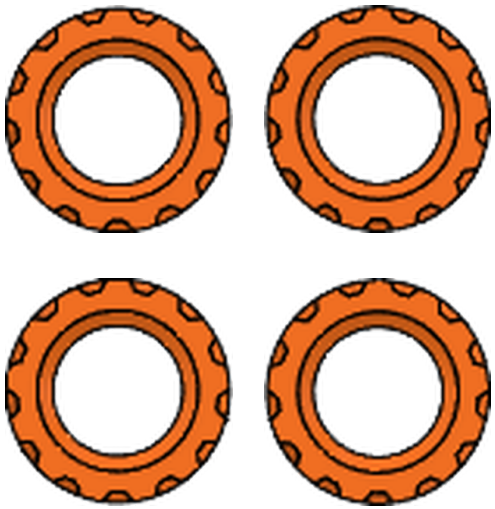
FIT THE FOUR HOUSING INSERTS

TFT50 DISPLAY HOUSING | DETAIL 1

1 BEFORE

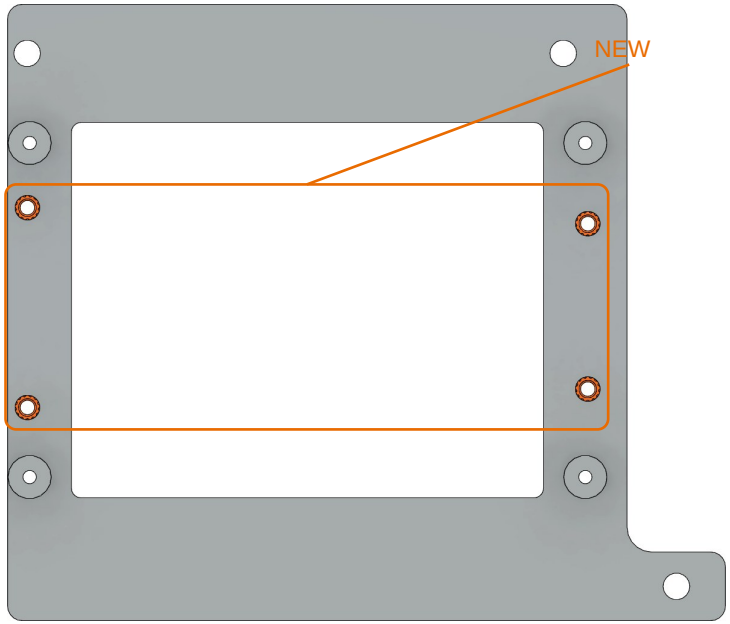


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Set the four M3 heat-set inserts into the highlighted pockets in the printed rear housing. Use a temperature suited to its printed material. Work with the display removed, keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

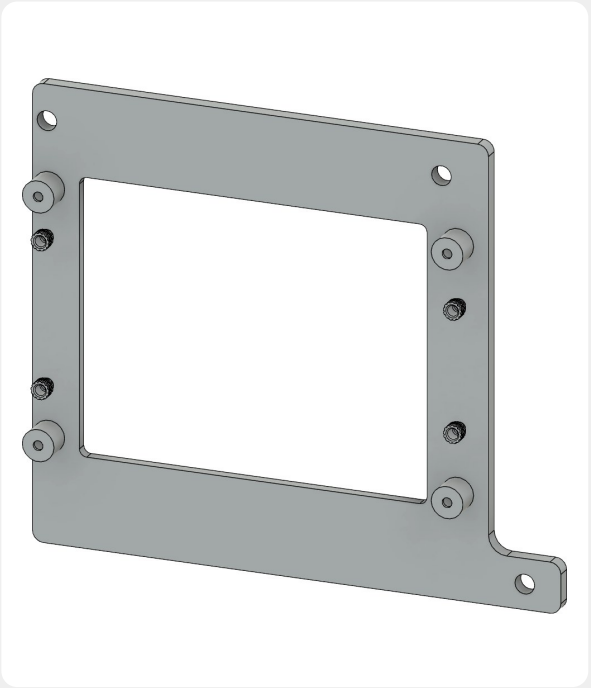
4 x M3 heat-set insert

CHECK BEFORE CONTINUING: Four inserts are seated without protruding into the display space. No electronic parts were present during heat-setting.

SEAT THE DISPLAY MODULE

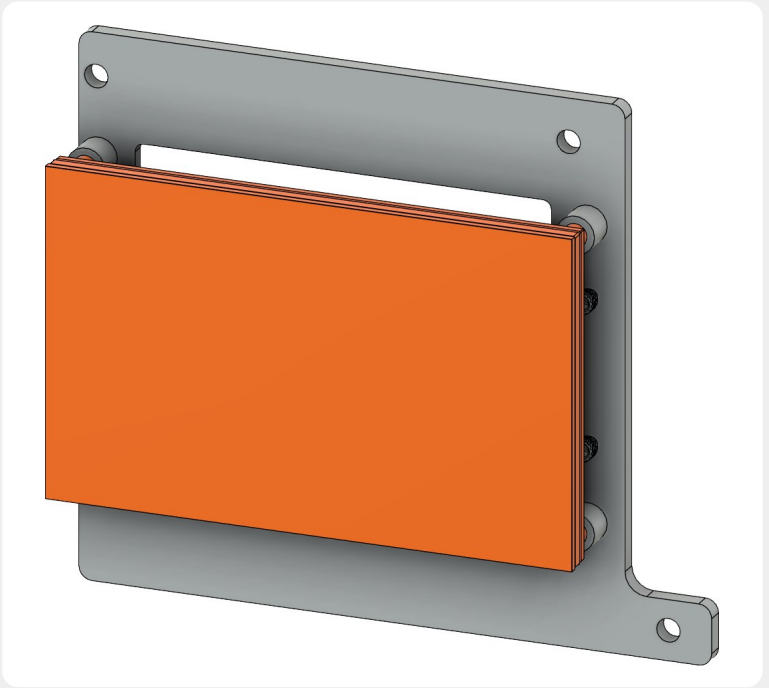
TFT50 DISPLAY HOUSING | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Handle the display as one assembled module. Place it into the housing in the modeled orientation, supporting its edges rather than pressing on the glass. Keep connectors and ribbon-cable areas clear of the housing.

PARTS IN THIS CAD STEP

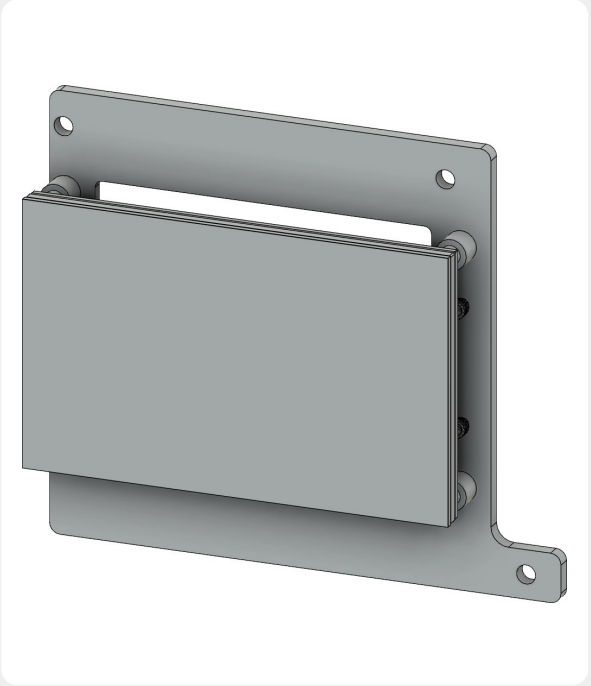
1 x assembled TFT50 display module; internal CAD pieces shown together, not separate installation parts

CHECK BEFORE CONTINUING: The module sits on its intended supports without being forced. Do not separate the internal display layers shown in the CAD.

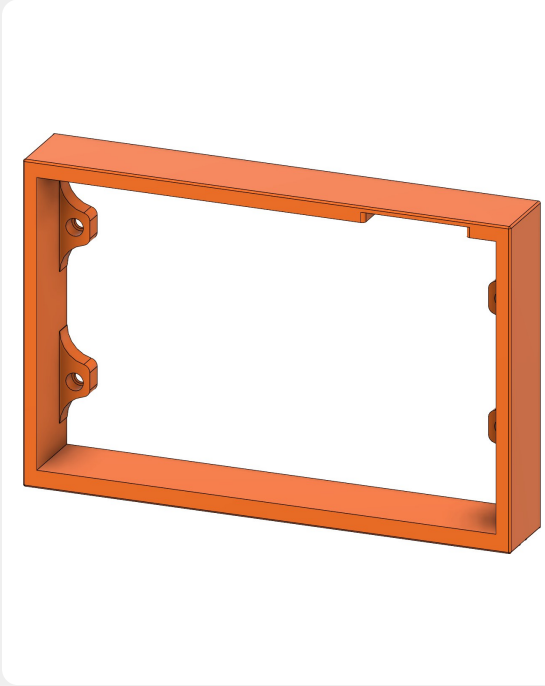
FIT THE FRONT BEZEL

TFT50 DISPLAY HOUSING | DETAIL 3

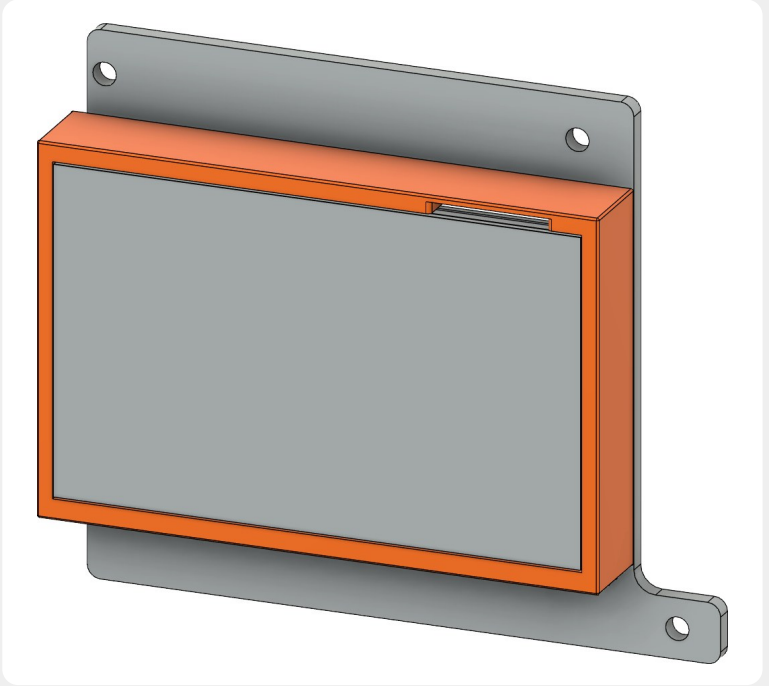
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Bring the front bezel over the display and align it with the rear housing. Check the glass opening and connector clearances before adding the retaining screws.

PARTS IN THIS CAD STEP

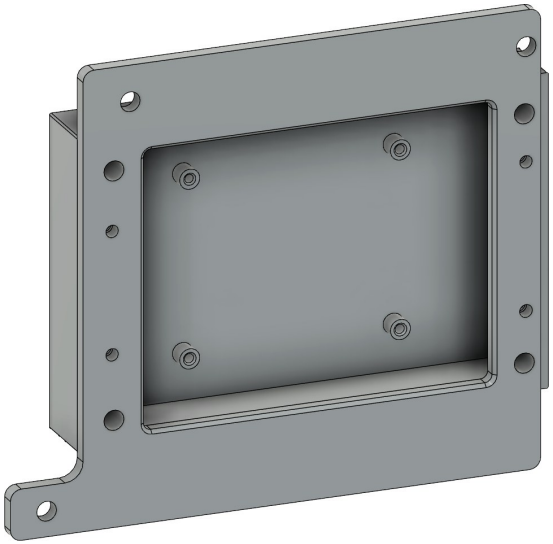
1 x pitft50 front

CHECK BEFORE CONTINUING: The housing closes without bending the display or trapping a cable.

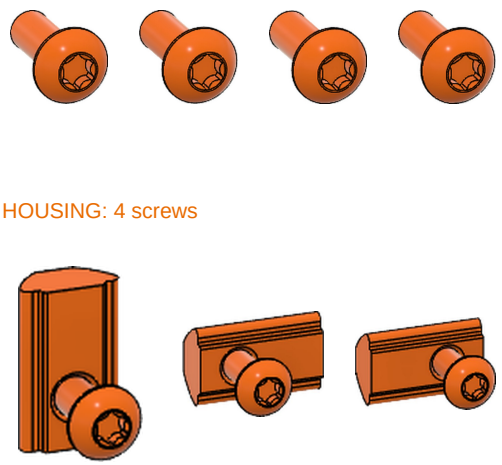
SECURE THE HOUSING AND FRAME MOUNTS

TFT50 DISPLAY HOUSING | DETAIL 4

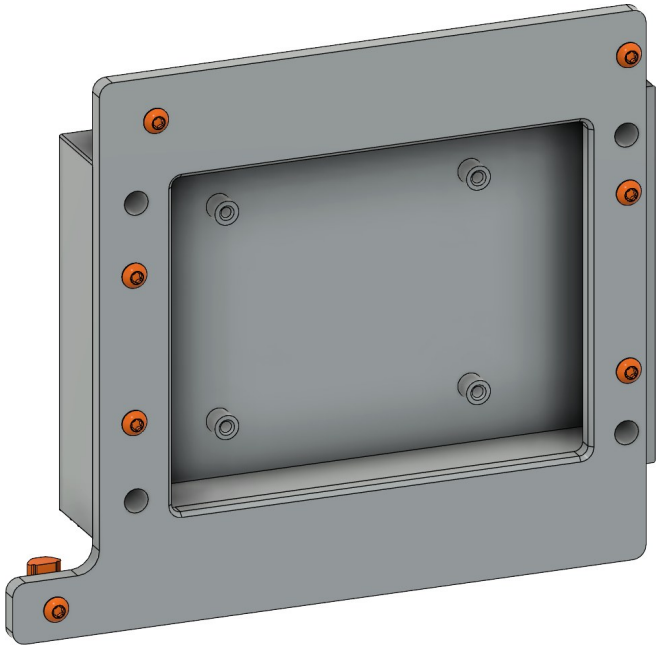
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the four housing screws and tighten lightly in a diagonal sequence. Then align the three frame T-nuts and install the three frame screws. Keep the housing square and leave connector access clear. Use the two labeled hardware groups shown above.

PARTS IN THIS CAD STEP

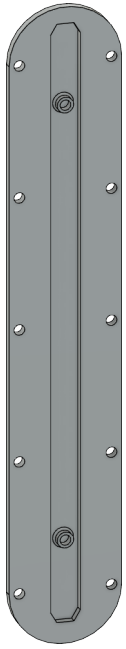
7 x M3 x 8 Torx button-head [90991A113]
3 x M3 Roll-In T Nut

CHECK BEFORE CONTINUING: Four screws close the housing; three screws and three T-nuts attach it to the frame. The display is not distorted and no cable is trapped.

FIT THE TEN PANEL INSERTS

PROGRESS-LIGHT PANEL | DETAIL 1

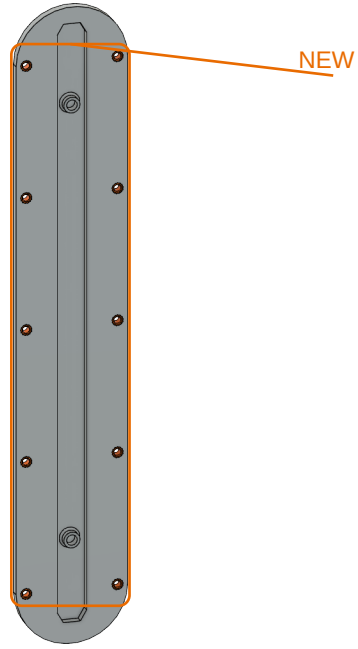
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

With the electronics removed, install the ten M3 heat-set inserts at the highlighted pockets in the printed progress-light panel. Keep each insert straight and let the panel cool before placing the board.

PARTS IN THIS CAD STEP

10 x M3 heat-set insert

CHECK BEFORE CONTINUING: Ten separate pockets contain one insert each. The board seating surfaces remain flat.

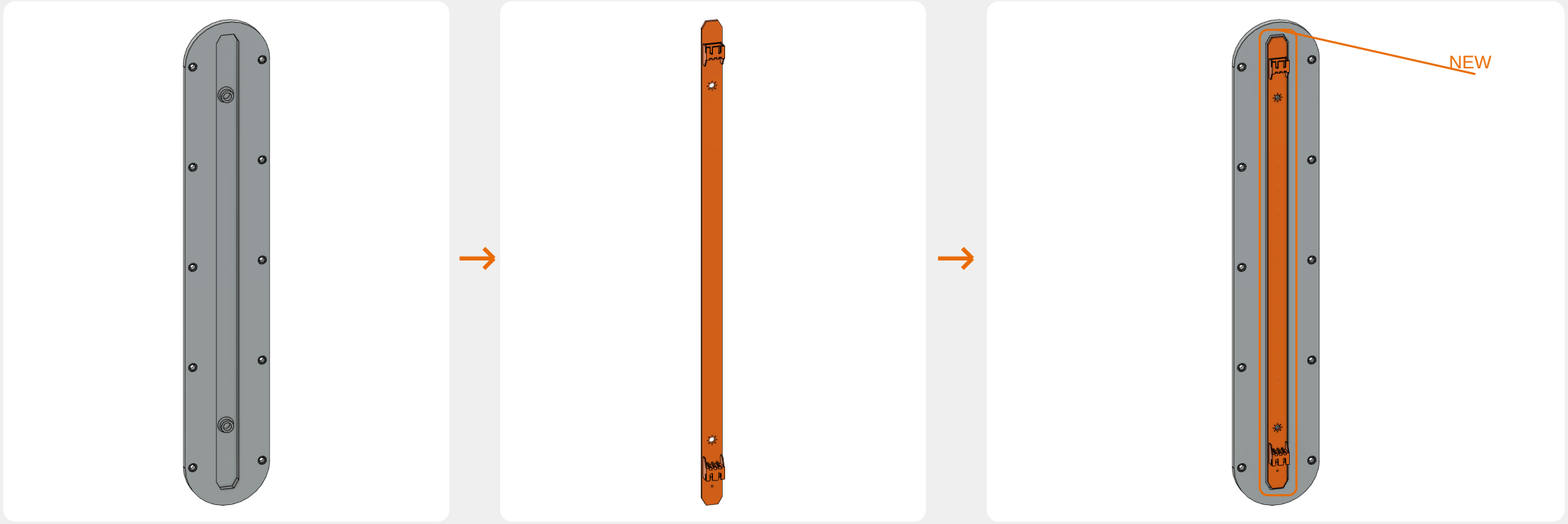
POSITION THE PROGRESS-LIGHT BOARD

PROGRESS-LIGHT PANEL | DETAIL 2

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the board against its matching panel supports. Align all fixing holes and check that the connector end matches the modeled orientation. Hold the board by its edges.

PARTS IN THIS CAD STEP

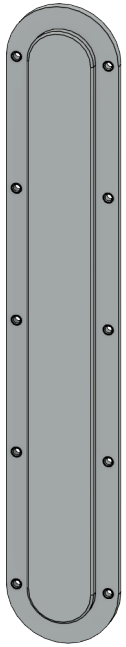
1 x assembled progress-light board; electronic components shown as part of the complete board

CHECK BEFORE CONTINUING: The board sits without rocking or bending; no component touches a mounting boss.

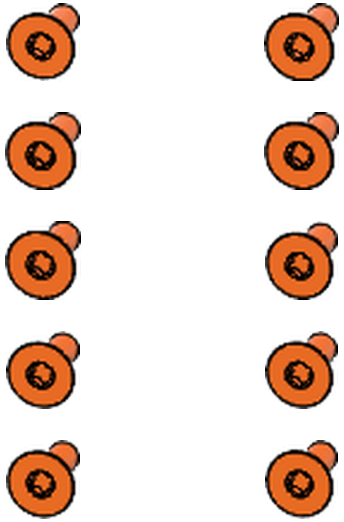
SECURE THE PROGRESS-LIGHT ASSEMBLY

PROGRESS-LIGHT PANEL | DETAIL 3

1 BEFORE

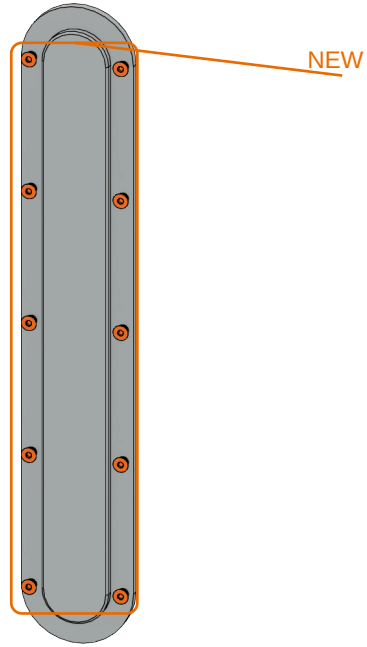


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the ten M3 x 8 Torx flat-head screws in the highlighted fixing positions. Work progressively along the panel and tighten only enough to retain the assembly without bowing it.

PARTS IN THIS CAD STEP

10 x M3 x 8 Torx flat-head [92703A149]

CHECK BEFORE CONTINUING: All fixing heads are seated in their intended recesses and the board remains flat.

ORIENT THE FAN

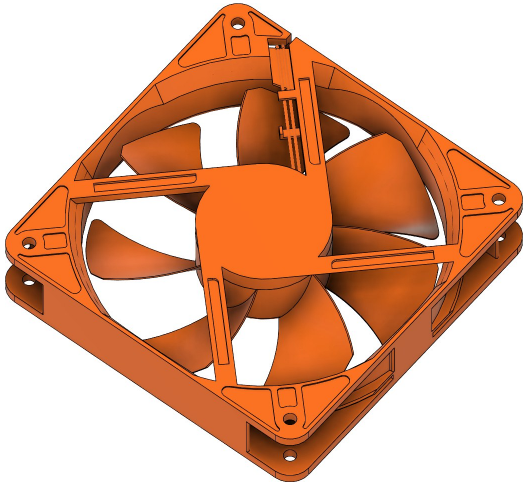
TOP ELECTRONICS FAN | DETAIL 1

1 BEFORE

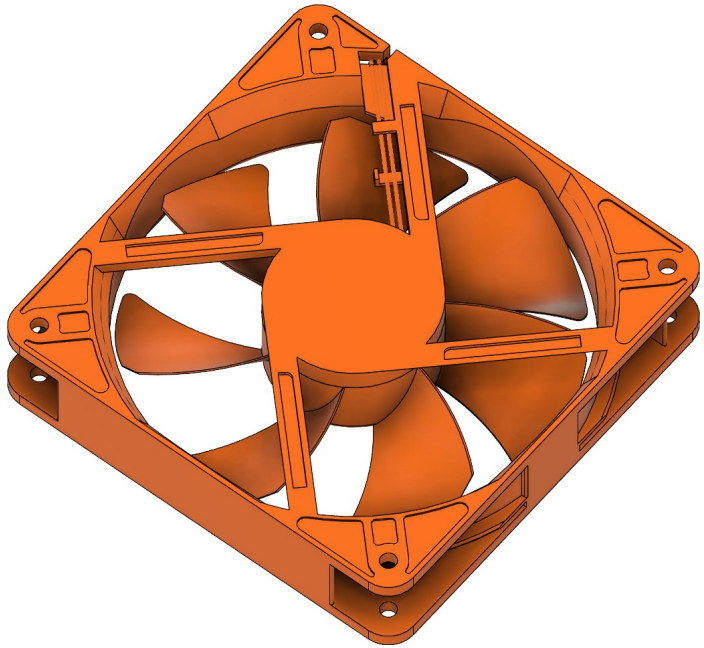
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify the fan location from the electronics overview. Match the mounting-hole pattern and keep its cable exit accessible. Confirm the required airflow direction from the system design and the arrows on the actual fan before fastening it.

PARTS IN THIS CAD STEP

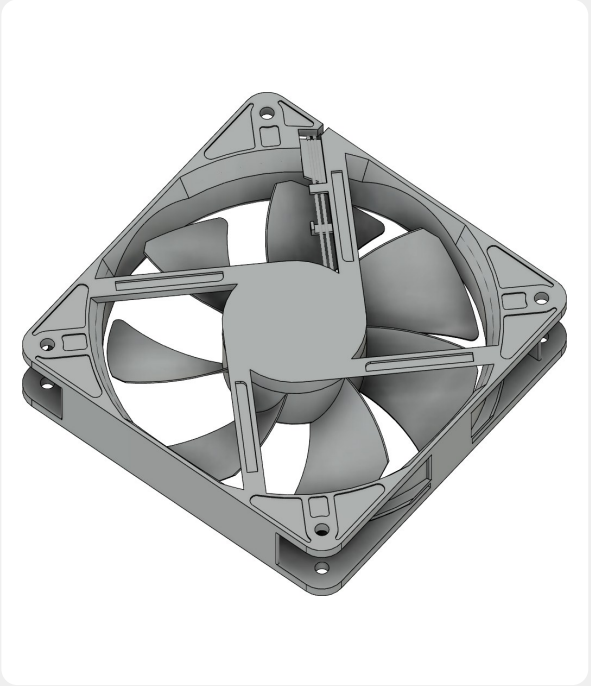
1 x 8774N18 DC Equipment-Cooling Fan

CHECK BEFORE CONTINUING: The fan is the correct size for this location. Airflow direction is confirmed separately; blade shape in CAD is not an airflow specification.

POSITION THE PROTECTIVE GUARD

TOP ELECTRONICS FAN | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the protective guard over the exposed fan face in the modeled position. Align all four mounting holes and keep the guard clear of the blades.

PARTS IN THIS CAD STEP

1 x 19155K46 Plastic Fan Guard for 4.69 and 4.72 Fans

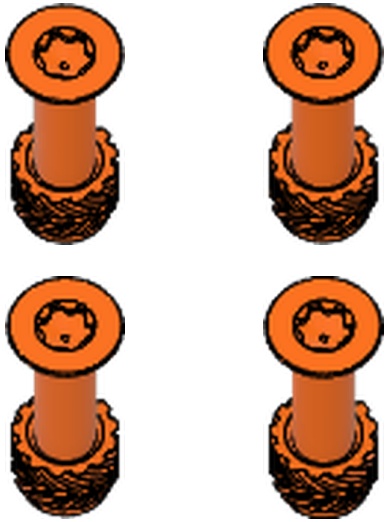
CHECK BEFORE CONTINUING: The guard sits squarely and does not contact the fan rotor.

CHECK AND FIT THE FOUR MOUNTING FIXINGS

TOP ELECTRONICS FAN | DETAIL 3

1 BEFORE

2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

The CAD shows four M3 inserts and four M3 x 12 Torx flat-head screws at this fan mount. Confirm the approved insert or threaded-mount arrangement before fastening. Do not heat-set inserts into the electronic fan unless its mounting design explicitly requires it.

PARTS IN THIS CAD STEP

4 x M3 heat-set insert
4 x M3 x 12 Torx flat-head [90236A122]

CHECK BEFORE CONTINUING: Mounting method is confirmed, all four fixings engage correctly, and neither guard nor fasteners touch the blades. This retention detail still needs physical review.

ORIENT THE FAN

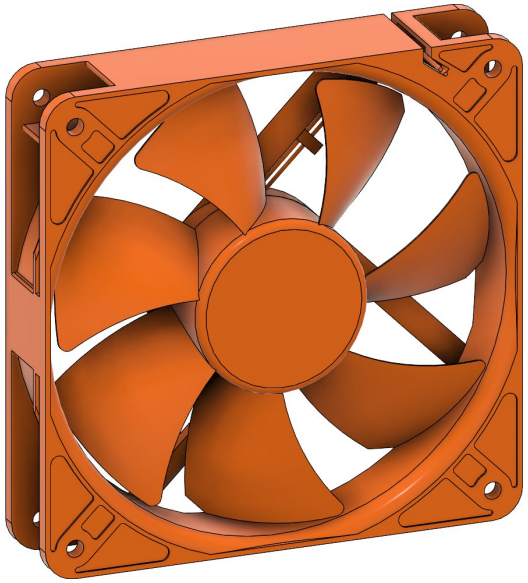
REAR ELECTRONICS FAN | DETAIL 1

1 BEFORE

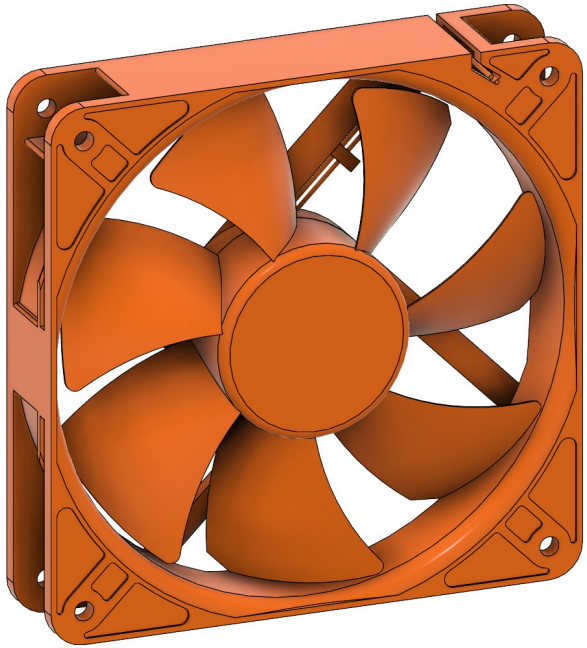
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify the fan location from the electronics overview. Match the mounting-hole pattern and keep its cable exit accessible. Confirm the required airflow direction from the system design and the arrows on the actual fan before fastening it.

PARTS IN THIS CAD STEP

1 x 8774N18 DC Equipment-Cooling Fan

CHECK BEFORE CONTINUING: The fan is the correct size for this location. Airflow direction is confirmed separately; blade shape in CAD is not an airflow specification.

POSITION THE PROTECTIVE GUARD

REAR ELECTRONICS FAN | DETAIL 2

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the protective guard over the exposed fan face in the modeled position. Align all four mounting holes and keep the guard clear of the blades.

PARTS IN THIS CAD STEP

1 x 19155K46 Plastic Fan Guard for 4.69 and 4.72 Fans

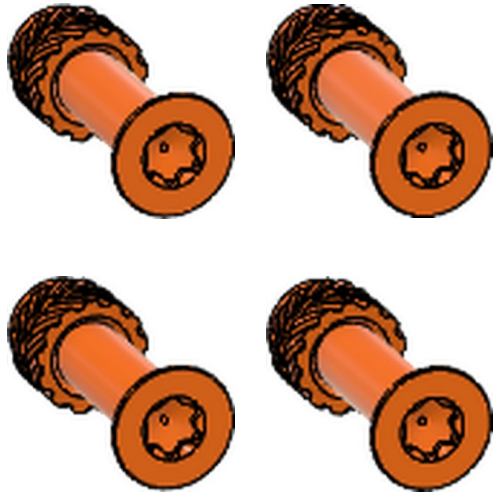
CHECK BEFORE CONTINUING: The guard sits squarely and does not contact the fan rotor.

CHECK AND FIT THE FOUR MOUNTING FIXINGS

REAR ELECTRONICS FAN | DETAIL 3

1 BEFORE

2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

The CAD shows four M3 inserts and four M3 x 12 Torx flat-head screws at this fan mount. Confirm the approved insert or threaded-mount arrangement before fastening. Do not heat-set inserts into the electronic fan unless its mounting design explicitly requires it.

PARTS IN THIS CAD STEP

4 x M3 heat-set insert
4 x M3 x 12 Torx flat-head [90236A122]

CHECK BEFORE CONTINUING: Mounting method is confirmed, all four fixings engage correctly, and neither guard nor fasteners touch the blades. This retention detail still needs physical review.

POSITION THE POWER SUPPLY

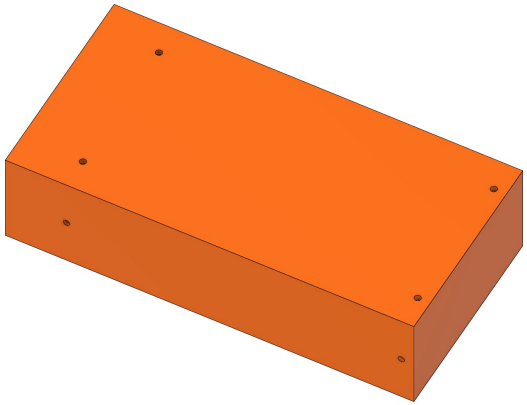
POWER SUPPLY MOUNTING | DETAIL 1

1 BEFORE

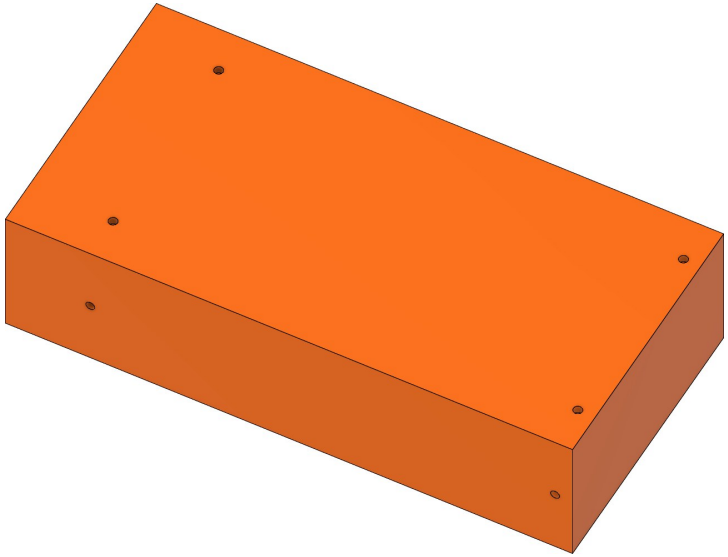
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

With power disconnected, orient the complete supply as shown. Match its four mounting positions to the supporting panel before inserting screws. Keep ventilation openings and terminal access clear.

PARTS IN THIS CAD STEP

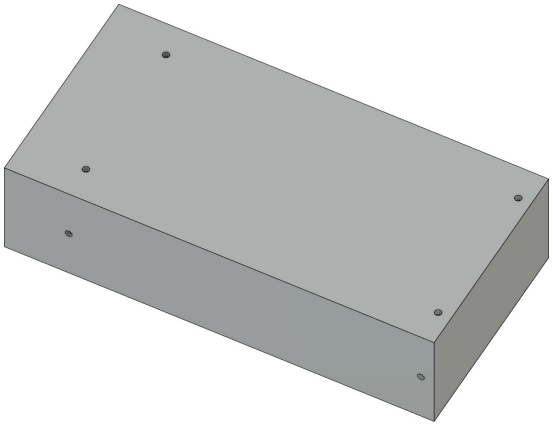
1 x complete XP Power X7 supply (CAD label)

CHECK BEFORE CONTINUING: The supply rests on its mounting surface without trapping wires. Confirm the panel and actual supply hole pattern before fastening.

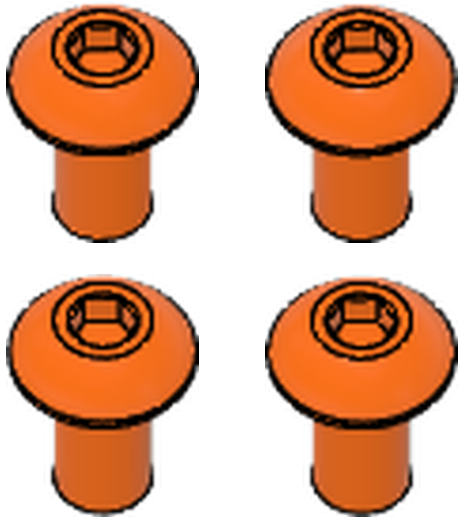
INSTALL THE FOUR SUPPLY SCREWS

POWER SUPPLY MOUNTING | DETAIL 2

1 BEFORE

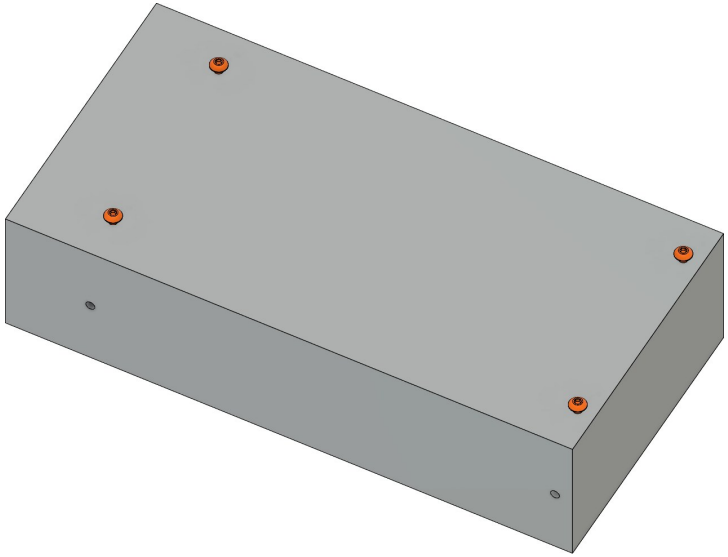


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Start the four highlighted button-head screws at the modeled supply mounting points. Confirm screw size and allowable insertion depth against the actual supply before tightening progressively. Do not open the supply housing.

PARTS IN THIS CAD STEP

4 x 92095A189 button-head screws; verify M4 x 8 specification before use

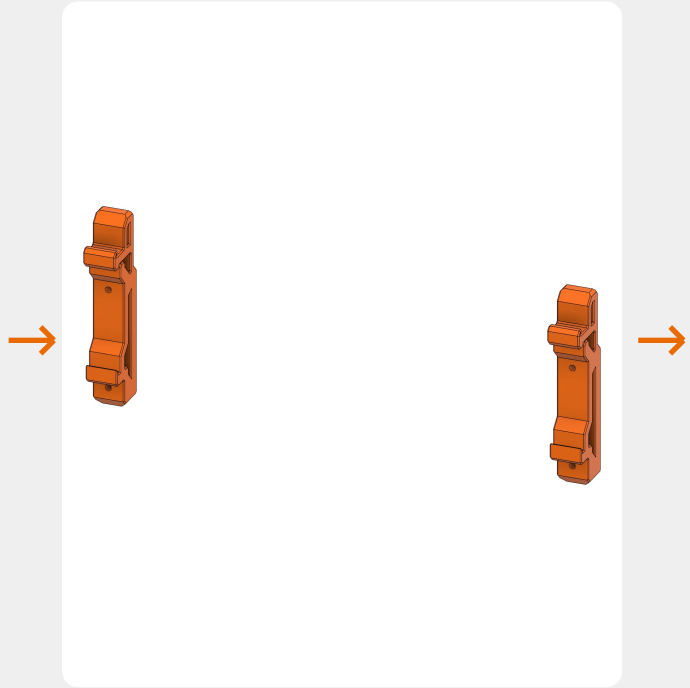
CHECK BEFORE CONTINUING: All four fixing points engage correctly. No screw projects into internal components; the supply cannot slide or rock.

LOCATE BOTH CONTROLLER DIN CLIPS

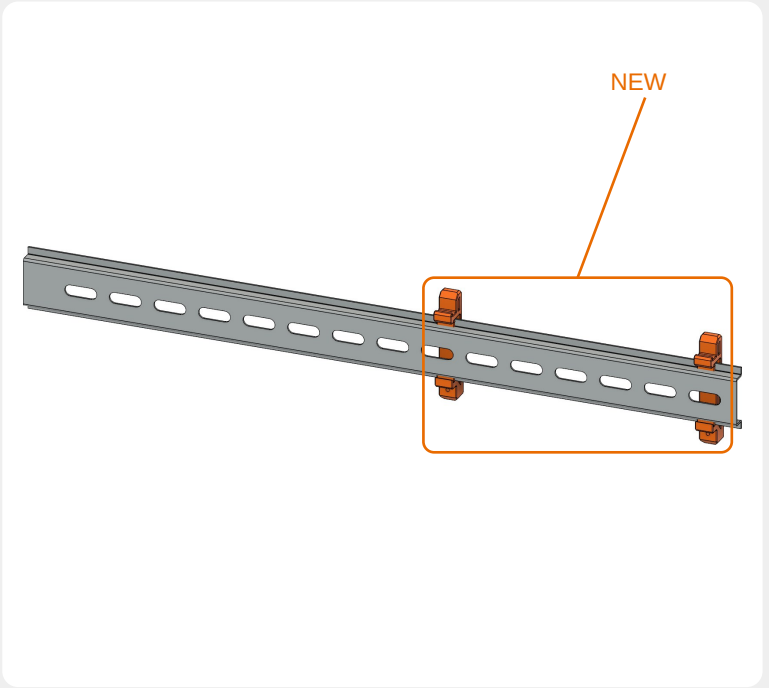
LEVIATHAN CONTROLLER MOUNTING | DETAIL 1

1 BEFORE

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match the two clips to the controller mounting positions on the upper rail. Check their rail engagement direction before fitting the carrier plates. Use this view to establish orientation; clip-to-plate fixing sizes still need confirmation.

PARTS IN THIS CAD STEP

2 x controller DIN clips

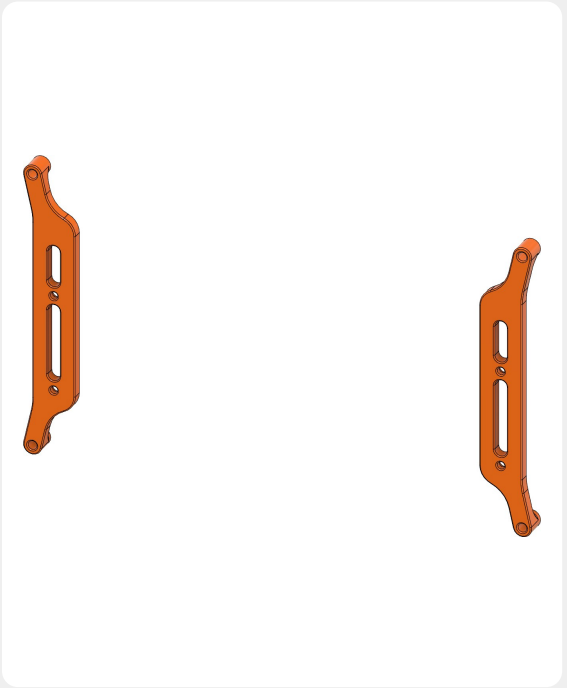
CHECK BEFORE CONTINUING: Both clips align with the carrier positions and their release features remain accessible.

ALIGN THE TWO CONTROLLER CARRIERS

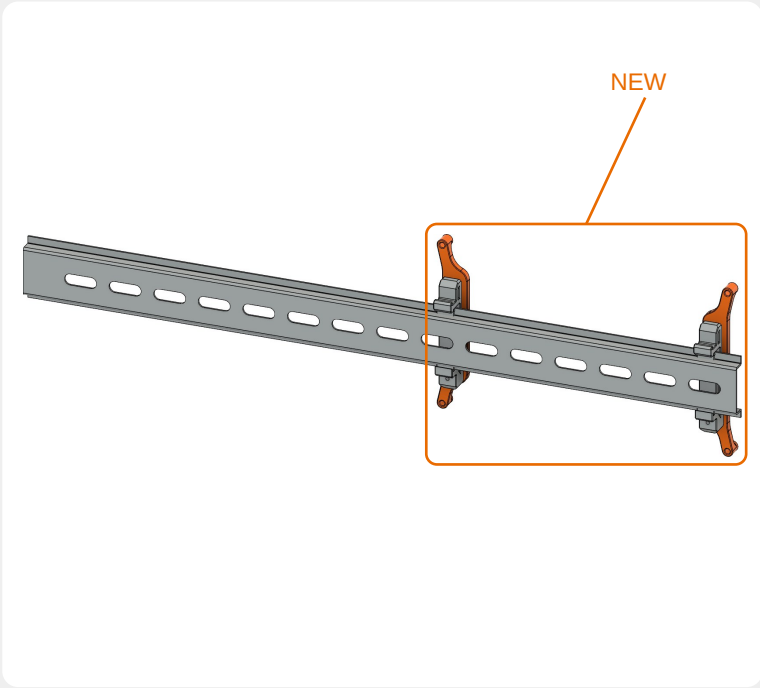
LEVIATHAN CONTROLLER MOUNTING | DETAIL 2

1 BEFORE

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the two printed carrier plates with the DIN clips and the controller mounting points. Confirm the clip-to-carrier fasteners from the approved parts list before securing this assembly.

PARTS IN THIS CAD STEP

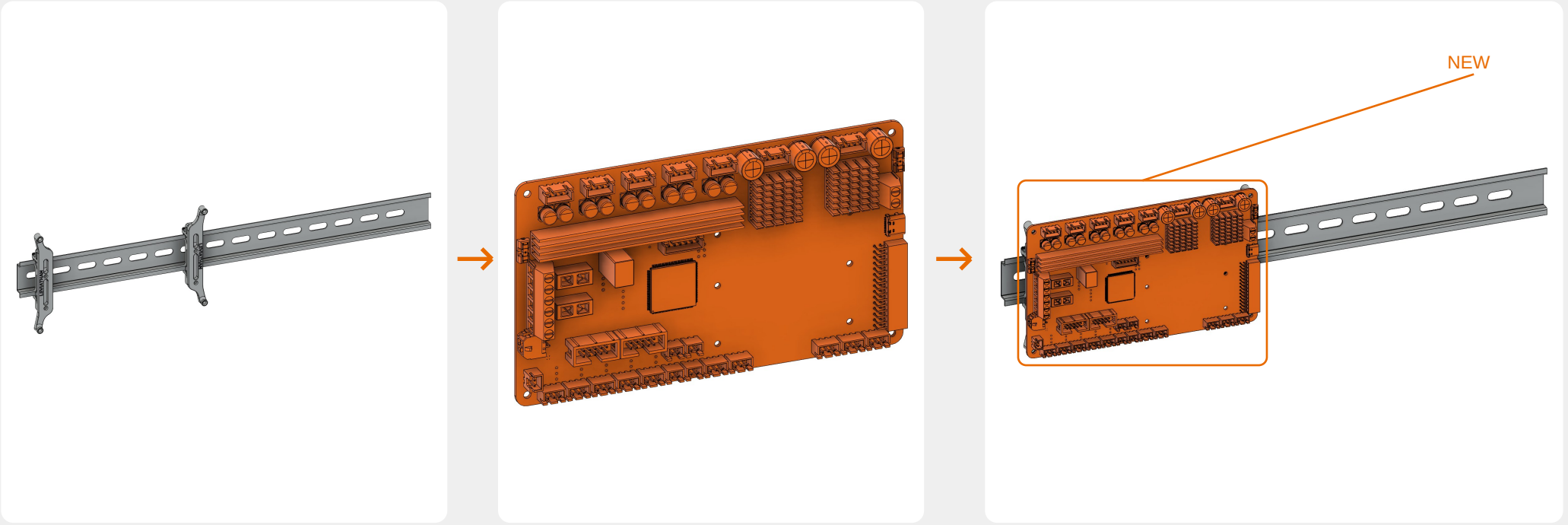
2 x controller carrier plates; retaining fasteners not identified in CAD

CHECK BEFORE CONTINUING: The two carrier faces line up. Do not force the controller to bridge twisted or mismatched supports.

POSITION THE ASSEMBLED CONTROLLER

LEVIATHAN CONTROLLER MOUNTING | DETAIL 3

1 BEFORE



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Handle the complete controller by its edges. Align it with the carrier supports in the shown orientation. Confirm and install the specified board fixings before loading or wiring the assembly; those fixings are not individually identified in this CAD.

PARTS IN THIS CAD STEP

1 x assembled Leviathan controller; internal board components stay assembled

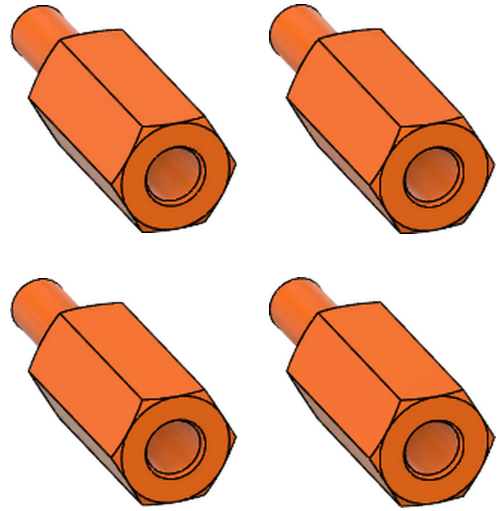
CHECK BEFORE CONTINUING: The PCB is supported without bending and no solder joints touch the mounting structure. Connector access is clear.

CHECK THE FOUR ACCESSORY STANDOFFS

LEVIATHAN CONTROLLER MOUNTING | DETAIL 4

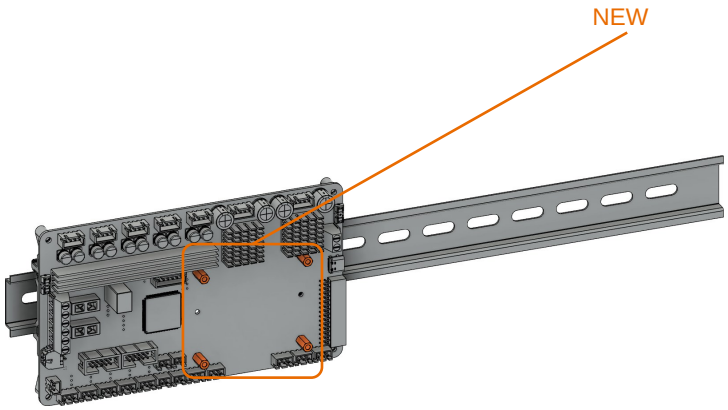
1 BEFORE

2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

The controller model includes four M2.5 male/female standoffs with 10 mm bodies. Check whether these are supplied fitted and which accessory they support. Fit only where required by the actual controller accessory arrangement; they are not substitutes for the board mounting screws.

PARTS IN THIS CAD STEP

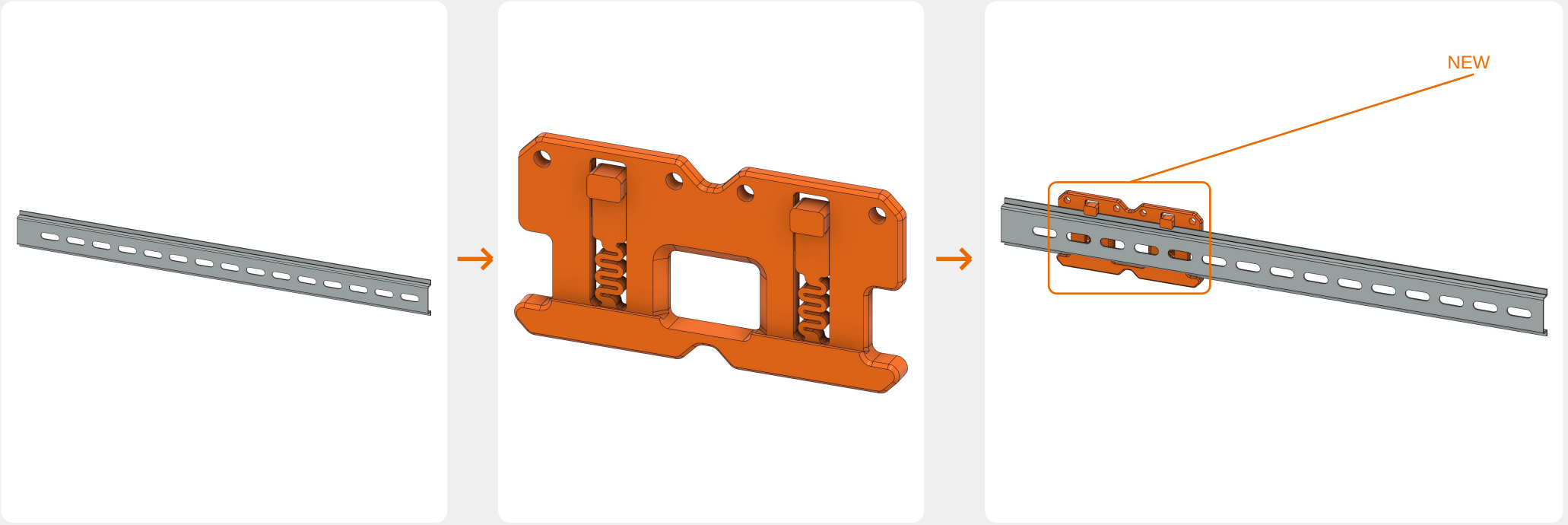
4 x M2.5 male/female standoffs, 10 mm body (CAD name)

CHECK BEFORE CONTINUING: All four positions match the actual board. Accessory identity and retention hardware are confirmed before fitting an accessory.

POSITION THE DRIVER CARRIER

DUAL 5160 DRIVER MOUNTING | DETAIL 1

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the dual-driver carrier on the lower electronics rail. Orient the two driver positions as shown and check the rail engagement. Confirm any separate retaining hardware before securing the carrier.

PARTS IN THIS CAD STEP

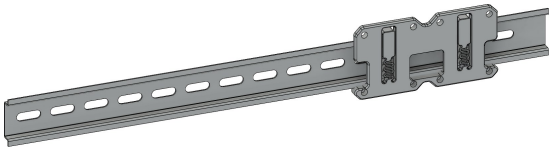
1 x dual 5160 carrier; separate fixings not identified

CHECK BEFORE CONTINUING: The carrier is retained on the rail and leaves both driver connector areas accessible.

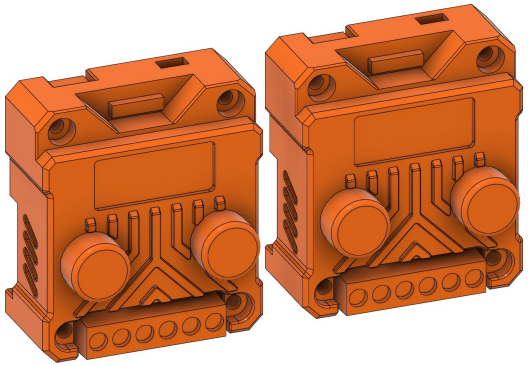
SEAT BOTH COMPLETE DRIVER MODULES

DUAL 5160 DRIVER MOUNTING | DETAIL 2

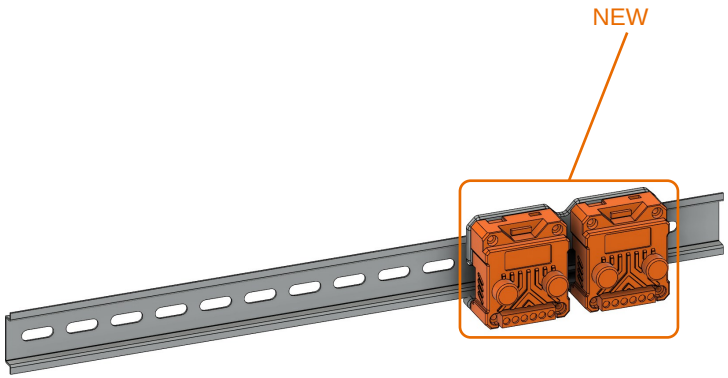
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Place the two complete driver modules in their matching carrier positions. Keep each module assembled. Verify the actual mounting or retention method before tightening any fixings; the CAD does not identify separate driver screws.

PARTS IN THIS CAD STEP

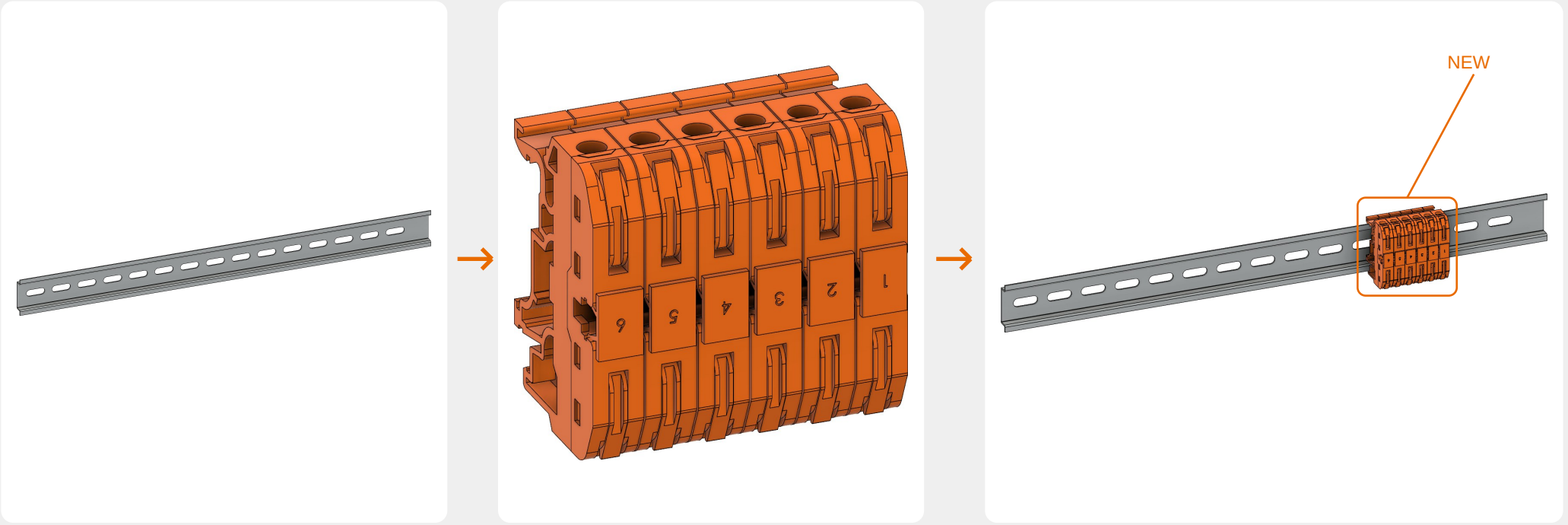
2 x complete 5160 driver modules

CHECK BEFORE CONTINUING: Both modules are secure without PCB distortion. Heatsinks and connectors clear the carrier.

POSITION THE SIX-BLOCK TERMINAL SET

UPPER TERMINAL-BLOCK MOUNTING | DETAIL 1

1 BEFORE



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the six-block set on the rail in the shown orientation. Engage the mounting feet using the terminal manufacturer's method and check each block is retained. Electrical bridges are omitted from this mounting view; fit bridges only to an approved wiring diagram.

PARTS IN THIS CAD STEP

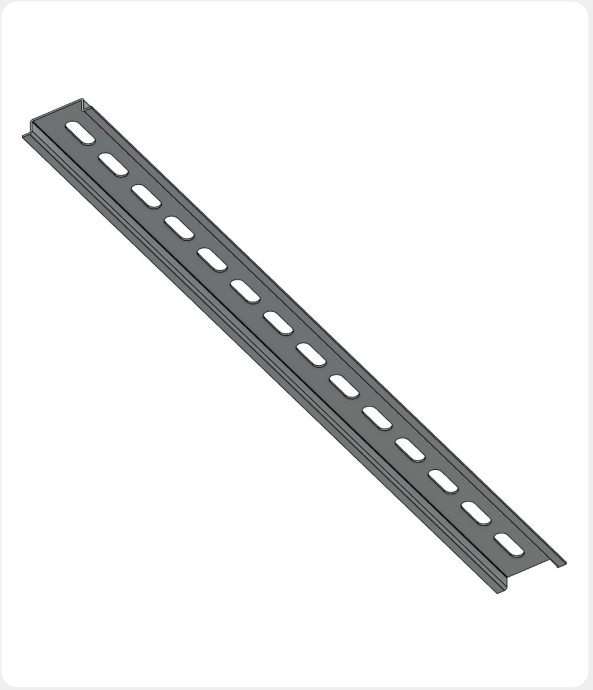
1 x set of six lever terminal blocks; electrical bridges excluded

CHECK BEFORE CONTINUING: All six blocks remain on the rail under a gentle pull. Confirm required end stops and wiring access.

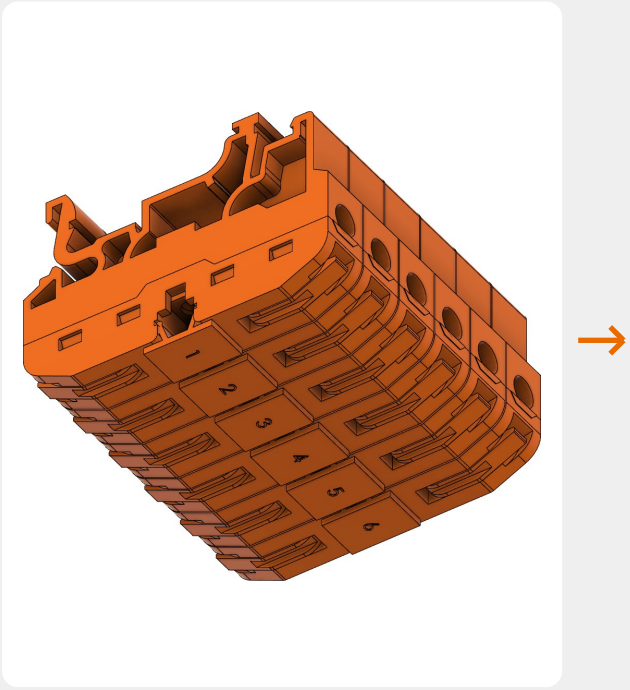
POSITION THE SIX-BLOCK TERMINAL SET

BASE TERMINAL-BLOCK MOUNTING | DETAIL 1

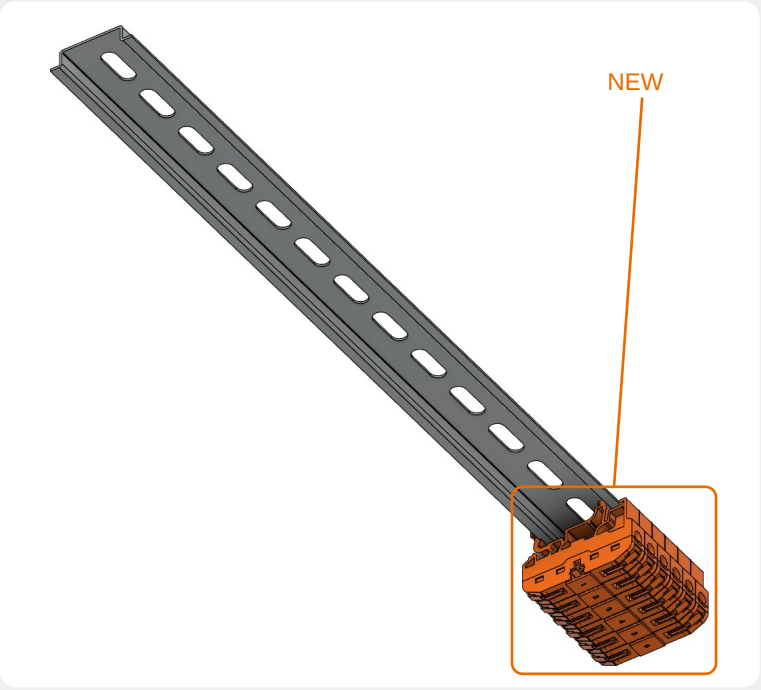
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the six-block set on the rail in the shown orientation. Engage the mounting feet using the terminal manufacturer's method and check each block is retained. Electrical bridges are omitted from this mounting view; fit bridges only to an approved wiring diagram.

PARTS IN THIS CAD STEP

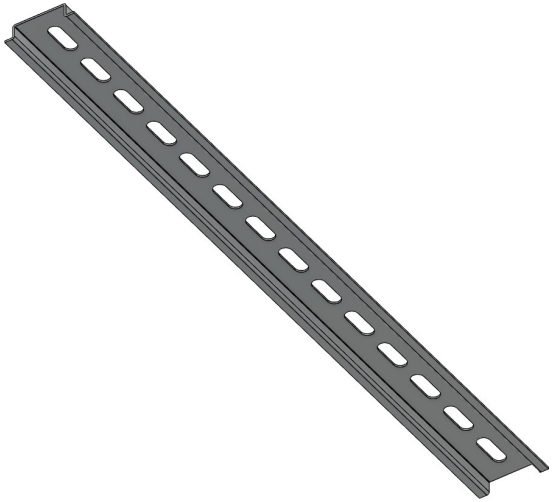
1 x set of six lever terminal blocks; electrical bridges excluded

CHECK BEFORE CONTINUING: All six blocks remain on the rail under a gentle pull. Confirm required end stops and wiring access.

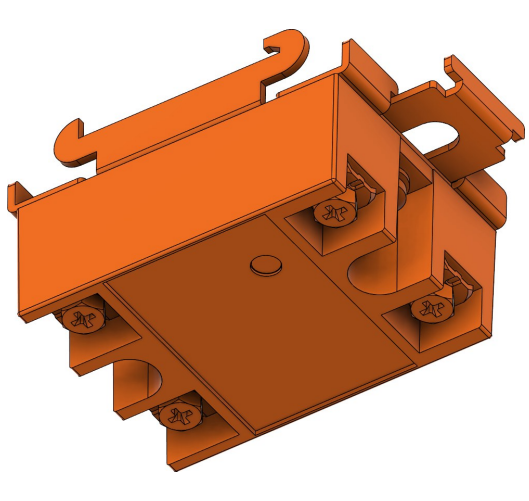
POSITION THE COMPLETE RELAY ASSEMBLY

RELAY 1 MOUNTING | DETAIL 1

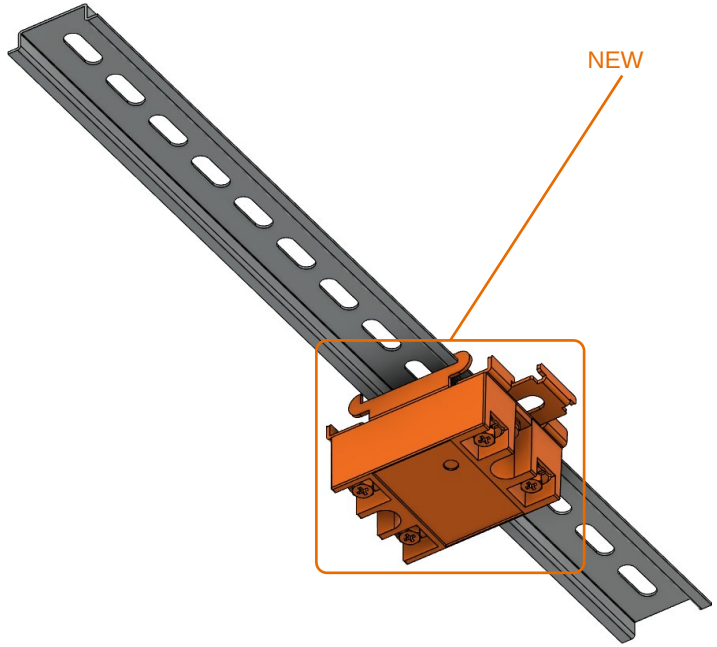
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Orient this complete relay assembly at the shown position on the base rail. Confirm its actual rail adapter and retention method before securing it. Keep the relay assembled and leave terminal access and cooling clearances available.

PARTS IN THIS CAD STEP

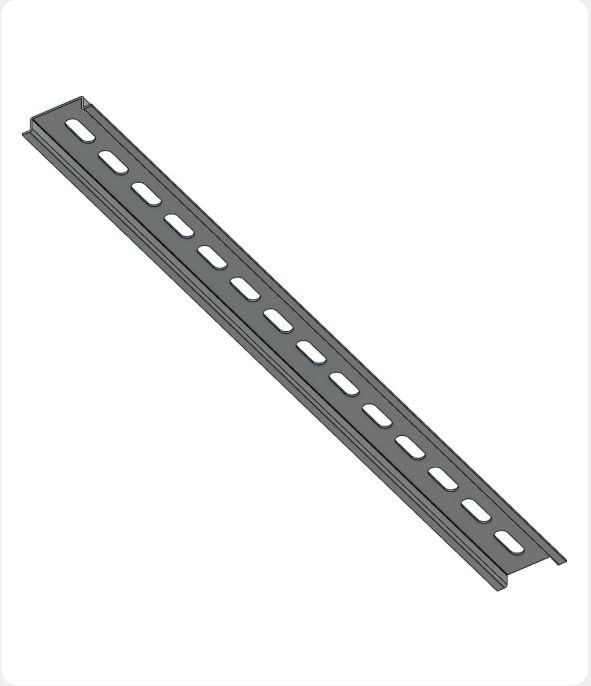
1 x complete solid-state relay assembly; adapter fixings require confirmation

CHECK BEFORE CONTINUING: The relay is securely retained. Confirm adapter hardware and required cooling clearance against the actual relay assembly.

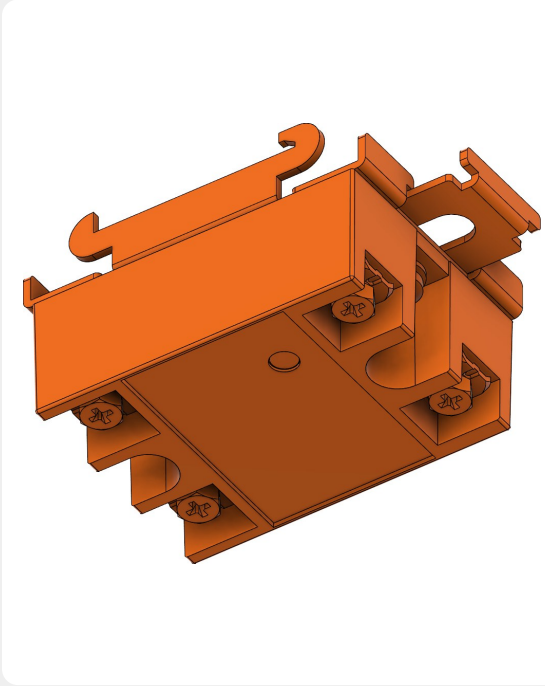
POSITION THE COMPLETE RELAY ASSEMBLY

RELAY 2 MOUNTING | DETAIL 1

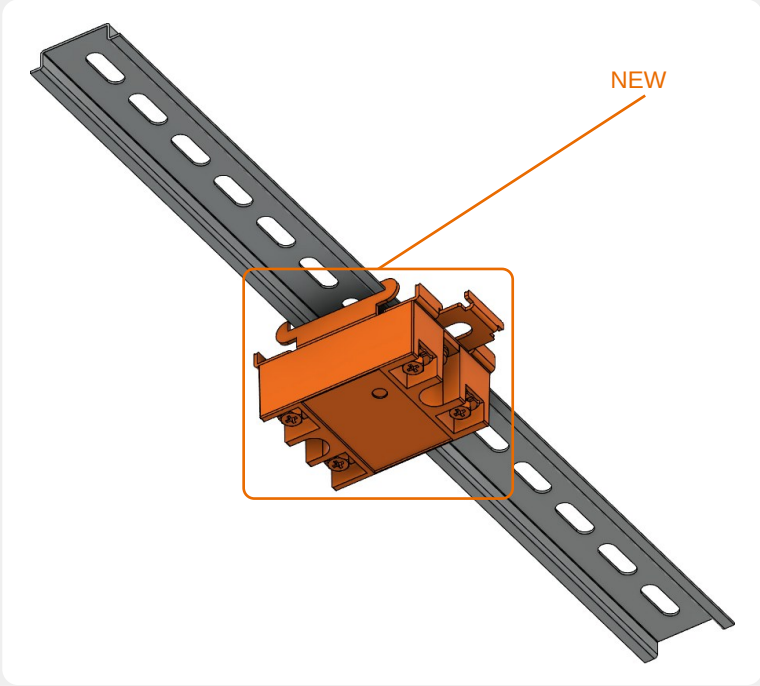
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Orient this complete relay assembly at the shown position on the base rail. Confirm its actual rail adapter and retention method before securing it. Keep the relay assembled and leave terminal access and cooling clearances available.

PARTS IN THIS CAD STEP

1 x complete solid-state relay assembly; adapter fixings require confirmation

CHECK BEFORE CONTINUING: The relay is securely retained. Confirm adapter hardware and required cooling clearance against the actual relay assembly.

POSITION BOTH CABLE CHANNELS

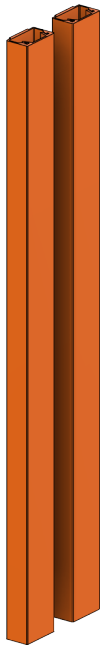
ELECTRONICS CABLE CHANNELS | DETAIL 1

1 BEFORE

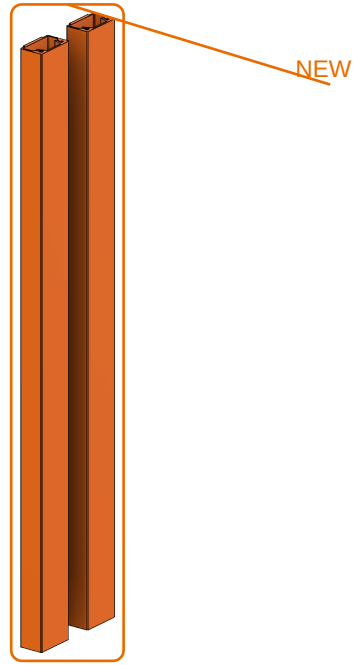
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match both channels to the modeled orientation beside the electronics bay. Check their mounting tabs or fixing holes against the supporting structure. Confirm the retention hardware before securing them; no separate fasteners are identified for these parts.

PARTS IN THIS CAD STEP

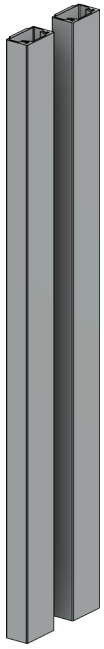
2 x cable channels; retention hardware requires confirmation

CHECK BEFORE CONTINUING: The channel is retained and its cable opening remains accessible. No edge can pinch a cable.

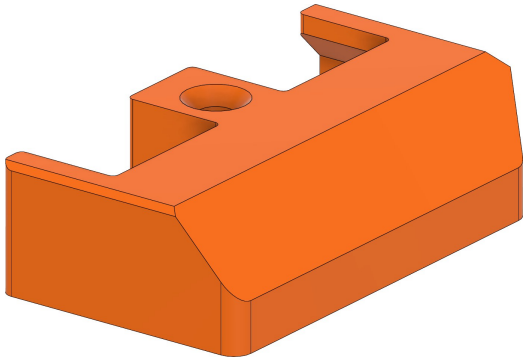
CHECK THE LOWER WIRE-COVER POSITION

ELECTRONICS CABLE CHANNELS | DETAIL 2

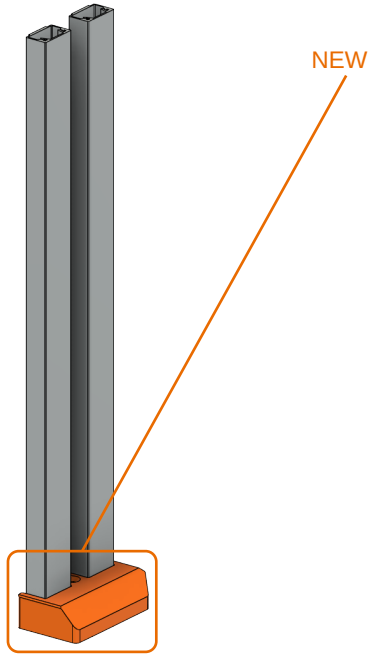
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the lower wire cover with the cable-channel area as shown. Check its attachment method before fitting. Leave the cover accessible until cable routing and inspection are complete.

PARTS IN THIS CAD STEP

1 x lower wire cover; retention hardware requires confirmation

CHECK BEFORE CONTINUING: The cover fits without forcing or trapping cables. Its final retention method is confirmed.

IDENTIFY THE TWO PRINTED HOLDER PARTS

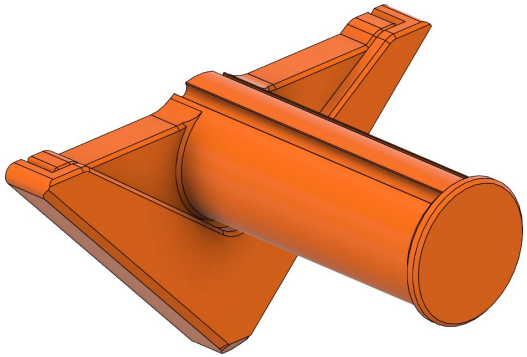
SPOOL HOLDER 1 | DETAIL 1

1 BEFORE

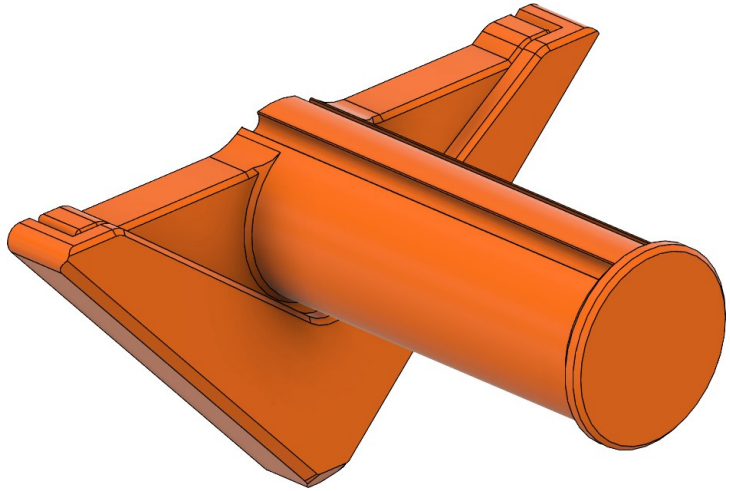
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify the two printed pieces for this holder. Match their mating faces and the three fixing positions to the shown orientation. Check that the spool-contact surfaces are smooth before assembly.

PARTS IN THIS CAD STEP

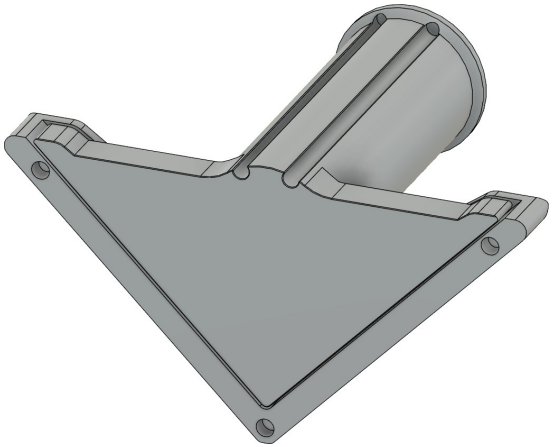
2 x printed spool-holder pieces

CHECK BEFORE CONTINUING: The pieces match without forcing; all three fixing positions align.

INSTALL THE THREE HOLDER INSERTS

SPOOL HOLDER 1 | DETAIL 2

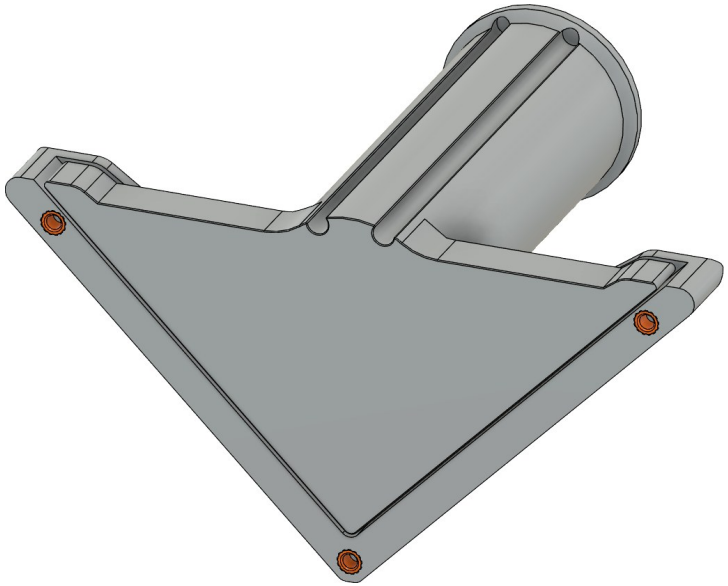
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Separate the printed pieces as needed to reach the three highlighted insert pockets. With all electronics and spool removed, heat-set one M3 insert in each pocket. Keep the inserts square and allow the plastic to cool before joining the pieces.

PARTS IN THIS CAD STEP

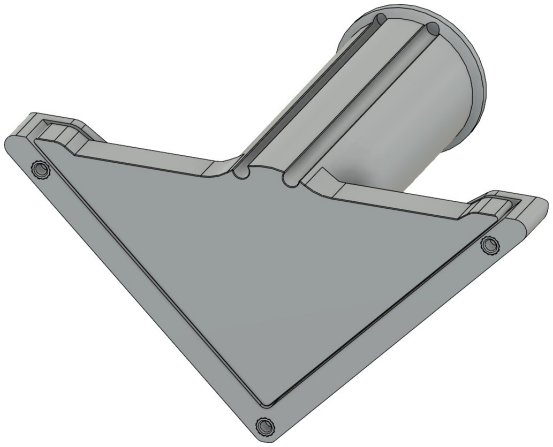
3 x M3 heat-set inserts

CHECK BEFORE CONTINUING: Three inserts are seated; their threaded openings remain clear. Confirm the pocket belongs to the correct printed piece before heating.

JOIN THE HOLDER WITH THREE SCREWS

SPOOL HOLDER 1 | DETAIL 3

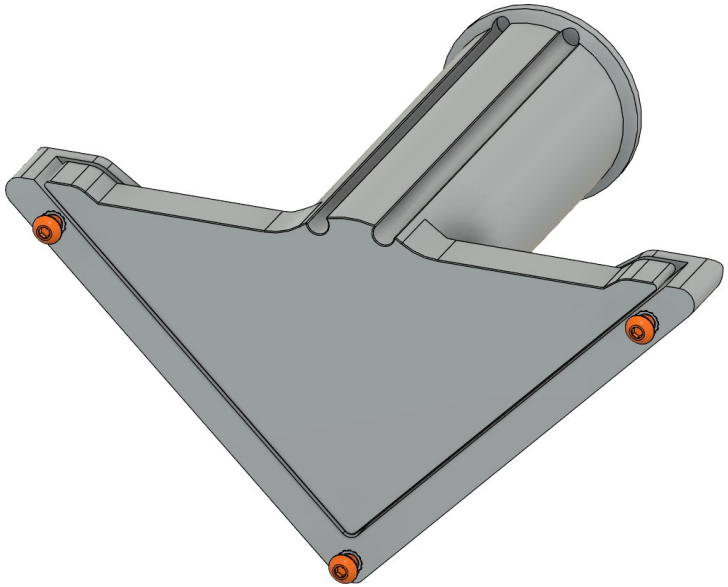
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Bring the printed pieces together in the shown orientation. Start the three M3 x 6 hex button-head screws by hand into the inserts. Tighten progressively while keeping the joint seated. Confirm any separate frame attachment before loading a spool.

PARTS IN THIS CAD STEP

3 x M3 x 6 hex button-head screws (92095A179)

CHECK BEFORE CONTINUING: All three screws engage without bottoming out. The holder is rigid; frame attachment and spool clearance still require physical confirmation.

IDENTIFY THE TWO PRINTED HOLDER PARTS

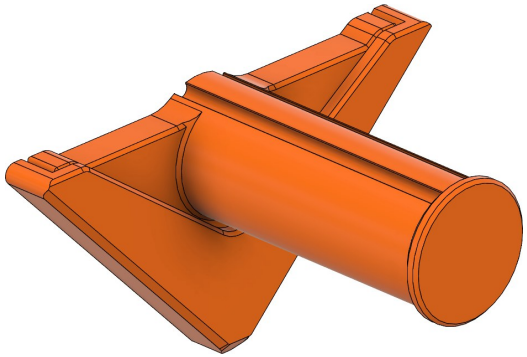
SPOOL HOLDER 2 | DETAIL 1

1 BEFORE

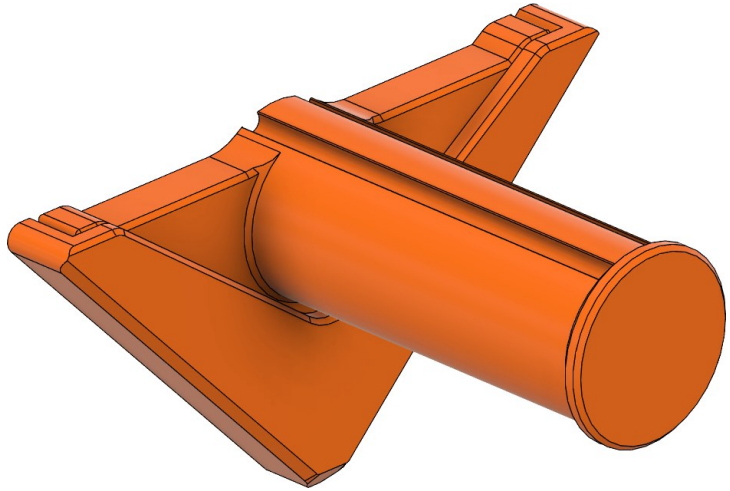
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify the two printed pieces for this holder. Match their mating faces and the three fixing positions to the shown orientation. Check that the spool-contact surfaces are smooth before assembly.

PARTS IN THIS CAD STEP

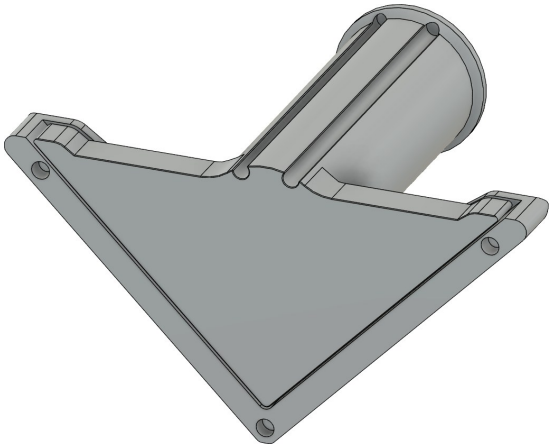
2 x printed spool-holder pieces

CHECK BEFORE CONTINUING: The pieces match without forcing; all three fixing positions align.

INSTALL THE THREE HOLDER INSERTS

SPOOL HOLDER 2 | DETAIL 2

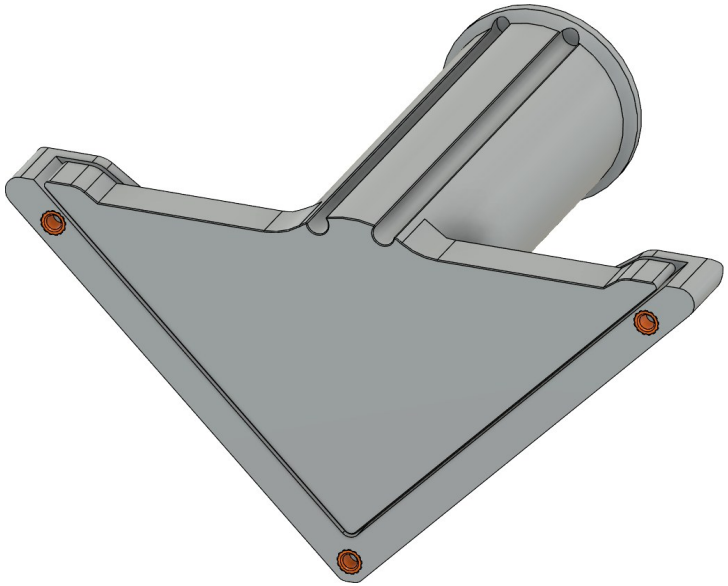
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Separate the printed pieces as needed to reach the three highlighted insert pockets. With all electronics and spool removed, heat-set one M3 insert in each pocket. Keep the inserts square and allow the plastic to cool before joining the pieces.

PARTS IN THIS CAD STEP

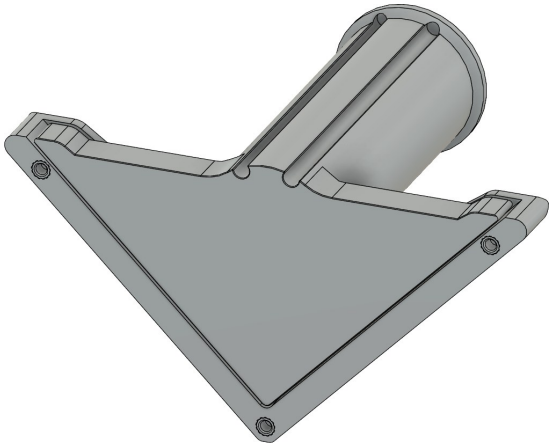
3 x M3 heat-set inserts

CHECK BEFORE CONTINUING: Three inserts are seated; their threaded openings remain clear. Confirm the pocket belongs to the correct printed piece before heating.

JOIN THE HOLDER WITH THREE SCREWS

SPOOL HOLDER 2 | DETAIL 3

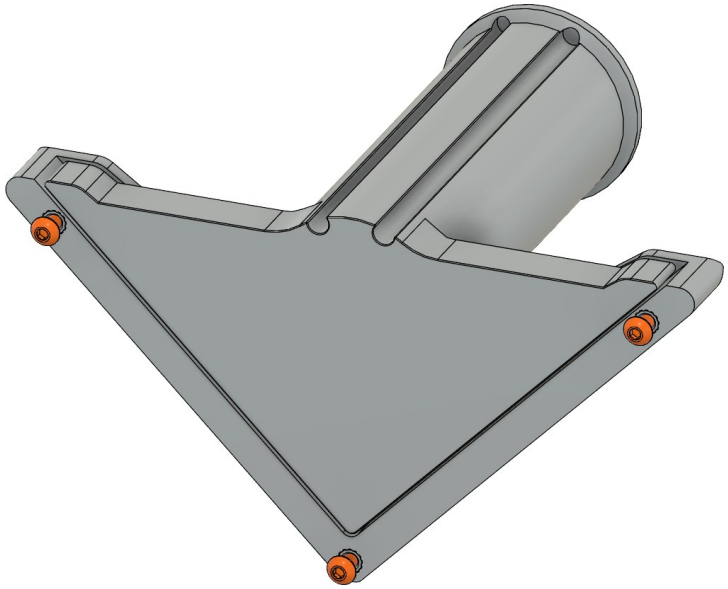
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Bring the printed pieces together in the shown orientation. Start the three M3 x 6 hex button-head screws by hand into the inserts. Tighten progressively while keeping the joint seated. Confirm any separate frame attachment before loading a spool.

PARTS IN THIS CAD STEP

3 x M3 x 6 hex button-head screws (92095A179)

CHECK BEFORE CONTINUING: All three screws engage without bottoming out. The holder is rigid; frame attachment and spool clearance still require physical confirmation.

ORIENT THE COMPLETE PANEL CONNECTOR

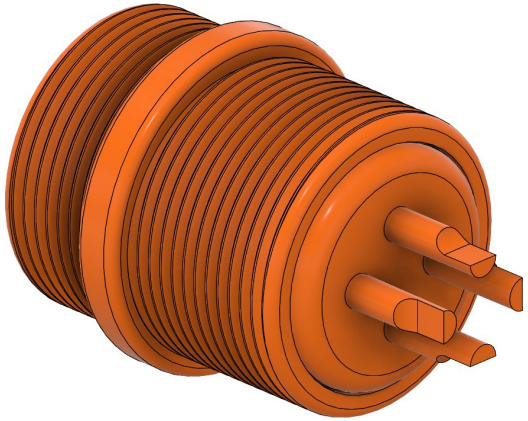
PANEL CONNECTOR 1 | DETAIL 1

1 BEFORE

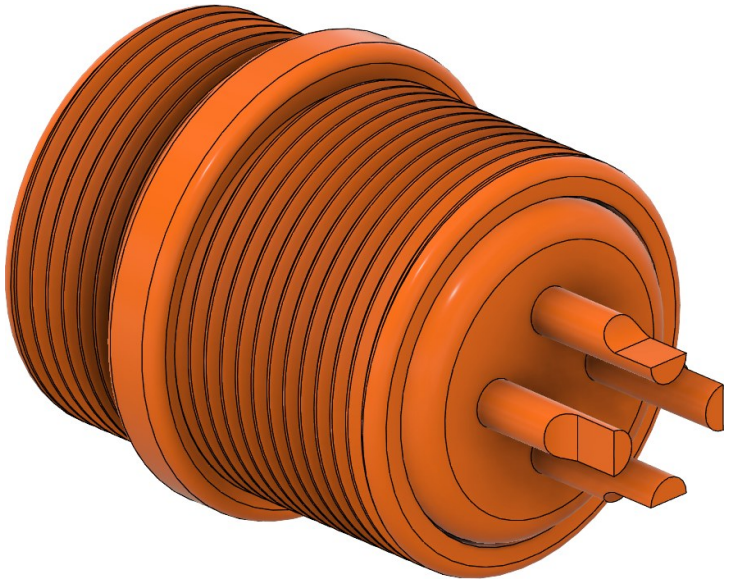
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Keep the connector body, insert and contacts assembled. Match its orientation to the intended panel opening. Check the actual connector and panel thickness before placing it through the opening.

PARTS IN THIS CAD STEP

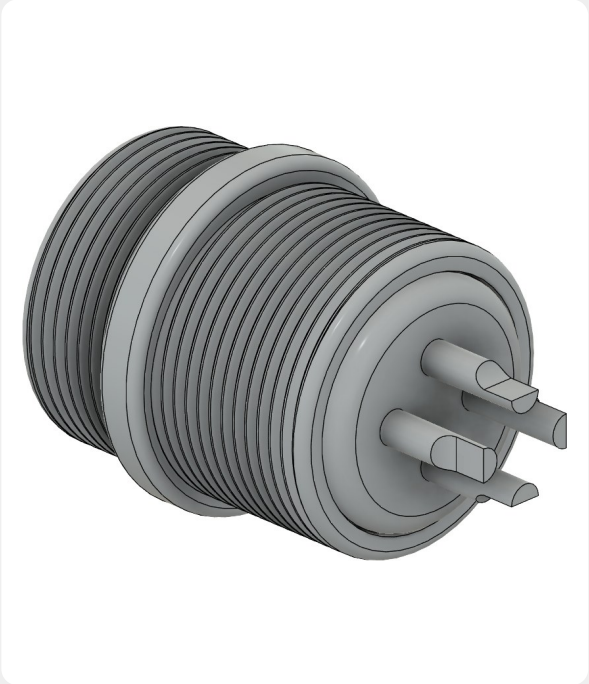
1 x assembled GX20-style panel connector (CAD label)

CHECK BEFORE CONTINUING: Connector identity, panel opening and access to the retaining nut are confirmed. Contact locations here do not specify wiring.

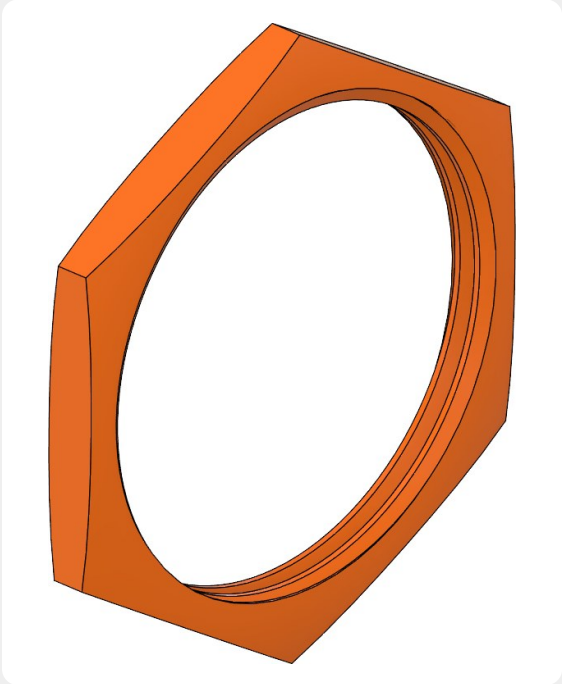
FIT THE CONNECTOR RETAINING NUT

PANEL CONNECTOR 1 | DETAIL 2

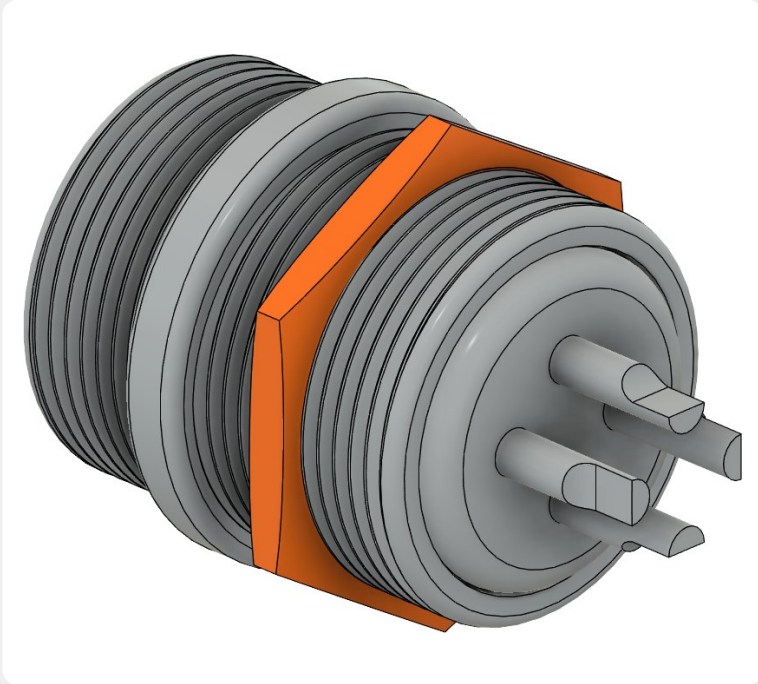
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

With the connector seated in its confirmed panel opening, fit the highlighted retaining nut. Hold the body against rotation and tighten to the connector manufacturer's requirement. The panel is omitted here for visibility.

PARTS IN THIS CAD STEP

1 x connector retaining nut

CHECK BEFORE CONTINUING: The connector cannot rotate or move axially. Confirm any supplied washer or seal required by the actual connector.

ORIENT THE COMPLETE PANEL CONNECTOR

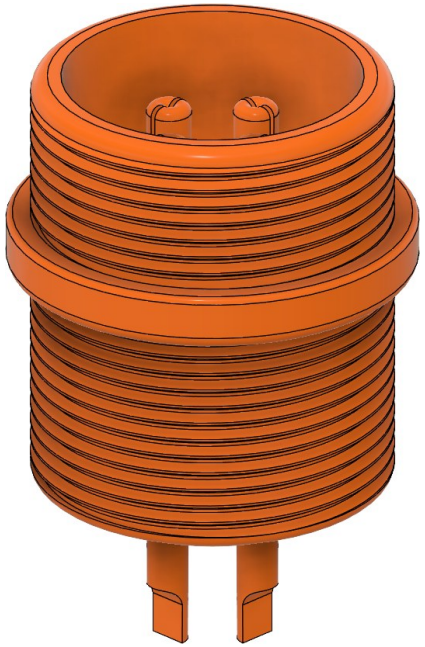
PANEL CONNECTOR 2 | DETAIL 1

1 BEFORE

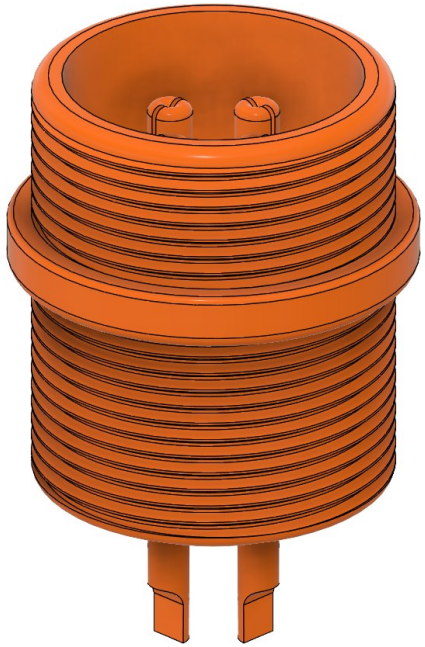
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Keep the connector body, insert and contacts assembled. Match its orientation to the intended panel opening. Check the actual connector and panel thickness before placing it through the opening.

PARTS IN THIS CAD STEP

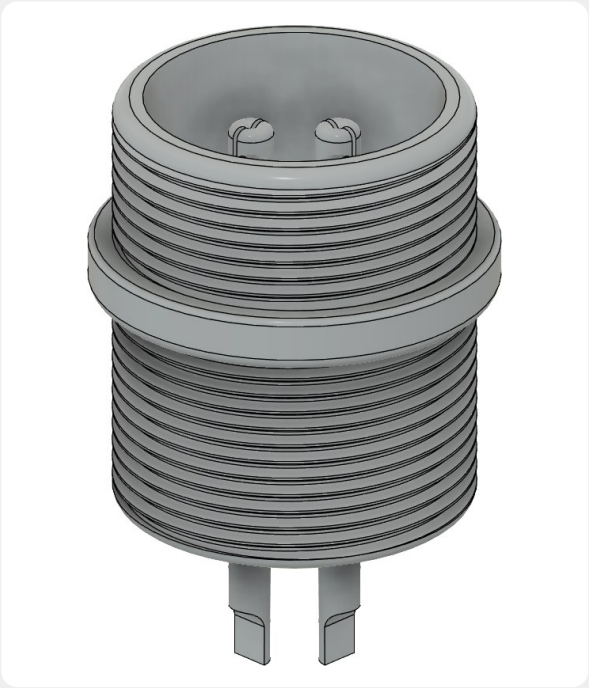
1 x assembled GX20-style panel connector (CAD label)

CHECK BEFORE CONTINUING: Connector identity, panel opening and access to the retaining nut are confirmed. Contact locations here do not specify wiring.

FIT THE CONNECTOR RETAINING NUT

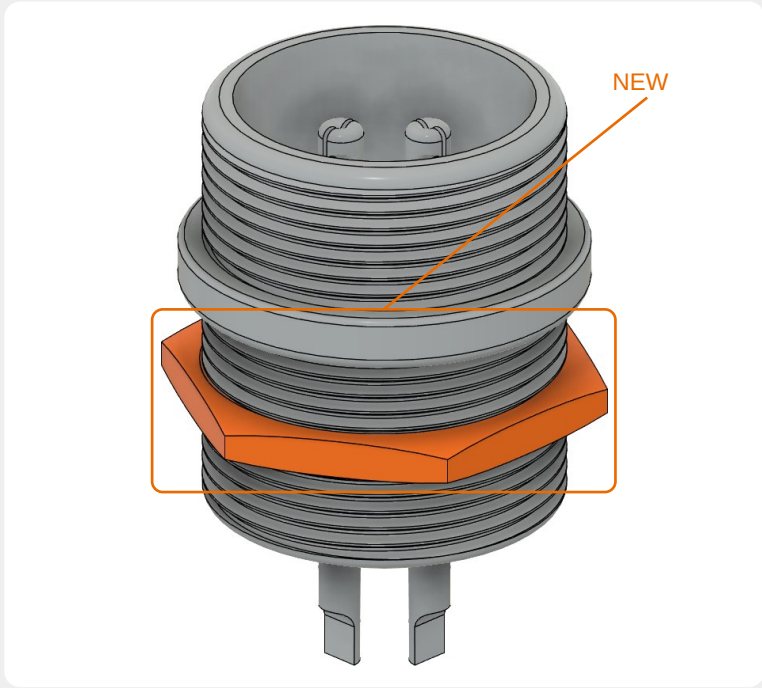
PANEL CONNECTOR 2 | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

With the connector seated in its confirmed panel opening, fit the highlighted retaining nut. Hold the body against rotation and tighten to the connector manufacturer's requirement. The panel is omitted here for visibility.

PARTS IN THIS CAD STEP

1 x connector retaining nut

CHECK BEFORE CONTINUING: The connector cannot rotate or move axially. Confirm any supplied washer or seal required by the actual connector.

ORIENT THE CHAMBER FAN

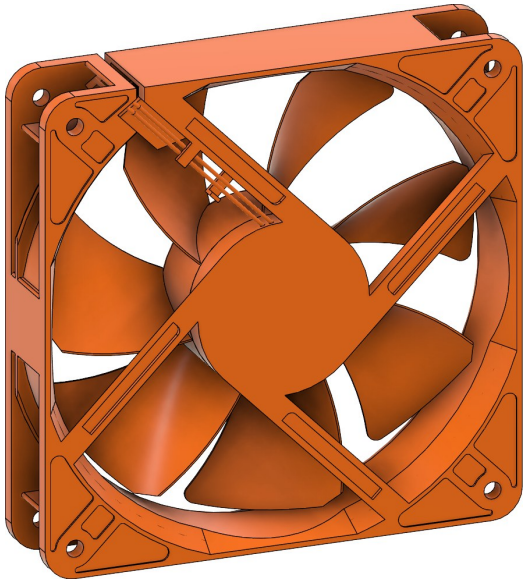
CHAMBER FAN 1 | DETAIL 1

1 BEFORE

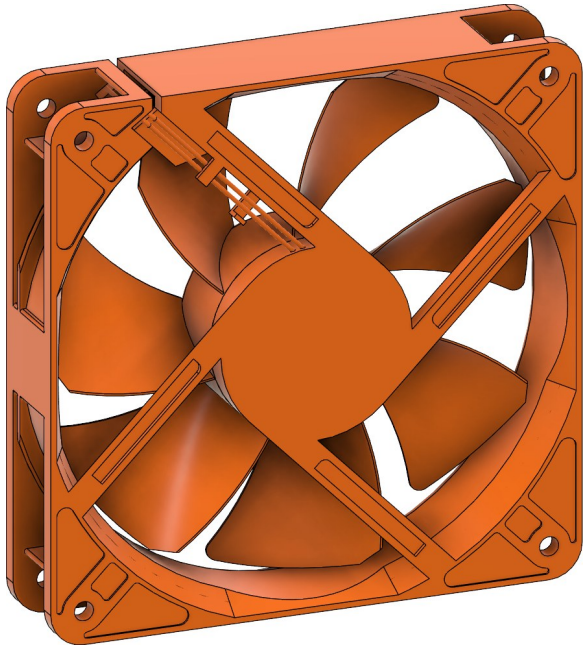
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this chamber fan from the overview and match its mounting pattern to the actual opening. Confirm airflow from the system design and the arrows on the actual fan. Keep the cable exit accessible.

PARTS IN THIS CAD STEP

1 x chamber fan

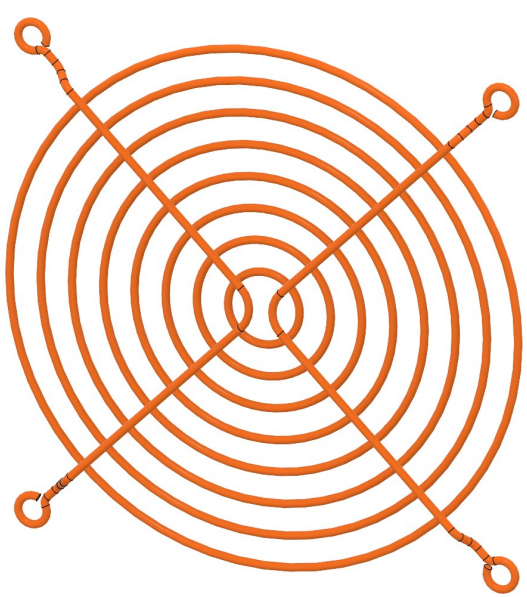
CHECK BEFORE CONTINUING: Fan identity, airflow direction and supporting mount are confirmed before fastening.

ALIGN THE STEEL FAN GUARD

CHAMBER FAN 1 | DETAIL 2

1 BEFORE

2 PARTS ADDED



3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the steel guard with the exposed fan face as modeled. Check the four mounting holes and blade clearance. Confirm the required retaining screws, nuts or threaded mount before securing it; separate fixings are not identified here.

PARTS IN THIS CAD STEP

1 x steel fan guard; mounting fasteners require confirmation

CHECK BEFORE CONTINUING: The guard clears the blades and is securely retained by the confirmed mounting arrangement.

ORIENT THE CHAMBER FAN

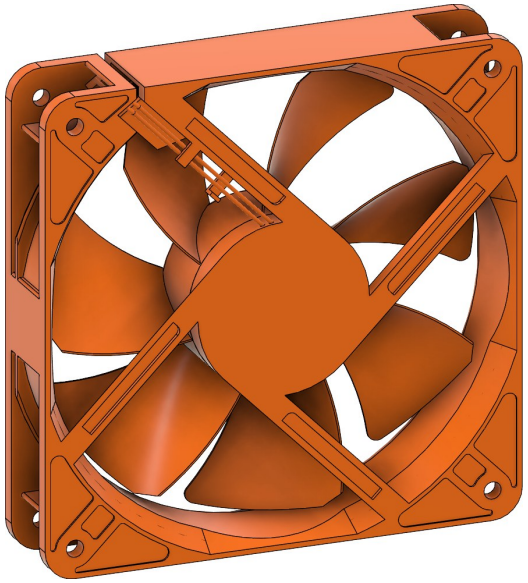
CHAMBER FAN 2 | DETAIL 1

1 BEFORE

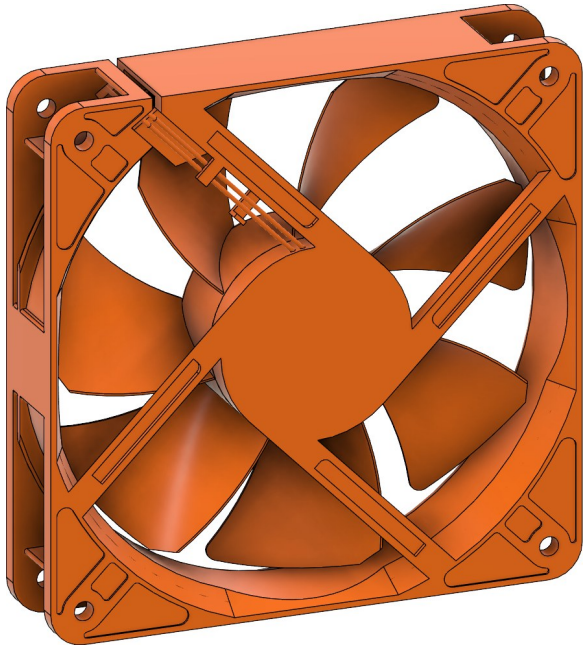
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this chamber fan from the overview and match its mounting pattern to the actual opening. Confirm airflow from the system design and the arrows on the actual fan. Keep the cable exit accessible.

PARTS IN THIS CAD STEP

1 x chamber fan

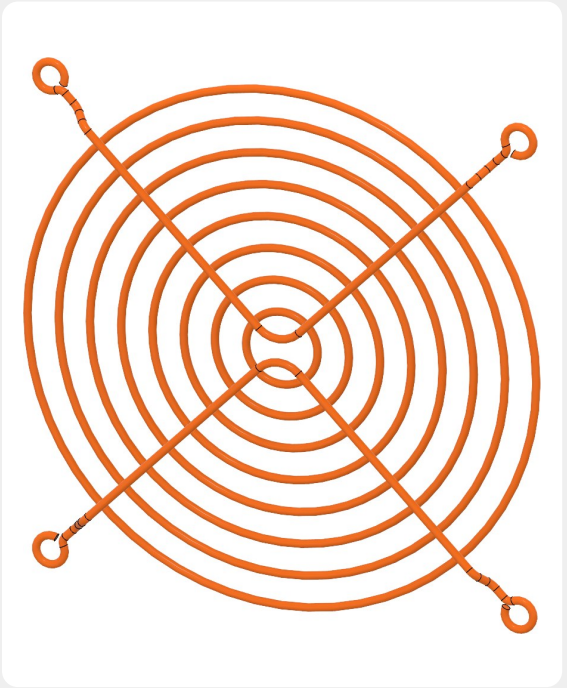
CHECK BEFORE CONTINUING: Fan identity, airflow direction and supporting mount are confirmed before fastening.

ALIGN THE STEEL FAN GUARD

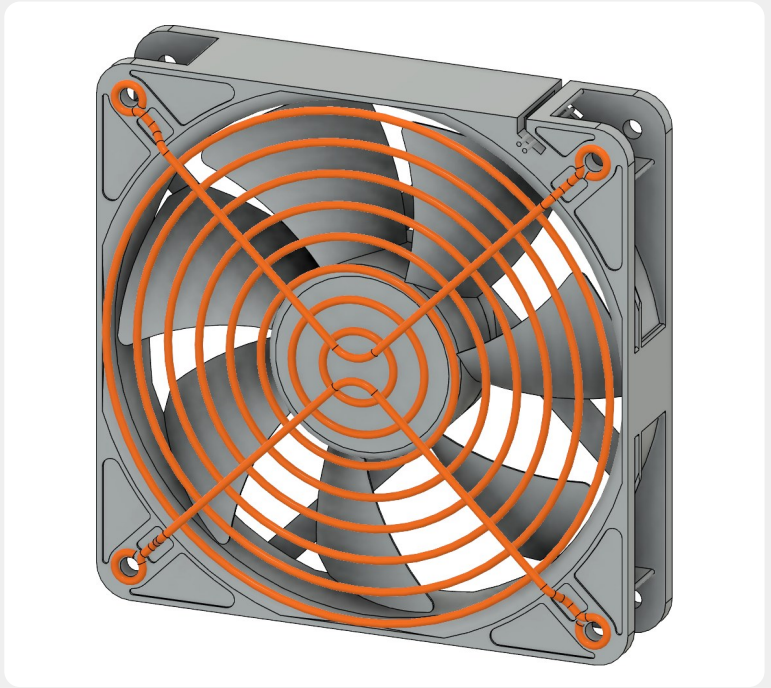
CHAMBER FAN 2 | DETAIL 2

1 BEFORE

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the steel guard with the exposed fan face as modeled. Check the four mounting holes and blade clearance. Confirm the required retaining screws, nuts or threaded mount before securing it; separate fixings are not identified here.

PARTS IN THIS CAD STEP

1 x steel fan guard; mounting fasteners require confirmation

CHECK BEFORE CONTINUING: The guard clears the blades and is securely retained by the confirmed mounting arrangement.

Required before wiring

Record the actual supply voltage, controller revision, driver types, motor ratings, heater ratings, temperature sensors, fans, probe, toolboard and display. CAD labels are not evidence of the installed electrical configuration.

Create the connection schedule

For every circuit, specify source and destination, connector and pin numbers, polarity, wire size, protection, grounding/bonding, strain relief and cable routing. Use the actual manufacturers' documentation.

Required before power

Have the completed wiring checked against that schedule and the equipment requirements. Verify protective earth, conductor retention and separation of mains and low-voltage circuits as applicable. Keep the system unpowered until those checks are complete.

Not supplied by this CAD export

No verified wiring schematic, fuse sizing, mains wiring sequence, firmware pins or heater configuration accompanies this edition. These are required additions to the final guide.

INSTALL THE SKIRTS AND FEET

Match each skirt to its side and opening. Prepare its confirmed insert hardware before mounting. Fit the feet and skirt sections while keeping access to frame and Z-drive fasteners.

CAD GROUP
Skirts, filtered sections and feet
M3 insert occurrences are modeled.

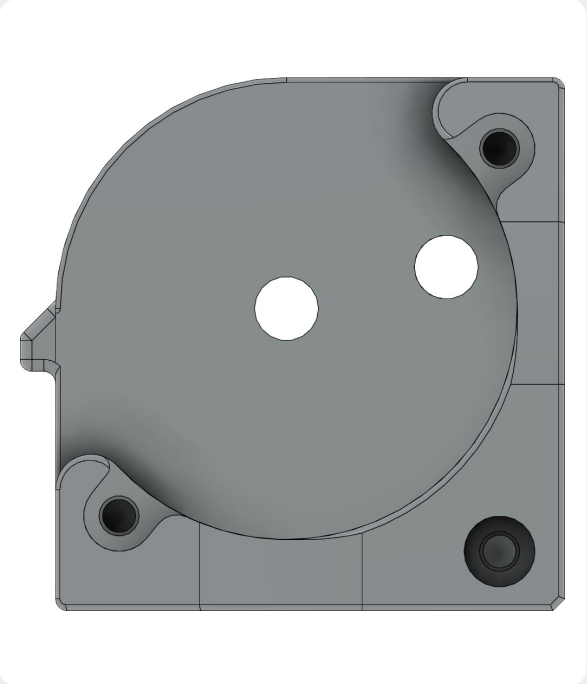
CHECK: The machine rests stably and the skirts clear moving parts and service access.

Current Hybrid assembly

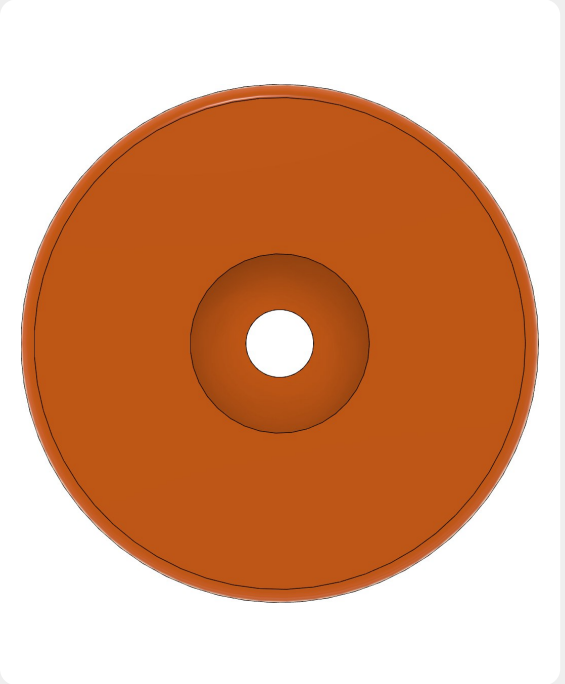
PLACE THE RUBBER FOOT

FRONT-LEFT FOOT | DETAIL 1

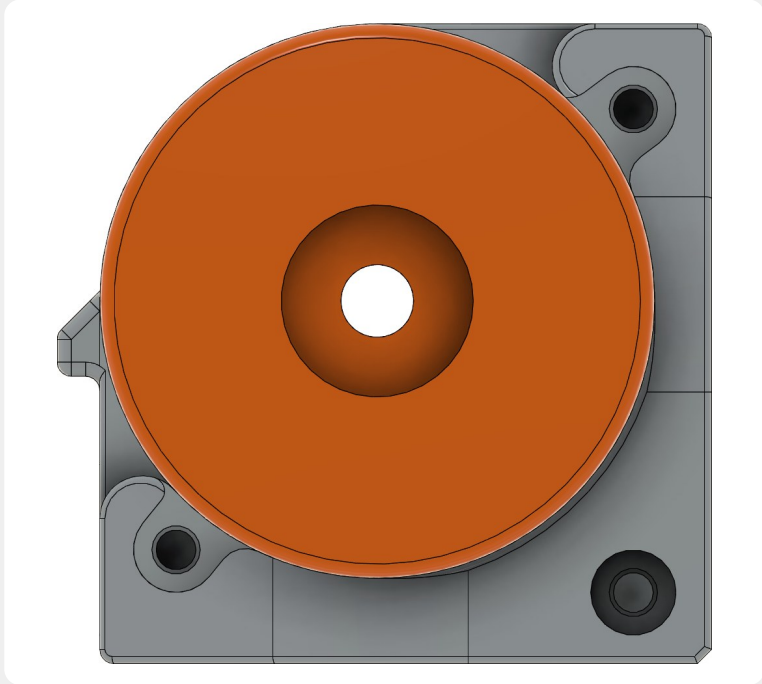
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match this corner to the location in the overview. Place the rubber foot under its circular seat, with the center hole aligned.

PARTS IN THIS CAD STEP

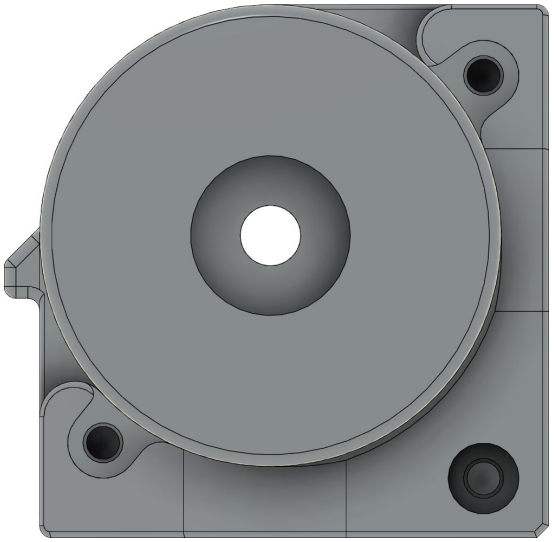
1 x Rubber Foot

CHECK BEFORE CONTINUING: The rubber foot is centered beneath the corner; its contact face points toward the table.

SECURE THE RUBBER FOOT

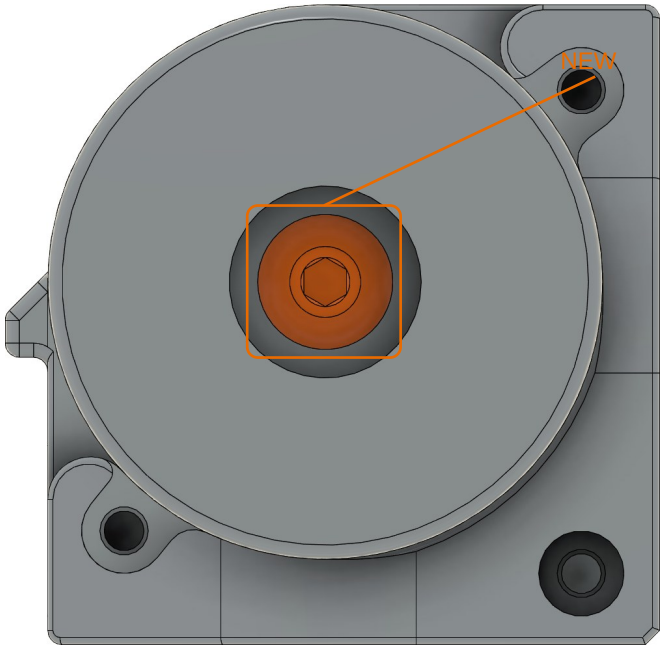
FRONT-LEFT FOOT | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the M5 nut into its seat. Pass the M5 x 16 button-head screw through the foot and start it into the nut. Tighten only enough to retain the foot.

PARTS IN THIS CAD STEP

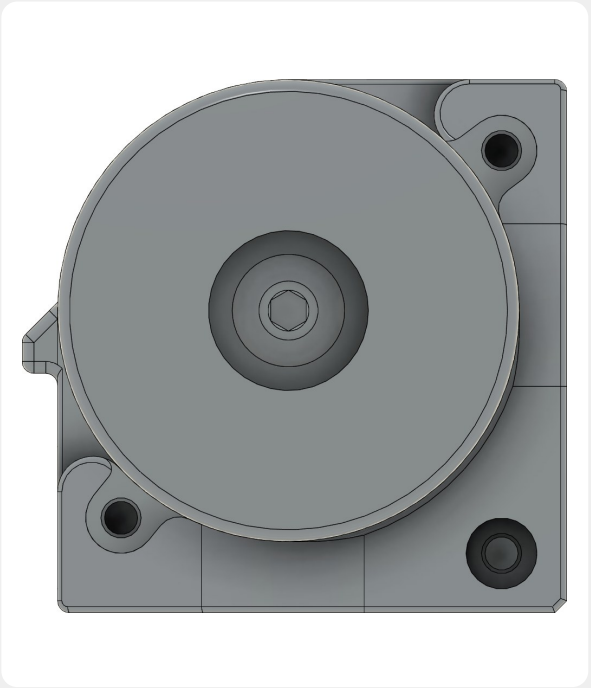
1 x M5x16 BHCS
1 x M5 Hex Nut

CHECK BEFORE CONTINUING: The foot is held securely without being crushed or pulled out of shape.

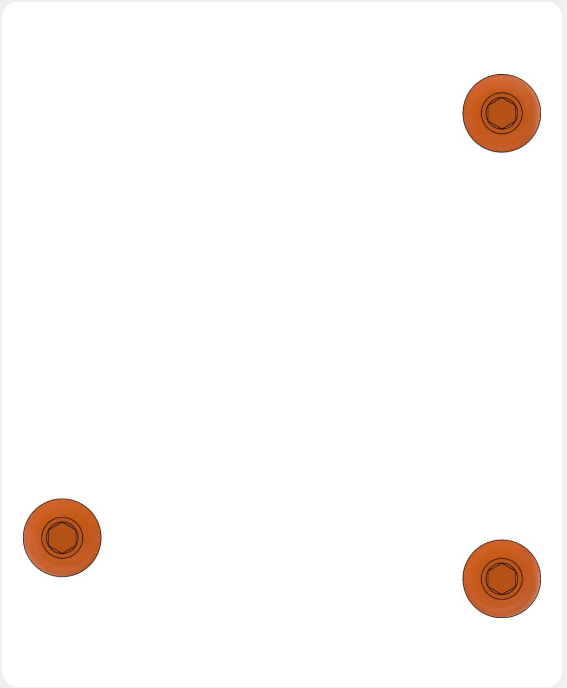
FIT THE THREE LOWER SCREWS

FRONT-LEFT FOOT | DETAIL 3

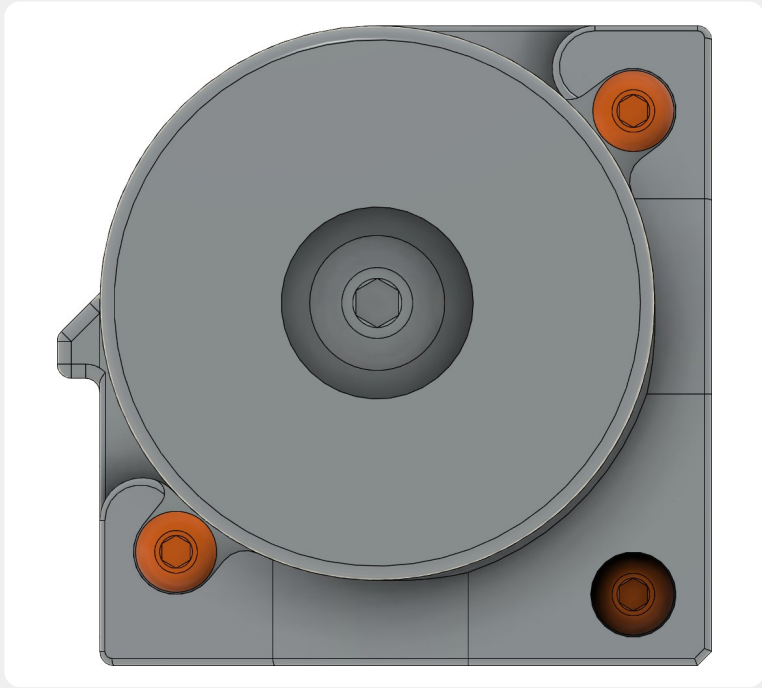
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Install the three M3 x 6 button-head screws at the highlighted lower fixing points. Start all three before gently tightening them.

PARTS IN THIS CAD STEP

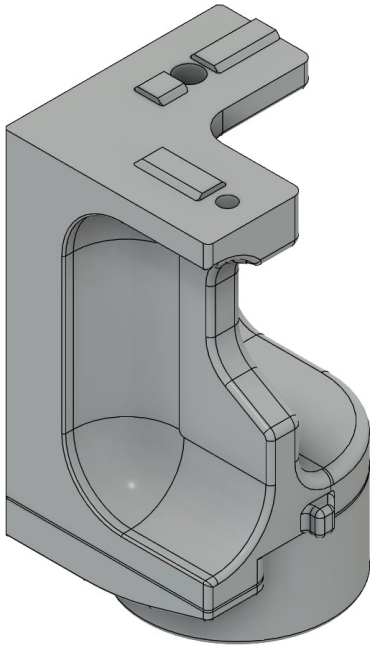
3 x M3x6 BHCS

CHECK BEFORE CONTINUING: Each screw enters its matching hole straight; all three heads are seated.

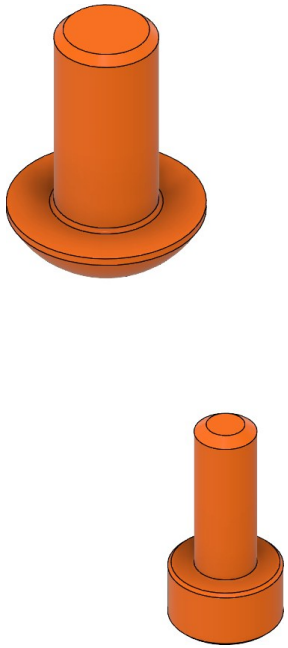
ATTACH THE CORNER TO THE FRAME

FRONT-LEFT FOOT | DETAIL 4

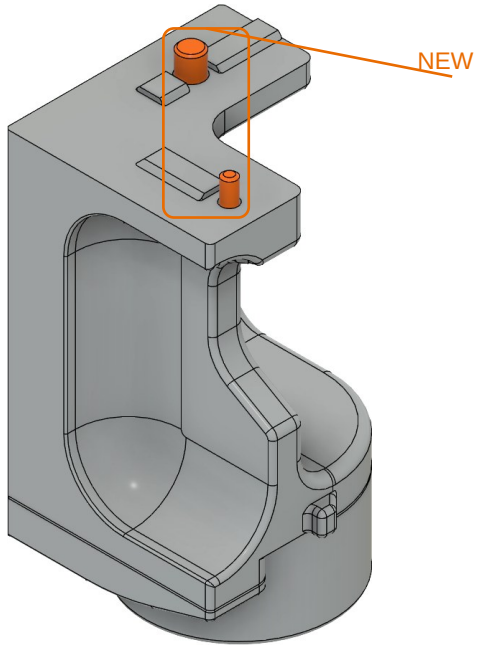
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the completed corner to its matching frame location. Fit the M5 x 10 button-head screw and M3 x 8 socket-head screw at the highlighted positions. Start both before tightening.

PARTS IN THIS CAD STEP

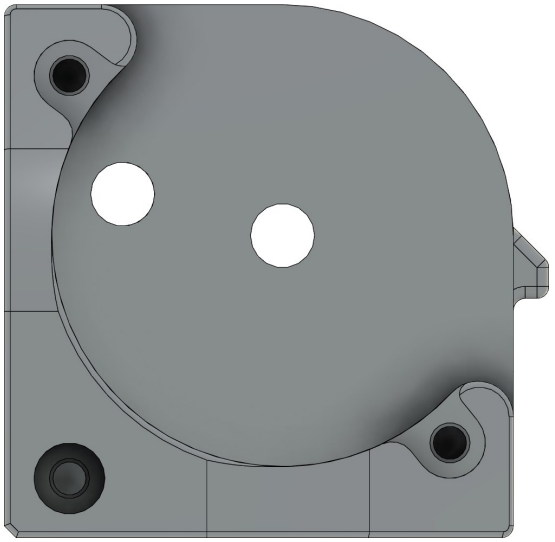
1 x M5x10 BHCS
1 x M3x8 SHCS

CHECK BEFORE CONTINUING: The corner sits against the frame without a gap. Check that the other feet can all touch a flat surface.

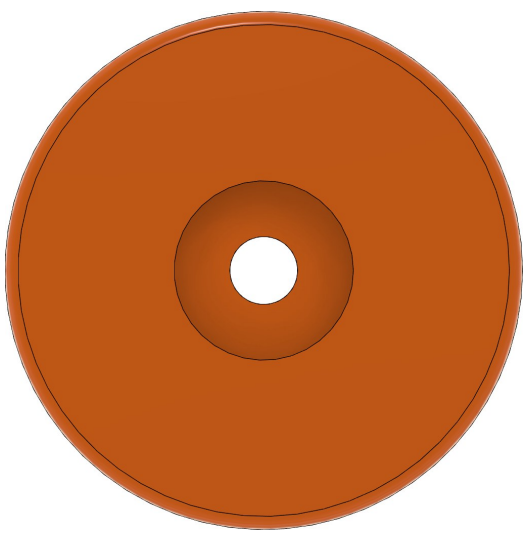
PLACE THE RUBBER FOOT

FRONT-RIGHT FOOT | DETAIL 1

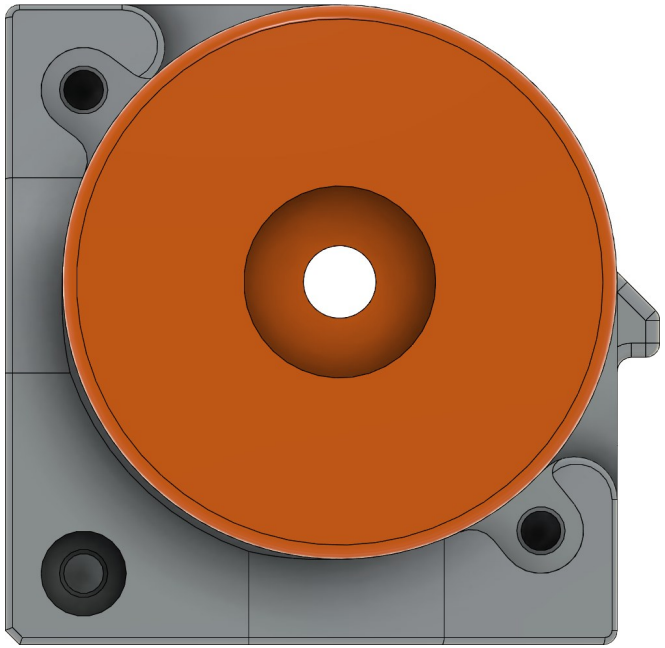
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match this corner to the location in the overview. Place the rubber foot under its circular seat, with the center hole aligned.

PARTS IN THIS CAD STEP

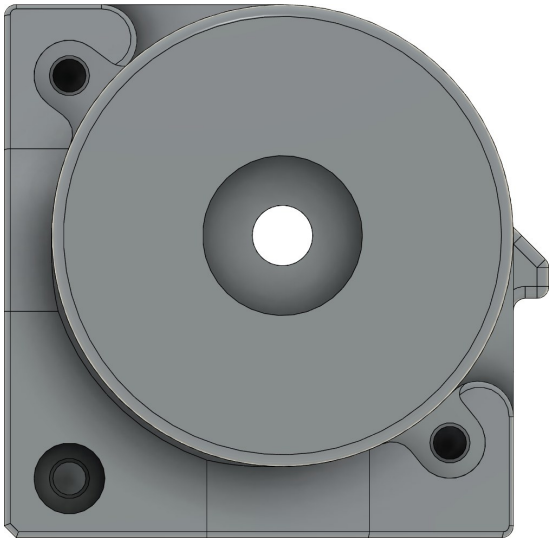
1 x Rubber Foot

CHECK BEFORE CONTINUING: The rubber foot is centered beneath the corner; its contact face points toward the table.

SECURE THE RUBBER FOOT

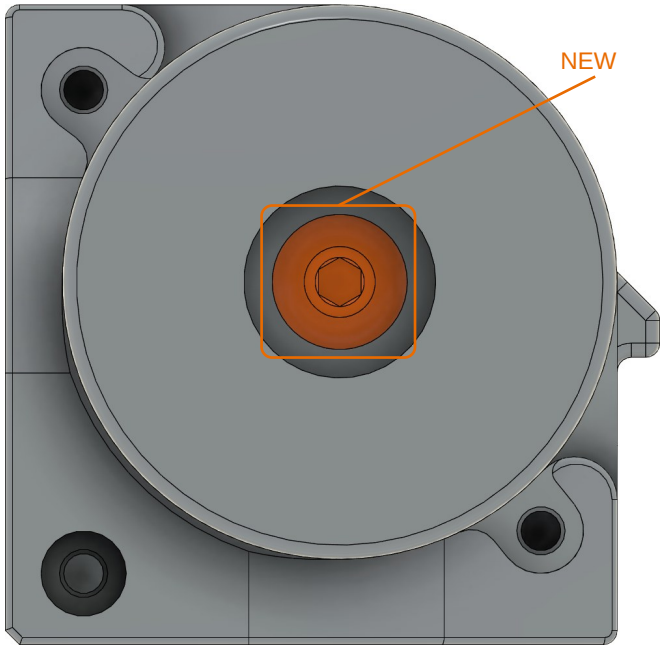
FRONT-RIGHT FOOT | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the M5 nut into its seat. Pass the M5 x 16 button-head screw through the foot and start it into the nut. Tighten only enough to retain the foot.

PARTS IN THIS CAD STEP

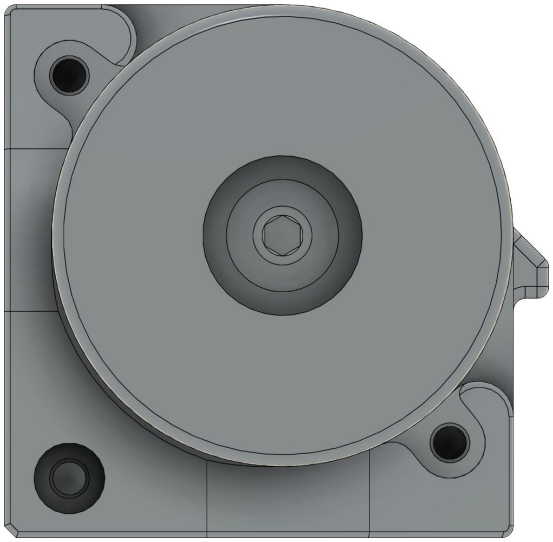
1 x M5x16 BHCS
1 x M5 Hex Nut

CHECK BEFORE CONTINUING: The foot is held securely without being crushed or pulled out of shape.

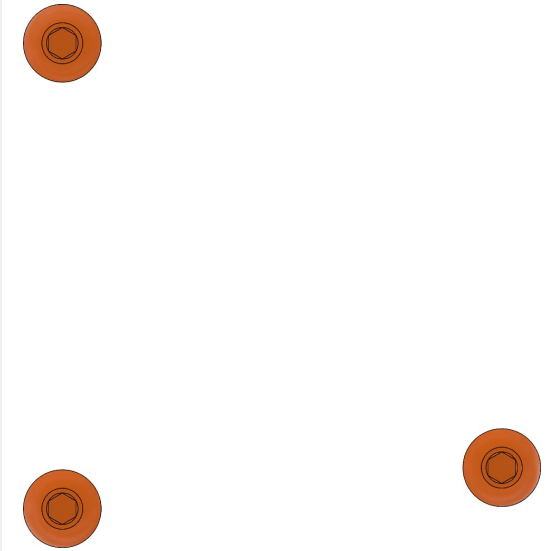
FIT THE THREE LOWER SCREWS

FRONT-RIGHT FOOT | DETAIL 3

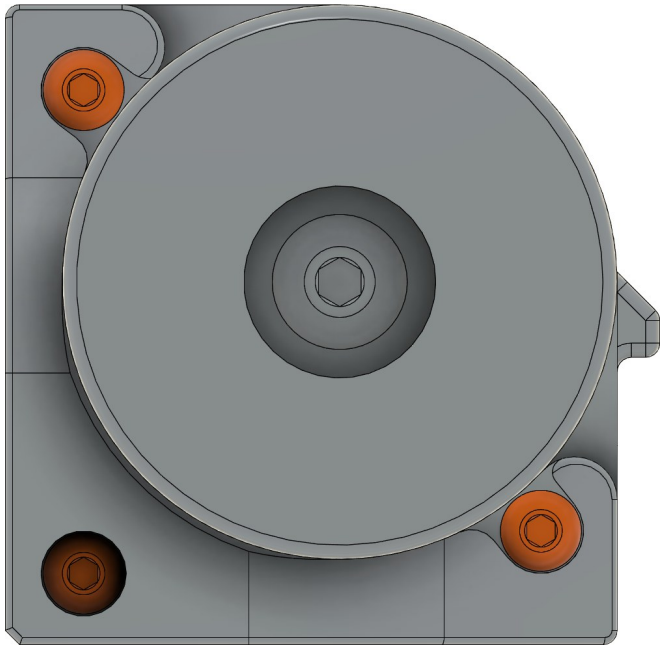
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Install the three M3 x 6 button-head screws at the highlighted lower fixing points. Start all three before gently tightening them.

PARTS IN THIS CAD STEP

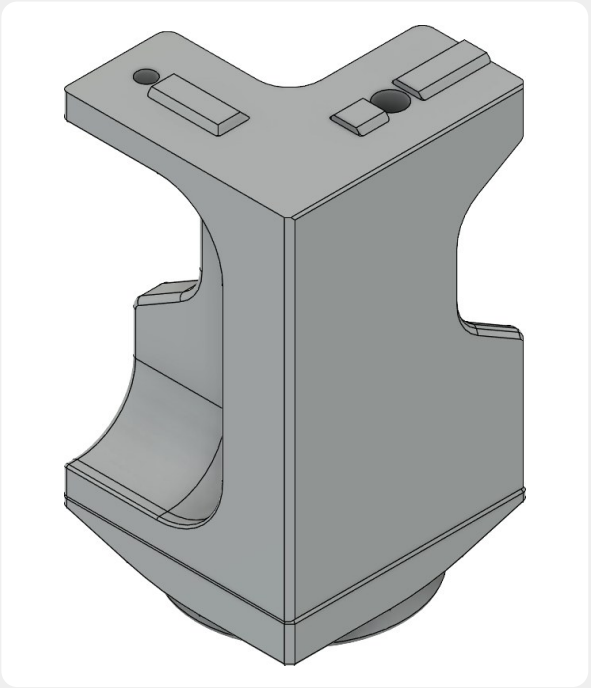
3 x M3x6 BHCS

CHECK BEFORE CONTINUING: Each screw enters its matching hole straight; all three heads are seated.

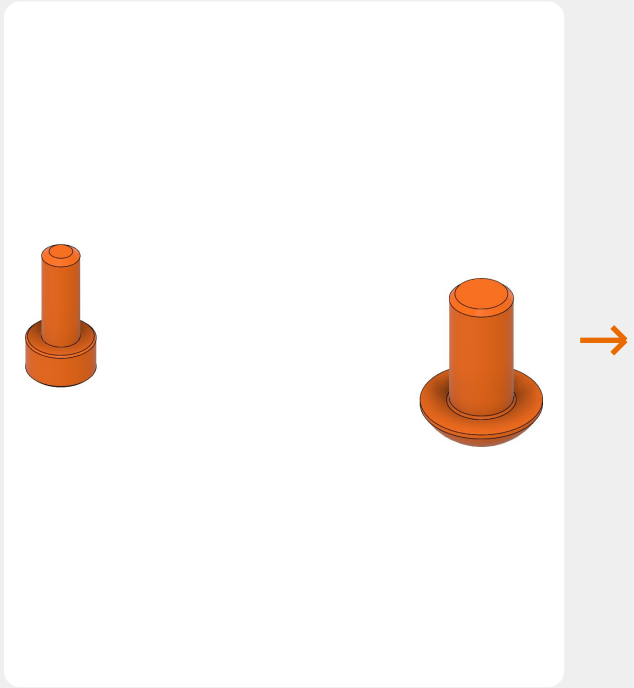
ATTACH THE CORNER TO THE FRAME

FRONT-RIGHT FOOT | DETAIL 4

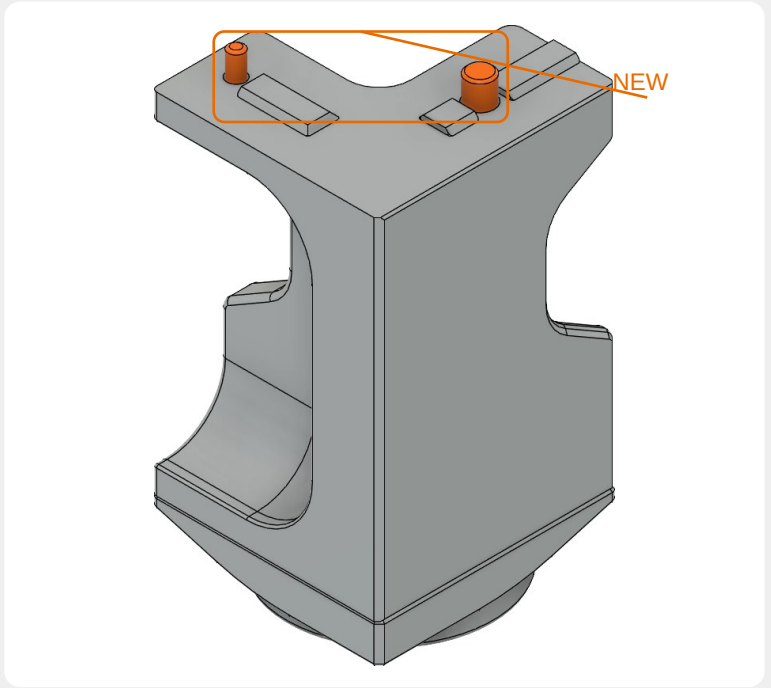
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the completed corner to its matching frame location. Fit the M5 x 10 button-head screw and M3 x 8 socket-head screw at the highlighted positions. Start both before tightening.

PARTS IN THIS CAD STEP

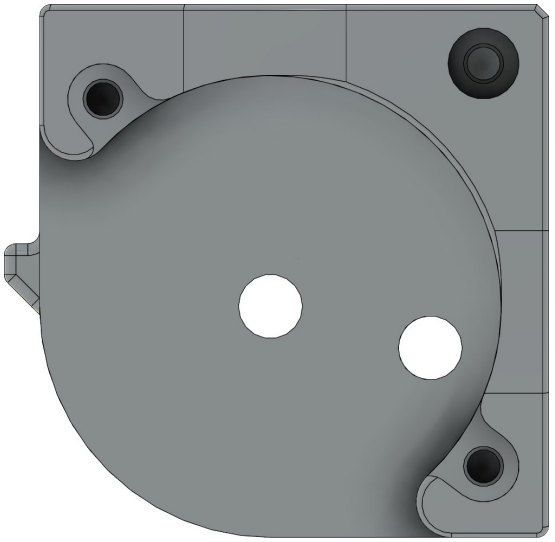
1 x M5x10 BHCS
1 x M3x8 SHCS

CHECK BEFORE CONTINUING: The corner sits against the frame without a gap. Check that the other feet can all touch a flat surface.

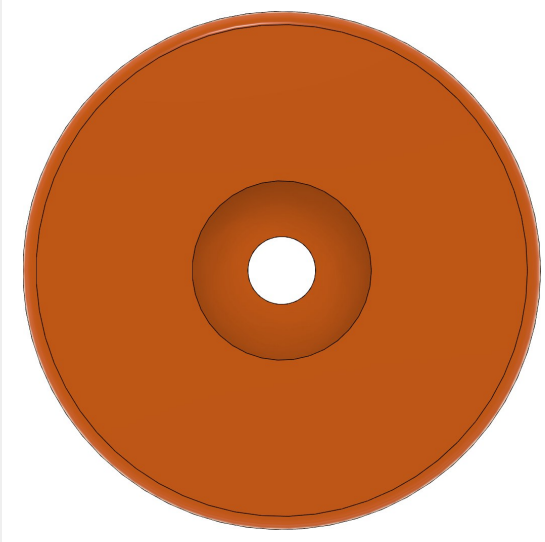
PLACE THE RUBBER FOOT

REAR-LEFT FOOT | DETAIL 1

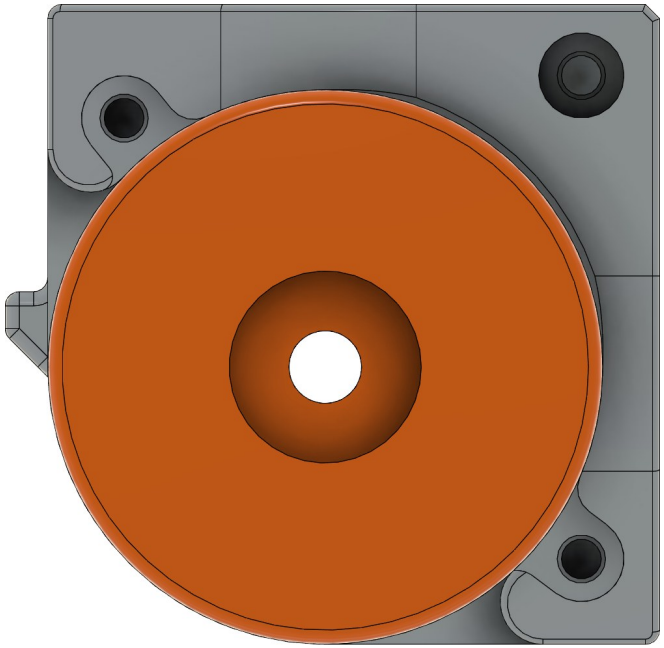
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match this corner to the location in the overview. Place the rubber foot under its circular seat, with the center hole aligned.

PARTS IN THIS CAD STEP

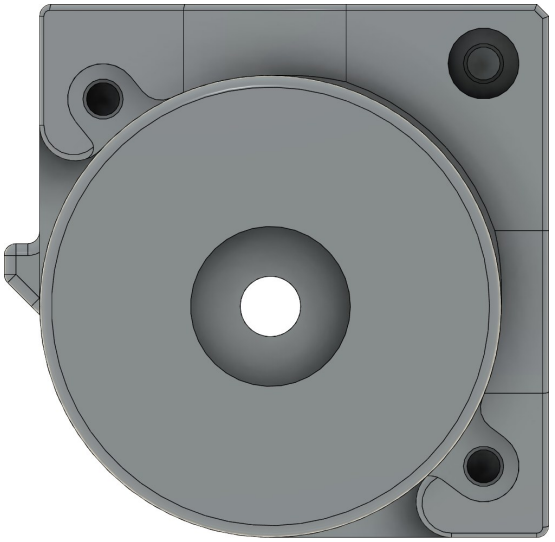
1 x Rubber Foot

CHECK BEFORE CONTINUING: The rubber foot is centered beneath the corner; its contact face points toward the table.

SECURE THE RUBBER FOOT

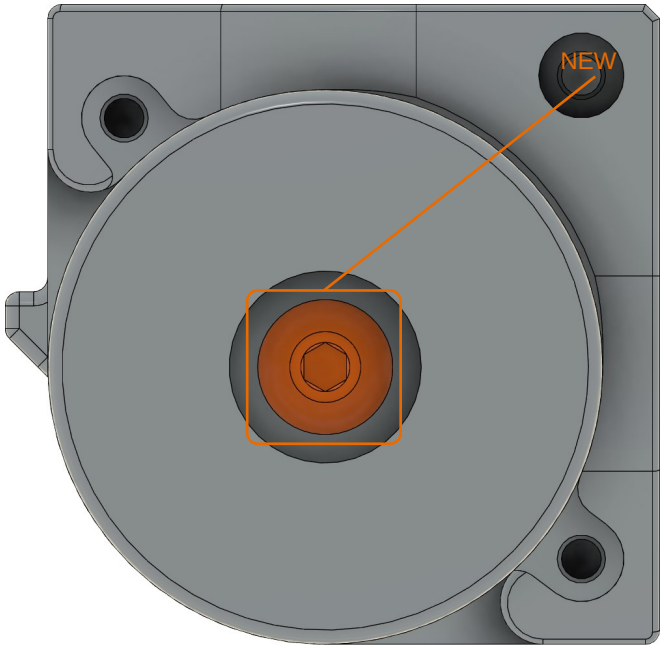
REAR-LEFT FOOT | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the M5 nut into its seat. Pass the M5 x 16 button-head screw through the foot and start it into the nut. Tighten only enough to retain the foot.

PARTS IN THIS CAD STEP

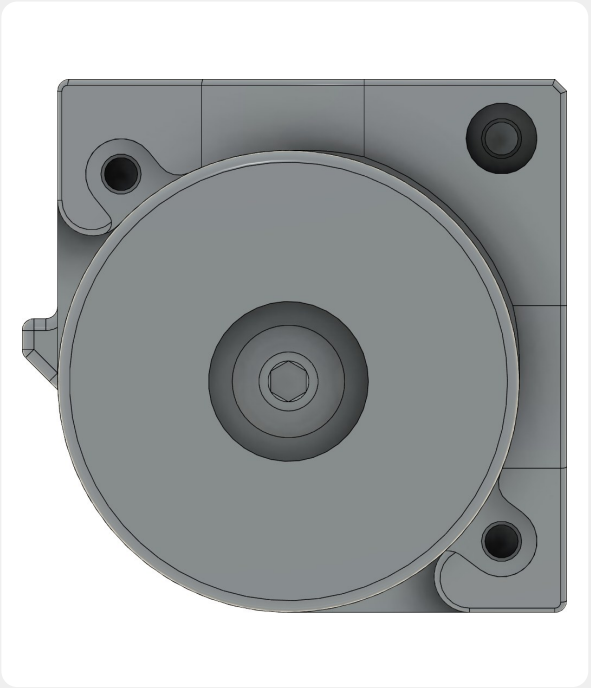
1 x M5x16 BHCS
1 x M5 Hex Nut

CHECK BEFORE CONTINUING: The foot is held securely without being crushed or pulled out of shape.

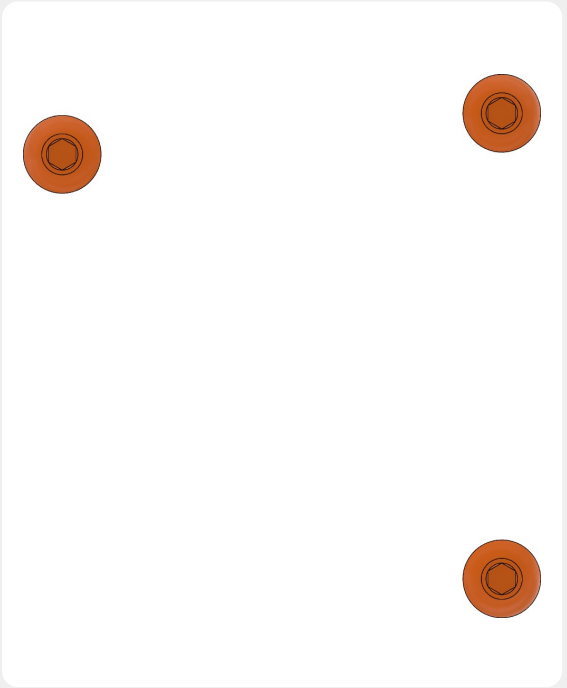
FIT THE THREE LOWER SCREWS

REAR-LEFT FOOT | DETAIL 3

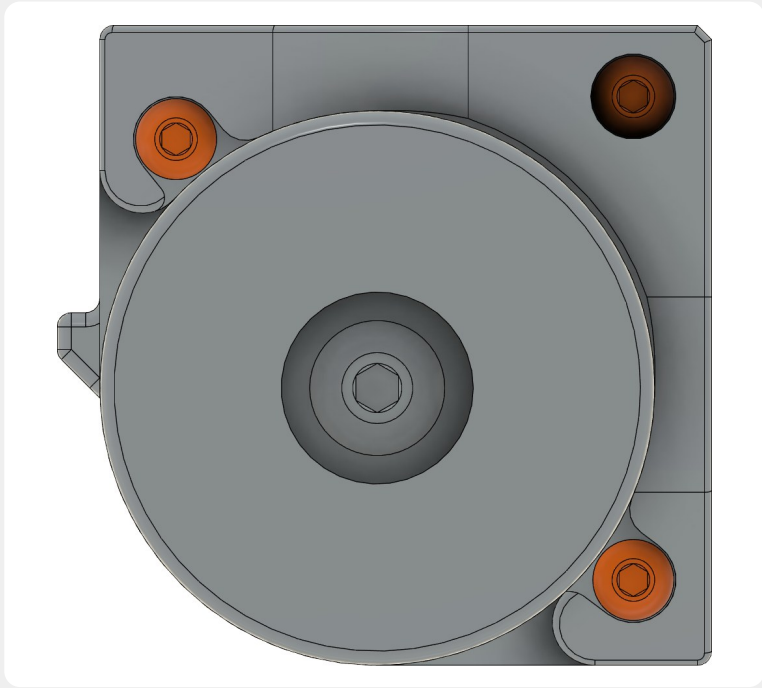
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Install the three M3 x 6 button-head screws at the highlighted lower fixing points. Start all three before gently tightening them.

PARTS IN THIS CAD STEP

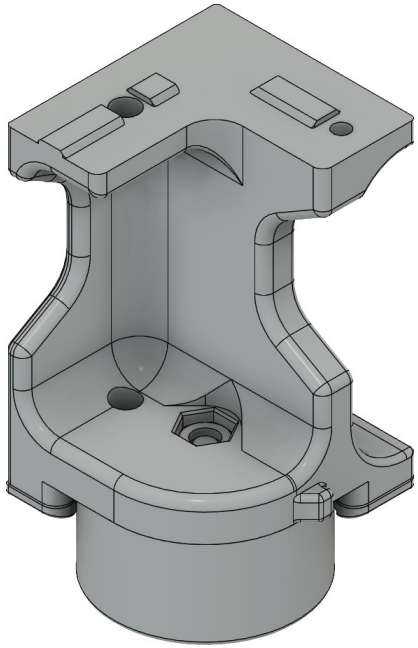
3 x M3x6 BHCS

CHECK BEFORE CONTINUING: Each screw enters its matching hole straight; all three heads are seated.

ATTACH THE CORNER TO THE FRAME

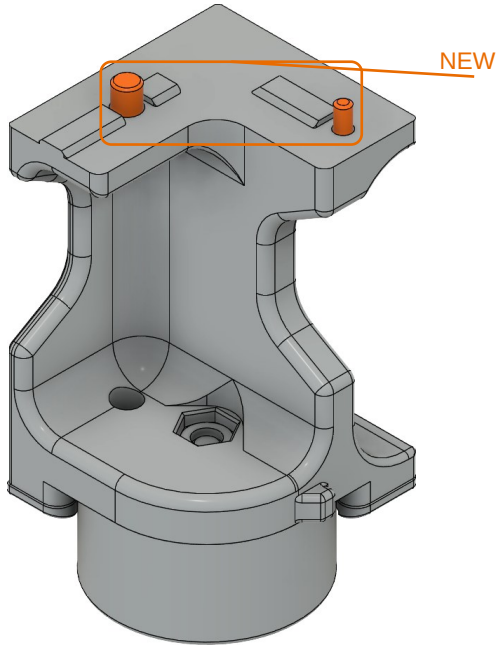
REAR-LEFT FOOT | DETAIL 4

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the completed corner to its matching frame location. Fit the M5 x 10 button-head screw and M3 x 8 socket-head screw at the highlighted positions. Start both before tightening.

PARTS IN THIS CAD STEP

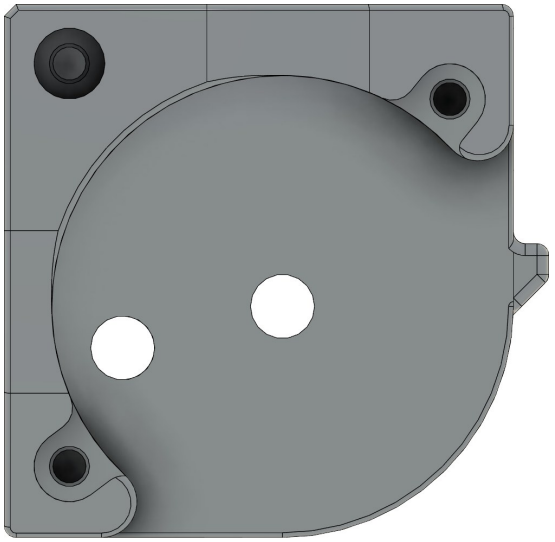
1 x M5x10 BHCS
1 x M3x8 SHCS

CHECK BEFORE CONTINUING: The corner sits against the frame without a gap. Check that the other feet can all touch a flat surface.

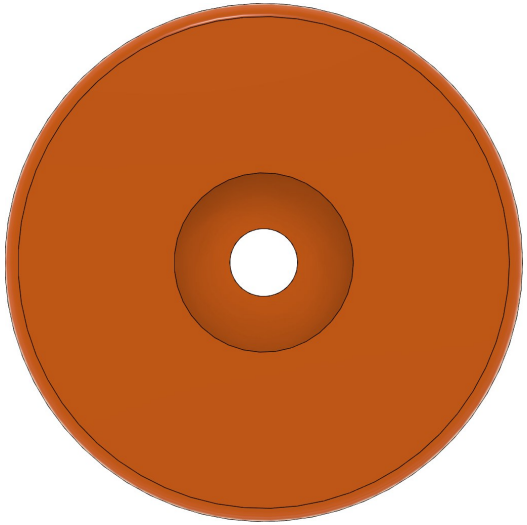
PLACE THE RUBBER FOOT

REAR-RIGHT FOOT | DETAIL 1

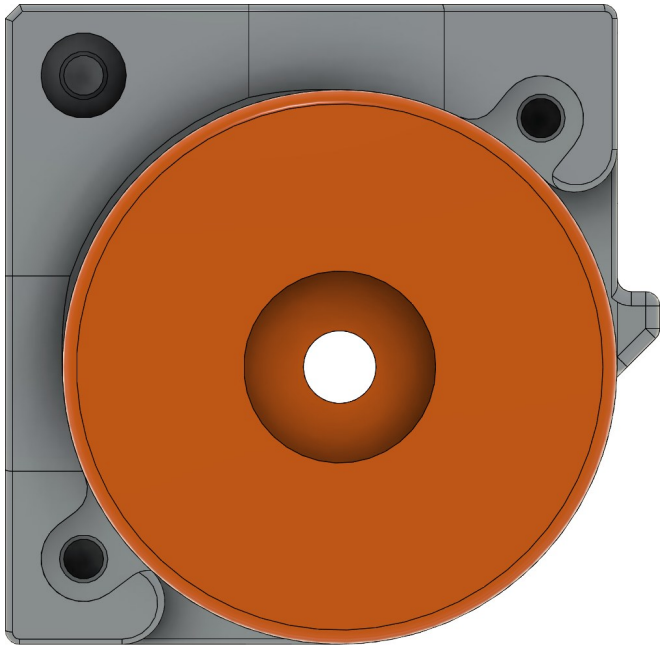
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match this corner to the location in the overview. Place the rubber foot under its circular seat, with the center hole aligned.

PARTS IN THIS CAD STEP

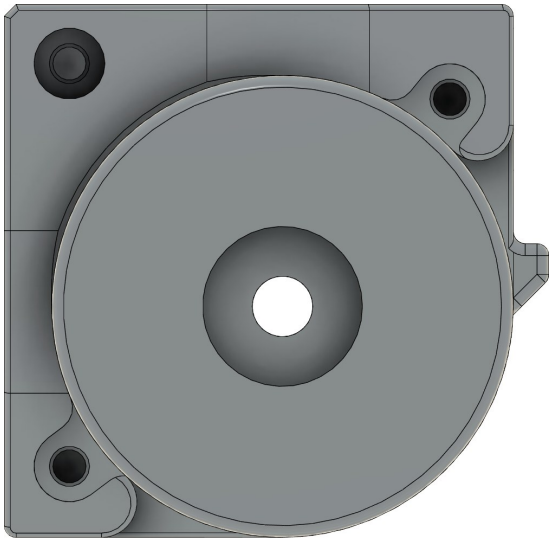
1 x Rubber Foot

CHECK BEFORE CONTINUING: The rubber foot is centered beneath the corner; its contact face points toward the table.

SECURE THE RUBBER FOOT

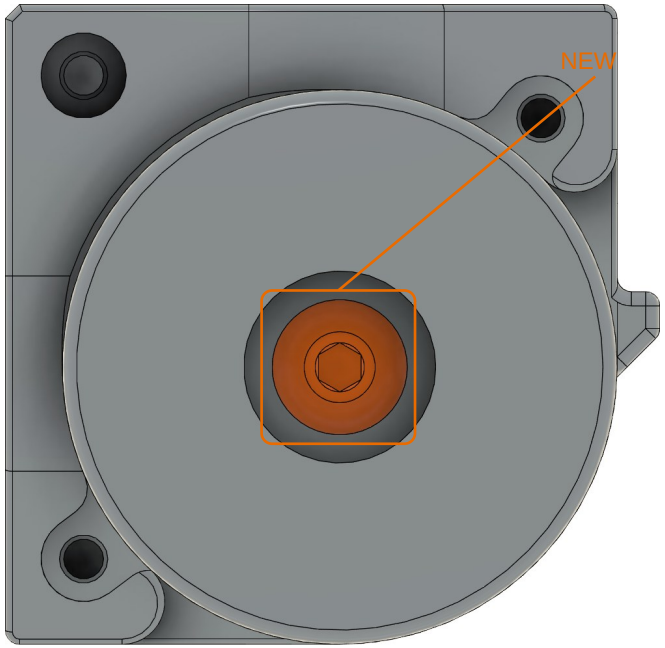
REAR-RIGHT FOOT | DETAIL 2

1 BEFORE



2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the M5 nut into its seat. Pass the M5 x 16 button-head screw through the foot and start it into the nut. Tighten only enough to retain the foot.

PARTS IN THIS CAD STEP

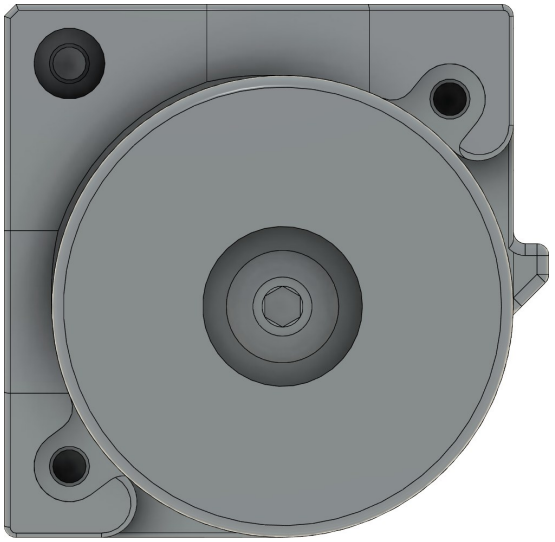
1 x M5x16 BHCS
1 x M5 Hex Nut

CHECK BEFORE CONTINUING: The foot is held securely without being crushed or pulled out of shape.

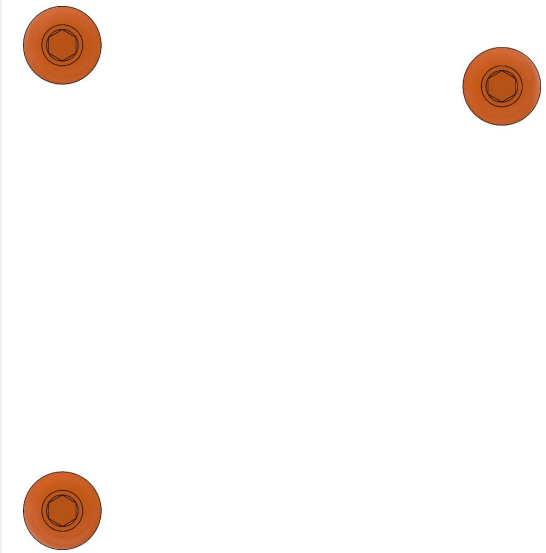
FIT THE THREE LOWER SCREWS

REAR-RIGHT FOOT | DETAIL 3

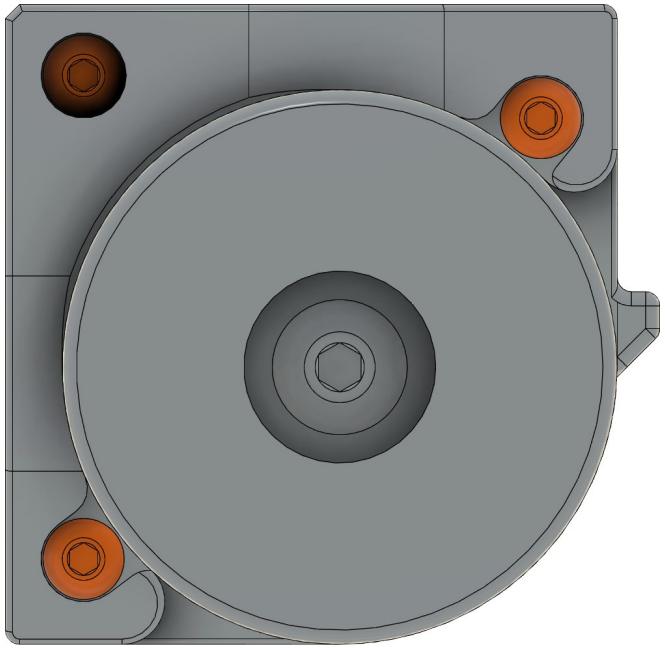
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Install the three M3 x 6 button-head screws at the highlighted lower fixing points. Start all three before gently tightening them.

PARTS IN THIS CAD STEP

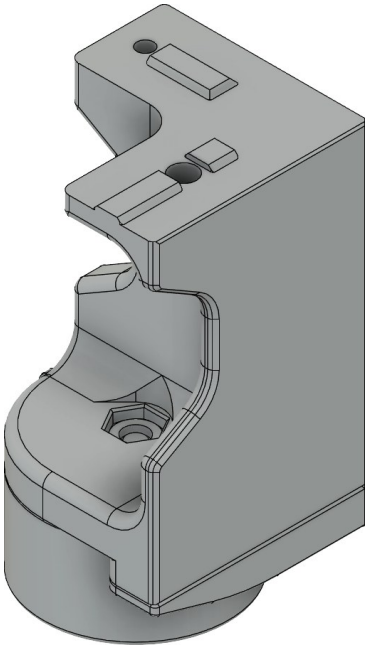
3 x M3x6 BHCS

CHECK BEFORE CONTINUING: Each screw enters its matching hole straight; all three heads are seated.

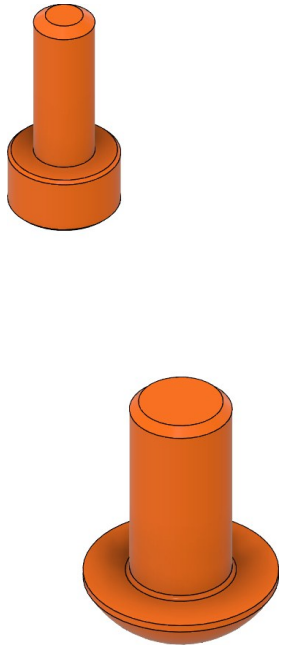
ATTACH THE CORNER TO THE FRAME

REAR-RIGHT FOOT | DETAIL 4

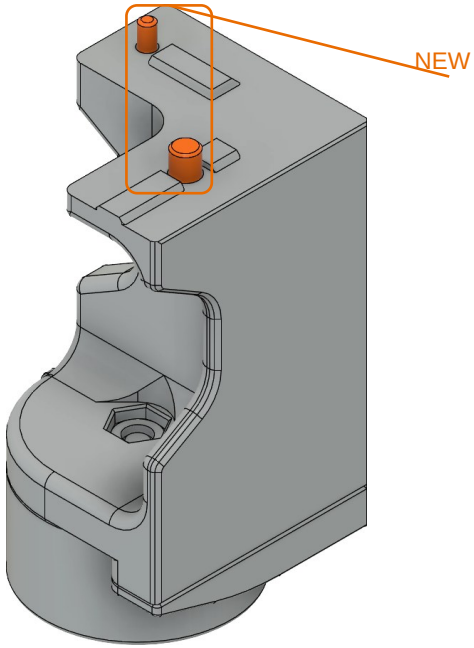
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the completed corner to its matching frame location. Fit the M5 x 10 button-head screw and M3 x 8 socket-head screw at the highlighted positions. Start both before tightening.

PARTS IN THIS CAD STEP

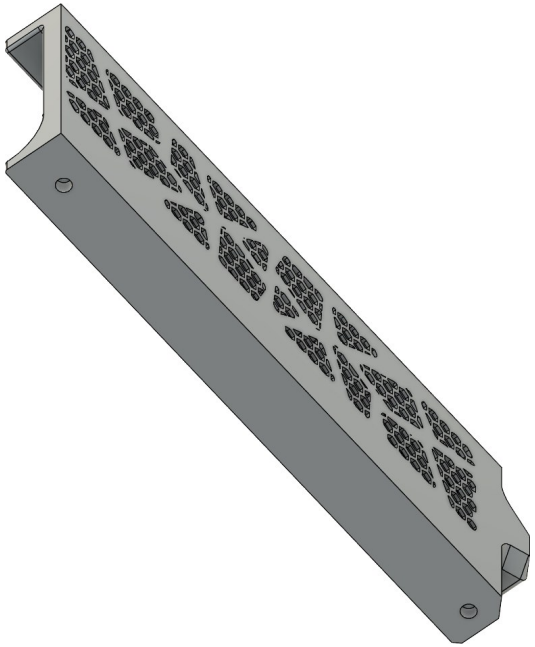
1 x M5x10 BHCS
1 x M3x8 SHCS

CHECK BEFORE CONTINUING: The corner sits against the frame without a gap. Check that the other feet can all touch a flat surface.

INSTALL THE TWO LOWER HEAT-SET INSERTS

LEFT REAR SIDE SKIRT | DETAIL 1

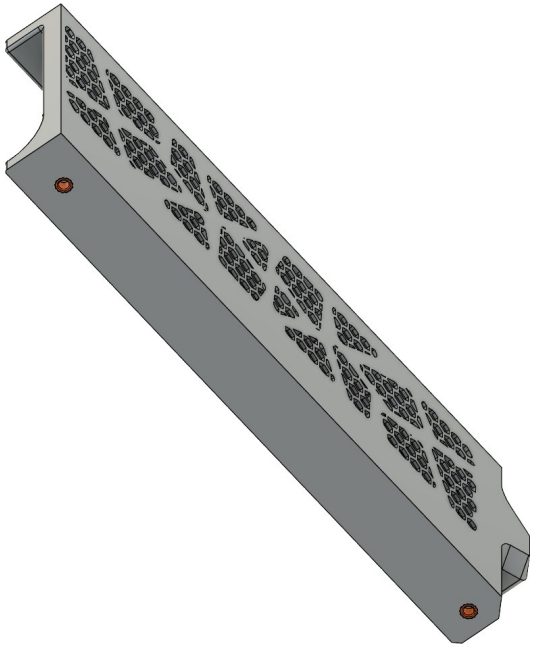
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

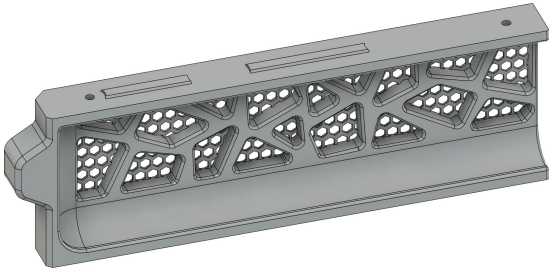
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

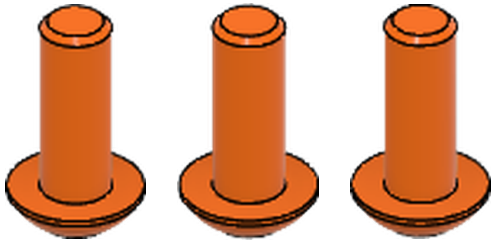
FIT THE SKIRT TO THE LOWER FRAME

LEFT REAR SIDE SKIRT | DETAIL 2

1 BEFORE

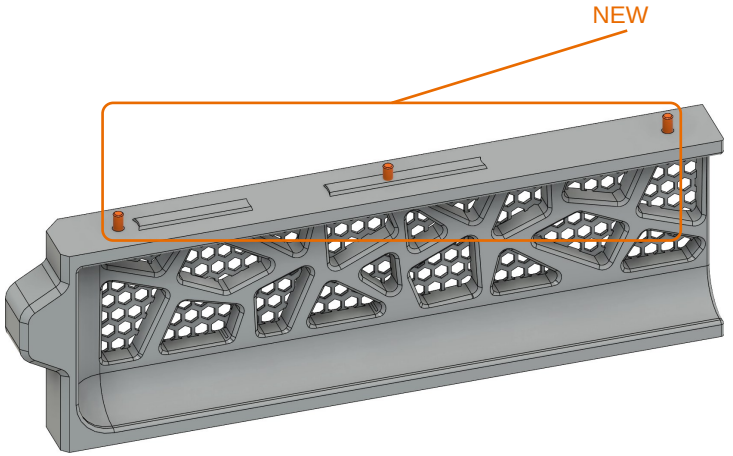


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

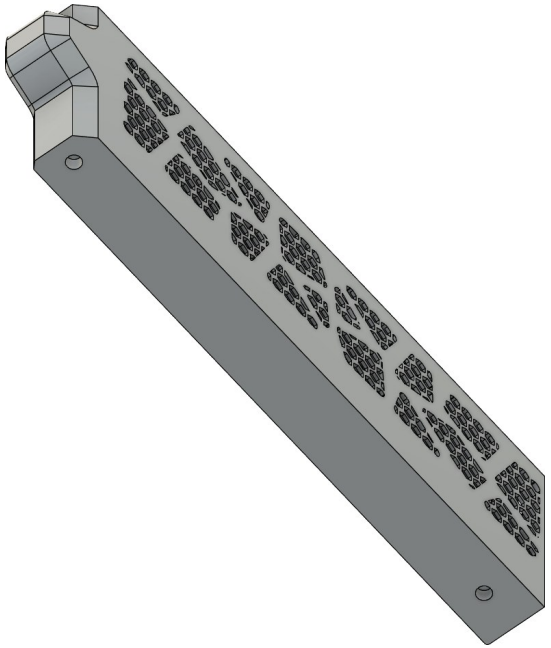
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

LEFT FRONT SIDE SKIRT | DETAIL 1

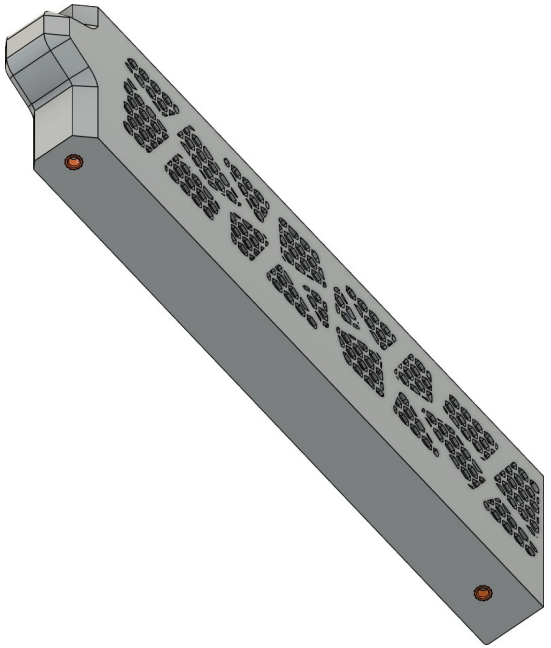
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

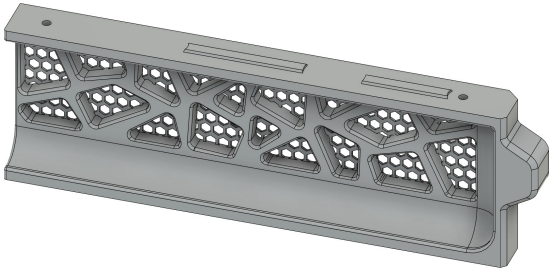
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

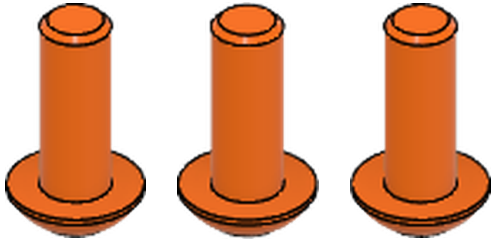
FIT THE SKIRT TO THE LOWER FRAME

LEFT FRONT SIDE SKIRT | DETAIL 2

1 BEFORE

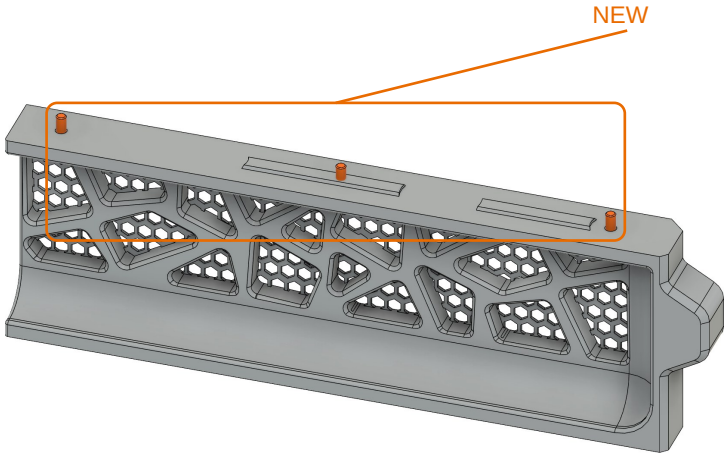


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

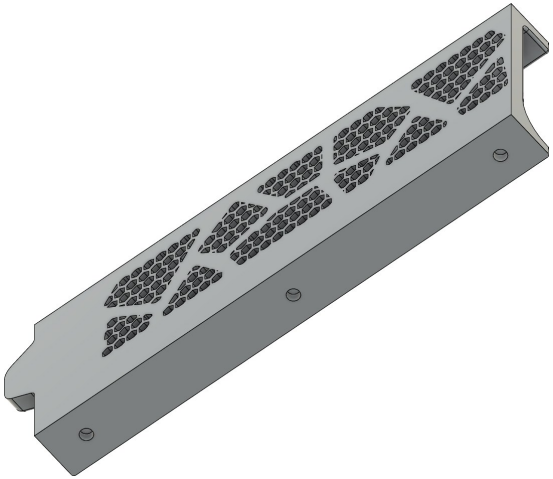
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

REAR LEFT SKIRT | DETAIL 1

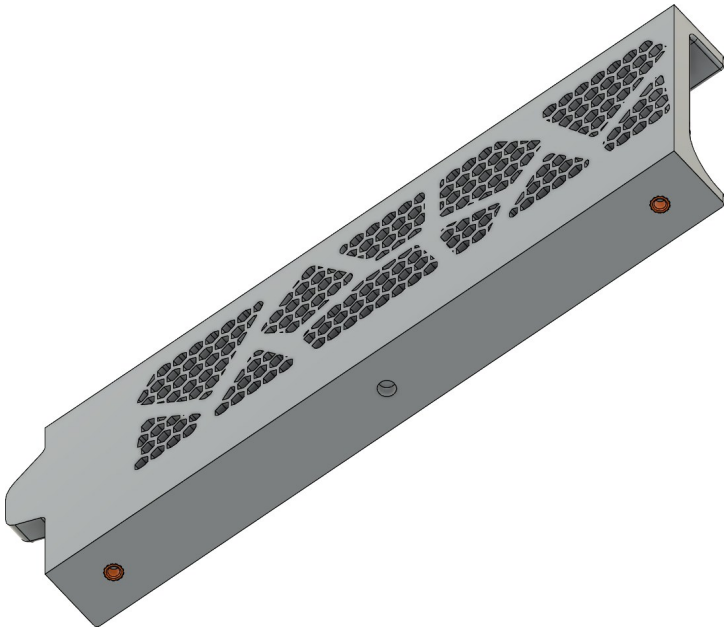
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

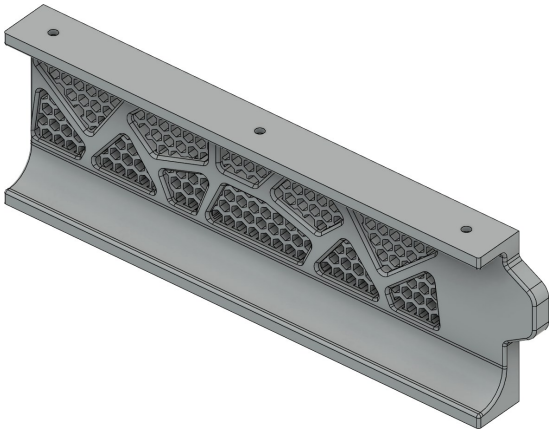
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

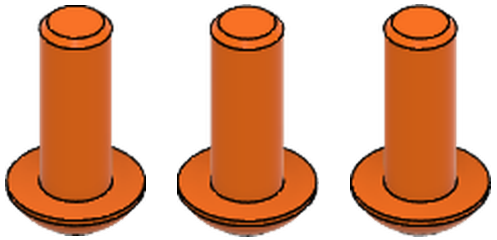
FIT THE SKIRT TO THE LOWER FRAME

REAR LEFT SKIRT | DETAIL 2

1 BEFORE

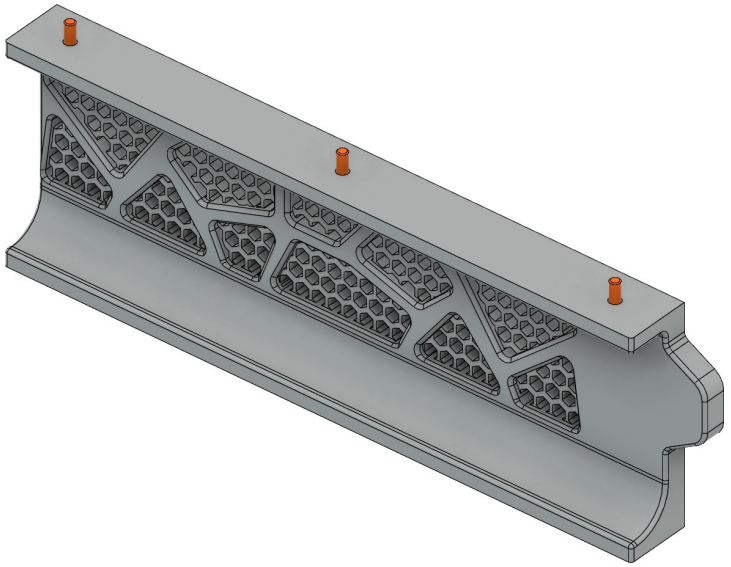


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

FRONT LEFT SKIRT | DETAIL 1

1 BEFORE

2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

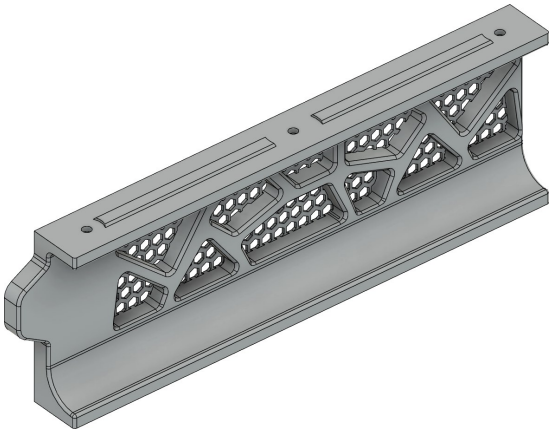
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

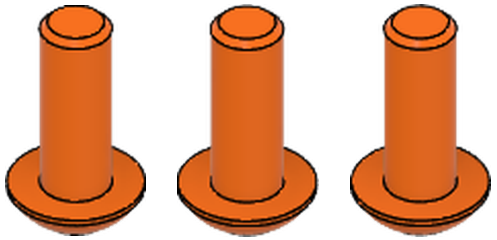
FIT THE SKIRT TO THE LOWER FRAME

FRONT LEFT SKIRT | DETAIL 2

1 BEFORE

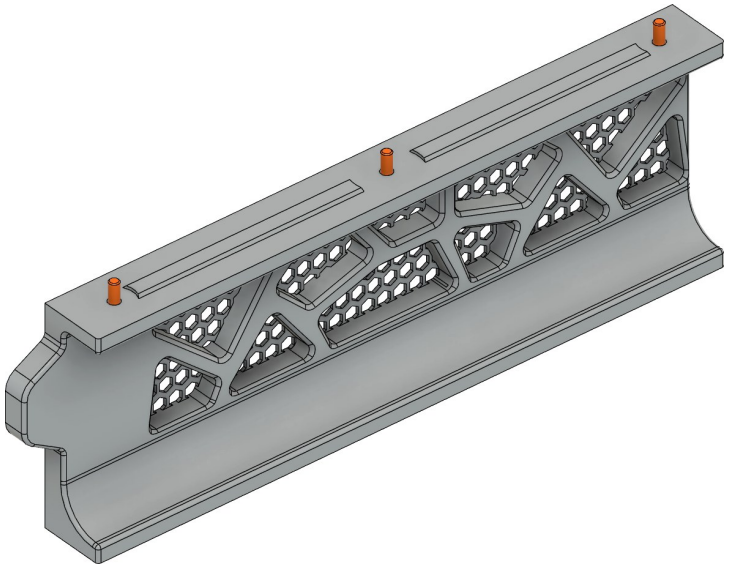


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

FRONT LOGO SKIRT | DETAIL 1

1 BEFORE

2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

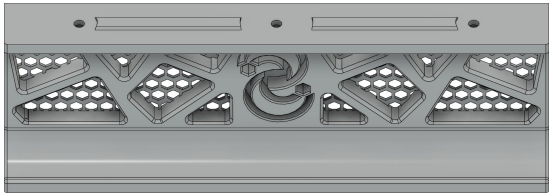
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

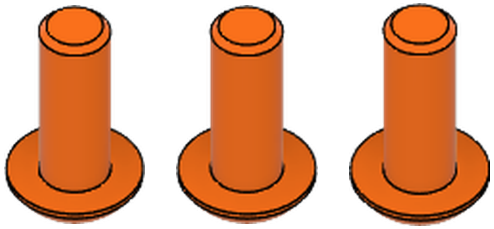
FIT THE SKIRT TO THE LOWER FRAME

FRONT LOGO SKIRT | DETAIL 2

1 BEFORE

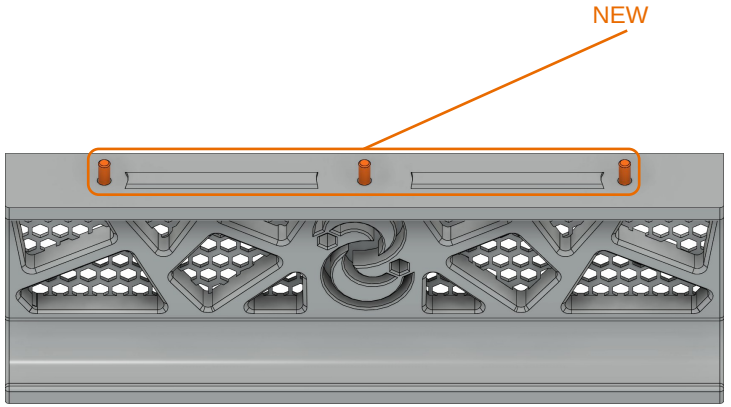


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

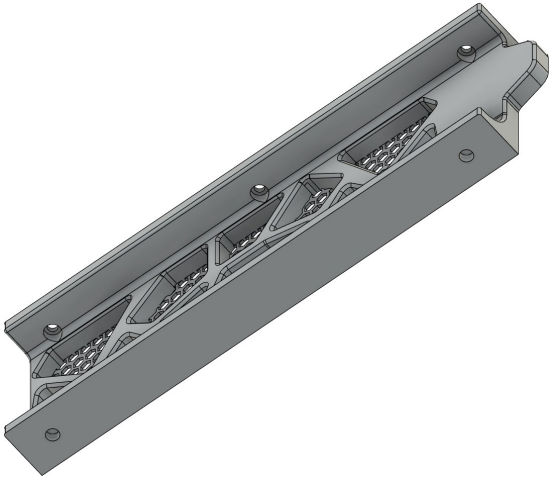
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

FRONT RIGHT SKIRT | DETAIL 1

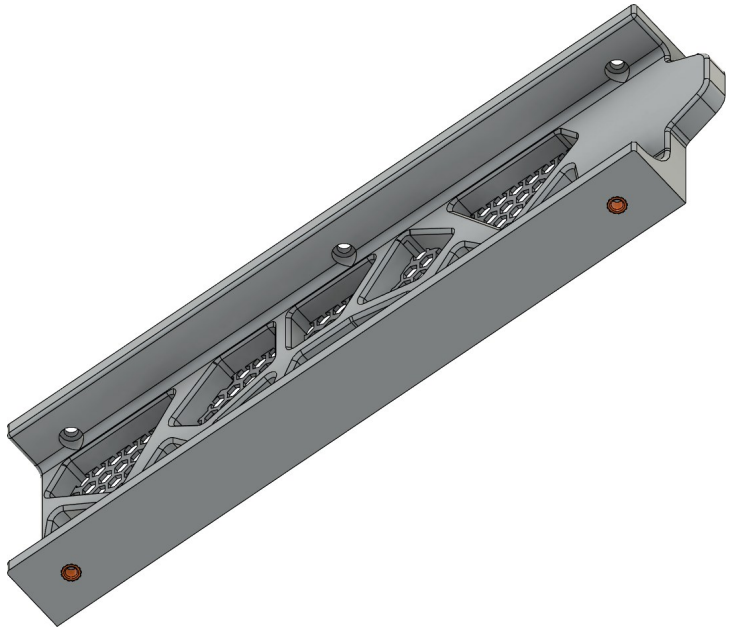
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

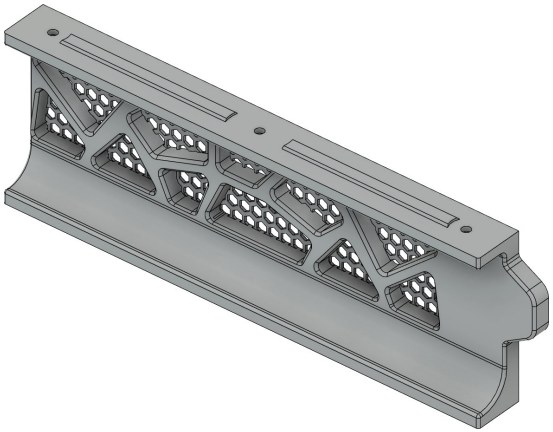
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

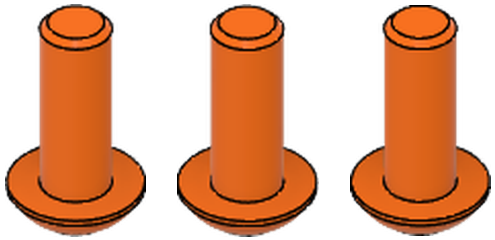
FIT THE SKIRT TO THE LOWER FRAME

FRONT RIGHT SKIRT | DETAIL 2

1 BEFORE

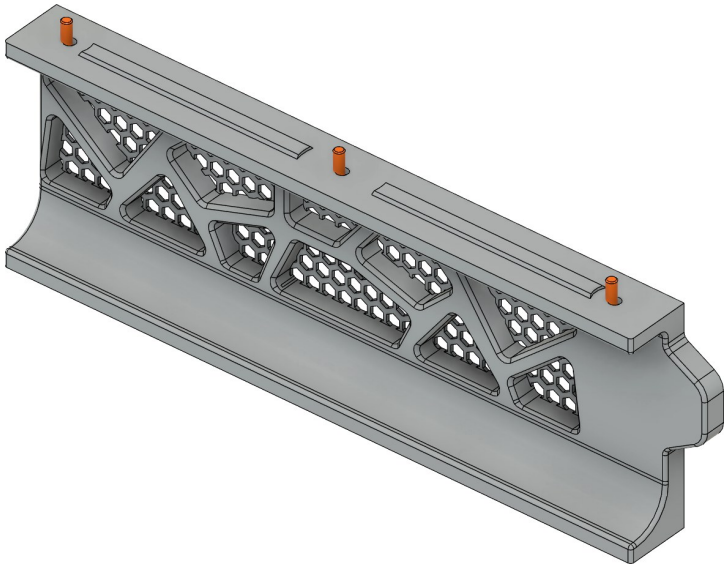


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

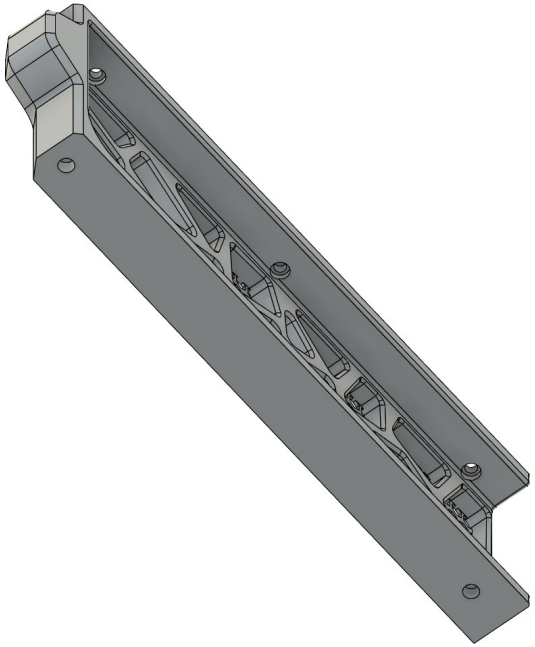
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

RIGHT FRONT SIDE SKIRT | DETAIL 1

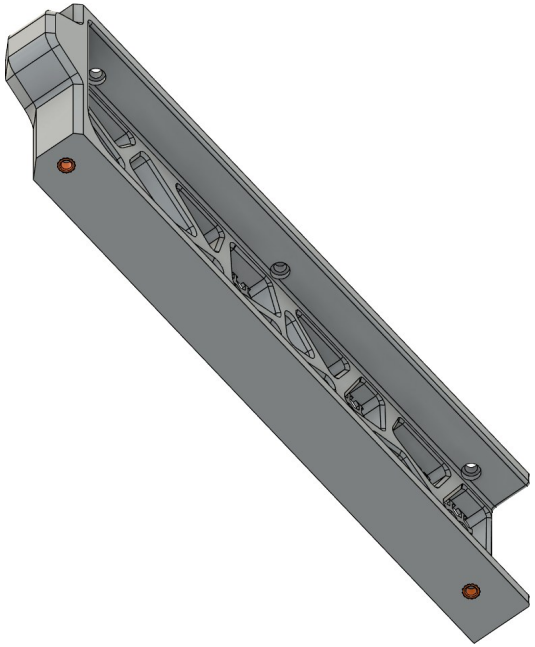
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

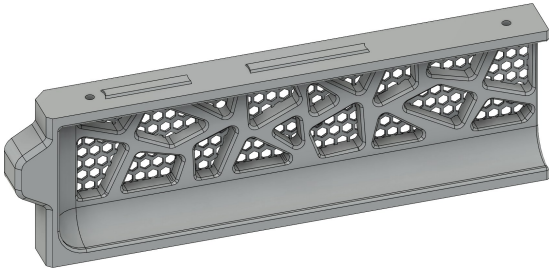
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

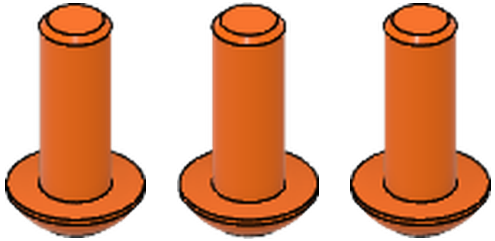
FIT THE SKIRT TO THE LOWER FRAME

RIGHT FRONT SIDE SKIRT | DETAIL 2

1 BEFORE

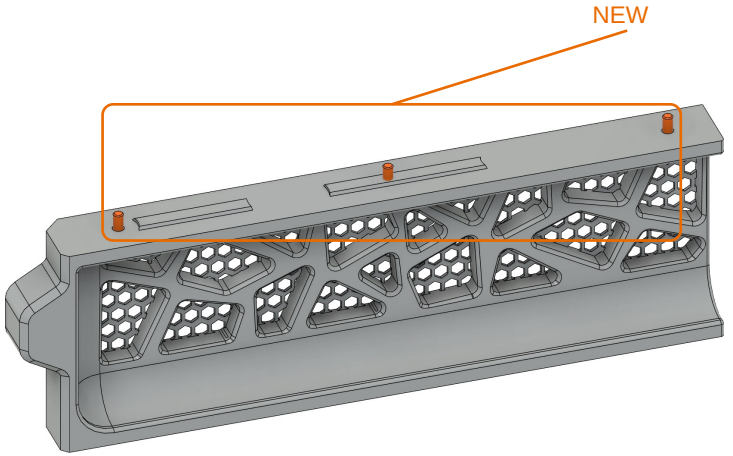


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

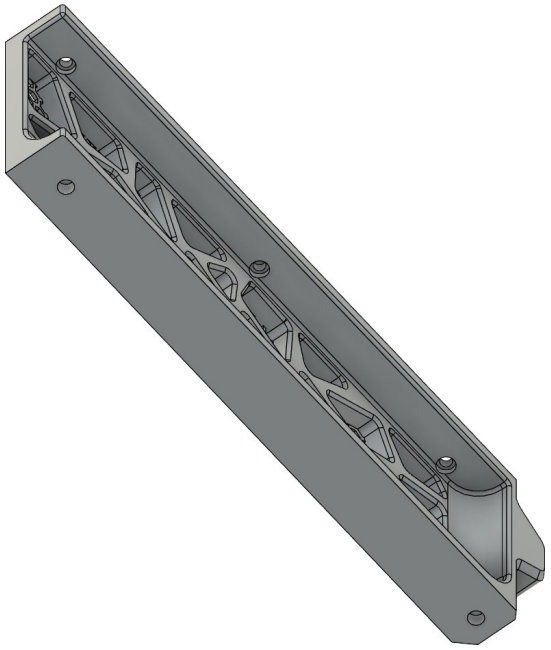
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

RIGHT REAR SIDE SKIRT | DETAIL 1

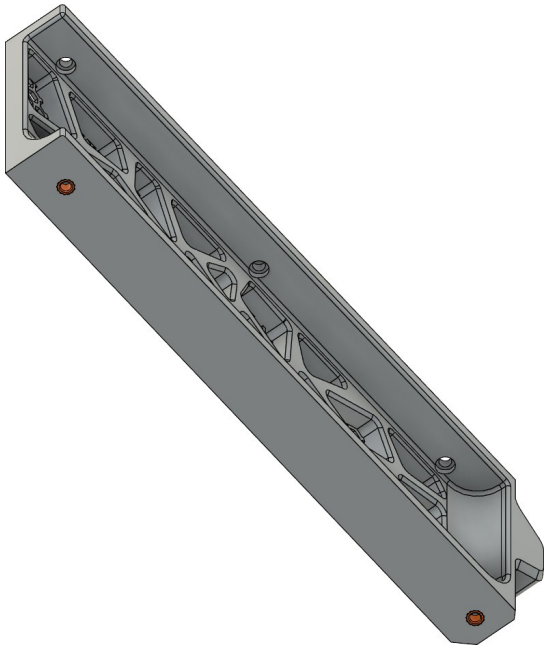
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

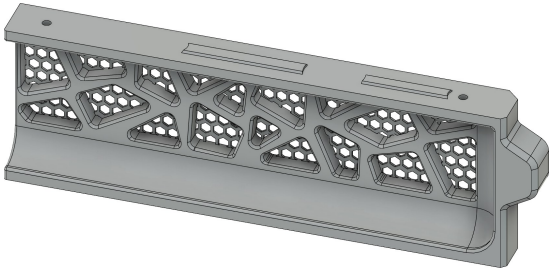
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

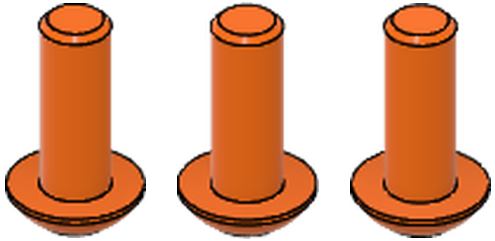
FIT THE SKIRT TO THE LOWER FRAME

RIGHT REAR SIDE SKIRT | DETAIL 2

1 BEFORE

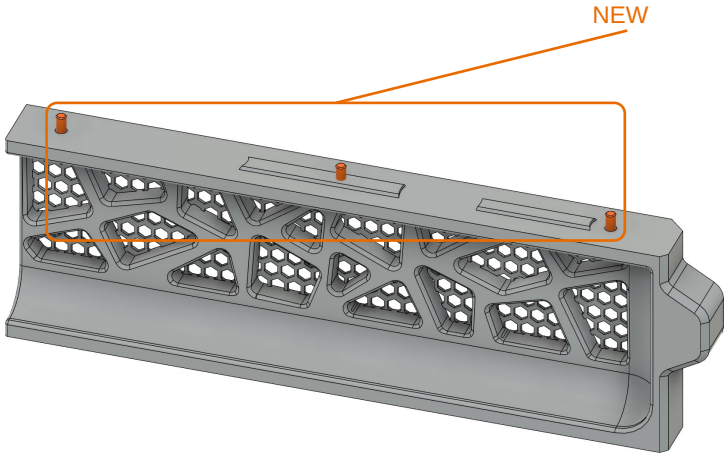


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

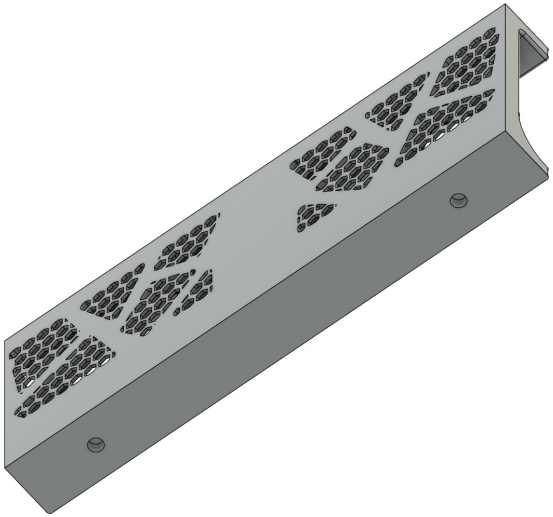
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

REAR LOGO SKIRT | DETAIL 1

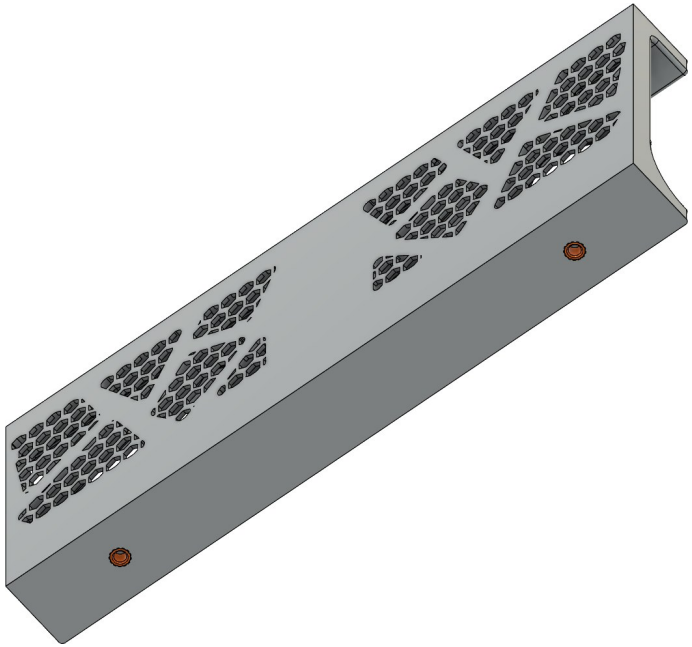
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

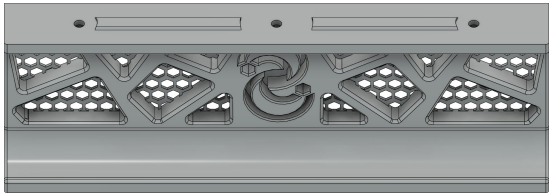
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

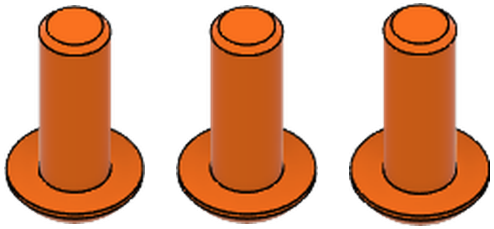
FIT THE SKIRT TO THE LOWER FRAME

REAR LOGO SKIRT | DETAIL 2

1 BEFORE

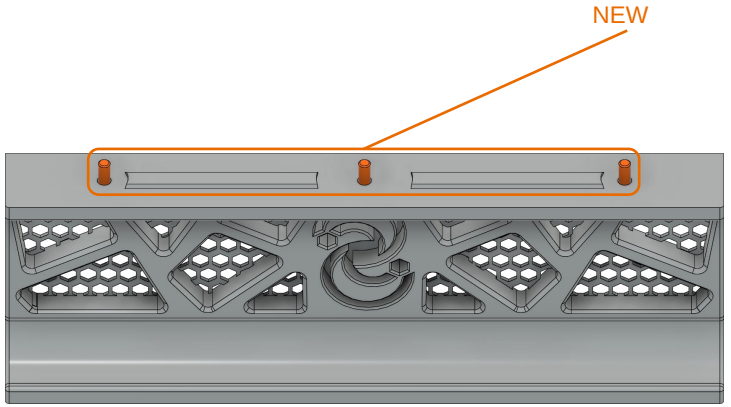


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

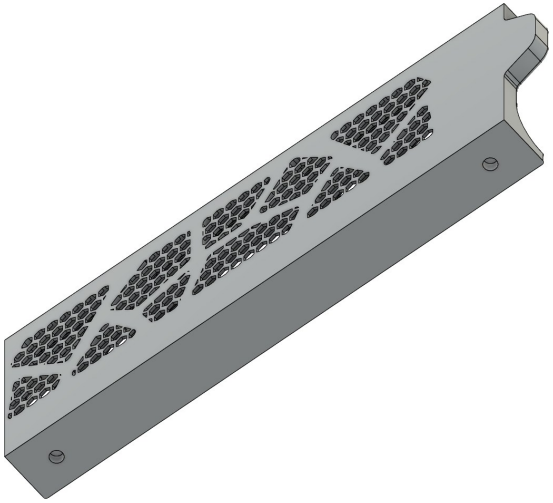
3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

INSTALL THE TWO LOWER HEAT-SET INSERTS

REAR RIGHT SKIRT | DETAIL 1

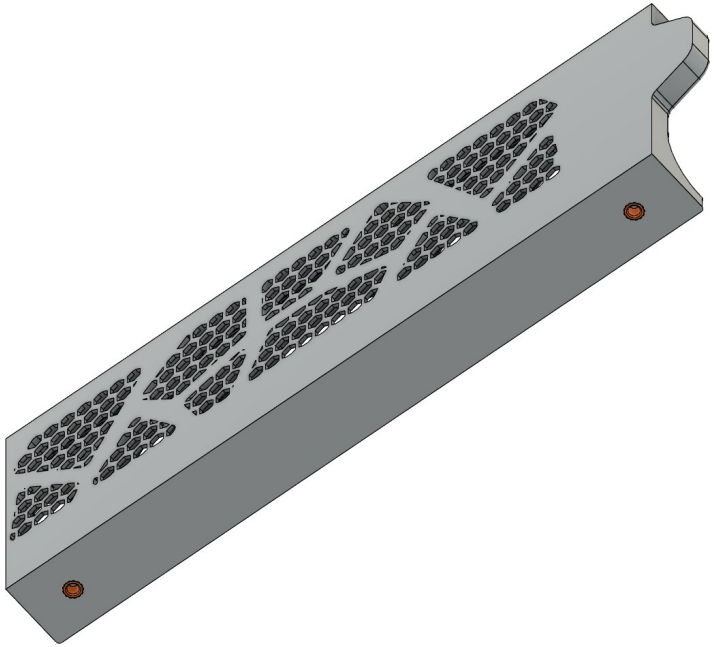
1 BEFORE



2 PARTS ADDED

Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Turn the skirt over and locate the two highlighted lower insert pockets. Set one M3 heat-set insert into each pocket with a temperature suited to the printed material. Keep the inserts straight and let them cool.

PARTS IN THIS CAD STEP

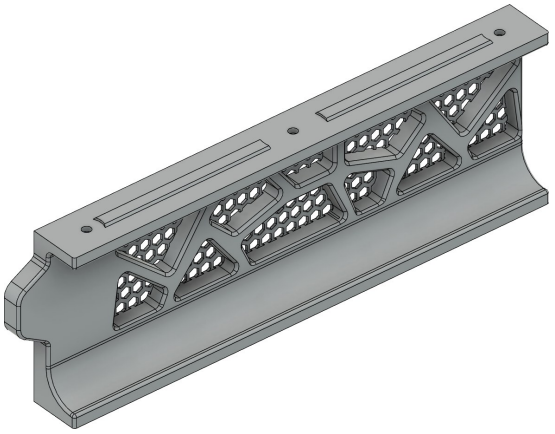
2 x M3 heat-set insert

CHECK BEFORE CONTINUING: Two inserts are seated in separate pockets. Repeated CAD occurrences at the same location count as one physical insert.

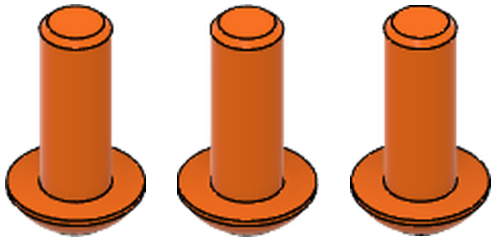
FIT THE SKIRT TO THE LOWER FRAME

REAR RIGHT SKIRT | DETAIL 2

1 BEFORE

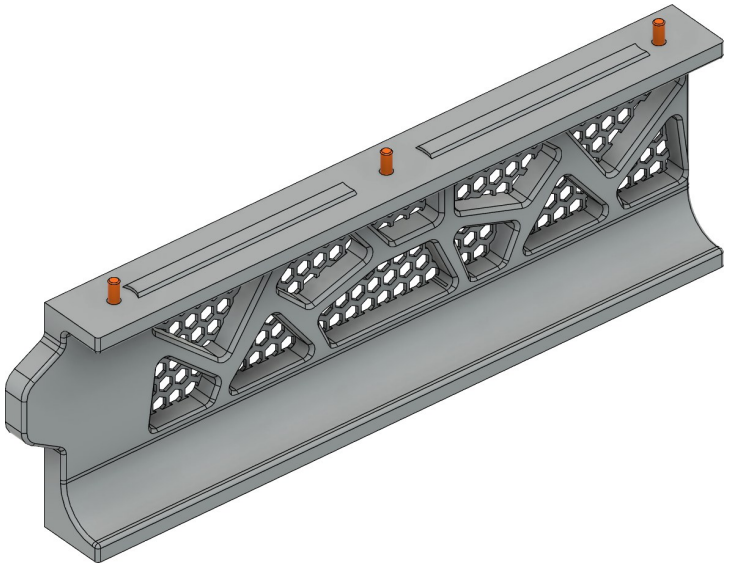


2 PARTS ADDED



Hardware enlarged; spacing not to scale.

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

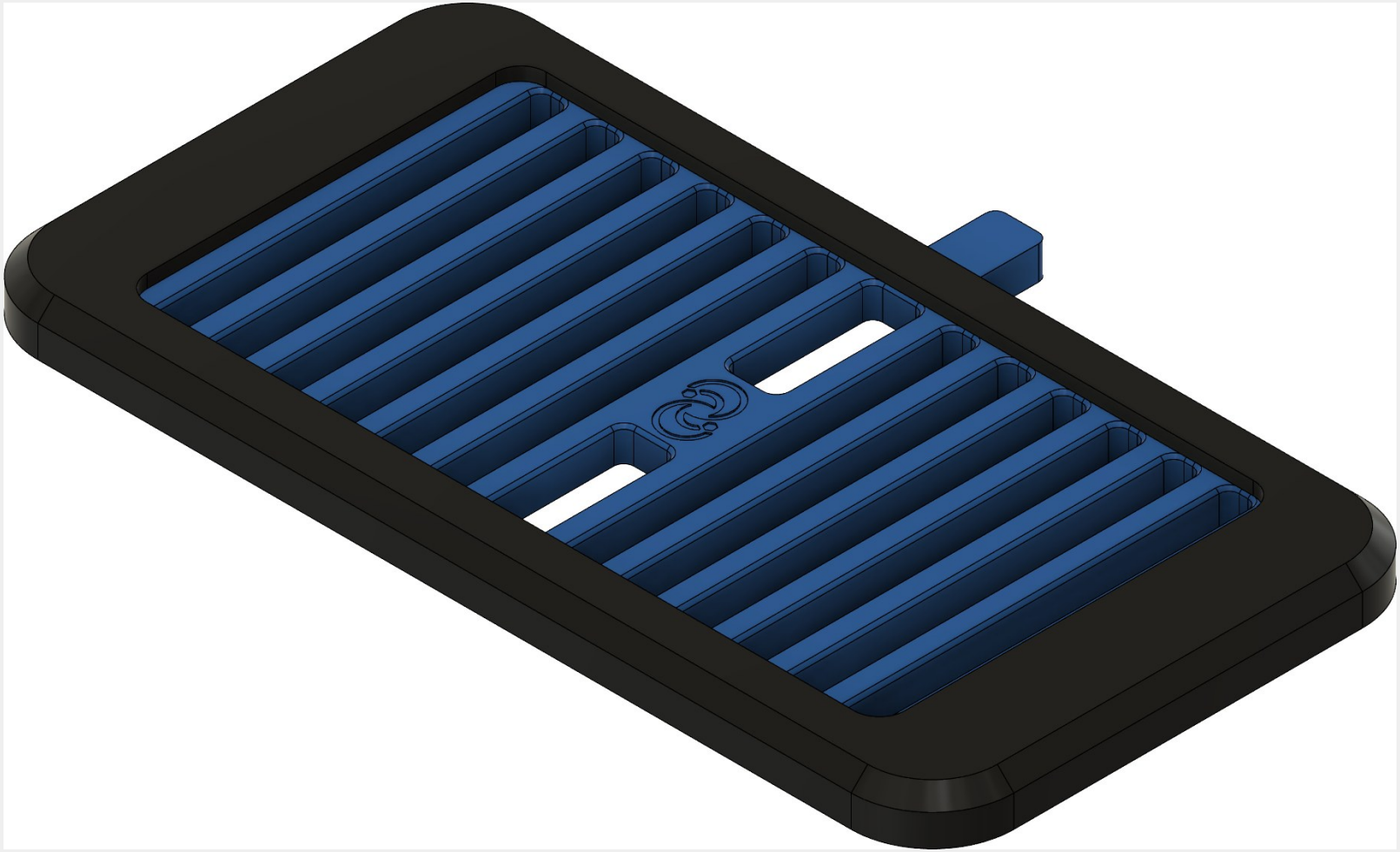
Offer the skirt to its matching lower frame location. From inside the skirt, insert the three M3 x 8 Torx button-head screws upward through the top flange. Confirm the frame-side mating threads or nuts, start all three screws, then tighten evenly.

PARTS IN THIS CAD STEP

3 x M3 x 8 Torx button-head [90991A113]

CHECK BEFORE CONTINUING: The skirt follows the frame edge and clears the neighboring corner. The lower heat-set inserts are separate fixings; do not confuse them with the three upper mounting holes.

FIT THE BOTTOM VENT



Fit the vent slider into its frame, then install the frame in the bottom opening. Check its range before tightening the mounting screws.

BOTTOM-VENT CAD ITEMS
One slider
Four M3 heat-set occurrences
Four mounting screw occurrences

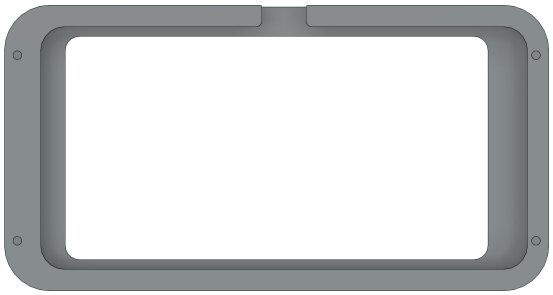
CHECK: The slider moves through its range without binding. Confirm the screw and insert dimensions against the supplied parts.

Current Hybrid assembly

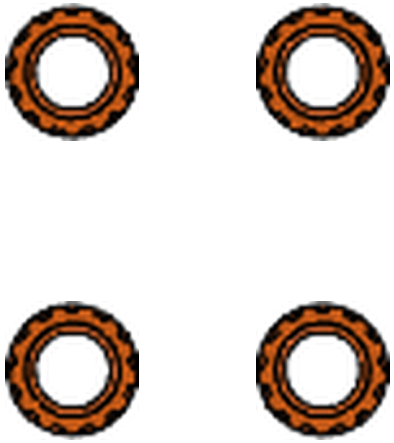
INSTALL THE THREADED INSERTS

VENT BOTTOM | DETAIL 1

1 BEFORE

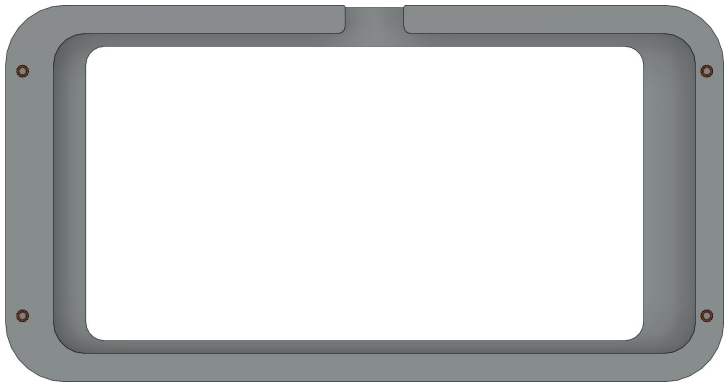


2 PARTS ADDED



4 separate parts - enlarged for identification

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the insert pockets marked in orange. Confirm insert length and the printed material before heat-setting; seat each insert straight and let it cool before using a screw.

PARTS IN THIS CAD STEP

4 x M3 heat-set insert

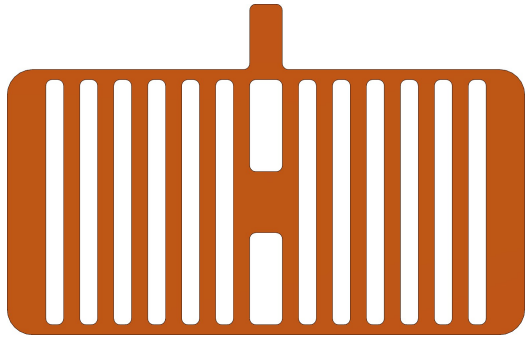
CHECK BEFORE CONTINUING: Each insert is square and at the intended seat depth. Installation temperature must come from the approved material/insert instructions.

INSTALL THE VENT SLIDER

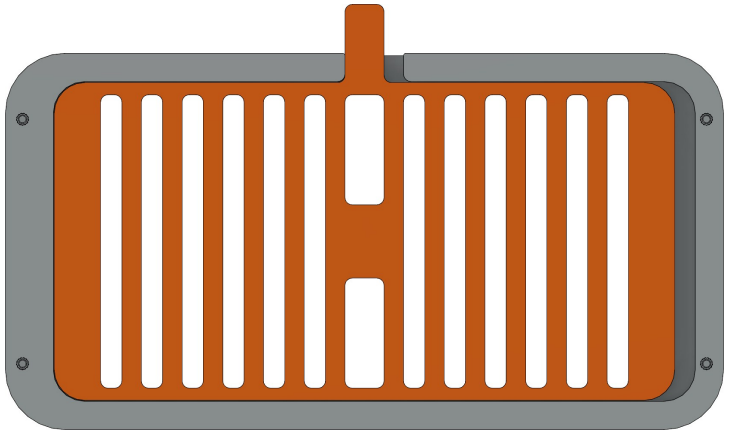
VENT BOTTOM | DETAIL 2

1 BEFORE

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Slide the moving panel into the matching frame guides. Check its full movement before mounting the vent to the enclosure.

PARTS IN THIS CAD STEP

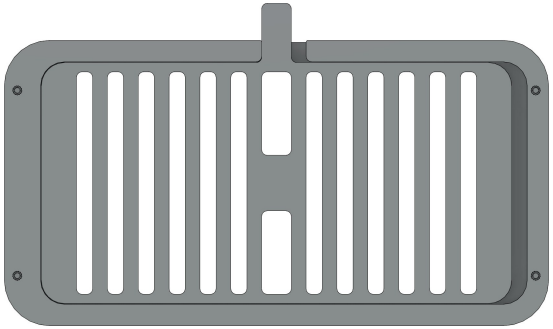
1 x Vent slider

CHECK BEFORE CONTINUING: The slider travels freely and stays retained. The next detail shows the four M3 x 6 mounting screws.

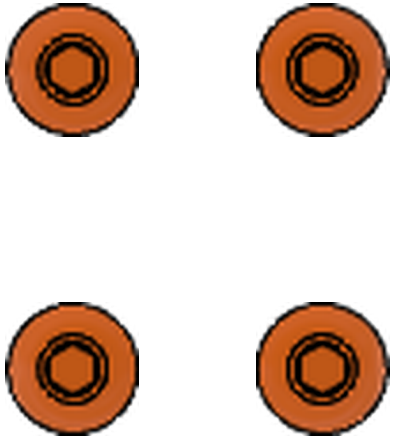
LOCATE THE FOUR VENT MOUNTING SCREWS

VENT BOTTOM | DETAIL 3

1 BEFORE

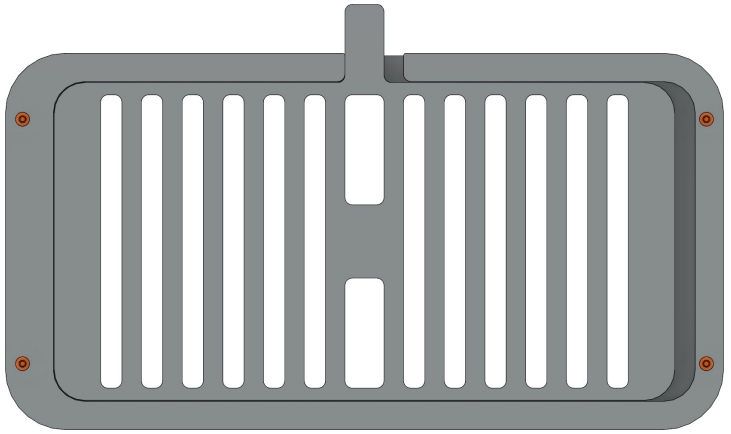


2 PARTS ADDED



4 separate parts - enlarged for identification

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

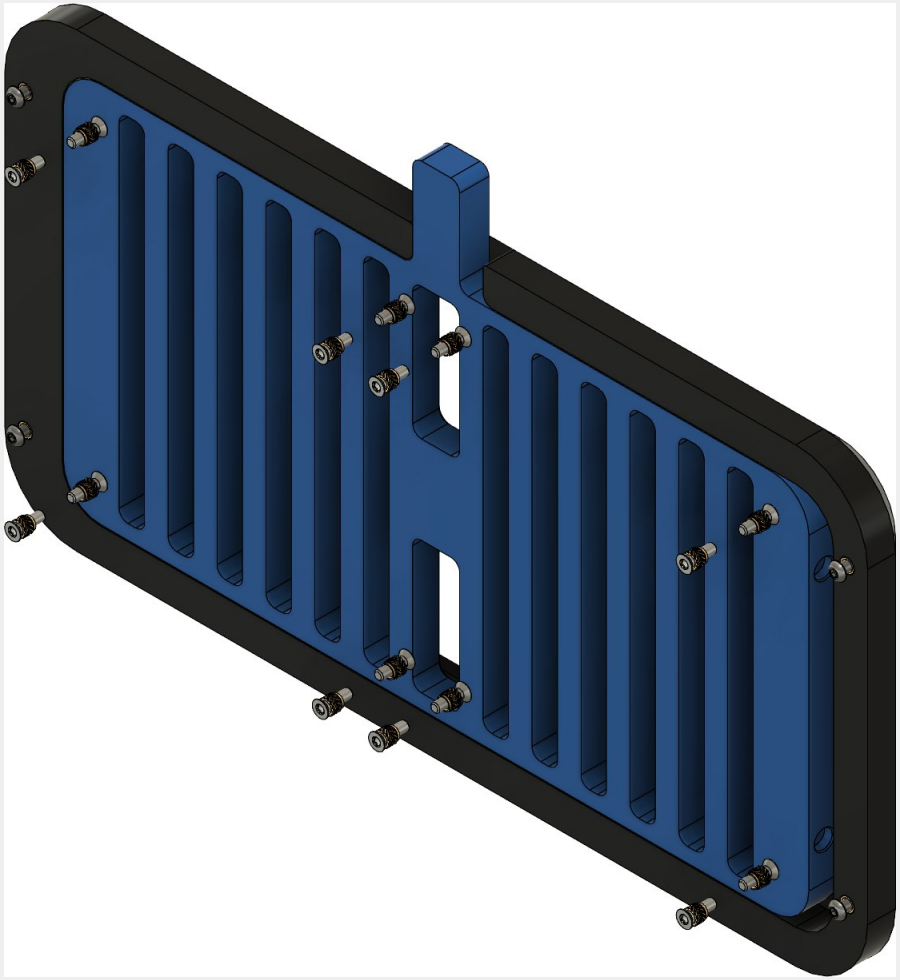
Fit the vent assembly at its panel opening. Locate the four M3 x 6 hex button-head screws shown in orange and start them into the matching inserts. The surrounding panel is omitted here to expose the screw locations.

PARTS IN THIS CAD STEP

4 x M3 x 6 hex button-head [92095A179]

CHECK BEFORE CONTINUING: Confirm screw engagement through the actual panel thickness. Tighten evenly without distorting the vent frame, then move the slider through its full travel.

FIT THE REAR VENT



Prepare the rear vent and its slider. Fit the assembly to the matching rear opening and tighten evenly so the mounting frame stays flat.

CAD GROUP
Vent_Frame rear
Use its own hardware layout.

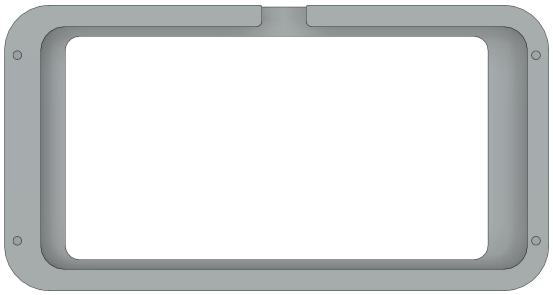
CHECK: The slider remains operable and the vent does not interfere with the rear cable chain or fans.

Current Hybrid assembly

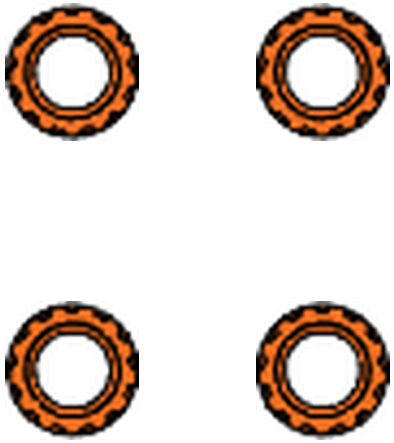
INSTALL THE THREADED INSERTS

VENT REAR | DETAIL 1

1 BEFORE



2 PARTS ADDED



4 separate parts - enlarged for identification

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the insert pockets marked in orange. Confirm insert length and the printed material before heat-setting; seat each insert straight and let it cool before using a screw.

PARTS IN THIS CAD STEP

4 x M3 heat-set insert

CHECK BEFORE CONTINUING: Each insert is square and at the intended seat depth. Installation temperature must come from the approved material/insert instructions.

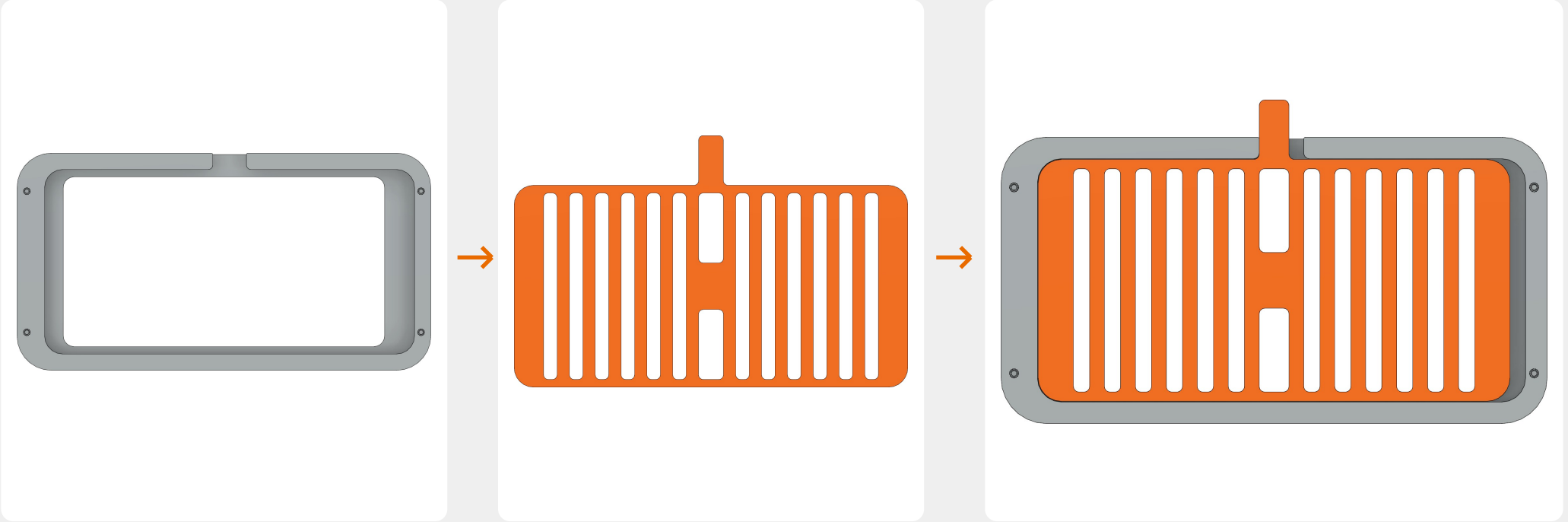
INSTALL THE VENT SLIDER

VENT REAR | DETAIL 2

1 BEFORE

2 PARTS ADDED

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Slide the moving panel into the matching frame guides. Check its full movement before mounting the vent to the enclosure.

PARTS IN THIS CAD STEP

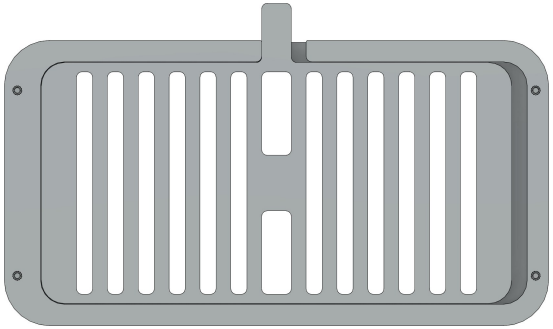
1 x Vent slider

CHECK BEFORE CONTINUING: The slider travels freely and stays retained. The next detail shows the four M3 x 6 mounting screws.

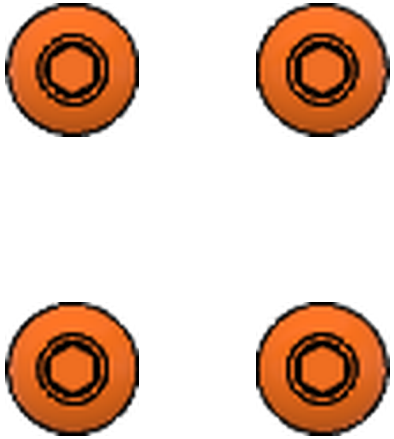
LOCATE THE FOUR VENT MOUNTING SCREWS

VENT REAR | DETAIL 3

1 BEFORE

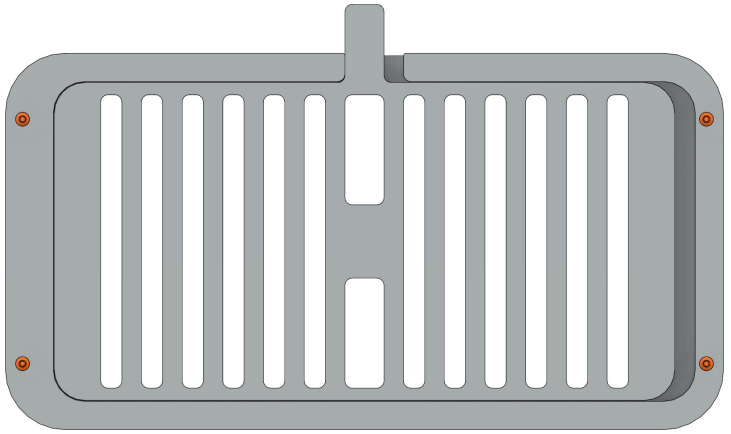


2 PARTS ADDED



4 separate parts - enlarged for identification

3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Fit the vent assembly at its panel opening. Locate the four M3 x 6 hex button-head screws shown in orange and start them into the matching inserts. The surrounding panel is omitted here to expose the screw locations.

PARTS IN THIS CAD STEP

4 x M3 x 6 hex button-head [92095A179]

CHECK BEFORE CONTINUING: Confirm screw engagement through the actual panel thickness. Tighten evenly without distorting the vent frame, then move the slider through its full travel.

INSTALL THE REMAINING ENCLOSURE PANELS

Fit the remaining rear, left, top and deck/bottom panels in their matching positions. Check access to the bay and motion system before securing the last removable panel.

PANEL CHECK
Match openings and hole patterns.
Keep service panels removable.

CHECK: No panel edge contacts belts, rails, leadscrews, fans or moving wiring.

Current Hybrid assembly

ORIENT AND DRY-FIT THE REAR ENCLOSURE PANEL

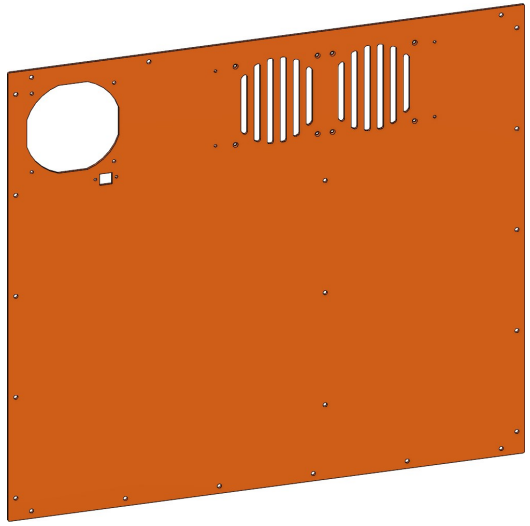
REAR ENCLOSURE PANEL | DETAIL 1

1 BEFORE

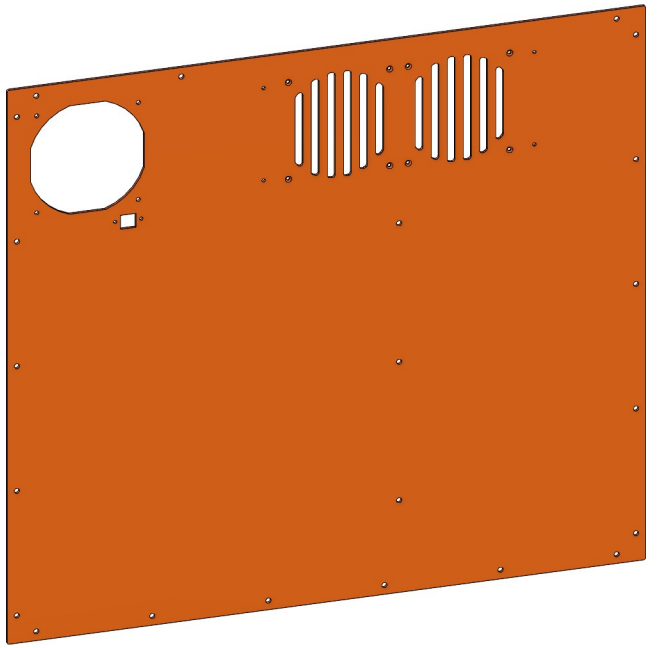
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x rear enclosure panel; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

ORIENT AND DRY-FIT THE LEFT ENCLOSURE PANEL

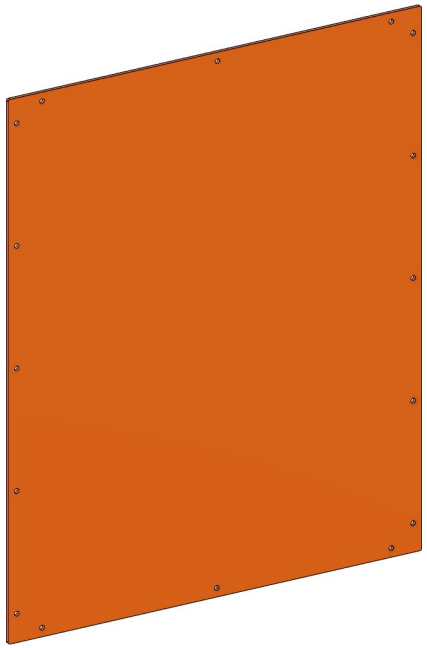
LEFT ENCLOSURE PANEL | DETAIL 1

1 BEFORE

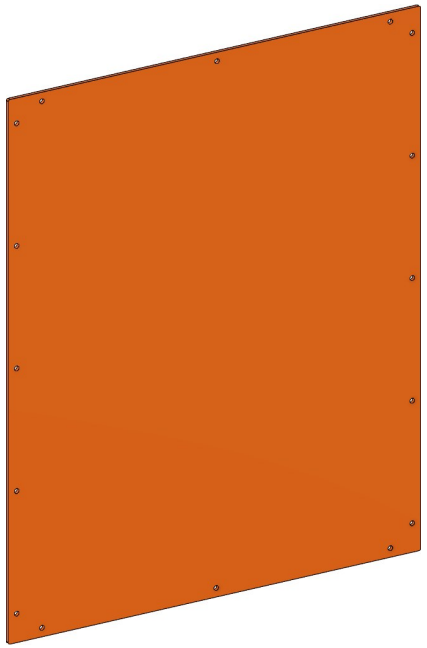
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x left enclosure panel; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

ORIENT AND DRY-FIT THE TOP ENCLOSURE PANEL

TOP ENCLOSURE PANEL | DETAIL 1

1 BEFORE

START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED

3 AFTER THIS STEP

Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x top enclosure panel; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

ORIENT AND DRY-FIT THE FRONT ELECTRONICS PANEL

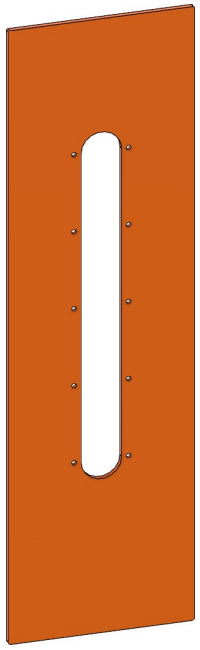
FRONT ELECTRONICS PANEL | DETAIL 1

1 BEFORE

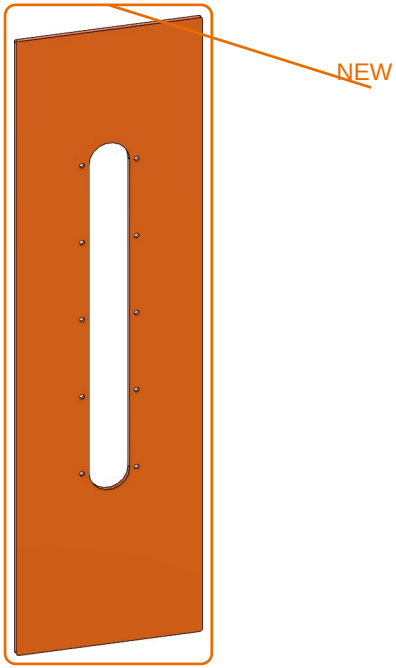
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x front electronics panel; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

ORIENT AND DRY-FIT THE LOWER ELECTRONICS PANEL

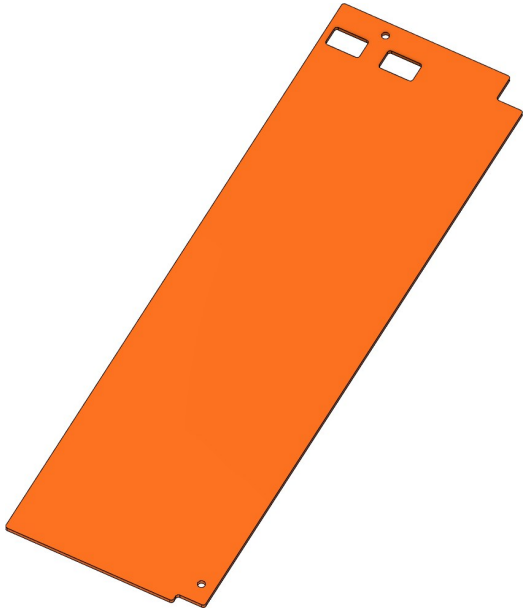
LOWER ELECTRONICS PANEL | DETAIL 1

1 BEFORE

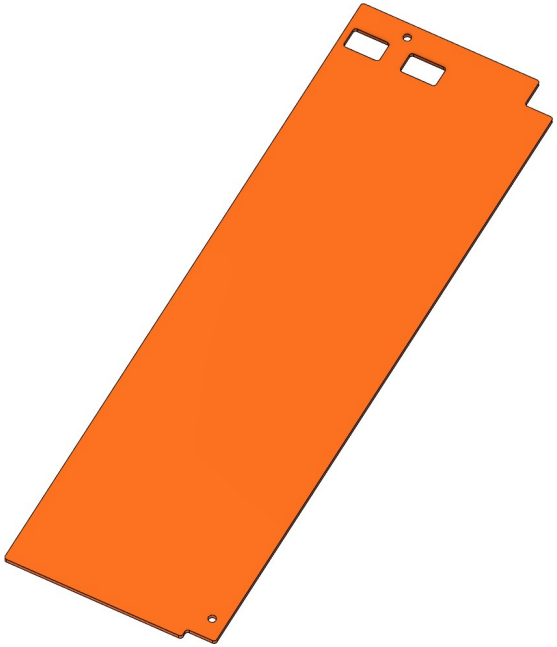
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x lower electronics panel; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

ORIENT AND DRY-FIT THE INNER ELECTRONICS SHELF

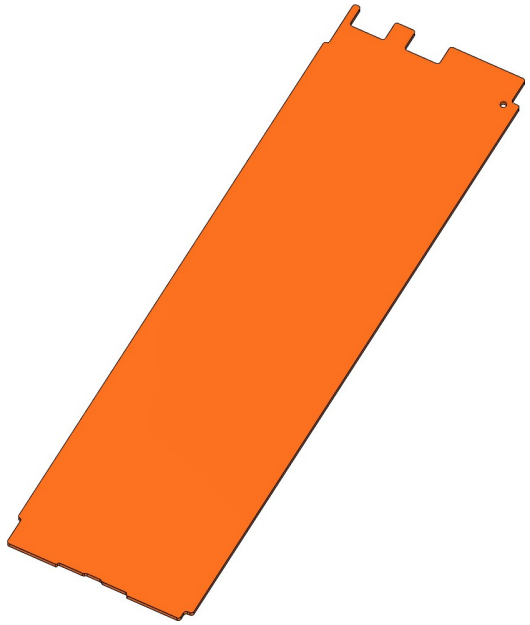
INNER ELECTRONICS SHELF | DETAIL 1

1 BEFORE

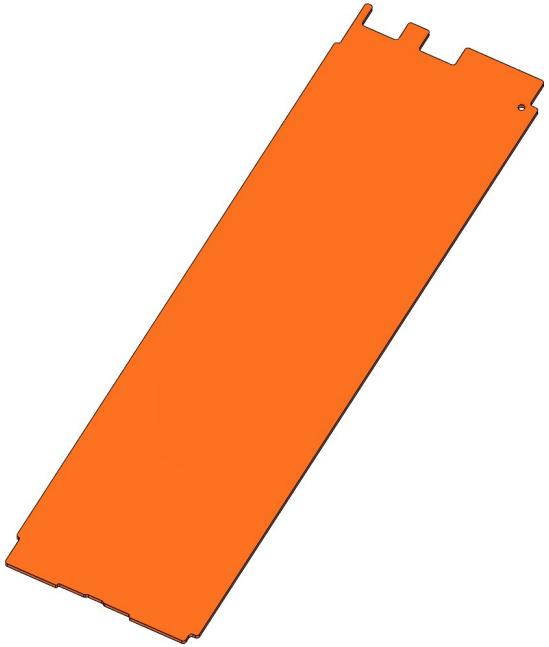
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x inner electronics shelf; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

ORIENT AND DRY-FIT THE SIDE ELECTRONICS PANEL

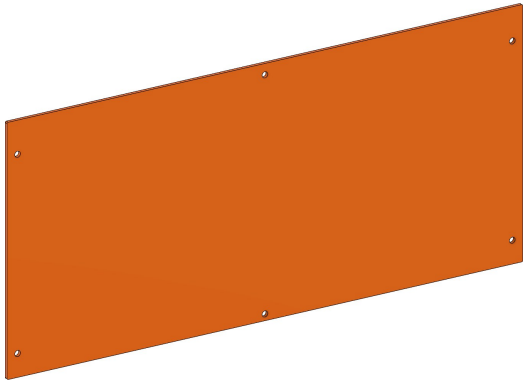
SIDE ELECTRONICS PANEL | DETAIL 1

1 BEFORE

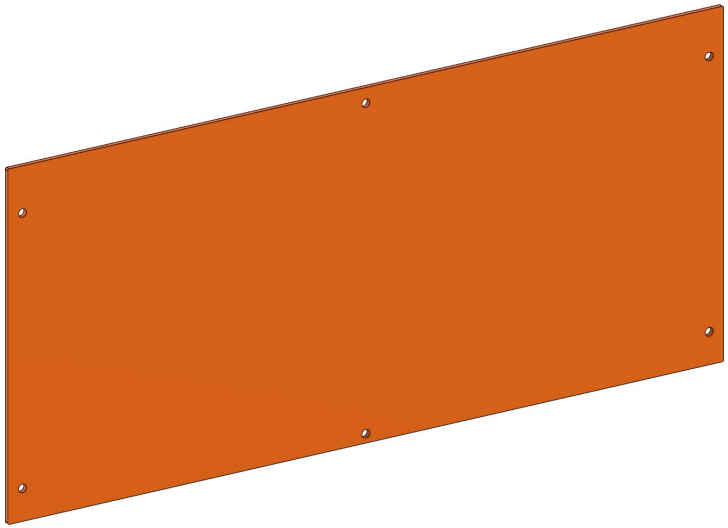
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

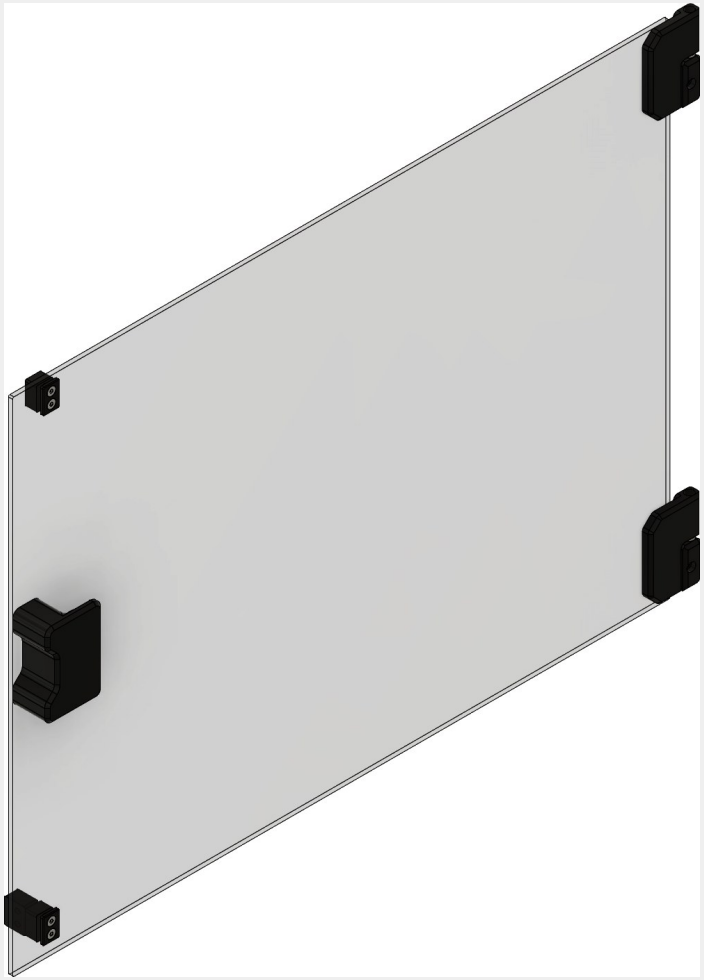
Identify this panel by its outline and openings. Dry-fit it at the matching location in the enclosure overview. Check edge clearance and access to the equipment behind it. Confirm the exact fixing positions, screws and mating threads before final attachment.

PARTS IN THIS CAD STEP

1 x side electronics panel; panel fixing assignment remains to be confirmed

CHECK BEFORE CONTINUING: The panel fits without bowing or pinching cables. Required hardware is confirmed and service access remains available.

INSTALL THE SPOOL-BAY DOOR



Fit the door and frame hinge mounts with the modeled pins. Align the door, then fit the magnetic closure hardware. Check spool access with the door fully open.

CAD ITEMS
Two door mounts
Two frame mounts
Two 28 mm pin-labelled occurrences

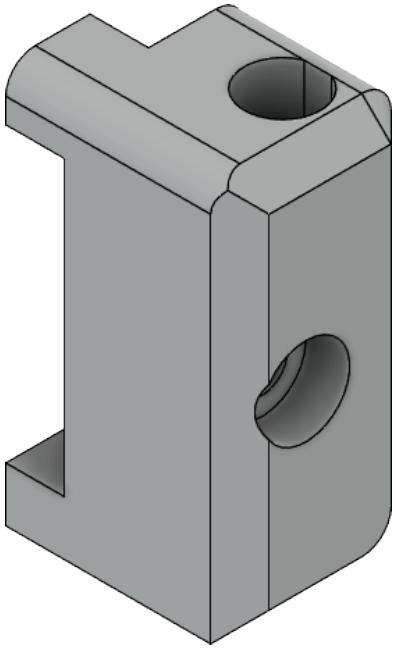
CHECK: The door swings freely, aligns with its closure and clears adjacent panels.

Current Hybrid assembly

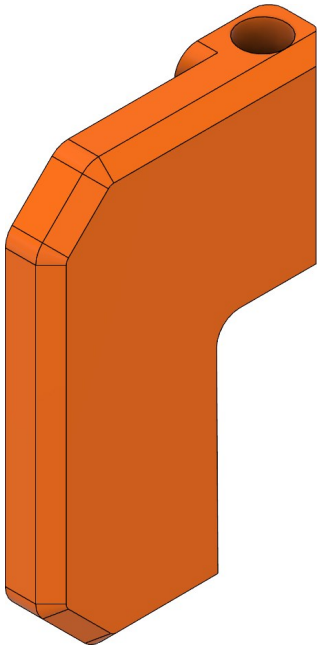
ALIGN THE DOOR-SIDE HINGE

UPPER SPOOL-DOOR HINGE | DETAIL 1

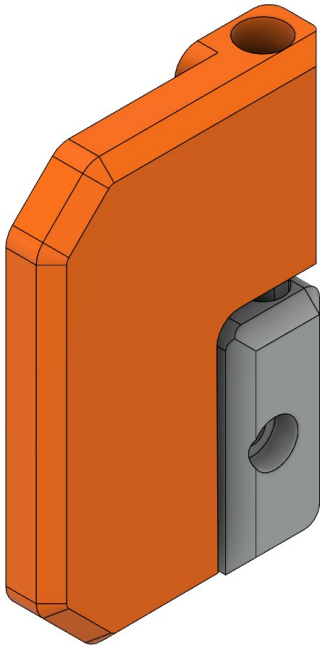
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Bring the door-side hinge to the frame-side hinge. Align the pin bores as shown. These isolated parts are shown at their final locations; mounting screws are not identified in this CAD group.

PARTS IN THIS CAD STEP

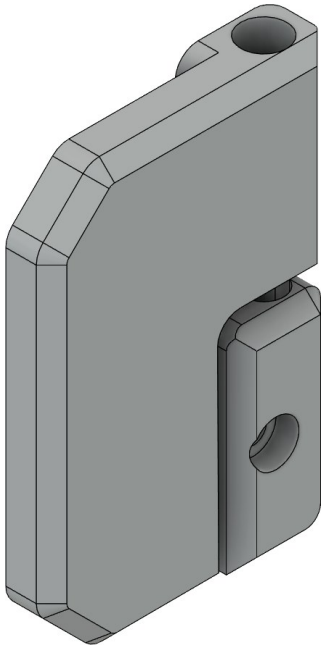
1 x door mount

CHECK BEFORE CONTINUING: The two bores share one axis. Confirm the hinge mounting fasteners before attaching it to the machine.

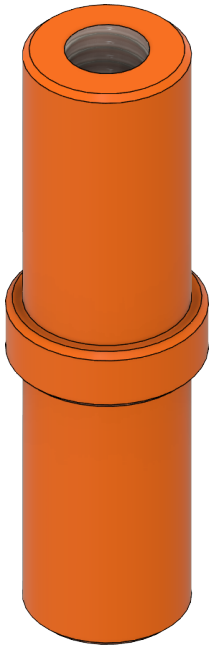
INSERT THE HINGE PIN

UPPER SPOOL-DOOR HINGE | DETAIL 2

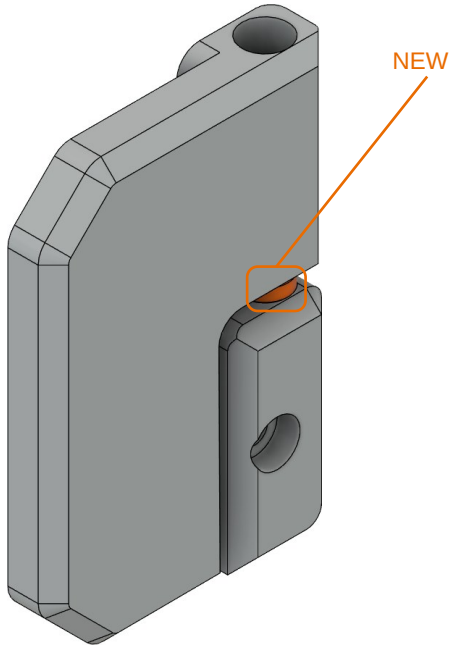
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Support both hinge halves and slide the 28 mm pin through their aligned bores. Do not use the pin to force misaligned halves together.

PARTS IN THIS CAD STEP

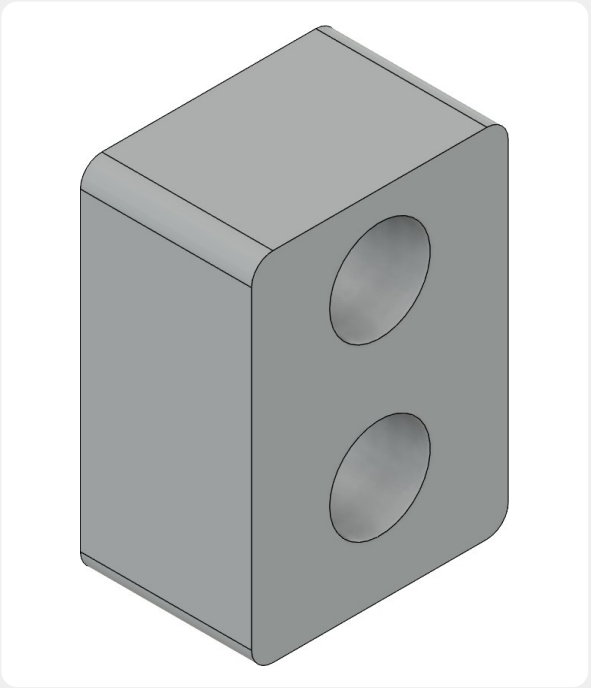
1 x 28 mm pin

CHECK BEFORE CONTINUING: The hinge rotates freely and the pin is retained by the intended physical arrangement. Confirm retention on the actual parts.

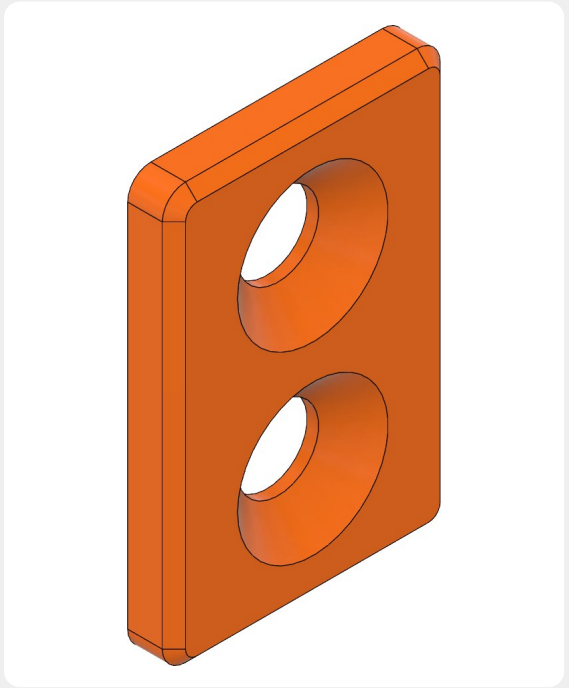
POSITION THE PANEL-SIDE LATCH PART

UPPER SPOOL-DOOR LATCH | DETAIL 1

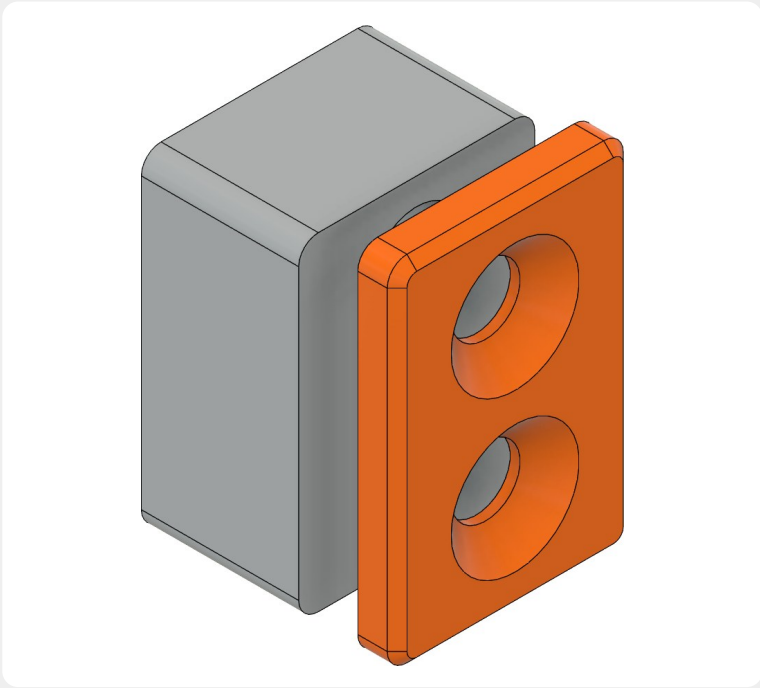
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match the panel-side part to the door-side latch location shown in orange. Dry-fit the latch before fastening. CAD names do not establish magnet material, polarity or adhesive requirements.

PARTS IN THIS CAD STEP

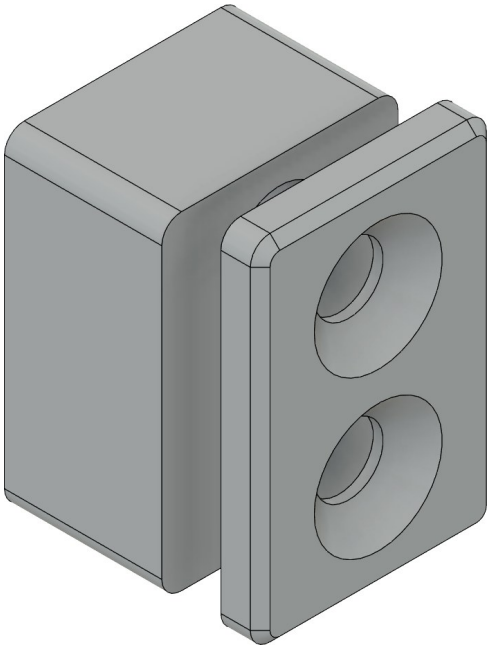
1 x panel magnet

CHECK BEFORE CONTINUING: The mating faces align. Verify the supplied magnet parts and attraction direction before permanent installation.

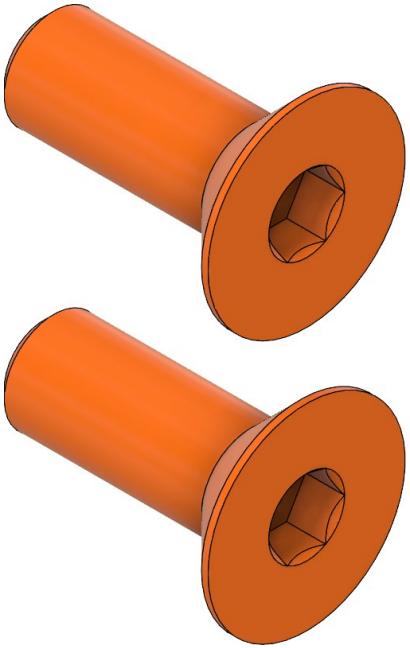
ADD THE TWO LATCH SCREWS

UPPER SPOOL-DOOR LATCH | DETAIL 2

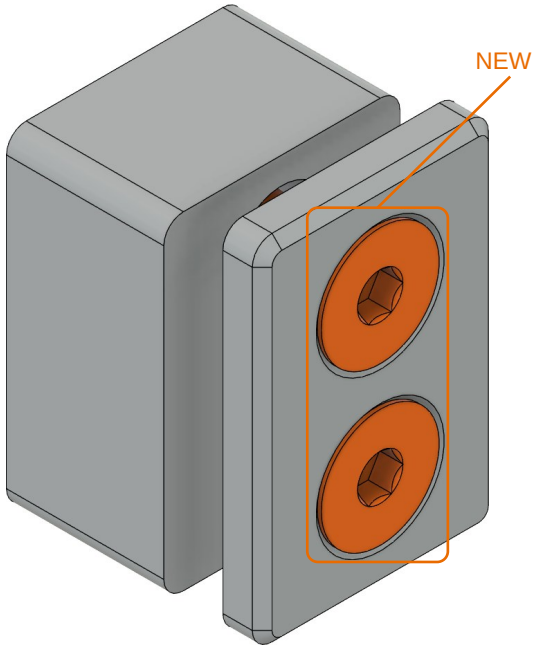
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the two M3 x 8 flat-head screws at this latch. Fit them through the real panel and matching latch part; the panel is omitted here to expose the screw positions. Start both threads before tightening.

PARTS IN THIS CAD STEP

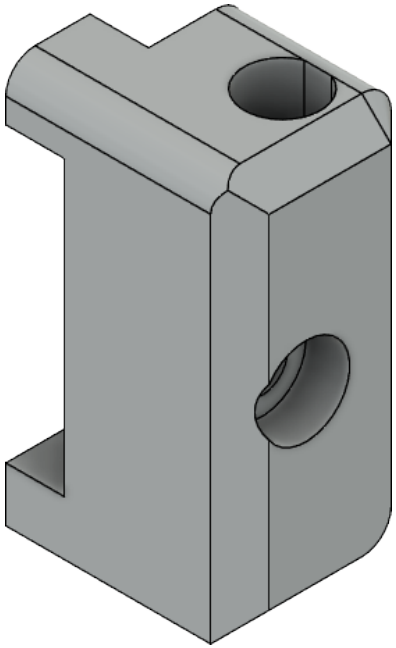
2 x M3x8 FHCS

CHECK BEFORE CONTINUING: Both heads seat in their intended recesses. Confirm thread engagement and avoid bending the panel.

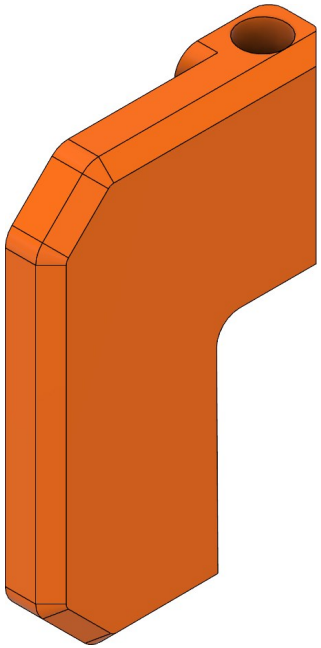
ALIGN THE DOOR-SIDE HINGE

LOWER SPOOL-DOOR HINGE | DETAIL 1

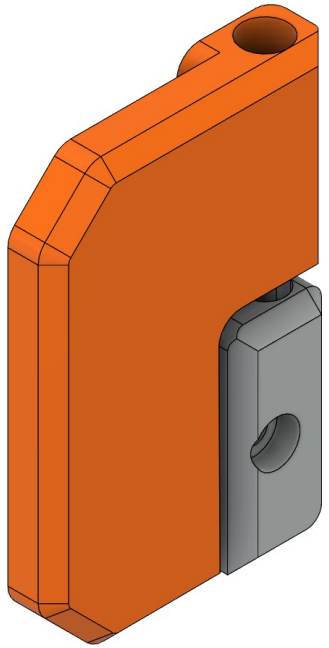
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Bring the door-side hinge to the frame-side hinge. Align the pin bores as shown. These isolated parts are shown at their final locations; mounting screws are not identified in this CAD group.

PARTS IN THIS CAD STEP

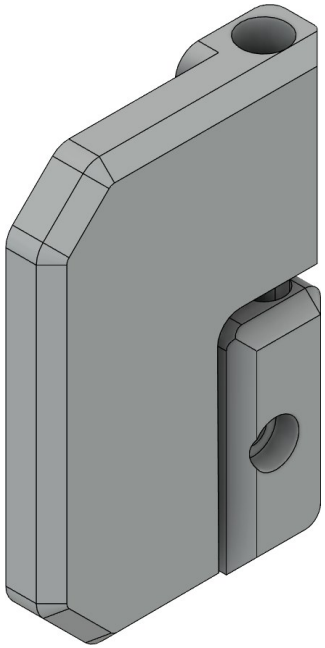
1 x door mount

CHECK BEFORE CONTINUING: The two bores share one axis. Confirm the hinge mounting fasteners before attaching it to the machine.

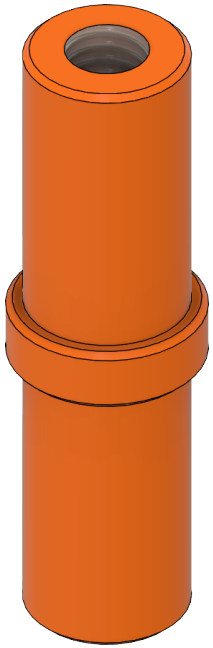
INSERT THE HINGE PIN

LOWER SPOOL-DOOR HINGE | DETAIL 2

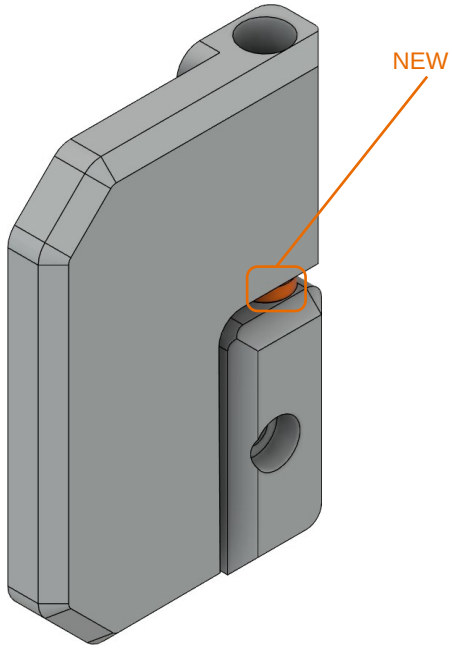
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Support both hinge halves and slide the 28 mm pin through their aligned bores. Do not use the pin to force misaligned halves together.

PARTS IN THIS CAD STEP

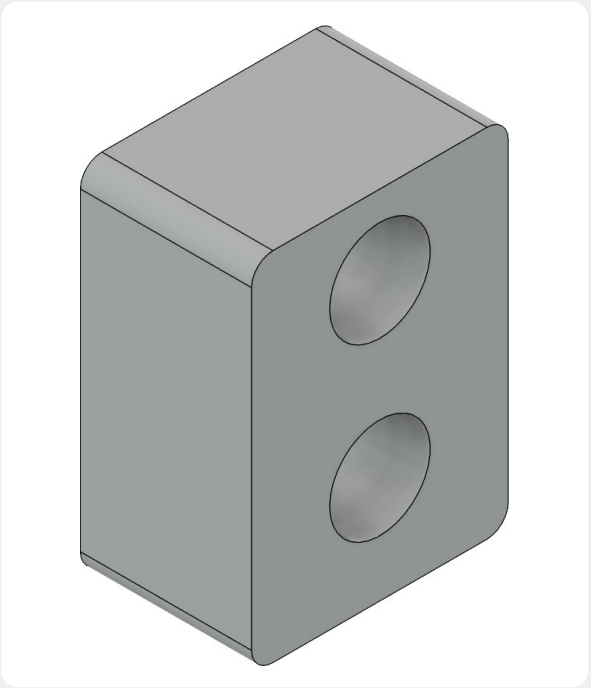
1 x 28 mm pin

CHECK BEFORE CONTINUING: The hinge rotates freely and the pin is retained by the intended physical arrangement. Confirm retention on the actual parts.

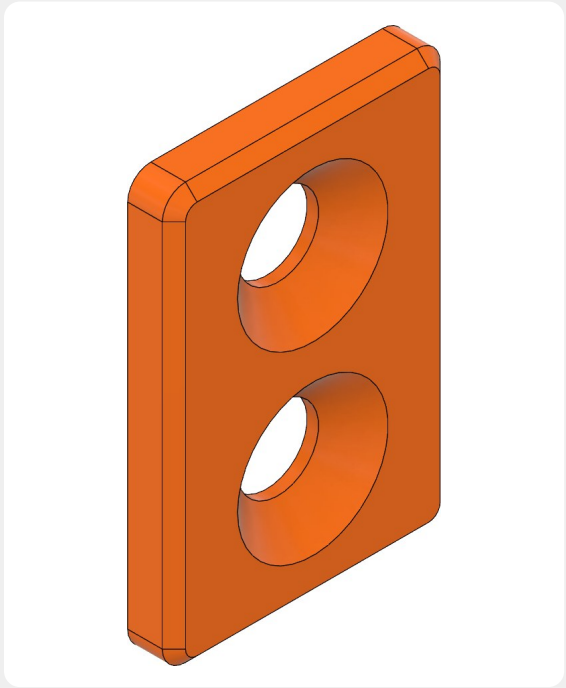
POSITION THE PANEL-SIDE LATCH PART

LOWER SPOOL-DOOR LATCH | DETAIL 1

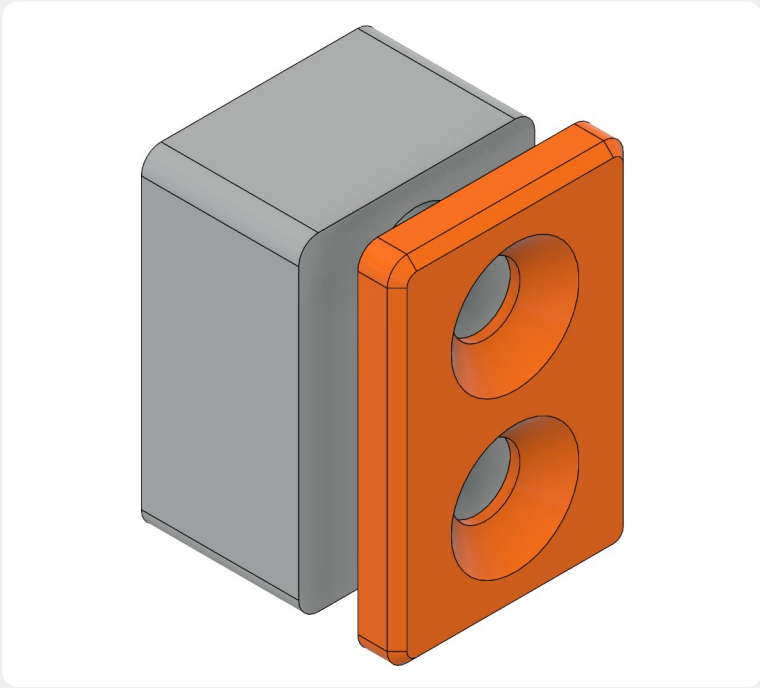
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Match the panel-side part to the door-side latch location shown in orange. Dry-fit the latch before fastening. CAD names do not establish magnet material, polarity or adhesive requirements.

PARTS IN THIS CAD STEP

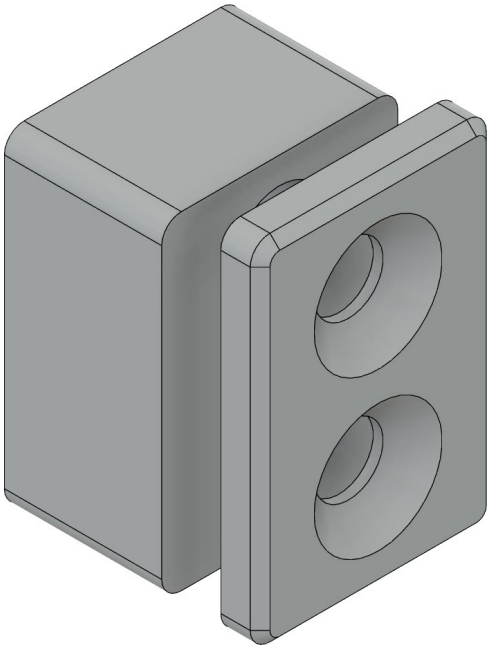
1 x panel magnet

CHECK BEFORE CONTINUING: The mating faces align. Verify the supplied magnet parts and attraction direction before permanent installation.

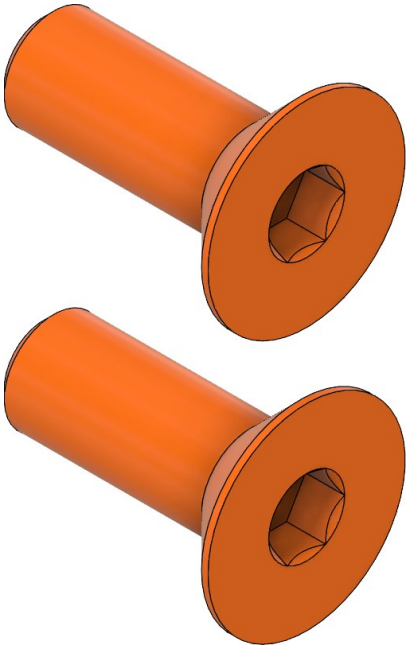
ADD THE TWO LATCH SCREWS

LOWER SPOOL-DOOR LATCH | DETAIL 2

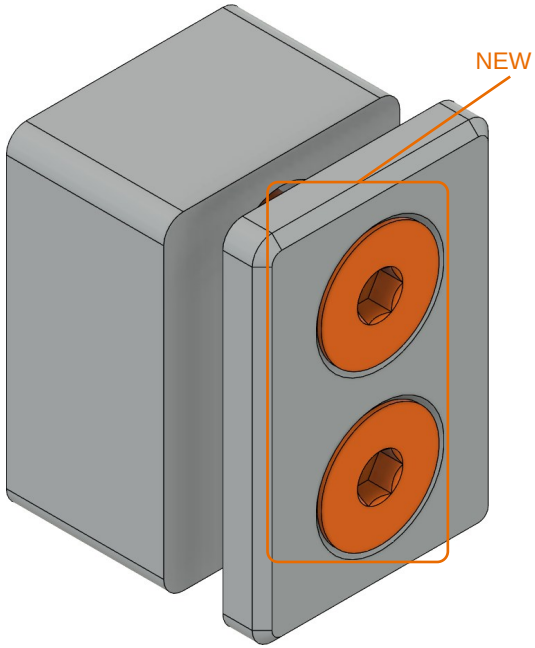
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Locate the two M3 x 8 flat-head screws at this latch. Fit them through the real panel and matching latch part; the panel is omitted here to expose the screw positions. Start both threads before tightening.

PARTS IN THIS CAD STEP

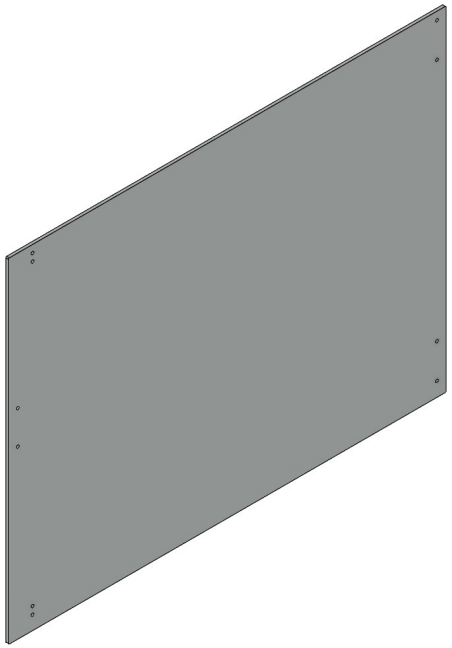
2 x M3x8 FHCS

CHECK BEFORE CONTINUING: Both heads seat in their intended recesses. Confirm thread engagement and avoid bending the panel.

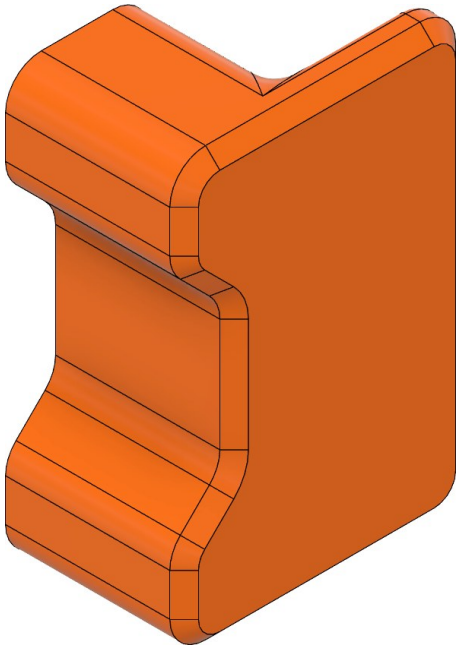
POSITION THE SPOOL-DOOR HANDLE

SPOOL-DOOR HANDLE | DETAIL 1

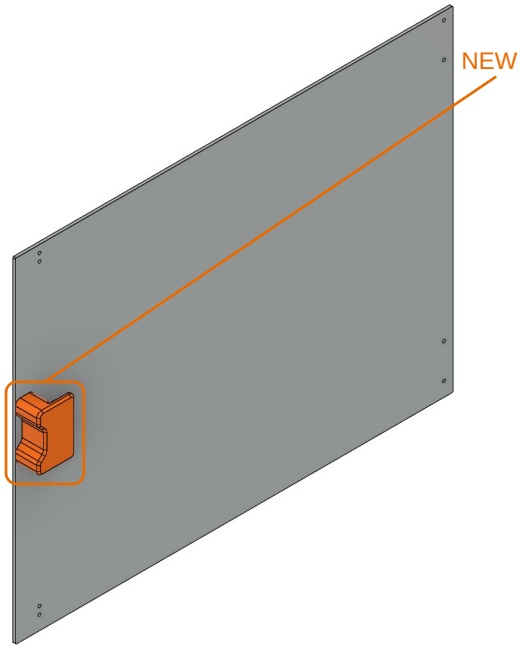
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

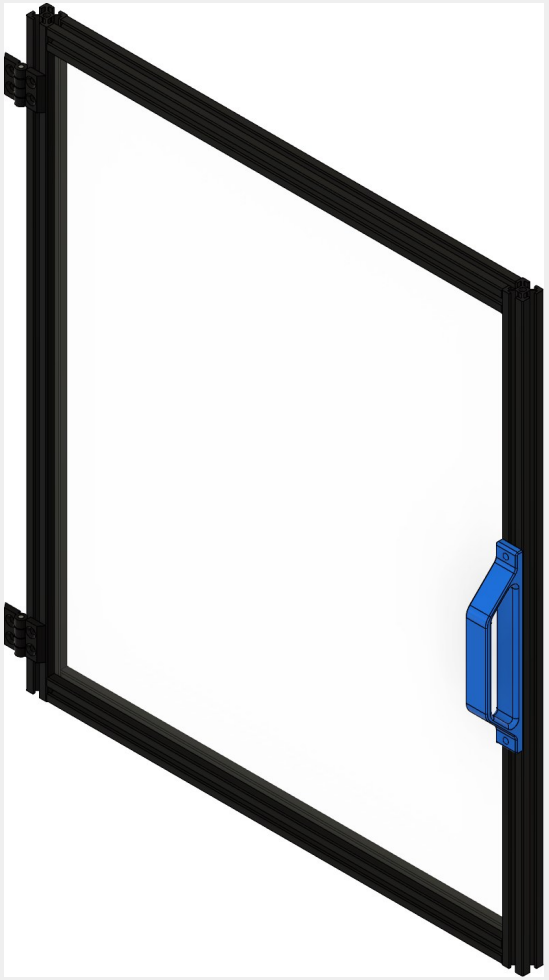
Place the handle at the highlighted panel edge, opposite the hinges. Support the panel while aligning the handle holes. The model does not identify the handle mounting fasteners.

PARTS IN THIS CAD STEP

1 x door handle

CHECK BEFORE CONTINUING: Confirm handle screw size and engagement from the actual handle before fastening. The closed door must clear the adjacent frame.

INSTALL THE FRONT DOOR



Assemble the front door with its frame members, hinges, panel and handle. Align it to the main chamber before securing the hinge hardware.

CAD GROUP
front_door
Shown isolated; hidden in the initial model view.

CHECK: The door opens and closes freely and its panel is retained without being distorted.

Current Hybrid assembly

ALIGN THE TWO HINGE LEAVES

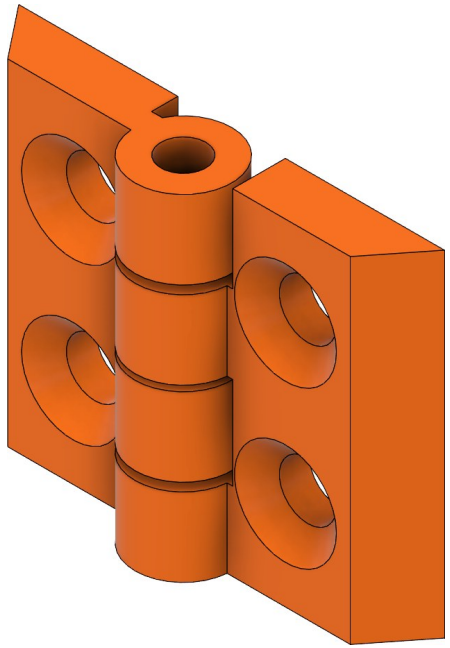
UPPER FRONT-DOOR HINGE | DETAIL 1

1 BEFORE

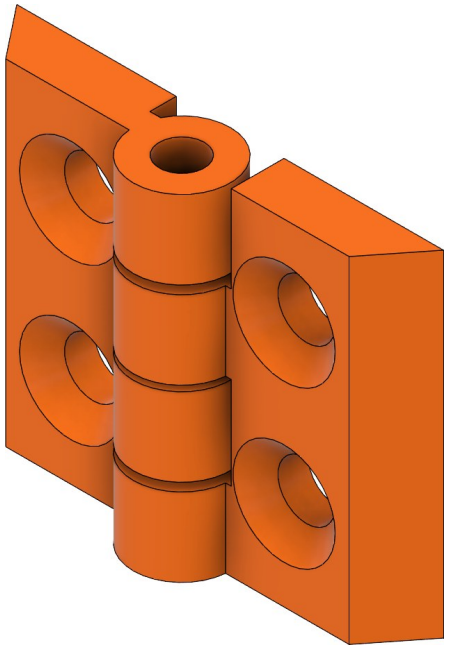
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the two hinge leaves with their pin bores concentric. Match the offset shown. The isolated view exposes the joint; verify mounting screws against the supplied hinge.

PARTS IN THIS CAD STEP

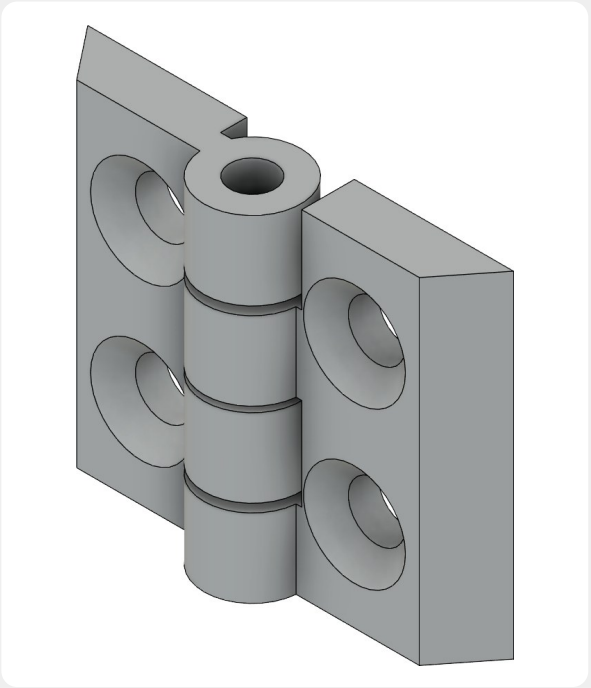
2 x Hinge Body^VSlotHinge

CHECK BEFORE CONTINUING: Both leaves face their corresponding frame surfaces. Do not fasten the hinge until the door can be supported and aligned.

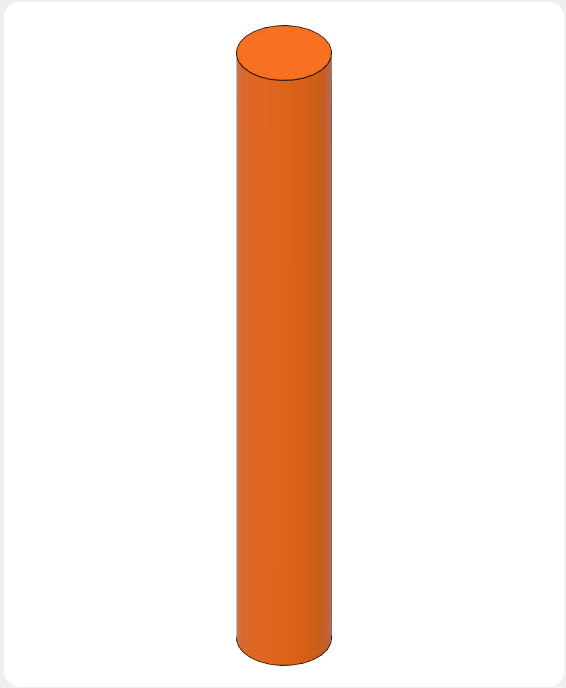
FIT THE HINGE PIN

UPPER FRONT-DOOR HINGE | DETAIL 2

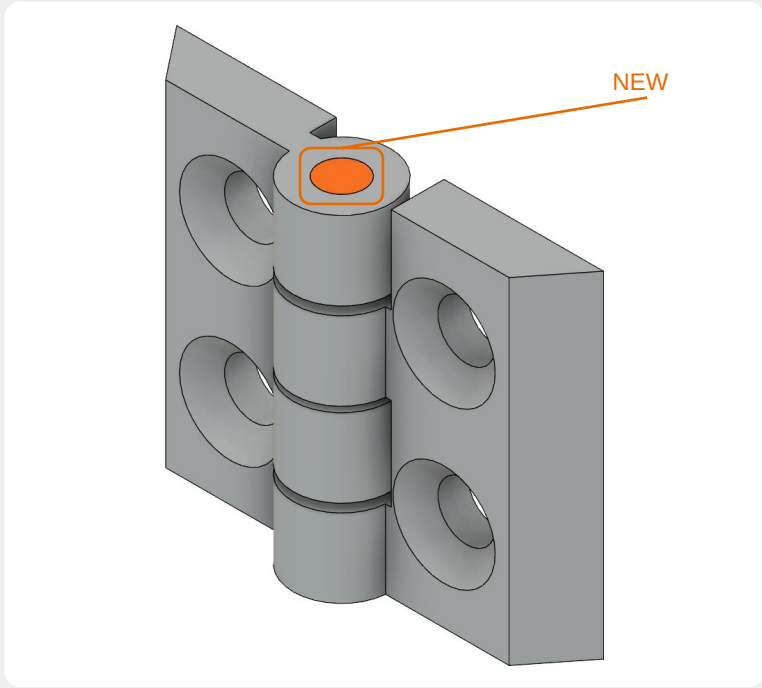
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Slide the hinge pin through the aligned leaves. If the supplied hinge arrives assembled, inspect this joint without removing a factory-retained pin.

PARTS IN THIS CAD STEP

1 x Hinge Pin^VSlotHinge

CHECK BEFORE CONTINUING: The hinge rotates smoothly through its required travel. Confirm the pin cannot fall out in the installed orientation.

ALIGN THE TWO HINGE LEAVES

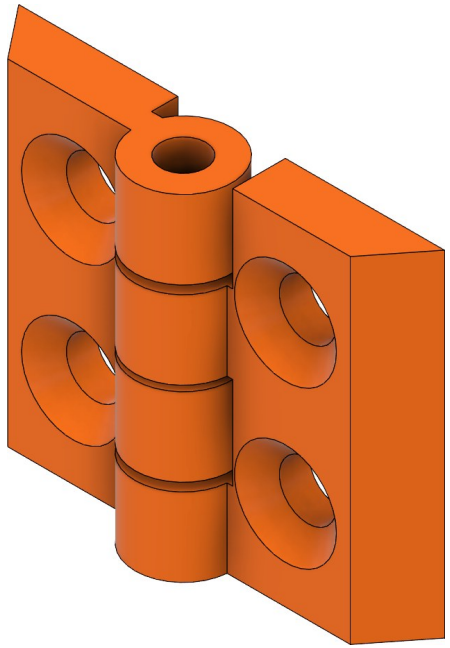
LOWER FRONT-DOOR HINGE | DETAIL 1

1 BEFORE

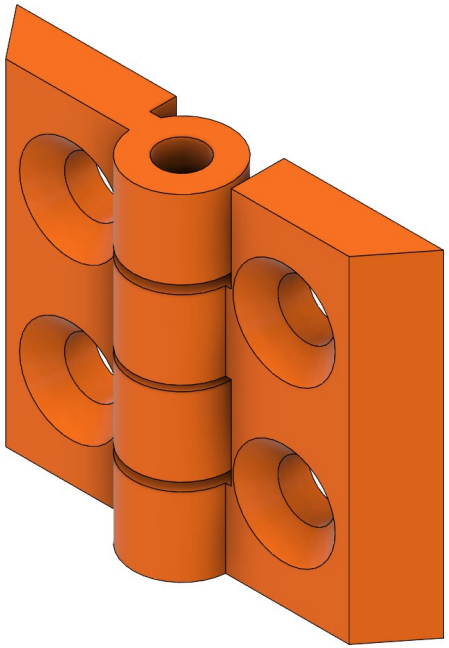
START WITH THE LOOSE PARTS

No assembly yet.

2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the two hinge leaves with their pin bores concentric. Match the offset shown. The isolated view exposes the joint; verify mounting screws against the supplied hinge.

PARTS IN THIS CAD STEP

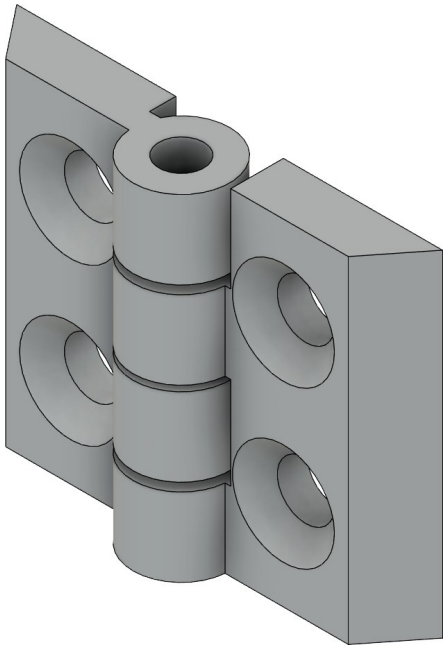
2 x Hinge Body^VSlotHinge

CHECK BEFORE CONTINUING: Both leaves face their corresponding frame surfaces. Do not fasten the hinge until the door can be supported and aligned.

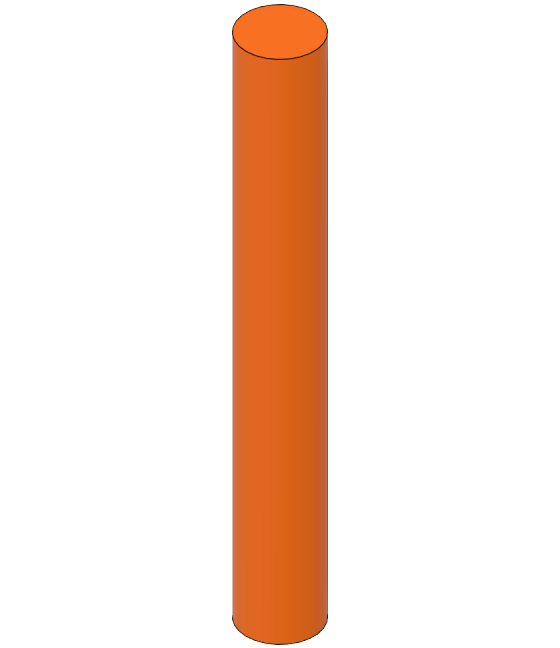
FIT THE HINGE PIN

LOWER FRONT-DOOR HINGE | DETAIL 2

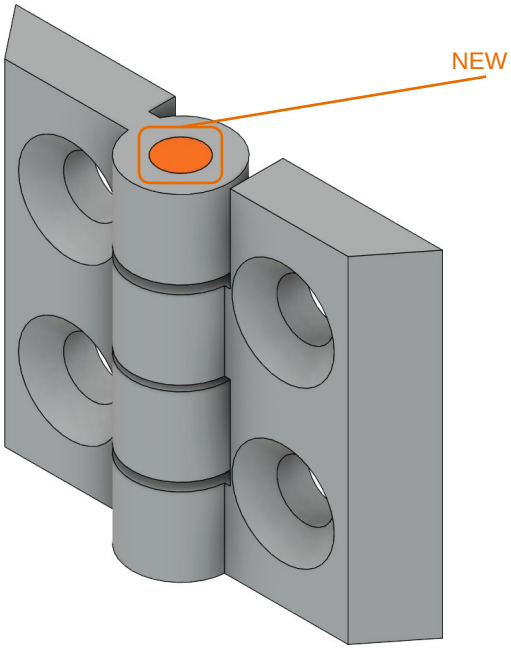
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Slide the hinge pin through the aligned leaves. If the supplied hinge arrives assembled, inspect this joint without removing a factory-retained pin.

PARTS IN THIS CAD STEP

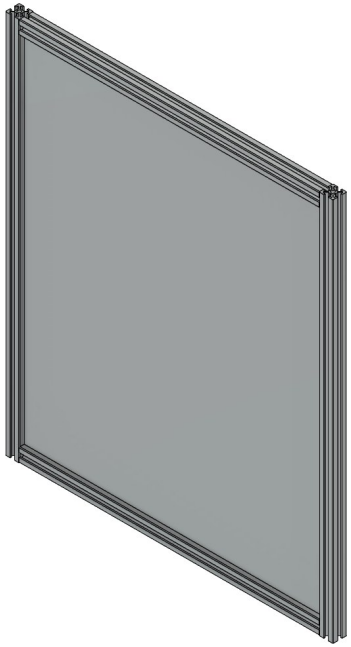
1 x Hinge Pin^VSlotHinge

CHECK BEFORE CONTINUING: The hinge rotates smoothly through its required travel. Confirm the pin cannot fall out in the installed orientation.

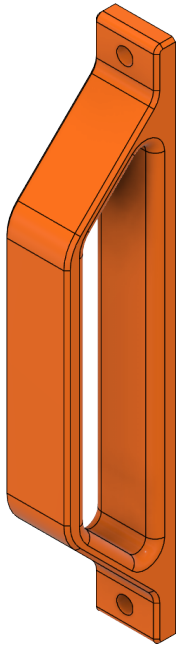
LOCATE THE FRONT-DOOR HANDLE

FRONT-DOOR HANDLE | DETAIL 1

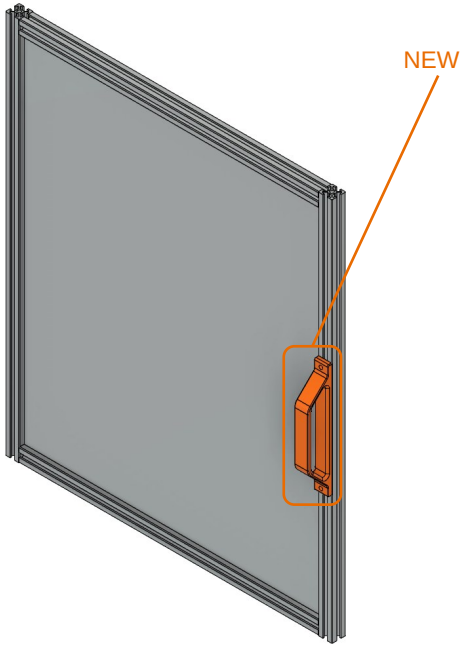
1 BEFORE



2 PARTS ADDED



3 AFTER THIS STEP



Orange = new in this step. Gray = previously assembled. Views are enlarged independently; not to scale.

DO THIS

Align the handle on the outer face of the right-hand door member at the highlighted location. The CAD component is named RANDLES. Its mounting hardware is not modeled in this group.

PARTS IN THIS CAD STEP

1 x RANDLES

CHECK BEFORE CONTINUING: Verify the handle hole spacing and hardware on the actual part. Support the door while fitting the handle; check clearance at full opening.

Frame and motion

Check frame squareness, all joint fasteners, rail alignment, pulley retention and idler rotation. Manually move XY through the intended range and inspect belt tracking at the extremes.

Z and bed

Support the bed while checking all three arm contacts. Verify leadscrew clearance, freedom of movement and chain behavior. Confirm the bed does not contact the frame or enclosure at either end of travel.

Enclosure and access

Check doors, magnets, panels, vent sliders, feet, fans and service access. Make sure no screw tip or panel edge reaches a moving part or cable.

Sign-off

Builder: _____ Date: _____
Mechanical reviewer: _____
Open issues recorded: _____

Toolhead and belt termination

Select and install the final toolhead, hotend, extruder, probe and belt clamps. The X-carriage subtree exported without usable solid geometry, so this edition cannot establish mounting or belt-end hardware.

Firmware and direction checks

Create a configuration for the actual electronics and motors. Verify inputs, motor direction, homing direction and travel limits in a controlled commissioning procedure before automated movement.

Thermal and print checks

Validate temperature sensing, heater control and protection using the installed equipment's procedure. Complete bed alignment, probe calibration and motion calibration before a supervised first print.

Release condition

The mechanical draft is ready for physical review. A finished customer manual needs the confirmed toolhead, electrical schematic, firmware setup and recorded first-build validation.

Frame cut list

The frame group has 21 extrusion occurrences. Some labels encode lengths that disagree with the modeled bounds. Confirm each stock length, machining code, access-hole location and quantity before ordering or cutting.

Rails and drives

The X assembly is named MGN12. Other rail labels retain earlier length and rail references. Verify rail type, length and carriage, plus pulley bore/tooth count, belt width, bearings, spacers and idler-pin fit.

Fasteners

Many screws are stored as supplier part numbers rather than plain metric sizes. Translate these into a reviewed step-by-step hardware list with quantities, head style, thread and length. Do not infer nominal screw length from a bounding box.

Bed and printed parts

Confirm bed drawing, mounting stack, heater details, approved printed materials and print settings. Check the left, right and rear Z-arm revisions as one matched set.

Primary assembly

HyperDrive Galaxy One hybrid - the model selected by the user and open in Fusion during this session. Its component tree and isolated views were exported on 10 November 2026. The older current v97 archive was not used to specify the Hybrid assembly.

Frame sequence

Existing local documentation: docs/printers/Build_guides/Hybrid/Frame_build.md, with its eleven Frame_build illustrations. These provide the established order for captive panels and frame closure.

Style reference

Hypernova Printhead build guide:
docs.hyperdrivedesign.com/products/Hypernova/hypernova_build_guide/
Landscape steps, large CAD views, blue headings and concise assembly callouts.

Revision 0.8 - spool holders, fans and enclosure closure

Left, right and rear Z-arm detail images refreshed from the corrected Fusion model on 11 November 2026. Adds 21 further steps for spool holders, panel connectors, chamber fans and guards, and enclosure panel orientation. Includes 193 hardware detail pages, orange highlights, before/after views and per-step CAD quantities. Not physically build-tested. Final hardware quantities, toolhead, belt clamps, wiring and firmware remain open. The rear vent group also contains 16 extra insert occurrences away from the four frame pockets; those are excluded from the frame-installation step pending review. Record revisions after the first physical build review.

Catalog sizes identified

McMaster-Carr catalog listings identify 90236A120 / A121 / A122 / A127 as M3 x 8 / 10 / 12 / 35 mm Torx flat-head screws. 92703A457 is M3 x 30 mm and 92703A161 is M5 x 10 mm. The detail lists now show these plain sizes beside the CAD supplier IDs.

Catalog sources

McMaster-Carr: Class 8.8 screw catalog
McMaster-Carr: stainless Torx flat-head size table
Checked 10 November 2026. Catalog identification does not validate fit in this assembly.

Front-idler discrepancy

The left front idler contains two 90304A337 screw occurrences at the same modeled location. Do not interpret these overlapping entries as two physical screws. The tension-adjuster stack and washer arrangement need designer confirmation; they are not prescribed by the new detail pages.

Additional catalog sizes

90991A113 is M3 x 8 mm Torx button-head (catalog). 92703A151 is M3 x 12 mm Torx flat-head; 92095A179 is M3 x 6 mm hex button-head. Sources: McMaster-Carr flat-head size table; 92095A179 product specification. Confirm physical engagement through the actual plate or panel before tightening.

Rail occurrence checks

The saved inventory contains overlapping copies of each side Z rail; the new attachment pages use one copy per side. Two Y rails and two side Z rails are illustrated. A separate rear Z rail could not be identified reliably from the exported component tree, so its attachment detail remains open. Align the actual rail holes and T-nuts; modeled bounding-box centers are not drilling dimensions.

Skirt hardware checks

Five duplicate insert occurrences were removed from the illustration selection, leaving two lower inserts per skirt section. The three upper screws and the lower inserts are separate fixing groups. One additional M3 x 8 screw near the right-front side skirt is excluded pending confirmation of its purpose. Confirm frame-side mating threads or nuts and any lower cover fasteners against the actual kit.

Cable-chain hardware remains open

The CAD supplies the lower mount, two end fittings and 26 chain links, but no identifiable end-fixing screws at those locations. The new pages show orientation and assembly of those parts. Confirm mount and end-fitting screw sizes, quantities and bed-side attachment before treating this sequence as a complete installation procedure.

Scope of the mounting pages

These pages document mechanical placement and modeled fasteners. They do not specify wiring, grounding, electrical ratings, controller configuration or commissioning. Finish mechanical review before using them as release instructions.

Fan retention needs confirmation

Each of the two illustrated electronics fan mounts contains four M3 insert occurrences and four M3 x 12 Torx flat-head screws. Confirm the actual insert carrier and approved fastening method. Do not treat these renders as approval to heat-set an insert into an electronic fan.

Mounts still requiring hardware identification

The new pages show the four power-supply screws and four controller accessory standoffs separately. Confirm the supply screw specification and insertion depth, controller and clip retaining screws, driver retention, terminal end stops, relay adapters and cable-channel retention. These unresolved details are labeled on their individual pages; placement alone does not establish the fixing method.

Additional catalog identification

92703A149 is an M3 x 8 mm Torx flat-head screw in the McMaster-Carr size table, checked 11 November 2026. It is used in the progress-light detail lists. Confirm actual engagement before tightening.